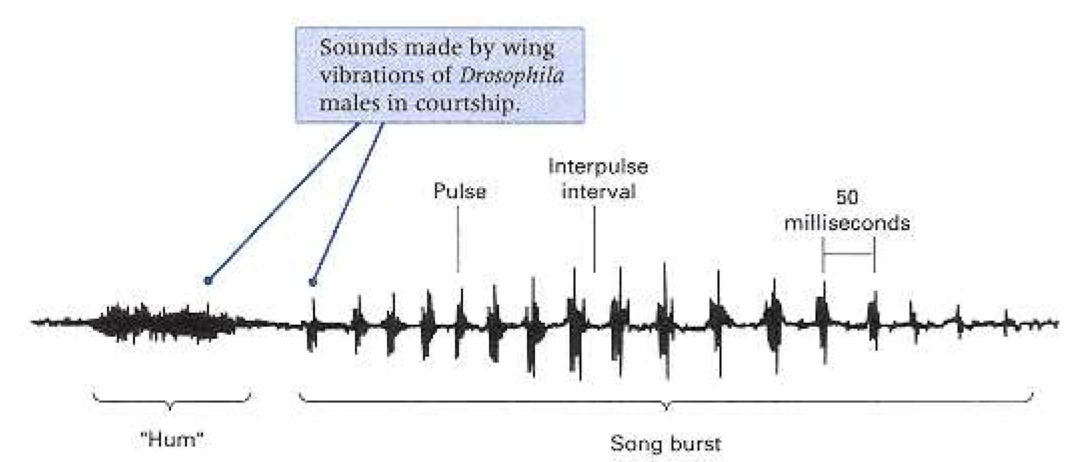



Figure 17.16
Typical song burst in *Drosophila* courtship. The burst is preceded by a "hum" of extremely rapid vibrations, followed by a series of discrete pulses. The time between each pulse is called an interpulse interval. [From C. P. Kyriacou, M. L. Greenacre, J. R. Thackeray, and J. C. Hall, in *Molecular Genetics of Biological Rhythms*, edited by M. W. Young (New York: Marcel Dekker, 1992), p. 172.]

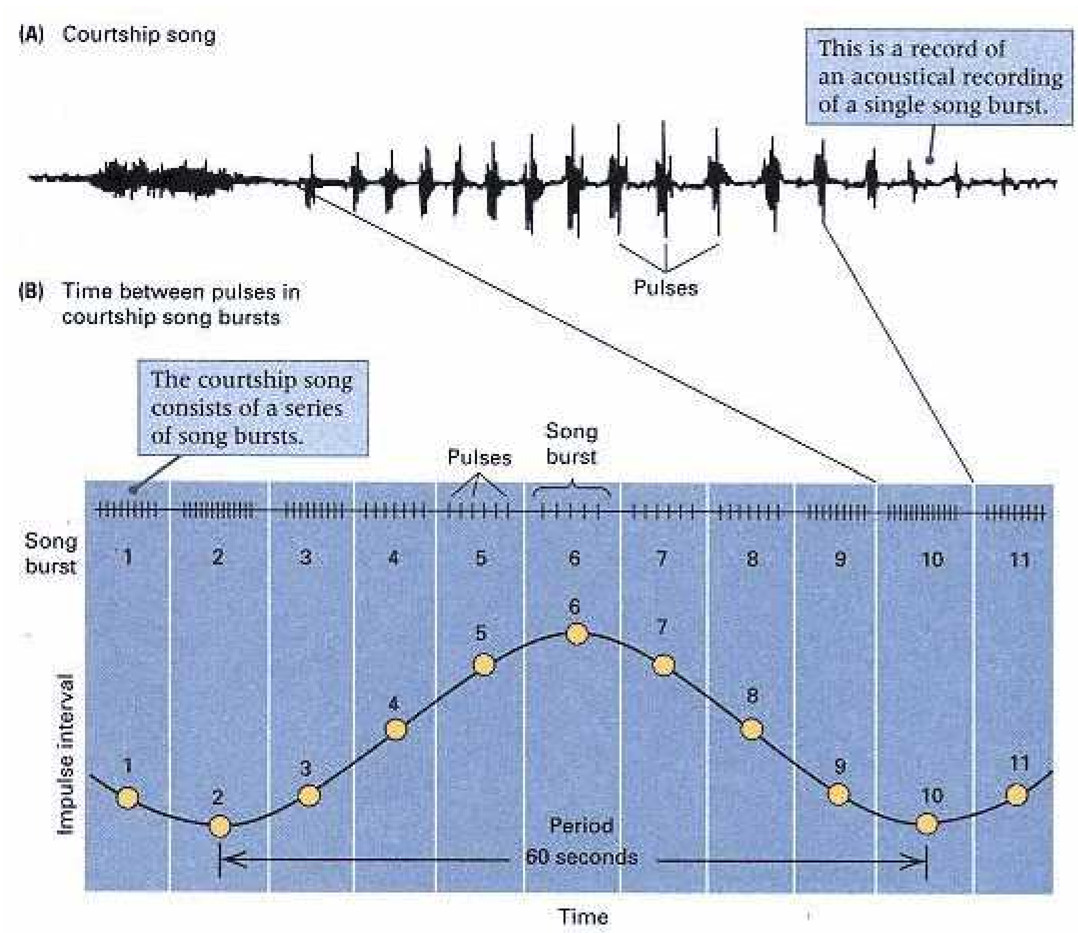



Figure 17.17
Rhythm in the *Drosophila* courtship song composed of wing vibrations. (A) Acoustical recording of a single burst. (B) Pattern of pulses in a sequence of 11 consecutive bursts. Note that the interpulse intervals expand and contract during the sequence. A plot (yellow dots) of interpulse intervals yields a rhythmic pattern, which in D. *melanogaster* has a period of approximately 60 seconds. [Data from C. P. Kyriacou, M. L. Greenacre, J. R. Thackeray, and J. C. Hall, in *Molecular Genetics of Biological Rhythms*, edited by M. W. Young (New York: Marcel Dekker, 1992), p. 171.

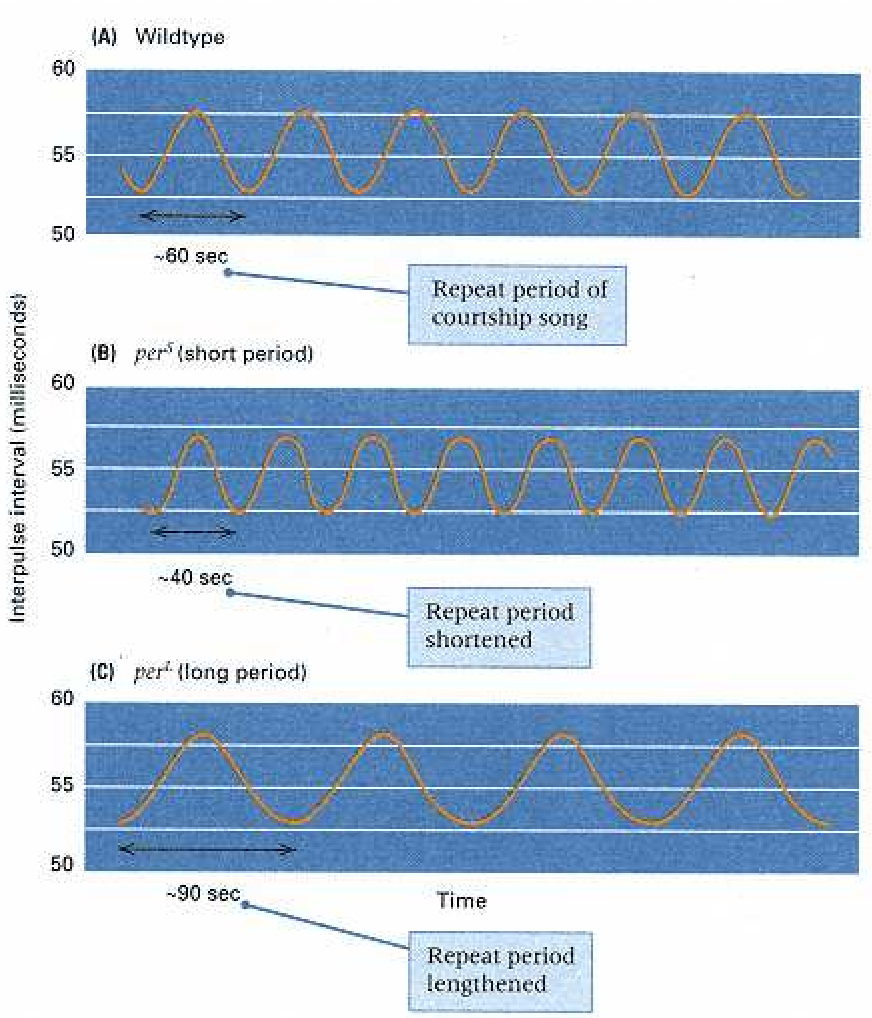



Figure 17.18
Effects of mutations in the period gene on the *period* of the *Drosophila* courtship song. (A) Wildtype flies have a courtship song with a period of approximately 60 seconds (see also Figure 17.17B). (B) Males carrying the *per*$^S$ allele have a shorter period, approximately 40 seconds. (C) Males carrying the *per*$^L$ allele have a longer period, approximately 90 seconds.

Therefore, the *per* gene affects the periods of both the courtship song and the circadian rhythm, even though the time scales are very different (60 seconds versus 24 hours). The *per* gene is also important in controlling differences in the period of the courtship song in different species. For example, in *Drosophila melanogaster* the period is about 60 seconds (Figure 17.17B), whereas in the closely related species *Drosophila simulans*, the period is about 35 seconds. The difference is apparently important in mate recognition: *D. melanogaster* females are maximally stimulated to mate when the courtship song has a 60-second period, whereas *D. simulans* females prefer the 35-second period. In any event, the role of the *per* gene in the difference between the species is indicated by the observation that transfer of the *per*$^+$ gene from *D. simulans* to *D. melanogaster* results in *D. melanogaster* males that exhibit the rhythm of a typical *D. simulans* courtship song.

### *Molecular Genetics of the Drosophila Clock*

Some years after the discovery of *per*, mutant screens in *Drosophila* turned up other mutations that were arrhythmic and still others with a shortened or lengthened period. Complementation tests showed that some of the new mutations were additional alleles of *per*. However, two of the mutations revealed a new complementation group that also controls the circadian clock. The gene defined by the new complementation group was designated *timeless (tim).* Unlike *per,* which is X-linked, *tim* is autosomal (chromosome 2).

Both *per* and *tim* have been cloned and sequenced, but each of the proteins, called PER and TIM, has an amino acid sequence that affords little insight into what the protein does. The cloned genes are important as probes, however, and show that the abundance of both *per* and *tim* mRNA cycle in a circadian manner and both peak after dark. The cyclic abundances of the mRNAs are illustrated by the dashed lines in Figure 17.19A. During the night, the abundance of the mRNAs plateaus and then begins to decrease until, by the next morning, both mRNAs are at low levels. The abundance of PER cycles also, with about an 8-hour time lag behind the *per* mRNA. The abundance of TIM cycles as well, but with a much shorter time lag behind its mRNA. The amount of TIM is at a low level in the daylight hours but increases rapidly to a peak after dark and diminishes thereafter.

If flies are reared in complete darkness, the TIM protein remains at a constant high level. When the lights are turned on, the TIM disappears within an hour. It has been noted that when wildtype $tim^{+}$ flies are reared in constant light, they become arrhythmic. The last observation suggests that TIM protein is destroyed by light.

In the circadian cycle, the PER protein is localized in the cytoplasm and gradually degraded as long as the level of TIM is low. After dark, when the amount of TIM begins to increase, there is a point at which PER moves into the nucleus. The movement of PER is associated with the increase in the level of TIM, and it has been suggested that TIM and PER molecules form a dimer that is transported into the nucleus. In any case, there is good evidence that both TIM and PER, once in the nucleus, exert negative feedback effects on the transcription of the genes that encode them.

Changes in the levels of *per* and *tim* mRNA and PER and TIM protein during the course of one day are shown in Figure 17.19A. At dawn, the levels of *per* and *tim* mRNA are low because transcription is repressed. When the sun comes up, the light destroys TIM, and both genes are derepressed. Transcription begins and the mRNA accumulates. Translation results in an accumulation of PER protein as well, but any TIM protein produced is immediately destroyed by the light. Not until the onset of darkness can TIM begin to accumulate, and it does so very rapidly because the amount of *tim* mRNA has already built up. Once TIM reaches an appreciable level, TIM and PER move into the nucleus and repress transcription. As the existing mRNA decays, no more is produced, and the protein levels drop as well until, in the morning, the situation is as it was 24 hours earlier.

The light sensitivity of TIM helps explain how the circadian clock is reset each day. No circadian rhythms have an exact 24-hour cycle. In the absence of light, the cycle time is either slightly shorter or slightly longer than 24 hours. To maintain synchrony with Earth's light-dark cycle, the circadian clock must be reset a small amount each day. The molecular changes that accompany the resetting are shown in Figures 17.19B and C. For clarity, both the "short-day" and "long-day" situations are exaggerated. Figure 17.19B shows what happens in a "short day," in which darkness comes on early; in this case, TIM accumulates at an earlier hour of the cycle than usual and so allows PER and TIM to enter the nucleus and repress transcription earlier than usual. The result is that the cycle is shortened (or, said another way, the clock advances). The response to a longer day is shown in Figure 17.19C. In this case, the longer day prevents TIM from accumulating at the normal time. It does not begin to accumulate until after dark, so PER and TIM enter the nucleus and repress transcription later than usual. The result is that the cycle is lengthened or, equivalently, the clock delayed.

Although the light sensitivity of TIM helps explain how the clock becomes reset each day to remain in synchrony with day length, it does not explain why, once the clock has been set, it continues to run even in complete darkness. Beyond the feedback repression of transcription of PER and TIM, the molecular mechanism by which the genes affect biological rhythms is not

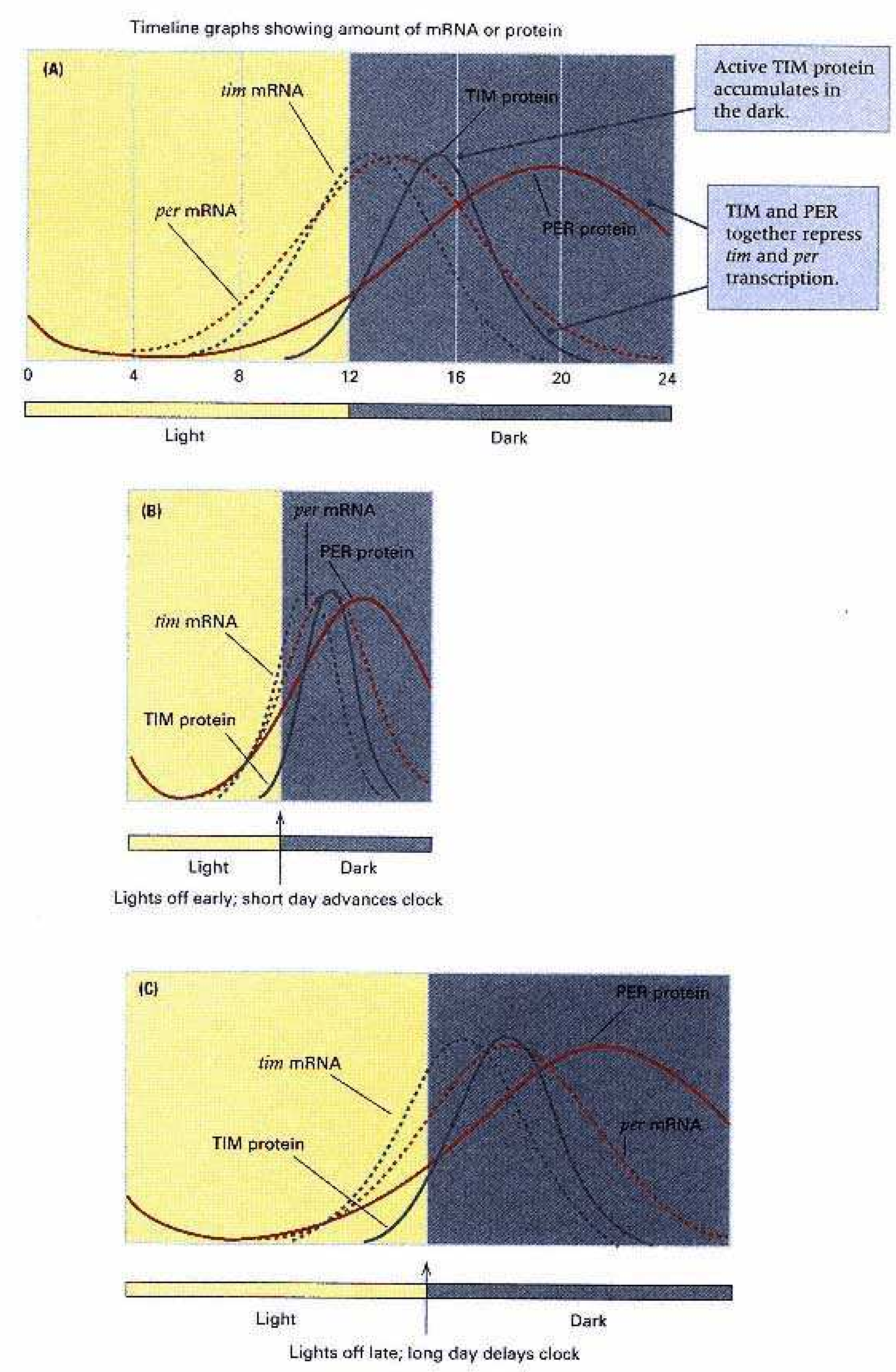



Figure 17.19
Circadian cycling of tim and *per* mRNA and of TIM and PER protein. The TIM protein is destroyed by light and so does not accumulate until after dark. When sufficiently abundant in the cytoplasm, TIM and PER move into the nucleus and repress *tim* and *per transcription*. (A) Normal cycles. (B) "short day" advances the clock by allowing TIM to accumulate sooner. (C) A "long day" delays the clock by delaying the accumulation of TIM.

known. Much remains to be explored, including the mechanisms by which the proteins affect the physiology and behavior of the organism. For example, it is not known how PER affects the courtship song. In any case, *per* and *tim* are only part of a larger story of biological rhythms, because various rhythms can be uncoupled. For example, rearing flies in continuous bright light completely obliterates the circadian rhythm of emergence of adults illustrated in Figure 17.13, whereas the rhythm of the courtship song remains unaltered.

### *The Mammalian Clock, Prion Protein, and Mad Cow Disease*

The TIM-PER mechanism in *Drosophila* is by no means universal in the control of circadian rhythms. Although circadian rhythms are extremely widespread, the mechanism of control differs from one species of organism to the next. For example, the fungus *Neurospora* has a circadian rhythm in growth rate. This rhythm is controlled by the gene *frq (frequency)* through its protein product FRQ. The cycling of FRQ is the opposite of that of TIM: The abundance of FRQ peaks in daylight and decreases at night. Although light controls the abundance of FRQ, the mechanism is through transcription of the *frq* gene, which is stimulated by light.

Mammals also have pronounced circadian rhythms that govern, among other physiological states, the sleep cycle. The phenomenon of jet lag results from the time needed to adjust the sleep cycle after a change of time zone. A key gene is called *CREM*, which encodes a transcriptional repressor protein, ICER, the level of which undergoes a circadian cycle in the pineal gland. At night, nerve input into the pineal gland from a region of the brain called the suprachiasmatic nucleus induces *CREM* transcription through a transcriptional activator protein called CREB. Induction of *CREM* causes an increase in the amount of *CREM* mRNA, and therefore of ICER protein, that is self-limiting and peaks during the night because ICER feedback represses *CREM* transcription. The regulatory system adjusts itself to changes in day length by altering the sensitivity of *CREM* to induction: In long-day situations, *CREM* is supersensitive to induction; in long-night situations, it is subsensitive to induction.

A number of mutations are known to affect the mammalian circadian rhythm. A mutation called *tau* in the hamster results in a short 20-hour cycle rather than the normal 23.5-hour cycle. In humans, there is a rare autosomal-dominant disease known as *familial fatal insomnia (FFI);* first appearing between the ages of about 35 and 60, the symptoms include progressive insomnia and degeneration of certain parts of the thalamus; the condition results in death about a year after onset. FFI is caused by a mutation in the gene for **prion protein,** located in chromosome 20pter-p12. The term *prion* stands for **protein infectious agent.** Prion proteins were originally discovered through their ability to act as infectious agents in the transmission of any of a number of serious neurological diseases. Produced initially as a precursor protein of 253 amino acids, the mature prion protein is found on the surface of neurons and glial cells in the brain. Its normal function is unknown. However, that it plays an important role in circadian regulation is supported by the observation that, in transgenic mice produced by genetic engineering (Section 9.4), a knockout mutation of the prion-protein gene results in altered circadian activity and sleep patterns resembling those in human FFI.

In human kindreds, other mutations in the prion-protein gene result in **Creutzfeldt-Jakob disease (CJD)** and **Gerstmann-Straussler disease (GSD).** Both of these conditions are marked by a mid-life age of onset and by progressive ataxia (loss of control of movement) and dementia (literally, "out of mind"), but their clinical symptoms differ somewhat. In both diseases, parts of the brain lose their normal histology and become full of holes (technically, *spongiform encephalopathy).*

Given the seemingly conventional genetics of FFI, CJD and GSD, each a manifestation of a different type of mutation in the prion-protein gene, it came as an almost unbelievable finding that prion-related diseases could also be transmitted from one organism to another by transfer of a protease-resistant form of the prion protein itself. For example, injection of a chimpanzee with prion protein from a

**Table 17.2** Diseases Associated with Defective Prion Protein

| | Organism | Disease |
|---|---|---|
| Inherited diseases | human | familial fatal insomnia (FFI) |
| | human | Creutzfeldt-Jakob disease (CJD) |
| | human | Gerstmann-Straussler disease (GSD) |
| "Infectious" diseases | human | kuru |
| | sheep and goat | scrapie |
| | cattle | mad cow disease |

human CJD patient results in a progressive spongiform encephalopathy indistinguishable from the human condition. Other infectious diseases once attributed to transmission of a "slow virus" are now known to have prions as the active agent (Table 17.2). One example is **kuru,** discovered originally among the Fore people of Papua New Guinea who practiced ritual cannibalism as a rite of mourning in which the brain of the dead was handled and eaten, particularly by the women and children. Another example is **scrapie,** the equivalent prion-protein disease in sheep and goats, so called because the diseased animal is so intolerably itchy that it rubs and scrapes against any convenient object. A third example is "mad cow disease" (technically, *bovine spongiform encephalopathy),* whose identification as a prion-protein defect resulted in an eruption of near hysteria over the importation of British beef into Europe. Although the transmissible prion agent is thought to be a protease-resistant, insoluble form of the protein derived from the normal protease-sensitive protein, it is not clear whether the transmissible protein should be regarded as a truly infectious agent or merely as a cytotoxic metabolite.

## 17.3— Learning

Higher animals have sophisticated nervous systems and correspondingly sophisticated behavioral repertoires. Birds and mammals are especially adept at modifying their behavior in response to environmental cues and previous experiences. In other words, they can learn. Although learning of sorts can be demonstrated in organisms such as *Drosophila,* the ability is most highly developed in birds and mammals-particularly in human beings and higher primates. Partly because of the importance of learning in human beings, and partly because of its intrinsic interest, psychologists and geneticists have tried to assess the genetic contribution to learning ability in various organisms.

### *Artificial Selection for Learning Ability*

Selection experiments for learning ability were some of the earliest studies in behavior genetics in animals. In these experiments, the animal is introduced into one end of a maze and must make its way to the other end to find a reward. One apparatus for testing the maze-learning ability of laboratory rats is illustrated in Figure 17.20A. For an animal that has previous experience in getting through the maze, the number of wrong turns ("errors") is a measure of the extent to which the animal has learned the layout of the maze. Artificial selection for better maze-learning ability is carried out by choosing superior learners as the parents in each generation; for poorer maze-learning ability, inferior learners are chosen as parents. The results of 21 generations of selection for better or poorer maze-learning ability are shown in Figure 17.20B. The initial population was a large, genetically heterogeneous strain of laboratory rats. In each generation, the rats were tested for

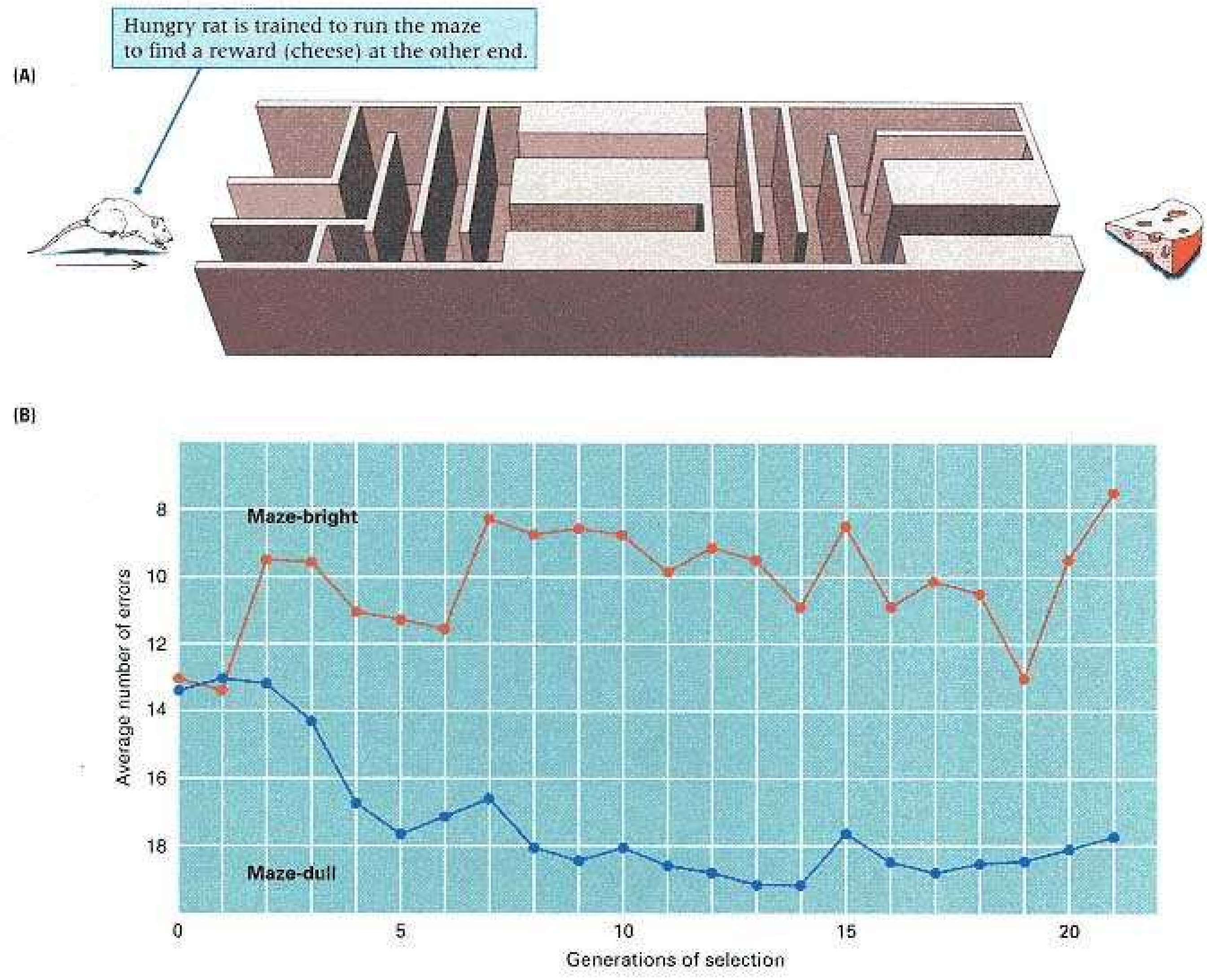



Figure 17.20
Maze-learning ability in laboratory rats. (A) A typical type of maze. Wrong turns are scored as "errors." (B) The results of 21 generations of artificial selection for maze-learning ability in laboratory rats. The "maze-bright" population was selected for better learning, and the "maze-dull" population for poorer learning. Each data point gives the average number of errors made by the animals in each population after the same number of learning trials. [Part A, after S. D. Porteus, *Porteus Maze Test*: *Fifty Years*' *Application* (Palo Alto: Pacific Books, 1965); part B, data from R. C. Tyron (1940) reported in G. E. McClearn, in *Psychology in the Making*, edited by L. Postman (New York: Knopf, 1962), p. 144.]

their ability to learn a maze. Those rats that performed best (that is, made the fewest errors after a given number of trials) were mated among themselves to form a "maze-bright" selected line, and in each generation, selection was carried out within this line for the brightest animals. Similarly, those rats that performed worst (made the most errors) were mated among themselves to form a "maze-dull" line, and in each generation, the dullest of the dull were selected to perpetuate the line.

As Figure 17.20B shows, selection in both directions was successful. By generation 2, there was virtually no overlap in learning ability between the two lines; the dullest "brights" were about equal in maze-learning ability to the brightest "dulls."

However, beyond generation 7, almost no further progress was made in either direction. Crosses between the "bright" and "dull" lines produced an $F^1$ generation that was intermediate in maze-learning ability. Additional crosses revealed that maze learning is inherited as a typical multifactorial trait of the sort discussed in Chapter 16.

The experiments represented in Figure 17.20B have a number of technical deficiencies. First, there was no control experiment consisting of a randomly mated population, unselected for maze-learning ability, established from the same initial population as the selected lines and main-tained with the same population size as the selected lines. An unselected control is necessary to detect possible systematic changes in the environment in the course of the selection. Second, the experiments in Figure 17.20B are unreplicated. A better set of experiments would have included at least two "bright" selected lines, at least two "dull" selected lines, and at least one unselected control line. The replication is necessary to assess the reproducibility of the result in independently selected populations. Nevertheless, in spite of the technical limitations, the experiments in Figure 17.20B are a classic in behavior genetics, because they do demonstrate hereditary variation affecting maze-learning ability in the rat population from which the selected lines were derived. In the absence of genetic variation, the divergence between the selected lines would be impossible. In quantitative terms, the response to selection in Figure 17.20B is of the magnitude that would be expected for a trait with a narrow-sense heritability of maze learning of 21 percent. (Narrow-sense heritability, discussed in Chapter 16, refers to the rate at which the mean of a trait can be changed by artificial selection. Roughly speaking, it is the proportion of the total variance in phenotype that is transmitted from parents to offspring.)

### *Genotype-Environment Interaction*

The maze-bright and maze-dull rats in Figure 17.20B are also of interest in demonstrating the complexity of the response to selection. Many behavioral differences were observed between the selected populations. For example, the selected populations differed in their maze-learning ability depending on whether the rats were motivated by hunger or by escape from water. The maze-bright animals learned better than the maze-dull animals when motivated by hunger (the condition in which the selection was carried out), but the maze-dull rats actually learned better than the maze-bright rats when tested in a situation in which getting through the maze allowed them to escape from immersion in water. Evidently, motivation makes a big difference, even to rats. Furthermore, the maze-bright rats were more active than the maze-dull rats when in mazes, but they were less active than the maze-dull rats when in exercise wheels. The differences between the strains were varied and complex, and it is not clear which of the differences may have been related to maze learning in a causal sense and which were fortuitous differences. Indeed, it is not even clear that the maze-bright rats were in any sense "smarter" than the maze-dull rats. The maze-bright rats may simply have responded better to the specific maze learning test situation.

The populations in Figure 17.20B can also demonstrate *genotype-environment interaction*, a complication that may arise with any multifactorial trait (Chapter 16) but particularly with complex behavioral traits. Genotype-environment interaction means that the relative performance of different genotypes changes drastically according to the environment. In another set of maze-learning experiments, the bright and dull rats from the thirteenth generation of an experiment like that in Figure 17.20B were reared separately in three different types of environments and then tested for their maze-learning ability. The environments were

1. The normal environment.

2. An "enriched" environment having cage walls decorated with designs in luminous paint and containing ramps, mirrors, swings, polished balls, marbles, barriers, slides, tunnels, bells, teeter-totters, and springboards-all constructed so that they could be easily shifted to new positions.

3. A "restricted" environment having cages surrounded by drab, gray walls and containing only food and water pans.

The maze-learning ability of the rats reared in these three environments is summarized in Figure 17.21. In the normal environment (Figure 17.21A), the rats from the bright strain performed much better than those from the dull strain. This difference is expected, because the normal environment is the one in which the selection experiments were carried out. The genotype-environment interaction is observed in the enriched and the restricted environments. Although the enriched environment (Figure 17.21B) produces a dramatic improvement in the maze performance of the dull rats, it hardly affects the performance of the bright rats. Therefore, the difference in maze-learning ability is much diminished in the enriched environment. In the restricted environment (Figure 17.21C), the performance of both strains is very adversely affected, but the bright strain is affected more than the dull strain. The result is that in the restricted environment, the maze-learning ability of both strains is approximately equal. The principal conclusion from these experiments is that, though the maze-bright and maze-dull strains certainly differ in genetic factors affecting maze learning, the magnitude of the difference between the strains results as much from their rearing environment as from their difference in genotype. This example of genotype-environment interaction should be kept in mind as a cautionary tale in interpreting experiments on the genetics of human behavioral variation.

## 17.4—
## Genetic Differences in Human Behavior

One can hardly imagine a subject more controversial than the inheritance of behavioral differences in human beings, particularly behavior that is considered socially undesirable. From the very beginning of modern genetics, commentators have been divided, some claiming the sim-

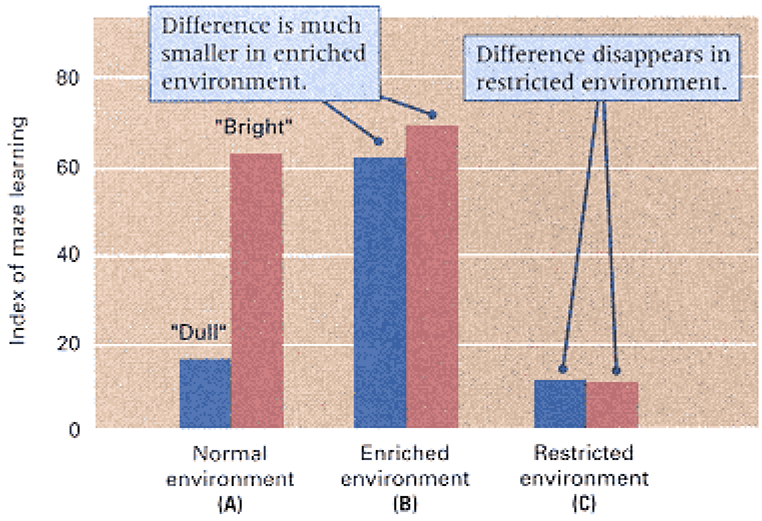



Figure 17.21
Maze-learning ability of maze-bright and maze-dull rats reared in three different environments. (A) In the normal environment, the bright rats are clearly better at learning the maze. (B) In an enriched environment, the difference is much smaller. (C) In a restricted environment, the difference disappears completely.
[Data from R. M. Cooper and J. P. Zubek. 1958. *Canad. J. Psychol*. 12: 159.]

ple and direct inheritance of virtually all forms of socially unacceptable behavior- "feeble-mindedness," habitual drunkenness, criminality, prostitution, and so on-and others arguing for environmental causation of such behavior. Extreme hered-itarian views are exemplified by the following statements:

> It may be stated that feeble-mindedness is generally of the inherited, not the induced, type; and that the inheritance is generally recessive. (R. R. Gates, 1933)

> In feeble-minded stocks, mental defect is interchangeable with imbecility, insanity, alcoholism, and a whole series of mental (and often physical) anomalies. (K. Pearson, 1931)

Such extreme hereditarian views were supported by studies of certain families with a high incidence of undesirable traits, the most famous being the Jukes family and the Kallikak family (both pseudonyms). Figure 17.22 is part of the Jukes pedigree published in 1902 and purporting to show the inheritance of "shiftlessness"

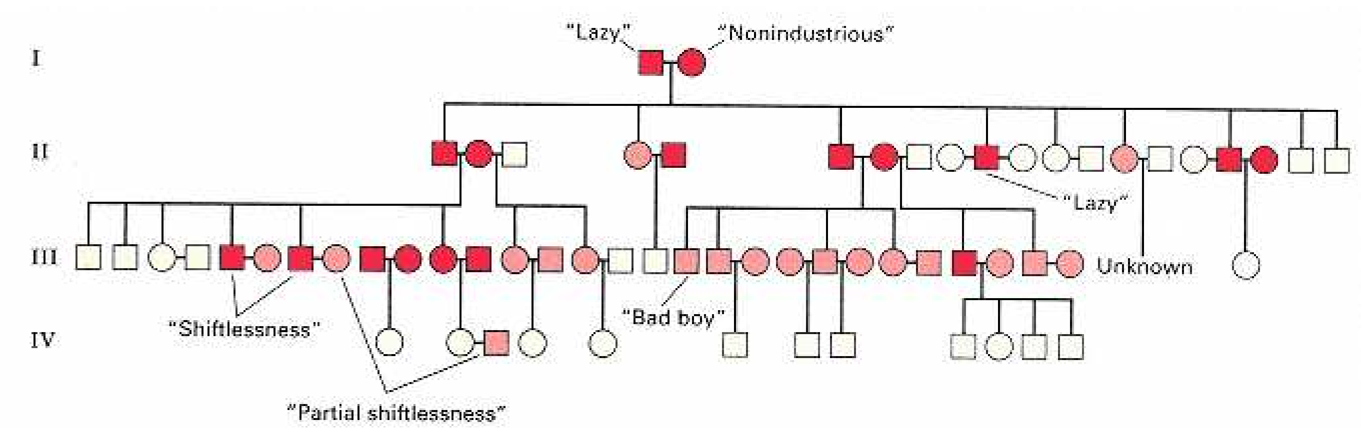



Figure 17.22
Part of the Jukes pedigree illustrating hereditarian prejudices of some early investigators. Red symbols purported to represent "shiftlessness" and pink symbols "partial shiftlessness."
[After R. L. Dugdale, *The Jukes: A Study in Crime, Pauperism, Disease and Heredity* (New York: Putnam, 1902); and C. B. Davenport, *Heredity in Relation to Eugenics* (New York: Holt, 1911).]

and "partial shiftlessness." The red-and pink-shaded symbols are as in the original report, and they give the false impression of objectivity. However, the diagnosis of "shiftlessness" or "partial shiftlessness" is entirely subjective, as indicated by the descriptions of some of the individuals in the pedigree. Individual I-1 is described as "a lazy mulatto," I-2 as "a nonindustrious harlot, but temperate," II-10 as "lazy, in poorhouse," and III-18 as a "bad boy." Today this sort of analysis is rejected as subjective and prejudicial, but at the time it was taken quite seriously as an element in the "inheritance" of poverty.

Even if we accept the pedigree at face value, all that it implies is that no family exists outside of its environment and that social environments, like genes, tend to be perpetuated from generation to generation. Familial association of certain characteristics (for example, poverty) and certain types of behavior is not evidence that the traits are genetically inherited, rather than socially transmitted. A clear example is English grammar and pronunciation. Children tend to speak with the same quirks and idioms as their parents and other relatives, yet nobody would presume that English usage is genetically transmitted. Nevertheless, some early behaviorists were extreme hereditarians

The opposite view was held by the extreme environmentalists, whose thinking is typified by the following example:

> So let us hasten to admit-yes, there are heritable differences in form, in structure . . .. These differences are in the germ plasm and are handed down from parent to child . . .. But do not let these undoubted facts of inheritance lead us astray as they have some of the biologists. The mere presence of these structures tells us not one thing about function . . .. Our hereditary structure lies ready to be shaped in a thousand different ways-the same structure-depending on the way in which the child is brought up . . .. We have no real evidence of the inheritance of behavioral traits. (J. B. Watson, 1925)

Watson went on to add,

> I should like to go one step further now and say, "Give me a dozen healthy infants, well formed, and my own specified world to bring them up in and I'll guarantee to take any one at random and train him to become any type of specialist I might select-doctor, lawyer, artist, merchant-chief and, yes, even beggar-man and thief, regardless of his ancestors." I am going beyond my facts and I admit it, but so have the advocates of the contrary . . ..

At the present time, environmentalist interpretations of human behavioral differences predominate. In the academic literature, you will not find any hereditarian views as extreme as those illustrated by the quotations from Gates and Pearson. However, apart from the use of the masculine pronoun, it is rather easy to find contemporary environmentalist statements that resemble the quotations from Watson. On the other hand, even though extreme hereditarian interpretations of human behavioral differences are strongly out of favor, most geneticists and psychologists are willing to concede the importance of both heredity *and* environment in human behavioral variation. The important questions involve the relative importance of "nature" (genetics) and "nurture" (environment) and how best to assess the situation experimentally.

### *Sensory Perceptions*

If "behavior" is interpreted in its widest sense to include sensory perception and processing, then genetic variation within the normal range is easy to demonstrate. Two examples will serve to illustrate this point. About 70 percent of North American whites, and 97 percent of North American blacks, are able to taste the chemical **phenylthiocarbamide (PTC)** when it is present in filter paper. Phenotypically, these people are "tasters." The remaining 30 and 3 percent, respectively, are unable to detect PTC and therefore have the "nontaster" phenotype. PTC tasting is a clear example of variation in sensory perception, and it is a simple Mendelian trait. The ability to taste PTC is due to a dominant allele, denoted *T*, located in chromosome 7q. The recessive allele is designated *t.* The genotypes *TT* and *Tt* are tasters of PTC, and *tt* genotypes are nontasters. Family data of the type illustrated in Table 17.3 provide an excellent fit to the simple two-allele hypothesis. However, a finer classification of people according to the *concentration* of PTC that can be tasted reveals subtler shades of tasting and nontasting ability that seem to have a multifactorial basis. This may account for the small number of unexpected taster offspring from nontaster × nontaster matings.

A second example of genetic variation in sensory perception concerns the ability to smell an odiferous substance, *methanethiol,* that some people excrete in the urine a few hours after eating asparagus. This trait is a good example of the difficulty of studying the genetics of human behavior. Simple as the trait may seem to be, there are two diametrically opposed explanations of its inheritance. One hypothesis is that the trait is a classic type of inborn error of metabolism in which about 40 percent of the population lacks an unidentified enzyme that results in the urinary excretion of methanethiol. The other hypothesis is that everyone excretes methanethiol in the urine but that the ability to smell the substance is a genetically determined hypersensitivity to the smell.

**Table 17.3** Distribution of PTC taster and nontaster offspring

| | | Observed [expected] number of offspring | | | | Proportion of nontasters among offspring | |
|---|---|---|---|---|---|---|---|
| | | Taster | | Nontaster | | | |
| **Mating** | **Number of families** | **Observed** | **Expected** | **Observed** | **Expected** | **Observed** | **Expected** |
| Taster × taster | 425 | 929 | 930 | 130 | 129 | 0.123 | 0.122 |
| Taster × nontaster | 289 | 483 | 495 | 278 | 266 | 0.366 | 0.349 |
| Nontaster × nontaster | 86 | 5 | 0 | 218 | 223 | 0.978 | 1.000 |

*Note:* The expected numbers are derived from the supposition of a dominant *T* and a recessive *t* allele with random mating. When the recessive has allele frequency *q*, it can be shown that the proportions expected in the right-hand column are $[q(1 + q)]^2$, $q(1 + q)$, and I, respectively. In this case $q = 0.537$. Although the fit with the single-locus hypothesis is excellent, the five taster offspring in the nontaster matings are unexplained. They could be due to misclassification of phenotype, to misassigned parentage, or perhaps to alleles at other loci that influence tasting ability. [From C. Sterm, *Principles of Human Genetics,* 3d ed. (San Francisco: Freeman, 1973).]

About 10 percent of people tested are able to smell the substance at low concentration. An important observation is that those people who *can* smell it can detect its presence in everyone's urine after ingestion of asparagus. This observation supports the second hypothesis: Every person excretes the substance, but not every person is able to smell it.

### *Severe Mental Disorders*

Examples of both genetic and environmental effects on mental function are easily enumerated. Down syndrome and the fragile-X syndrome are major genetic causes of mental retardation (Section 7.5). Another inherited form of mental retardation is **phenylketonuria,** an inborn error of metabolism caused by an autosomal recessive gene at 12q24.1 that results in a deficiency of the enzyme phenylalanine hydroxylase and an accumulation of the amino acid phenylalanine. The metabolic defect causes profound mental deficiency unless the dietary intake of phenylalanine is rigorously controlled. On the environmental side, severe mental impairment of genetically normal infants can result from prenatal infection with rubella virus (German measles), excessive exposure of the fetus to alcohol *(fetal alcohol syndrome),* insufficient oxygen or brain damage at

**Table 17.4** Risk of schizophrenia: family and adoption data

| Relationship | Number of persons | Risk (%) |
|---|---|---|
| Familial data | | |
| Identical twins, one affected | 210 | 46 |
| Same-sex fraternal twins, one affected | 309 | 14 |
| Siblings, one affected | 9921 | 10 |
| Children with one affected parent | 1577 | 13 |
| Children with two affected parents | 134 | 46 |
| Genetically unrelated controls | 399 | 2 |
| Adoption data | | |
| Adopted child of affected mother | 47 | 17 |
| Adopted child of nonaffected mother | 50 | 0 |
| Normal child adopted by future schizophrenic | 28 | 11 |
| Biological parent of affected adoptee | 66 | 12 |
| Adopted parent of affected adoptee | 63 | 2 |

Data are the combined results of various studies differing somewhat in diagnostic criteria. Data from I. I. Gottesman and J. Shields, *Schizophrenia: The Epigenetic Puzzle* Cambridge, England: Cambridge University Press, 1982)

birth, and many other factors. These examples illustrate that assignment of the primary cause of mental retardation to genetics or environment must be done on a case-by-case basis. As might be expected, in many instances, the precise cause is unas-signable, so it remains unclear what overall proportion of mental retardation is genetic in origin and what proportion is environmental.

Genetic factors can also influence personality. Table 17.4 summarizes family and adoption data on the risk of schizophrenia, a classic form of "madness" characterized by delusions, auditory hallucinations, and other progressively worsening forms of dementia. In view of its severity, the trait is remarkably common, affecting about 1 person in 100. The genetic contribution to the trait is seen in the high concordance (46 percent) in identical twins, in the high risk among children with two affected parents, and in the substantial risk among adopted children when their biological mother is affected. Indeed, regarding schizophrenia, adopted children resemble their biological parents more than their adoptive parents. The genetic risk factors for schizophrenia have not been identified, but it is clear that there are multiple genetic factors and that the relative importance of the genetic factors differs from one kindred to the next. Genetic mapping studies in families with schizophrenia have implicated genes at least in chromosomes 5q, 6p, and 22q.

On the other hand, Table 17.4 also shows that environmental influences are important in schizophrenia. The environmental component is evidenced by the incomplete concordance of identical twins (when one twin is affected, there is a 46 percent chance that the other will also be affected) and by the increased risk in normal children adopted into families in which one partner later develops schizophrenia. Most psychiatric geneticists currently favor the view that genotype is an important predisposing factor toward schizophrenia but that environmental stress is an essential trigger for actual onset of the disease.

Severe mental retardation, senile dementia (such as Alzheimer disease), and personality disorders (such as schizophrenia) lie well outside the range of normal intellectual function. Hence the relative importance of genetic and environmental contributions to these conditions may not yield an accurate picture of behavioral differences among people. Among any random collection of people, there is enormous variation in personality, emotional traits, and intellectual abilities. How much of this variation can be attributed to genotypic differences among people and how much can be attributed to differences in their environments? This is the key issue in understanding how genes influence human behavioral traits.

### *Genetics and Personality*

The possible role of hereditary factors in influencing human personality traits within the normal range of variation in personality is difficult to assess. These traits are affected by environmental as well as genetic factors. If the genetic factors consist of more than a handful of genes, each with a relatively small effect, then the genetic factors are very difficult to detect, even using the methods outlined in Chapter 16 for genetic mapping of quantitative-trait loci. Even though the genes cannot be identified individually, it is nevertheless possible to estimate how important genetic factors are, in the aggregate, by separating the observed variance in the trait into a genetic variance component (estimating the importance of genetic factors in causing the total variance in phenotype) and an environmental component (estimating the importance of environmental factors in causing the total variance in phenotype). The concepts of genetic and environmental variance were discussed in Chapter 16. One overall measure of the importance of genetic factors in determining the variance of a trait is the broad-sense heritability, which is the ratio obtained by dividing the variance in phenotype attributable to differences in genotype by the total variance in phenotype from all causes together. The uses and limitations of this concept are also discussed in Chapter 16.

The broad-sense heritability of personality traits has been estimated experimentally from studies of identical twins reared together and reared apart in comparison with fraternal twins and ordinary siblings reared together and reared apart. For each of five broad personality traits, the broad-sense heritability ranges from 30 percent to 50 percent (Figure 17.23). Each of these broad personality traits is expressed not in an all-or-none fashion but quantitatively. After examination by trained personnel, each person can be assigned a score on a continuous scale of measurement reflecting the degree to which the trait is expressed in the person. The traits are

• *Agreeableness: a trait associated with friendliness, likeability, and pleasantness* People who strongly express this trait are sympathetic, warm, kind, and good-natured, and they tend not to take advantage of others. The opposite end of the scale is associated with aggressiveness, pugnacity, coldness, and vindictiveness.

• *Conscientiousness: associated with conformity and dependability* People who are high in conscientiousness are well organized; they plan ahead; they are responsible and practical. People who score strongly negative for this trait are impulsive, careless, and undependable.

• *Extroversion: associated with social dominance* Extroverts are outgoing, decisive, and persuasive. At the other end of the scale are introverts, who tend to be reserved and withdrawn and to avoid attention.

• *Neuroticism: associated with anxiety and emotional instability* People high on the neuroticism scale tend to be nervous, self-conscious, tense, and prone to worry, and they try to avoid situations that they perceive as harmful. People at the opposite end of the scale are easy-going, emotionally stable, and not prone to high anxiety.

• *Openness: associated with receptiveness to novel ideas and stimuli* People high on the openness scale are curious, imaginative, insightful, and original. People at the negative end of the openness scale are described as shallow or lacking in curiosity.

A knowledge that the broad-sense heritability of each of these personality traits is from 30 percent to 50 percent tells us nothing about the number of genes that are

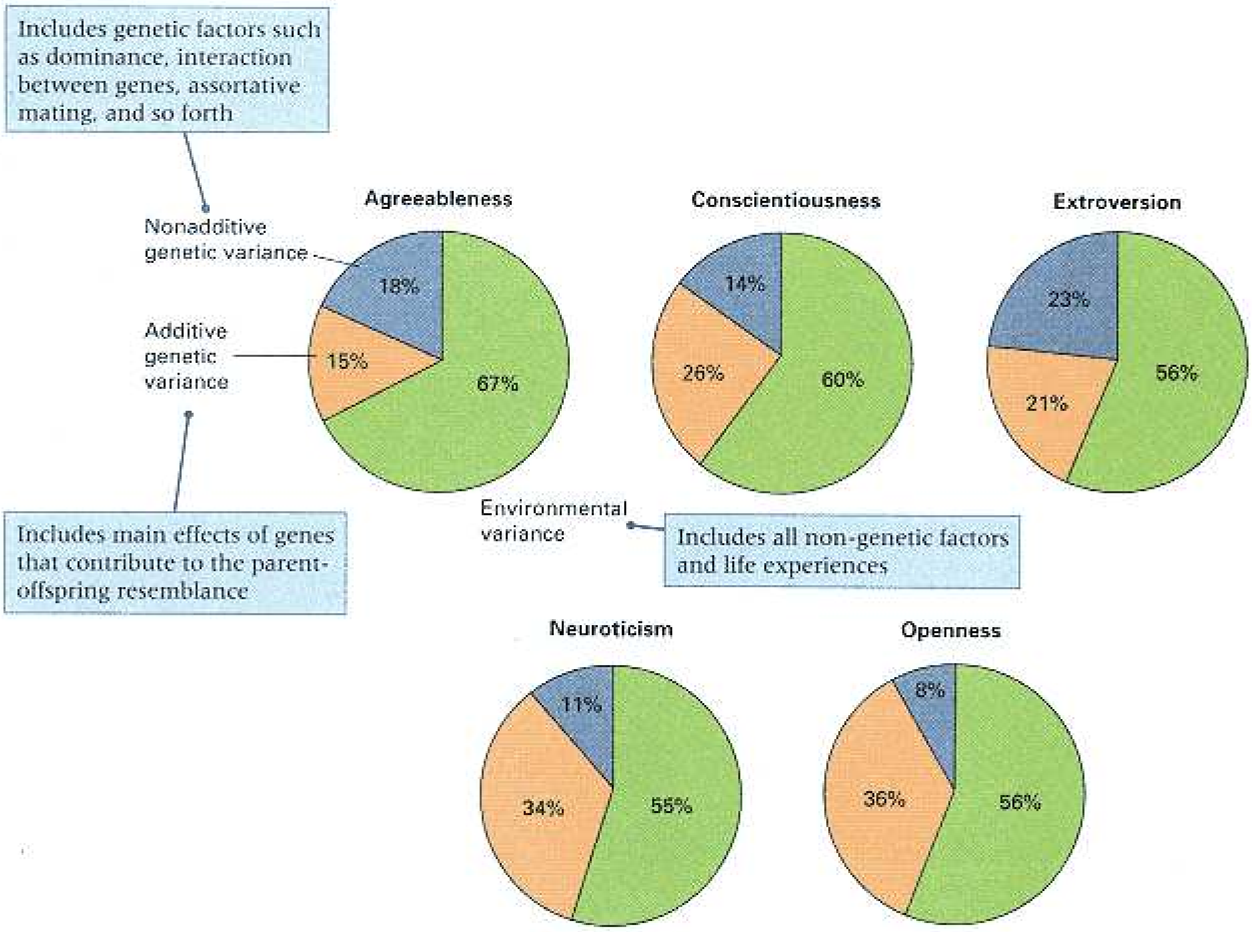



Figure 17.23
Percentages of the total phenotypic variance attributable to additive gene effects, nonadditive gene effects, and environmental effects for each of five broad categories of personality traits. The broad-sense heritability equals the sum of the additive and nonadditive percentages.
[Data from T. J. Bouchard, Jr. 1994. *Science* 264: 1700.]

involved or the magnitude of the effects of the genes, let alone what the genes are at the molecular level. Because many of the genes affecting personality may have relatively small effects, they are likely to remain undetected in the usual types of pedigree studies for genetic mapping.

Another approach to the identification of genes that influence personality and behavior is through candidate genes. A **candidate gene** for a trait is a gene for which there is some a *priori* basis for suspecting that the gene may affect the trait. In human behavior genetics, for example, if a pharmacological agent is known to affect a personality trait, and the molecular target of the drug is known, then the gene that codes for the target molecule and any gene whose product interacts with the drug or with the target molecule are all candidate genes for affecting the personality trait.

One example of the use of candidate genes in the study of human behavior genetics is in the identification of a naturally occurring genetic polymorphism that affects neuroticism and, in particular, the anxiety component of neuroticism. The neurotransmitter substance serotonin (5-hydroxytryptamine) is known to influence a variety of psychiatric conditions, such as anxiety and depression. Among the important components in serotonin action is the serotonin transporter protein. Neurons that release serotonin to stimulate other neurons also take it up again through the serotonin transporter. The uptake terminates the stimulation and also recycles the molecule for later use.

The serotonin transporter became a strong candidate gene for anxiety-related personality traits when it was discovered that the transporter is the target of a class of antidepressants known as selective serotonin reuptake inhibitors. The widely prescribed antidepressant Prozac is an example of such a drug.

Motivated by the strong suggestion that the serotonin transporter might be involved in anxiety-related traits, researchers looked for evidence of genetic polymorphisms affecting the transporter gene in human populations. Such a polymorphism was found in the promoter region of the transporter. About 1 kb upstream from the transcription start site is a series of 16 tandem repeats of a nearly identical DNA sequence of about 15 base pairs. This is the *l* (*long)* allele, which has an allele frequency of 57 percent among Caucasians. There is also an *s* ( *short)* allele, in which three of the repeated sequences are not present. The *s* allele has an allele frequency of 43 percent. The genotypes *l l*, *l s*, and *s s* are found in the Hardy-Weinberg proportions of 32 percent, 49 percent, and 19 percent, respectively (Chapter 15). Further studies revealed that the polymorphism does have a physiological effect. In cells grown in culture, *l l* cells had approximately 50 percent more mRNA for the transporter than *l s* or *s s* cells, and *l l* cells had approximately 35 percent more membrane-bound transporter protein than *l s* or *s s* cells.

On the basis of these results, the researchers predicted that the serotonin transporter polymorphism would be found to be associated with anxiety-related personality traits. The predicted result was observed in a study of 505 people who were genotyped for the transporter polymorphism and whose personality traits were classified by means of a self-report questionnaire. Significant associations were found between the transporter genotype and the overall neuroticism score, and the highest correlations were with the anxiety-related traits "tension" and "harm avoidance." A comparison between the genotypes with respect to neuroticism score is shown in Figure 17.24. Because the *s* allele is dominant to *l* with respect to gene expression and personality score, the L

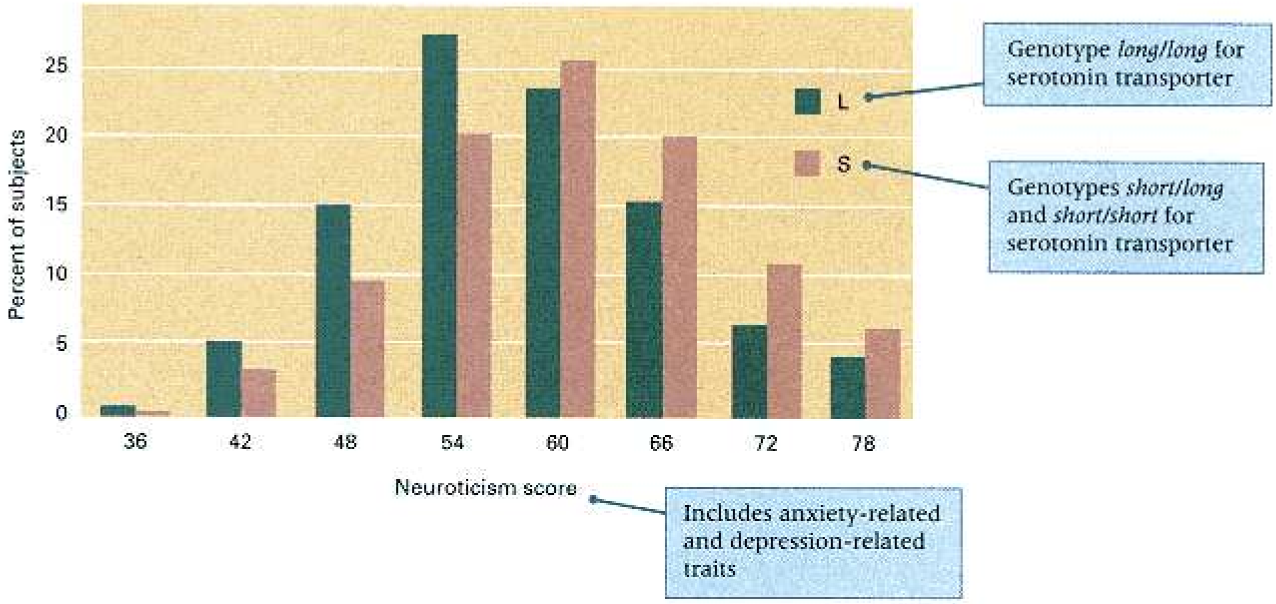



Figure 17.24
Distribution of neuroticism scores among different genotypes for the serotonin transporter polymorphism. The genotype L includes only *l/l*, whereas S includes *s/s* and *s/l*. The total is 163 subjects for the L genotype and 342 for the S genotypes.
[From K.-P. Lesch, et al. 1996. *Science* 274: 1527.]

category includes only the *l l* genotype whereas the S category includes both *s s* and *l s.*

Note in Figure 17.24 that there is a great deal of overlap in the L and S distributions. The overlap means that the transporter polymorphism accounts for a relatively small proportion of the total variance in neuroticism score in the whole population. In quantitative terms, the transporter polymorphism accounts for 3 to 4 percent of the total variance in neuroticism score and 7 to 9 percent of the genotypic variance in neuroticism score. If all genes affecting neuroticism had the same magnitude of effect as the serotonin transporter polymorphism, approximately 10 to 15 genes would be implicated in the 45 percent of the variance that can be attributed to genetic factors.

***Genetic and Cultural Effects in Human Behavior***

Human beings exhibit many important behavioral traits for which no candidate genes have been identified and for which the effects of individual genes are probably too small to detect with genetic linkage studies. For these traits, one measure of the overall importance of genetic factors is the broad-sense heritability, which is the ratio of the genotypic variance in the trait to the total phenotypic variance in the trait.

Estimating the heritability of any human behavioral trait requires that special attention be given to cultural transmission of nongenetic factors that potentially affect the behavior. These familial factors increase the resemblance between relatives, and, unless properly taken into account, they inflate the apparent genetic heritability. Most conventional genetic studies of human behavior are biased in that they include cultural transmission in the estimate of heritability, treating the situation as though cultural differences were genetic in origin.

Another difficulty in applying the concepts of quantitative genetics to human behavior is that the concept of heritability is prone to being misinterpreted. The most common mistake is to interpret the heritability as applying to genetic differences *between* populations. For essentially the same reasons that identical twins reared in different environments may have different phenotypes, two genetically identical populations that differ in their environments may differ in average phenotypes. The error in this interpretation is that

> The heritability of a trait *within* groups tells one nothing about the genetic or environmental causes of a difference in the average of a trait *between* groups.

Both genetic and cultural factors are relevant to interpreting the variation in phenotypes within a specified population. Use of these quantities for comparison between populations is unjustified and may often lead to incorrect conclusions. Because the emotional overtones of human behavioral differences are so great, the central issue may become clearer by means of analogy. Instead of a human behavioral difference, one might just as well ask why farmer Brown's hens lay an average of 230 eggs per year but farmer Smith's average only 210. Lacking any knowledge of the genetic relationship between the flocks, we cannot specify the reason for the difference. This is true even though we might know with certainty that the genetic heritability within each flock is 30 percent, because heritability within flocks is irrelevant to the comparison. On the one hand, farmer Brown's hens could be genetically superior in egg laying. On the other hand, farmer Smith's chicken feed or husbandry could be inferior. In order to determine the reason for the difference, we would have to rear some of Brown's chickens on Smith's farm, and vice versa. If the egg laying of the transplanted hens did not change, then the difference between the two groups of hens would have been shown to be entirely genetic. If the egg laying of the transplanted hens did change, then the difference between the groups would have been shown to be environmental. Because experiments of this type cannot be carried out with humans, the issue of genetic versus environmental causes of behavioral differences among human groups is essentially unresolvable.

**Chapter Summary**

Chemotaxis refers to movement toward attractants and away from repellents. Cells of *E. coli* propel themselves by means of flagella, counterclockwise rotation of which forms a bundle that propels the cell forward and clockwise rotation of which causes tumbling. Chemotactic movements of *E. coli* are the result of a bias toward swimming when the cell is headed toward an attractant or away from a repellent. Most attractants for *E. coli* are nutrients, and most repellents are harmful substances.

Mutations affecting chemotaxis fall into a number of distinct categories. Mutants that are specifically nonchemotactic show no response to a particular attractant or repellent, usually because of a nonfunctional chemosensor protein responsible for binding the particular attractant or repellent. Mutants that are multiply nonchemotactic show no response to a group of chemically unrelated substances, usually because of a nonfunctional chemoreceptor that interacts with a particular subset of chemosensors. All of the chemoreceptors feed into a common biochemical pathway of chemotaxis. Mutants with defects in genes for the components of the common pathway are generally nonchemotactic and fail to respond to any attractant or repellent. Mutations that affect chemotaxis also may cause defects in the structural components of the flagella or in the motor components that control rotational switching.

The molecular basis of chemotaxis is mediated by phosphate transfer and methylation. Chemoreceptor proteins repeatedly initiate a chain of phosphorylations, resulting in clockwise flagellar rotation (tumbling). When a chemoreceptor becomes bound with a chemosensor, the phosphoryl groups are rapidly removed, restoring swimming, and the rate of phosphorylation is decreased, so the episodes of swimming get longer. There is also a feedback mechanism because phosphorylation results in removal of certain methyl groups from the chemoreceptors. Demethylation decreases the rate of phosphorylation and, hence, also promotes swimming. Methylation of the chemoreceptors is the basis of sensory adaptation, in which the cells become adjusted to a particular level of an attractant because each additional methyl group increases the level of attractant needed for excitation. Chemoreceptors are also called methyl-accepting chemotaxis proteins (MCPs) because of the role of methylation in their function.

Circadian rhythms are cycles synchronized with the 24-hour cycle of day and night. Several mutations in *Drosophila* alter the period of circadian rhythms and affect such behaviors as locomotor activity, the time of day at which the newly developed adults emerge from the pupa cases, and the courtship song, which refers to the rhythmic patterns of wing vibrations that the male directs toward the female during courtship. In *Drosophila,* circadian rhythms are controlled by two proteins, TIM and PER. TIM is the product of the gene *timeless* (*tim),* and PER is the product of the gene *period* (*per).* TIM is light-sensitive and so cannot accumulate during daylight hours. The mRNA of both genes accumulates in the daytime, as does PER protein. TIM begins to accumulate after nightfall, and when it is sufficiently abundant, enters the nucleus along with PER, where transcription of both *tim* and *per* is repressed, resulting in a progressive decrease in *tim* and *per* mRNA, as well as TIM and PER proteins, throughout the night.

Other organisms have different molecular mechanisms that control circadian rhythms. In *Neurospora,* transcription of the gene *frequency* (*frq)* is stimulated by light. In mammals, prion protein is implicated in the control of circadian rhythms. Mutations in the prion-protein gene result in inherited forms of familial fatal insomnia (FFI), Creutzfeldt-Jakob disease (CJD), and other conditions. In certain of these conditions, there is also produced a protease-resistant form of the prion protein, which can be transmitted from one organism to another as though it were an infectious agent. Transmissible prion protein has been implicated in kuru in human beings, scrapie in sheep and goats, and mad cow disease (bovine spongiform encephalopathy).

Selection experiments for maze-learning ability in laboratory rats demonstrate that a substantial amount of genetic variation affects the trait. The experiments also demonstrate the importance of genotype-environment interaction in complex behaviors. The maze-bright rats were superior to maze-dull rats in learning the maze when motivated by hunger (the condition of the selection), but the maze-dull rats were superior to the maze-bright rats in learning the maze when motivated by escape from water. Furthermore, although rats from the bright strain performed much better than those from the dull strain in a normal environment, the difference was much reduced in an enriched environment and virtually disappeared in a restricted environment.

Although a few differences in human behavior have a relatively simple genetic basis (for example, the ability to taste phenylthiocarbamide, or PTC), most human behavior is influenced by multiple genes and by environmental factors. For example, a genetic predisposition to schizophrenia is indicated by the increased risk in close relatives of affected persons, but environmental factors also are important, as shown by the increased risk in normal children adopted into families in which one partner later develops schizophrenia. With a behavioral trait that can be measured on a continuous scale, the heritability of the trait can be estimated by analysis of the correlation between relatives. Such heritability estimates often include cultural transmission in the "genetic" component of variation, so the heritability estimate is inflated. In any case, the heritability of a human behavioral trait applies only to variation in the trait within the relevant population; it provides no information about the genetic basis of possible differences in the trait among diverse racial or ethnic groups.

## Key Terms

| | | |
|---|---|---|
| attractant | flagella | prion protein |
| behavior | kuru | PTC tasting |
| candidate gene | locomotor activity | repellent |
| chemoreceptor<br>chemosensor | methyl-accepting chemotaxis protein (MCP) | sensory adaptation<br>taster |
| chemotaxis | nontaster | |
| circadian rhythm | phenylketonuria | |
| courtship song | phenylthiocarbamide (PTC) | |

## Review the Basics

• When cells of *E. coli* are placed in medium containing lactose, they begin to transcribe the lactose operon and to produce the proteins needed for lactose transport and degradation. Is this an example of behavior? Why or why not?

• How is rotation of the flagella related to chemotaxis in *E. coli?*

• What is the molecular basis of sensory adaptation in bacterial chemotaxis?

• How do the *Drosophila* mutations *period* and *timeless* demonstrate that circadian rhythms are under genetic control?

• What does it mean to say that the broad-sense heritability of a personality trait is 50 percent?

• What is a candidate gene for a trait and how are candidate genes identified?

• What is sensory adaptation? How does sensory adaptation in *E. coli* take place in the presence of a repellent?

## Guide to Problem Solving

**Problem 1:** Three strains of *E. coli* were tested for chemotactic response to the amino acids serine and leucine. Strain 1 swims toward serine and away from leucine; strain 2 swims toward serine but shows no response to leucine; and strain 3 shows no response to either serine or leucine. Additional tests showed that each strain had a normal chemotactic response in the presence of other attractants and repellents. Which strain is wildtype, which specifically nonchemotactic, and which multiply nonchemotactic? Which genes and gene products are likely to be defective in the mutant strains?

***Answer:*** Strain 1 is wildtype with respect to serine and leucine chemotaxis. Strain 2 is specifically nonchemotactic with respect to leucine; it is likely to have a mutation in the gene encoding the chemosensor for leucine. Strain 3 is multiply nonchemotactic. It is likely that strain 3 has a mutation in the gene for the chemoreceptor that is common to the chemotaxis pathway stimulated by both serine and leucine; alternatively, strain 3 may have two mutations-one in the serine chemosensor and the other in the leucine chemosensor.

**Problem 2:** What types of genetic crosses would you do, and what tests would you carry out, to determine whether $per^O$, $per^S$, and $per^L$ are alleles of the same gene affecting locomotor activity?

**Answer:** A complementation test is necessary, which means that females heterozygous for pairs of the alleles need to be examined for monitoring locomotor activity. Therefore, cross $per^O/per^O$ females with $per^S$ males and examine the locomotor activity of the $per^O/per^S$ daughters. Likewise, cross $per^O/per^O$ females with $per^L$ males and examine the locomotor activity of the *per*$^O$/*per*$^L$ daughters. (The required female genotypes could also be produced by carrying out the reciprocal of each cross.)

The result of the cross is that *per*$^O$/*per*$^S$ females have a short-period rhythm and *per*$^O$/*per*$^L$ females have a long-period rhythm. Therefore, *per*$^S$ is an allele of *per*$^O$, and *per*$^L$ is an allele of *per*$^O$. By inference, *per*$^S$ is an allele of *per*$^L$. (The direct test of the allelism between *per*$^S$ and *per*$^L$ by examining *per*$^S$/*per*$^L$ heterozygotes is ambiguous, because *per*$^S$ shortens the period and *per*$^L$ lengthens it. The effects of the mutant alleles are opposite, with the result that the period of *per*$^S$/*per*$^L$ heterozygotes is approximately normal.)

**Problem 3:** Studies of pedigrees, twins, and adoptions suggest that some cases of the psychiatric disorder called bipolar affective disorder have a genetic basis. It has been difficult to identify genetic markers associated with this disorder. For example, one group of researchers reported genetic linkage with markers on chromosome 11, only to report at a later date that additional data eliminated the statistical signifi-

## Chapter 17 GeNETics on the web

*GeNETics on the web* will introduce you to some of the most important sites for finding genetic information on the Internet. To complete the exercises below, visit the Jones and Bartlett home page at

**http://www.jbpub.com/genetics**

Select the link to ***Genetics: Principles and Analysis*** and then choose the link to **GeNETics on the web.** You will be presented with a chapter-by-chapter list of highlighted keywords.

### GeNETics EXERCISES

Select the highlighted keyword in any of the exercises below, and you will be linked to a web site containing the genetic information necessary to complete the exercise. Each exercise suggests a specific, written report that makes use of the information available at the site. This report, or an alternative, may be assigned by your instructor.

**1** In the control of the *Drosophila* **circadian rhythm,** it has been suggested that the TIM and PER proteins form a dimer that enables them to enter the nucleus together and act to repress transcription of the *tim* and *per* genes. The evidence for this model can be found by requesting a *full report* on the gene *tim* at the keyword site. If assigned to do so, write one paragraph describing the nature of the evidence.

**2** You can learn more about **mad cow disease** and related diseases at this site. Particularly interesting is the link to *History*. If assigned to do so, write one paragraph identifying Victoria Rimmer and explaining her significance in the story.

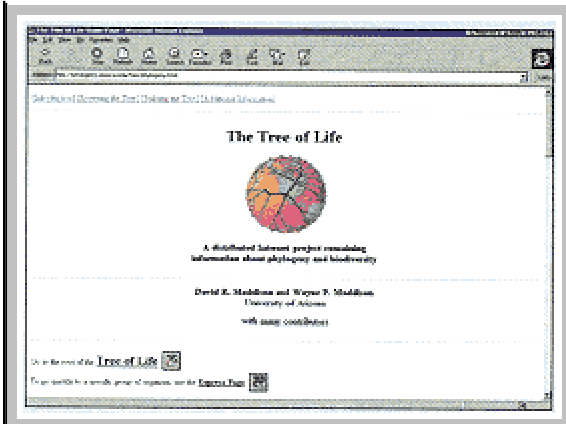
**3** A personality trait known as **novelty seeking** can be found by searching at this keyword site. As with anxiety-related traits, some of the genetic variation in novelty seeking has been traced to a genetic polymophism in a candidate gene. If assigned to do so, write a 200-word report about the candidate gene, specifying its function and why it seemed like a good candidate.

### MUTABLE SITE EXERECISES

The Mutable Site Exercise changes frequently. Each new update includes a different exercise that makes use of genetics resources available on the World Wide Web. Select the **Mutable Site** for Chapter 17, and you will be linked to the current exercise that relates to the material presented in this chapter.

### PIC SITE

The Pic Site showcases some of the most visually appealing genetics sites on the World Wide Web. To visit the showcase genetics site, select the **Pic Site** for Chapter 17.




cance of their original results. Other researchers, studying different pedigrees, observed weak but suggestive linkage with markers on chromosomes other than chromosome 11. Give at least two reasons why difficulties in genetic analysis might be expected with psychiatric traits such as bipolar affective disorder.

***Answer:*** First, psychiatric disorders are difficult phenotypes to identify and classify, and there is some subjectivity even when the examination is done by trained psychiatrists. Second, psychiatric disorders are complex, multifactorial traits that are likely to be influenced by several different genes. Even a single pedigree may have genetic variation at more than one locus, and the occurrence and severity of the disorder may depend on the presence of particular combinations of alleles of several genes. (Distinct genes influencing the trait may also explain why different pedigrees may indicate genetic linkage to markers in different chromosomes.) Third, environmental factors, such as personal experiences, diet, and alcohol or drug abuse may influence the expression or severity of the trait. Misclassification of the phenotype, multiple genes influencing the trait, and environmental effects on expression-each of these complicates the detection of genetic linkage.

**Analysis and Applications**

**17.1** Why are strains of *E. coli* with mutations that affect chemoreceptors multiply nonchemotactic rather than specifically nonchemotactic or generally nonchemotactic?

**17.2** CheW is necessary for autophosphorylation of CheA. Knowing this, what would you predict for the phenotype of a *cheW* deletion mutation?

**17.3** Extreme *cheB* mutations have a phenotype of continuous tumbling but swim in response to very strong attractant stimuli. Why? Extreme *cheR* mutations have a phenotype of continuous swimming but tumble in response to very strong repellent stimuli. Why?

**17.4** Why is phosphorylation of proteins important in bacterial chemotaxis? If the transfer of phosphate groups were inhibited, would the bacteria be specifically nonchemotactic, multiply nonchemotactic, or generally nonchemotactic? Explain.

**17.5** Are mutations in the *per* gene of *Drosophila* more likely to affect the mating success of males or females? Explain.

**17.6** How could you use an attached-X stock of *Drosophila* to produce a series of populations in each of which all of the males carry replicas of a single mutagen-treated X chromosome?

**17.7** In an experiment to obtain mutants affecting courtship behavior in *D. melanogaster,* several strains with abnormal male courtship songs were recovered. Most of the strains had wing defects that were probably responsible for the song abnormalities, but one strain was morphologically normal. In males of this strain, the song starts out normally but eventually becomes louder than the normal courtship song, with pulses much longer than wildtype. The circadian rhythm of these flies is also abnormal. How would you determine whether this strain of flies has a mutation in the *per* gene? Is it possible that the mutant gene in these flies is different from *per?* Explain.

**17.8** The mutation *transformer (tra)* in *Drosophila* affects the sexual phenotype of females (chromosomally XX), transforming the flies into phenotypic males. Although the males are sterile, their sexual behavior is that of normal males, and they readily court females and mate with them. What period of courtship song would you expect in each of the following genotypes?

**(a)** $per^{s}/per^{s}$; *tra*/*tra*

**(b)** $per^{L}/per^{L}$; *tra*/*tra*

**(c)** $per^{s}/per^{L}$; *tra*/*tra*

**17.9** Adults of *D. melanogaster* may be attracted to light of 480 nanometers (nm), which is colored blue to blue-green, or to light of 370 nm, which is ultraviolet, depending on the intensity of the light and the physiological state of the flies. Mutations in several different genes can eliminate either the response to 480-nm light or the response to 370-nm light. Describe these behavioral traits using categories of phototactic behavior analogous to the categories of chemotactic behavior described for *E. coli.* What types of molecules or cellular processes might be affected in the *Drosophila* mutants? (*Hint:* The perception of 370-nm light and that of 480-nm light make use of different visual pigments present in different photoreceptor cells.)

**17.10** If extremely bright light of 480 nm is flashed into the eyes of wildtype adults of *D. melanogaster*, subsequent attempts to elicit physiological or behavioral responses to this wavelength of light meet with no response for several seconds after the flash. Would you regard this phenomenon as an example of sensory adaptation? Explain.

**17.11** Suppose that the maze-learning experiments in rats described in the text were repeated using an inbred, genetically uniform strain of rats as the initial population. Would you expect to be able to select for maze-bright and maze-dull lines? Explain.

**17.12** The detection of odors by mammals is mediated by odorant-binding proteins, membrane-bound olfactory receptors, and a host of cellular proteins that work together to stimulate olfactory neurons to respond to specific chemicals in the environment. Mutations in genes for the odorantbinding proteins or the olfactory-receptor proteins might affect the sense of smell and may affect an animal's ability to detect food, potential mates, dangerous predators, and so forth. How might the olfactory system affect the performance of rats in a maze-learning experiment where the goal is to find food at the end of the maze? If the initial population were known to have heterogeneity in olfactory receptors, would maze learning be an effective measure of the rat's "intelligence"? How might you control for this type of variability in a maze-learning experiment?

**17.13** After several generations of selection for maze-learning ability in rats, the maze-bright rats were superior to mazedull rats in learning the maze when motivated by hunger (the condition of the selection). However, the maze-dull rats were superior to the maze-bright rats in learning the maze when motivated by escape from water. Is this an example of genotype-environment interaction? Explain.

**17.14** Although most behaviors are complex traits influenced by several genes, some behavioral differences among individuals can be ascribed to alleles of a single gene (for example, the *period* gene in *Drosophila).* Mice that are mutant for the *waltzer* gene run around in circles, shake their heads both horizontally and vertically, and are very irritable. Four matings among animals isolated from a small population containing *waltzer* yielded the following litters:

| | |
|---|---|
| wildtype × waltzer | 3 wildtype, 2 waltzer |
| wildtype × wildtype | 7 wildtype |
| wildtype × wildtype | 6 wildtype, 2 waltzer |
| wildtype × waltzer | 5 waltzer |

Judged on the basis of these data, do you think it is likely that the *waltzer* allele is dominant or recessive to the wildtype allele?

**17.15** The mammalian visual system is very complex, but the ability to detect colors is under relatively simple genetic control. The genes that are defective in color blindness code for a family of photoreceptor proteins known as opsins. Normal people have four opsin genes that code for four different photoreceptor proteins: an opsin (called rhodopsin) constituting the major visual pigment, an opsin that is sensitive to blue light, an opsin that is sensitive to green light, and an opsin that is sensitive to red light. Under what conditions might a simple genetic trait, such as color blindness, affect a person's behavior?

**17.16** Huntington disease is an autosomal-dominant neurodegenerative disease marked by motor disturbance, cognitive loss, and psychiatric disorders; it has a typical age of onset of between 30 and 40 years. The function of the wildtype gene product is not known. However, the gene shares an interesting feature with a number of other disease genes in human beings: It contains a region with several repeats of the trinucleotide CAG. Wildtype alleles of the gene typically have 11 to 34 CAG repeats; among mutant alleles, the number of CAG repeats ranges from 40 to 86. What molecular test could be used to detect the mutant Huntington disease allele in the child of a person who develops the disease at the age of 39? If the affected person's spouse is homozygous normal for the gene, what proportion of their children would be expected to be affected?

### Challenge Problem

**17.17** Recurring behavioral disorders were observed in some male members of a large pedigree extending over several generations. The males were mildly mentally retarded and, especially when under stress, were prone to repeated acts of aggression-including sex offenses, attempted murder, and arson. An X-linked gene, *MAO,* coding for the enzyme monoamine oxidase, which participates in the breakdown of neurotransmitters, was found to be defective in the affected men in this pedigree. Other researchers found abnormal levels of monoamine oxidase in some unrelated men with similar disorders, even though the *MAO* gene was not defective in these cases. Does this evidence support the hypothesis that defective monoamine oxidase is responsible for the behavioral disorder? Explain.

### Further Reading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## Answers to Chapter-End Problems

### *Chapter 1*

**1.1** It could be considered either way. On the one hand, the deficiency of the enzyme G6PD is hereditary; on the other hand, the disease itself has an environmental trigger (eating broad beans). Both answers are too simple, because the disease actually results from an interaction of a particular genetic constitution with a particular factor in the environment.

**1.2** Replication results in two daughter DNA duplexes, each identical in base sequence to the parental molecule (except for possible mutations in the sequence). Each of the daughter molecules contains one of the original intact parental strands.

**1.3** The messenger RNA carries the genetic information in DNA (in the form of a specific base sequence) to the ribosome. The ribosome is the physical structure in the cell on which a polypeptide chain is formed. The transfer RNA molecules are adapters that enable each codon in the messenger RNA to specify the presence of a particular amino acid at the corresponding position in the polypeptide chain. There is only one type of ribosome, but there are many different types of transfer RNA (typically, more than one for each amino acid).

**1.4** A mixture of heat-killed S cells and living R cells causes pneumonia in mice, but neither heat-killed S cells nor living R cells alone do so.

**1.5** The substance was destroyed by DNA-degrading enzymes but not by protein-degrading enzymes.

**1.6** The inside of the head is mainly DNA, and the rest of the phage is mainly protein. When a bacterial cell is infected, the head contents are transferred into the cell, but the outer "ghost" of the phage remains attached to the external surface of the cell.

**1.7** A small amount of protein was also transferred into the bacterial cell during infection, and some of this was even recovered in the progeny phage. A die-hard advocate of protein as the genetic material could claim that the transmitted protein, though small in amount, was critical in the hereditary process.

**1.8** Because the amount of A equals that of T, and the amount of G equals that of C, it can be inferred that the DNA in this bacteriophage is double-stranded; 56 percent of the base pairs are A–T pairs, and 44 percent are G-C pairs.

**1.9** Because, in this case, the amount of A does not equal that of T, and the amount of G does not equal that of C, it can be concluded that the DNA molecule is single-stranded, not double-stranded.

**1.10** Because A pairs with T, and G pairs with C, the base composition of the other strand must be 24 percent T, 28 percent A, 22 percent C, and 26 percent G.

**1.11** Yes, it is possible, because in some viruses the genetic material is RNA.

**1.12** Because of A–T and G-C base pairing, the complementary strand has the sequence

```
3'-TAGCATACGTGAAATGGGCC-5'
```

Note that the paired strands have opposite 5'-to-3' polarity.

**1.13** In this region the complementary strand has the sequence

```
3'-AGCAGCAGCAGCAGC-5'
```

**1.14** The probability that four particular bases have the sequence 5'-GGCC-3' is $(1/4)^4 = 1/256$, so 256 base pairs is the average spacing between consecutive occurrences; similarly, the average spacing for the sequence 5'-GAATTC-3' is 4096 base pairs.

**1.15** The corresponding region of RNA will contain no U (uracil), because U in RNA pairs with A in DNA.

**1.16** The top strand is the RNA because it contains U instead of T; the mismatched base pair is the sixth from the right.

**1.17** The initial part of the RNA transcript matches the DNA template starting at the tenth nucleotide. Hence the completed transcript has the sequence

```
5'-UAGCUACGAUCGCGUUGGA-3'
```

**1.18** The complementary sequence is 3'-UAUGCUAU-5'.

**1.19** The codon for phenylalanine must be 5'-UUU-3'.

**1.20** The codon for leucine must be 5'-UUA-3'.

**1.21** Three with *in vitro* translation:

```
5'-CGC/UUA/CCA/CAU/GUC/GCG/AAC/UCG-3'
5'-C/GCU/UAC/CAC/AUG/UCG/CGA/ACU/CG-3'
5'-CG/CUU/ACC/ACA/UGU/CGC/GAA/CUC/G-3'
```

With *in vivo* translation, there is only one

```
5'-CGCUUACCAC > AUG/UCG/CGA/ACU/CG-3'
```

where the symbol > means "start translation with next codon."

**1.22** Six. Either DNA strand could be transcribed, and each transcript could be translated in any one of three reading frames.

**1.23** The amino acids alternate, because with a triplet code, the codons alternate: 5'-UCU/CUC/UCU/CUC/UCU/CUC-3'. From this result we can conclude that Ser and Leu are encoded by 5'-UCU-3' and 5'-CUC-3'; we cannot specify which codon corresponds to which amino acid, because *in vitro* translation begins with either 5'-UCU-3' or 5'-CUC-3'.

**1.24** The result means that an mRNA is translated in nonoverlapping groups of three nucleotides: the genetic code is a triplet code.

**1.25** There are 64 possible codons and only 20 amino acids; hence some amino acids (most, in fact) are specified by two or more codons. A mutation that changes a codon for a particular amino acid into a synonymous codon for the same amino acid does not change the amino acid sequence.

***Chapter 2***

**2.1** The strain or variety must be homozygous for all genes that affect the trait.

**2.2** *Aa* yields *A* and *a* gametes, *Bb* yields *B* and *b* gametes, and *Aa Bb* yields *A B, A b, a B,* and *a b* gametes.

**2.3** Multiply the number of possible gametes formed for each gene (one for each homozygous gene and two for each heterozygous gene). In the case of *AA Bb Cc Dd Ee*, there are a total of 1 × 2 × 2 × 2 × 2 = 16 possible gametes. With *n* homozygous genes and *m* heterozygous genes, the number of possible gametes is $1^n \times 2^m$, which equals $2^m$.

**2.4** What Mendel apparently means is that the gametes consist of equal numbers of all possible combinations of the alleles present in the true-breeding parents; in other words, the cross *AA BB* × *aa bb* produces $F_1$ progeny of genotype *Aa Bb,* which yields the gametes *A B, A b, a B,* and *a b* in equal numbers. Segregation is illustrated by the 1 : 1 ratio of *A : a* and *B : b* gametes, and independent assortment is illustrated by the equal numbers of *A B, A b, a B,* and *a b* gametes.

**2.5** The round seeds in the $F_2$ generation consist of 1/3 *AA* and 2/3 *Aa.* Only the latter produce *a*-bearing pollen, so the fraction of *aa* seeds expected is 2/3 × 1/2 = 1/3.

**2.6** With dominance, there are two phenotypes, corresponding to the genotypes *RR* or *Rr* (dominant) and *rr* (recessive), in the ratio 3 : 1. With no dominance, there are three phenotypes, corresponding to the genotypes *RR, Rr,* and *rr,* in the ratio 1 : 2 : 1.

**2.7** The production of a homozygous genotype would require that the pollen and egg contain identical self-sterility alleles, but this is impossible because pollen cannot function on plants that contain the same self-sterility allele.

**2.8** 1/2; 1/2. The probability argument is that each birth is independent of all the previous ones, so the number of girls and boys already born has no influence on the sexes of future children.

**2.9** Because the probability of each child's being a girl is 1/2, and the two children are independent, the probability of two girls in a row is (1/2) × (1/2) = 1/4. The probability of a girl and a boy (not necessarily in that order) equals 2 × (1/4) × (1/4) = 1/2. The reason for the factor of 2 in this case is that the sexes may occur in either of two possible orders: girl-boy or boy-girl. Each of these has a probability of 1/4, so the total is 1/4 + 1/4 = 1/2.

**2.10 (a)** The parent with the dominant phenotype must carry one copy of the recessive allele and hence must be heterozygous. **(b)** Because some of the progeny are homozygous recessive, both parents must be heterozygous. **(c)** The parent with the dominant phenotype could be either homozygous dominant or heterozygous; the occurrence of no homozygous recessive offspring encourages the suspicion that the parent may be homozygous dominant, but because there are only two offspring, heterozygosity cannot be ruled out.

**2.11** An *Ab* gamete can be formed only if the parent is *AA Bb*, which has probability 1/2, and in this case, 1/2 of the gametes are *Ab;* overall, the probability of an *Ab* gamete is (1/2) × (1/2) = 1/4. *AB* gametes derive from *AA BB* parents with probability 1 and from *AA Bb* parents with probability 1/2; overall, the probability of an *AB* gamete is (1/2) × 1 + (1/2) × (1/2) = 3/4. (Alternatively, the probability of an *AB* gamete may be calculated as 1 - 1/4 = 3/4, because *Ab* and *AB* are the only possibilities.)

**2.12 (a)** Two phenotypic classes are expected for the *A*, *a* pair of alleles (*A*- and *aa),* two for the *B, b* pair (*B*- and *bb),* and three for the *R*, *r* pair (*RR*, *Rr*, and *rr),* yielding a total number of phenotypic classes of 2 × 2 × 3 = 12. **(b)** The probability of an *aa bb* RR offspring is 1/4 (*aa)* × 1/4 (*bb)* × 1/4 (*RR)* = 1/64. **(c)** Homozygosity may occur for either allele of each of the three genes, yielding such combinations as *AA BB rr*, *aa bb RR*, *AA bb rr*, and so forth. Because the probability of homozygosity for either allele is 1/2 for each gene, the proportion expected to be homozygous for all three genes is (1/2) × (1/2) × (1/2) = 1/8.

**2.13** The probability of a heterozygous genotype for any one of the genes is 1/2, and so for all four together, the probability is $(1/2)^4$= 1/16.

**2.14** Because both parents have solid coats but produce some spotted offspring, they must be *Ss.* With respect to the *A*, *a* pair of alleles, the female parent (tan) is *aa*, and because there are some tan offspring, the genotype of the black male parent must be *Aa.* Thus the parental genotypes are *Ss aa* (solid tan female) and *Ss Aa* (solid black male).

**2.15** Because one of the children is deaf (genotype *dd*, in which *d* is the recessive allele), both parents must be heterozygous *Dd.* The progeny genotypes expected from the mating *Dd* × *Dd* are 1/4 *DD*, 2/4 *Dd*, and 1/4 *dd.* The son is not deaf and hence cannot be *dd.* Among the nondeaf offspring, the genotypes *DD* and *Dd* are in the proportions 1 : 2, so their relative probabilities are 1/3 *DD* and 2/3 *Dd.* Therefore, the probability that the normal son is heterozygous is 2/3.

**2.16 (a)** Because the trait is rare, it is reasonable to assume that the affected father is heterozygous *HD*/*hd*, where *hd* represents the normal allele. Half of his gametes contain the *HD* allele, so the probability is 1/2 that the son received the allele and will later develop the disorder. **(b)** We do not know whether the son is heterozygous *HD*/*hd*, but the probability is 1/2 that he is; if he is heterozygous, half of his gametes will contain the *HD* allele. Therefore, the overall probability that his child has the *HD* allele is (1/2) × (1/2) = 1/4.

**2.17** Both parents must be heterozygous (*Aa)* because each had an albino (*aa)* parent. Therefore, the probability of an albino child is 1/4, and the probability of two homozygous recessive children is (1/4) × (1/4) = 1/16. The probability that at least one child is an albino is the probability that exactly one is albino plus the probability that both are albinos. The probability that exactly one is an albino is 2 × (3/4) × (1/4) = 6/16, in which the factor of 2 comes from the fact that there may be two birth orders: normal-albino or albino-normal. Therefore, the overall probability of at least one albino child equals 1/16 + 6/16 = 7/16. Alternatively, the probability of at least one albino child may be calculated as 1 minus the probability that both are nonalbino, or $1 - (3/4)^2 = 7/16$.

**2.18** Compatible transfusions are A donor with A or AB recipient, B donor with B or AB recipient, AB donor with AB recipient, and O donor with any recipient.

**2.19 (a)** The cross *RR BB* × *rr bb* yields $F_1$ progeny of genotype *Rr Bb*, which have red kernels. **(b)** Because there is independent segregation (independent assortment), the $F_2$ genotypes are *R-B-*, *R-bb*, *rr B-*, and *rr bb.* These genotypes are in the proportions 9 : 3 : 3 : 1, and they produce kernels that are red, brown, brown, and white, respectively. Therefore, the $F_2$ plants have red, brown, or white seeds in the proportions 9/16 : 6/16 : 1/16.

**2.20** Because the genes segregate independently, they can be considered separately. For the *Cr*, *cr* pair of alleles, the genotypes of the zygotes are *Cp Cp. Cp cp*, and *cp cp* in the proportions 1/4, 1/2, and 1/4, respectively. However, the *Cp Cp* zygotes do not survive, so among the survivors the genotypes are *Cp cp* (creeper) and *cp cp* (noncreeper), in the proportions 2/3 and 1/3, respectively. For the *W*, *w* pair of alleles, the progeny genotypes are W- (white) and *ww* (yellow), in the proportions 3/4 and 1/4, respectively. Altogether, the phenotypes and proportions of the surviving offspring can be obtained by multiplying (2/3 creeper + 1/3 noncreeper) × (3/4 white + 1/4 yellow), which yields 6/12 creeper, white + 2/12 creeper, yellow + 3/12 noncreeper, white + 1/12 noncreeper, yellow. Note that the proportions sum to 12/12 = 1, which serves as a check.

**2.21** Colored offspring must have the genotype *C- ii;* otherwise, color could not be produced (as in *cc)* or would be suppressed (as in *I-*). Among the $F_2$ progeny, 3/4 of the offspring have the *C-* genotype, and 1/4 of the offspring have the *ii* genotype. Because the genes undergo independent assortment, the overall expected proportion of colored offspring is (3/4) × (1/4) = 3/16.

**2.22** The 9 : 3 : 3 : 1 ratio is that of the genotypes *A- B-*, *A- bb*, *aa B-*, and *aa bb.* The modified 9 : 7 ratio implies that all of the last three genotypes have the same phenotype. The $F_1$ testcross is between the genotypes *Aa Bb* × *aa bb*, and the progeny are expected in the proportions 1/4 *Aa Bb*, 1/4 *Aa bb*, 1/4 *aa Bb*, and 1/4 *aa bb.* Again, the last three genotypes have the same phenotype, so the ratio of phenotypes among progeny of the testcross is 1 : 3.

**2.23** For the child to be affected, both III-1 and III-2 must be heterozygous *Aa.* For III-1 to be *Aa*, individual II-2 must be *Aa* and must transmit the recessive allele to III-1. The probability that II-2 is *Aa* equals 2/3, and then the probability of transmitting the *a* allele to III-1 is 1/2. Altogether, the probability that III-1 is a carrier equals (2/3) × (1/2) = 1/3. This is also the probability that III-2 is a carrier. Given that both III-1 and III-2 have genotype *Aa*, the probability of an *aa* offspring is 1/4. Overall, the probability that both III-1 and III-2 are carriers and that they have an *aa* child is (1/3) × (1/3) × (1/4) = 1/36. The numbers are multiplied because each of the events is independent. (The reason for the 2/3 is as follows: Because II-3 is affected, the genotypes of I-1 and I-2 must be *Aa.* Among nonaffected individuals from this mating (individuals II-2 and II-4), the ratios of *AA* and *Aa* are 1/4 : 2/4, so the probability of *Aa* is 2/3.)

**2.24** For any individual offspring, the probabilities of black *B-* and white *bb* are 3/4 and 1/4, respectively, **(a)** The order white-black-white occurs with probability (1/4) × (3/4) × (1/4) = 3/64. The order black-white-black occurs with probability (3/4) × (1/4) × (3/4) = 9/64. Therefore, the probability of either the order white-black-white or the order black-white-black equals 3/64 + 9/64 = 3/16. The probabilities are summed because the events are mutually exclusive. **(b)** The probability of exactly two white and one black equals $3 \times (1/4)^2 \times (3/4)^1 = 9/64$. The factor of 3 is the number of birth orders of two white and one black (there are three, because the black offspring must be the first, second, or third), and the rest of the expression is the probability that any one of the birth orders occurs (in this case, 3/64).

**2.25** The probability of all boys is $(1/2)^4 = 1/16$. The probability of all girls is also 1/16, so the probability of either all boys or all girls equals 1/16 + 1/16 = 1/8. The probability of equal numbers of the two sexes is $6 \times (1/2)^4 = 3/8$. The factor of 6 is the number of possible birth orders of two boys and two girls (BBGG, BGBG, BGGB, GGBB, GBGB, and GBBG).

**2.26** Black and splashed white are the homozygous genotypes and slate blue the heterozygote. The probabilities of each of these phenotypes from the mating between two heterozygotes are 1/4 black, 1/2 slate blue, and 1/4 splashed white. The probability of occurrence of the particular birth order black, slate blue, and splashed white is (1/4) × (1/2) × (1/4) = 1/32, but altogether there are six different orders in which exactly one of each type could be produced. (The black offspring could be first, second, or third, and in each case, the remaining phenotypes could occur in either of two orders.) Consequently, the overall probability of one of each phenotype, in any order, is 6 × (1/32) = 3/16.

**2.27** The probability that one or more offspring are *aa* equals 1 minus the probability that all are *A*-. In the mating *Aa* × *Aa*, the probability that all of *n* offspring will be *Aa* equals $(3/4)^n$, and the question asks for the value of *n* such that $1 - (3/4)^n \geq 0.95$. Solving this inequality yields $n \geq \log(1 - 0.95)/\log(3/4) = 10.4$. Therefore, 11 is the smallest number of offspring for which there is a greater than 95 percent chance that at least one will be *aa*. As a check, note that $(3/4)^{10} = 0.056$ and $(3/4)^{11} = 0.042$, so with 10 offspring, the chance of one or more *aa* equals 0.944, and with 11 offspring, the chance of one or more *aa* equals 0.958.

**2.28** The ratio of probabilities that the sire is *AA : Aa* is 1/3 : 2/3. The ratio of the probabilities of producing *n* pups, all *A*-, for *AA : Aa* sires is $1 : (1/2)^n$. Hence, for *n A*- pups, the ratio of probabilities of *AA : Aa* sires is $1/3 : (2/3)(1/2)^n$. Therefore, the probability that the sire is *AA*, given that he had a litter of *n A*- pups, equals $(1/3)/[(1/3) + (2/3)(1/2)^n]$. This is the degree of confidence you can have that the sire is *AA*. For *n* = 1 to 15, the answer is shown in the accompanying table. It is interesting that 6 progeny are required for 95 percent confidence and 8 for 99 percent confidence.

| | | | | | |
|---|---|---|---|---|---|
| 1 | 0.5000 | 6 | 0.9697 | 11 | 0.9990 |
| 2 | 0.6667 | 7 | 0.9846 | 12 | 0.9995 |
| 3 | 0.8000 | 8 | 0.9922 | 13 | 0.9998 |
| 4 | 0.8889 | 9 | 0.9961 | 14 | 0.9999 |
| 5 | 0.9412 | 10 | 0.9981 | 15 | 0.9999 |

**2.29** Make a Punnett square with the ratio of *D : d* along each margin as 3/4 : 1/4. **(a)** The progeny genotypes *DD, Dd,* and *dd* therefore occur in the proportions 9/16 : 6/16 : 1/16. **(b)** If *D* is dominant, the ratio of dominant: recessive phenotypes is 15 : 1. **(c)** Among the *D*- genotypes, the ratio *DD : Dd* is 9 : 6. **(d)** If meiotic drive happens in only one sex, then the Punnett square has 3/4 : 1/4 along one margin and 1/2 : 1/2 along the other. The progeny genotypes *DD, Dd,* and *dd* are in the ratio 3/8 : 1/2 : 1/8. The ratio of dominant: recessive phenotypes is 7 : 1. Among *D*-genotypes, the ratio *DD : Dd* is 3 : 4.

***Chapter 3***

**3.1** Chromosome replication takes place in the S period, which is in the interphase stage of mitosis and the interphase I stage of meiosis. (Note that chromosome replication does *not* take place in interphase II of meiosis.) Chromosomes condense and first become visible in the light microscope in prophase of mitosis and prophase I of meiosis.

**3.2** Each chromosome present in telophase of mitosis is identical to one of a pair of chromatids present in the preceding metaphase that were held together at the centromere. In this example, there are 23 pairs of chromosomes present after telophase, which implies 23 × 2 = 46 chromosomes altogether. Therefore, at the preceding metaphase, there were 46 × 2 = 92 chromatids.

**3.3** In leptotene, the chromosomes first become visible as *thin threads.* In zygotene, they become *paired threads* because of synapsis. The chromosomes continue to condense to become *thick threads* in pachytene. In diplotene, the sister chromatids become evident, and each chromosome is seen to be a *paired thread* consisting of the sister chromatids. (Early signs of the bipartite nature of each chromosome are evident in pachytene.) In diakinesis, the homologous chromosomes repulse each other and *move apart;* they would fall apart entirely were they not held together by the chiasmata.

**3.4** The number of chromosomes. Each centromere defines a single chromosome; two chromatids are considered part of a single chromosome as long as they share a single centromere. As soon as the centromere splits in anaphase, each chromatid is considered a chromosome in its own right. Meiosis I reduces the chromosome number from diploid to haploid because the daughter nuclei contain the haploid number of chromosomes (each chromosome consisting of a single centromere connecting two chromatids.) In meiosis II, the centromeres split, so the numbers of chromosomes are kept equal.

**3.5 (a)** 20 chromosomes and 40 chromatids; **(b)** 20 chromosomes and 40 chromatids; **(c)** 10 chromosomes and 20 chromatids.

**3.6 (a)** For each centromere, there are two possibilities (for example, A or a); so altogether there are $2^7 = 128$ possible gametes. **(b)** The probability of a gamete's receiving a particular centromere designated by a capital letter is 1/2 for each centromere, and each centromere segregates independently of the others; hence the probability for all seven simultaneously is $(1/2)^7 = 1/128$.

**3.7** The wheat parent produces gametes with 14 chromosomes and the rye parent gametes with 7, so the hybrid plants have 21 chromosomes.

**3.8** The crisscross occurs because the X chromosome in a male is transmitted only to his daughters. The expression is misleading because any X chromosome in a female can be transmitted to a son or a daughter.

**3.9** Because there is an affected son, the phenotypically normal mother must be heterozygous. This implies that the daughter has a probability of 1/2 of being heterozygous.

**3.10** The *Bar* mutation is an X-linked dominant. Because the sexes are affected unequally, some association with the sex chromosomes is suggested. Mating (a) provides an important clue. Because a male receives his X chromosome from his mother, the wildtype phenotype of the sons suggests that the *Bar* gene is on the X chromosome. The fact that all daughters are affected is also consistent with X-linkage, provided that the *Bar* mutation is dominant. Mating (b) confirms the hypothesis, because the females from mating (a) would have the genotype *Bar*/+ and so would produce the indicated progeny.

**3.11 (a)** Cross is *v*/Y × +/+, in which Y represents the Y chromosome; progeny are 1/2 +/Y males (wildtype) and 1/2 *v*/+ females (phenotypically wildtype, but heterozygous. **(b)** Mating is *v*/*v* × +/Y; progeny are 1/2 *v*/Y males (vermilion) and 1/2 *v*/+ females (phenotypically wildtype, but heterozygous). **(c)** Mating is *v*/+ × +/Y; progeny are 1/4 *v*/+ females (phenotypically wildtype), 1/4 ++/+ females (wildtype), 1/4 *v*/Y males (vermilion), and 1/4 +/Y males (wildtype). **(d)** Mating is *v*/+ × *v*/ Y; progeny are 1/4 *v*/*v* females (vermilion), 1/4 *v*/+ females (wildtype), 1/4 *v*/Y males (vermilion), and 1/4 +/Y males (wildtype).

**3.12** All the females will be $v/v^+$; *bw*/*bw* and will have brown eyes; all the males will be *v*/Y; *bw*/*bw* and will have white eyes.

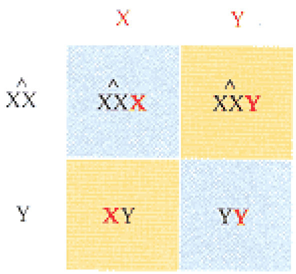
**3.13** Because the sex-chromosome situation is the reverse of that in mammals, female chickens are ZW and males ZZ, and females receive their Z chromosome from their father. Therefore, the answer is to mate a gold male (*ss)* with a silver female (*S).* The male progeny will all be *Ss* (silver plumage), the female progeny will be *s* (gold plumage), and these are easily distinguished.

**3.14 (a)** The mother is a heterozygous carrier of the mutation, so the probability that a daughter is a carrier is 1/2. The probability that both daughters are carriers is $(1/2)^2 = 1/4$. **(b)** If the daughter is not heterozygous, the probability of an affected son is 0, and if the daughter is heterozygous, the probability of an affected son is 1/2; the overall probability is therefore 1/2 (that is, the chance that the daughter is a carrier) × 1/2 (that is, the probability of an affected son if the daughter is a carrier) = 1/4.

**3.15** Let *A* and *a* represent the normal and mutant X-linked alleles. Y represents the Y chromosome. The genotypes are as follows: I-1 is *Aa* (because there is an affected son), I-2 is *A*Y, II-1 is *A*Y II-2 is *a*Y, II-3 is *Aa* (because there is an affected son), II-4 is *A*Y, and III-1 is *a*Y.

**3.16** 20, because in females there are 5 homozygous and 10 heterozygous genotypes, and in males there are 5 hemizygous genotypes.

**3.17** The accompanying Punnett square shows the outcome of the cross. The X and Y chromosomes from the male are denoted in red, the attached-X chromosomes (yoked Xs) and Y from the female in black. The red X chromosome also contains the *w* mutation. The zygotes in the upper left and lower right corners (shaded) fail to survive because they have three X chromosomes or none. The surviving progeny consist of white-eyed males (XY) and red-eyed females (XXY) in equal proportions. Attached-X inheritance differs from the typical situation in that the sons, rather than the daughters, receive their fathers' X chromosome.




**3.18** The genotype of the female is *Cc,* where *c* denotes the allele for color blindness. In meiosis, the *c*-bearing chromatids undergo nondisjunction and produce an XX-bearing egg of genotype *cc.* Fertilization by a normal Y-bearing sperm results in an XXY zygote with genotype *cc,* which results in color blindness.

**3.19** The female produces 1/2 X-bearing eggs and 1/2 O-bearing eggs ("O" means no X chromosome), and the male produces 1/2 X-bearing sperm and 1/2 Y-bearing sperm. Random combinations result in 1/4 XX, 1/4 XO, 1/4 XY, and 1/4 YO, and the last class dies because no X chromosome is present. Among the survivors, the expected progeny are 1/3 XX females, 1/3 XO females, and 1/3 XY males.

**3.20 (a)** II-2 must be heterozygous, because she has an affected son, which means that half of her sons will be affected. The probability of a nonaffected child is therefore 3/4, and the probability of two nonaffected children is $(3/4)^2 = 9/16$. **(b)** Because the mother of II-4 is heterozygous, the probability that II-4 is heterozygous is 1/2. If she is heterozygous, half of her sons will be affected. Therefore, the overall probability of an affected child is $1/2 \times 1/4 = 1/8$.

**3.21 (a)** This question is like asking for the probability of the sex distribution GGGBBB (in that order), which equals $(1/2)^6 = 1/64$. **(b)** This question is like asking for the probability of either sex distribution GGGBBB or BBBGGG, which equals $2(1/2)^6 = 1/32$.

**3.22** With 1 : 1 segregation, the expected numbers are 125 in each class, so the $X^2$ value equals $(140 - 125)^2/125 + (110 - 125)^2/125 = 3.6$, and there is 1 degree of freedom because there are two classes of data. The associated *P* value is approximately 6 percent, which means that a fit at least as bad would be expected 6 percent of the time even with Mendelian segregation. Therefore, on the basis of these data, there is no justification for rejecting the hypothesis of 1 : 1 segregation. (On the other hand, the observed *P* value is close to 5 percent, which is the conventional level for rejecting the hypothesis, so the result should not inspire great confidence.)

**3.23** The total number of progeny equals 800, and on the assumption of a 9 : 3 : 3 : 1 ratio, the expected numbers are 450, 150, 150, and 50, respectively.

The $X^2$ equals $(462 - 450)^2/450 + (167 - 150)^2/150 + (127 - 150)^2\ 150 + (44 - 50)^2/50 = 6.49$ and there are 3 degrees of freedom (because there are four classes of data). The *P* value is approximately 0.09, which is well above the conventional rejection level of 0.05. Consequently, the observed numbers provide no basis for rejecting the hypothesis of a 9 : 3 : 3 : 1 ratio.

***Chapter 4***

**4.1** Because the progeny are wildtype, the mutations complement each other, and both $m_1$ and $m_2$ must be heterozygous. This indicates that the genotype of the progeny is $m_1$ +/+ $m_{2,}$ which implies that $m_1$ and $m_2$ are mutations in different genes.

**4.2** **(a)** *A b, a B, A B,* and *a b.* **(b)** *A b* and *a B.*

**4.3** The nonrecombinant gametes are *A B* and *a b,* and the recombinant gametes are *A b* and *a B.* The nonrecombinant gametes are always more frequent than the recombinant gametes.

**4.4** The gametes from the *A B*/*a b* parent are *A B* and *a b;* those from the other parent are *A b* and *a B.* The offspring genotypes are expected to be *A B*/*A b, A B*/*a B, a b*/*A b,* and *a b*/*a B,* in equal numbers. The phenotypes are *A- B-, A- bb,* and *aa B-,* which are expected in the ratio 2 : 1 : 1.

**4.5** The *cis* configuration is *a b*/*A B.*

**4.6** There is no crossing-over in male *Drosophila,* so the gametes will be *A B* and *a b* in equal proportions. In the female, each of the nonrecombinant gametes *A B* and *a b* is expected with a frequency of (1 - 0.05)/2 = 0.475, and each of the recombinant gametes *A b* and *a B* is expected with a frequency of 0.05/2 = 0.025.

**4.7** **(a)** 17; **(b)** 1; **(c)** in diploids there is one linkage group per pair of homologous chromosomes, so the number of linkage groups is 42/2 =21.

**4.8** Recombination at a rate of 6.2 percent implies that double crossovers (and other multiple crossovers) occur at a negligible frequency; in such a case, the distance in map units equals the frequency of recombination, so the map distance between the genes is 6.2 map units.

**4.9** Each meiotic cell produces four products: namely, *A B* and *a b,* which contain the chromatids that do not participate in the crossover, and *A b* and *a B,* which contain the chromatids that do take part in the crossover. Therefore, the frequency of recombinant gametes is 50 percent. Because one crossover occurs in the region between the genes in every cell, the occurrence of multiple crossing-over does not affect the recombination frequency as long as the chromatids participating in each crossover are chosen at random.

**4.10** The *aa bb* genotype requires an *a b* gamete from each parent. From the *A b*/*a B* parent, the probability of an *a b* gamete is 0.16 / 2 = 0.08, and from the *A B*/*a b* parent, the probability of an *a b* gamete is (1 - 0.16)/2 = 0.42; therefore, the probability of an *aa bb* progeny is 0.08 × 0.42 = 0.034, or 3.4 percent.

**4.11** All the female gametes are *bz m.* The male gametes are $bz^+M$ with frequency (1 - 0.06)/2 = 47 percent, *bz m* with frequency 47 percent, $bz^+m$ with frequency 0.06/2 = 3 percent, and *bz M* with frequency 3 percent. The progeny are therefore black males ($bz^+$ *M*/*bz m)* with frequency 47 percent, bronze females (*bz m*/*bz m)* with frequency 47 percent, black females ($bz^+$ *m*/*bz m)* with frequency 3 percent, and bronze males (*bz M*/*bz m)* with frequency 3 percent).

**4.12** The most frequent gametes from the double heterozygous parent are the nonrecombinant gametes, in this case *Gl ra* and *gl Ra.* The genotype of the parent was therefore *Gl ra*/*gl Ra.* The recombinant gametes are *Gl Ra* and *gl ra,* and the frequency of recombination is calculated as the ratio (6 + 3)/(88 + 103 + 6 + 3) = 4.5 percent.

**4.13** There is no recombination in the male, so the male gametes are ++ and *b cn,* each with a probability of 0.5. A map distance of 8 units means that the recombination frequency between the genes is 0.08. The female gametes are ++, *b cn* [the nonrecombinants, each of which occurs at a frequency of (1 - 0.08)/2 = 0.46] and + *cn* and *b* + [the recombinants, each of which occurs at a frequency of 0.08/2 = 0.04]. When the male and female gametes are combined at random, their frequencies are multiplied, and the progeny and their frequencies are as follows:

| | | | |
|---|---|---|---|
| ++/++ | 0.23 | ++/*b cn* | 0.23 |
| *b cn*/++ | 0.23 | *b cn*/*b cn* | 0.23 |
| *b*+/++ | 0.02 | *b*+/*b cn* | 0.02 |
| +*cn*/++ | 0.02 | +*cn*/*b cn* | 0.02 |

Note that *b cn*/+ + is the same as + + *b cn*, so the overall frequency of this genotype is 0.46. With regard to phenotypes, the progeny are wildtype (73 percent), black cinnabar (23 percent), black (2 percent), and cinnabar (2 percent).

**4.14** The most frequent classes of gametes are the nonrecombinants, and so the parental genotype was *A B c*/*a b C.* The least frequent gametes are the double recombinants, and the allele that distinguishes them from the nonrecombinants (in this case, *C* and *c)* is in the middle. Therefore, the gene order is *A*-*C*-*B.*

**4.15** Comparison of the first three numbers implies that *r* lies between *c* and *p*, and the last two numbers imply that *s* is closer to *r* than to *c.* The genetic map is therefore *c*-10-*r*-3-*p*-5-*s.* From this map, you would expect the recombination frequency between *c* and *p* to be 13 percent and that between *c* and *s* to be 18 percent. The observed values are a little smaller because of double crossovers.

**4.16 (a)** The nonrecombinant gametes (the most frequent classes) are *y* $v^+$ *sn* and $y^+$ *v* $sn^+$, and the double recombinants (the least-frequent classes) indicate that *sn* is in the middle. Therefore, the correct order of the genes is *y-sn-v,* and the parental genotype was *y sn* $v^+$/$y^+sn^+$ *v.* The total progeny is 1000. The recombination frequency in the *y-sn* interval is (108 + 5 + 95 + 3)/1000 = 21.1 percent, and the recombination frequency in the *sn-v* interval is (53 + 5 + 63 + 3)/1000 = 12.4 percent. The genetic map is *y*-21.1-*sn*-12.4-*v.* **(b)** On the assumption of independence, the expected frequency of double crossovers is 0.211 × 0.124 = 0.0262, whereas the observed frequency is (5 + 3)/1000 = 0.008. The coincidence is the ratio 0.008/0.0262 = 0.31, so the interference is 1 - 0.31 = 0.69, or 69 percent.

**4.17** The nonrecombinant gametes are *c wx Sh* and *C Wx sh,* and the double recombinants indicate that *sh* is in the middle. The parental genotype was therefore *c Sh wx*/*C sh Wx.* The frequency of recombination in the *c-sh* region is (84 + 20 + 99 + 15)/6708 = 3.25 percent, and that in the *sh-wx* region is (974 + 20 + 951 + 15)/6708 = 29.2 percent. The genetic map is therefore *c*-3.25-*sh*-29.2-*wx.*

**4.18** Let *x* be the observed frequency of double crossovers. Then single crossovers in the *a-b* interval occur with frequency 0.15 - *x,* and single crossovers in the *b-c* interval occur with frequency 0.20 - *x.* The observed recombination frequency between *a* and *c* equals the sum of the single-crossover frequencies (because double crossovers are undetected), which implies that 0.15 - *x* + 0.20 - *x* = 0.31, or *x* = 0.02. The expected frequency of double crossovers equals 0.15 × 0.20 = 0.03, and the coincidence is therefore 0.02/0.03 = 0.67. The interference equals 1 - 0.67 = 33 percent.

**4.19** *b* and *st* are unlinked, as are *hk* and *st.* The evidence is the 1 : 1 : 1 : 1 segregation observed. For *b* and *st,* the comparisons are 243 +10 (black, scarlet) versus 241 + 15 (black, nonscarlet) versus 226 +12 (nonblack, scarlet) versus 235 + 18 (nonblack, nonscarlet). For *hk* and *st,* the comparisons are 226 + 10 (hook, scarlet) versus 235 + 15 (hook, nonscarlet) versus 243 + 12 (nonhook, scarlet) versus 241 + 18 (nonhook, nonscarlet). On the other hand, *b* and *hk* are linked, and the frequency of recombination between them is (15 + 10 + 12 + 18)/1000 = 5.5 percent.

**4.20** The frequency of *dpy-21 unc-34* recombinants is expected to be 0.24/2 = 0.12 among both eggs and sperm, so the expected frequency of *dpy-21 unc-34* / *dpy-21 unc-34* zygotes is $0.12^2$ = 1.44 percent.

**4.21** The genes must be linked, because with independent assortment, the frequency of dark eyes (*R- P-*) would be 9/16 in both experiments. The first experiment is a mating of *R P*/*r p* × *r p r p.* The dark-eyed progeny represent half the nonrecombinants, so the recombination frequency *r* can be estimated as (1 - r)/2 = 628/(628 + 889), or *r* = 0.172. The second experiment is a mating of *R p*/*r P* × *r p*/*r p.* In this case, the dark-eyed progeny represent half the recombinants, so *r* can be estimated as *r*/2 = 86/(86 + 771), or *r* = 0.201. It is not uncommon for repeated experiments or crosses carried out in different ways to yield somewhat different estimates of the recombination frequency—in this problem, 0.17 and 0.20. The most reliable value is obtained by averaging the experimental values—in this case the average is (0.171 + 0.201)/2 = 0.186.

**4.22** Consider row 1: there are - entries for 3 and 7, and so 1, 3, and 7 form one complementation group. Row 2: all +, and so 2 is the only representative of a second complementation group. Row 3: mutation already classified. Row 4: there is a - for 6 only and no other - in column 4, and so 4 and 6 make up a third complementation group. Row 5: - only for 9 and no other - in column 5, and so 5 and 9 form a fourth complementation group. Rows 6 and 7: mutations already classified. Row 8: all +, and so 8 is the only member of a fifth complementation group. In summary, there are five complementation groups as follows: group 1 (mutations 1, 3, and 7); group 2 (mutation 2); group 3 (mutations 4 and 6); group 4 (mutations 5 and 9); group 5 (mutation 8).

**4.23** Consider each gene in relation to first- and second-division segregation. Gene *a* gives 1766 asci with first-division segregation and 234 asci with second-division segregation; the frequency of second-division segregation is 234/(1766 + 234) = 0.117, which implies that the distance between *a* and the centromere is 0.117/2 = 5.85 percent. Gene *b* gives 1780 first-division and 220 second-division segregations, for a frequency of second-division segregation of 220/(1780 + 220) = 0.110. The distance between *b* and the centromere is therefore 0.110/2 = 5.50 map units. If we consider *a* and *b* together, there are 1986 PD asci, 14 TT asci, and no NPD. Because NPD << PD, genes *a* and *b* are linked. The frequency of recombination between *a* and *b* is [(1/2) × 14]/2000 = 0.35 percent. Comparing this distance with the gene-centromere distances calculated earlier results in the map *a*-0.35-*b*-5.50—centromere.

**4.24 (a)** On the assumption of independence, (0.30 × 0.10)/2 = 0.015. (Division by 2 is necessary because *sh wx gl* gametes represent only one of the two classes of double recombinants.) **(b)** If the interference is 60 percent, then the coincidence equals 1 - 0.60 = 0.40. The observed frequency of doubles is therefore 40 percent as great as with independence, so the expected frequency of *sh wx gl* gametes is 0.015 × 0.40 = 0.006.

**4.25** Considered individually, gene *c* yields 90 first-division segregation asci and 10 (1 + 9) second-division segregation asci, so the frequency of recombination between *c* and the centromere is (10/100) × 1/2 = 5 percent. Gene *v* yields 79 first-division and 21 (20 + 1) second-division asci, so the frequency of recombination between *v* and the centromere is (21/100) × 1/2 = 10.5 percent. If the genes are taken together, the asci include 34 PD, 36 NPD, and 30 (20 + 1 + 9) TT. Because NPD = PD, the genes are unlinked.

In summary, *c* and *v* are in different chromosomes, *c* being 5 map units from the centromere and *v* being 10.5 map units from the centromere.

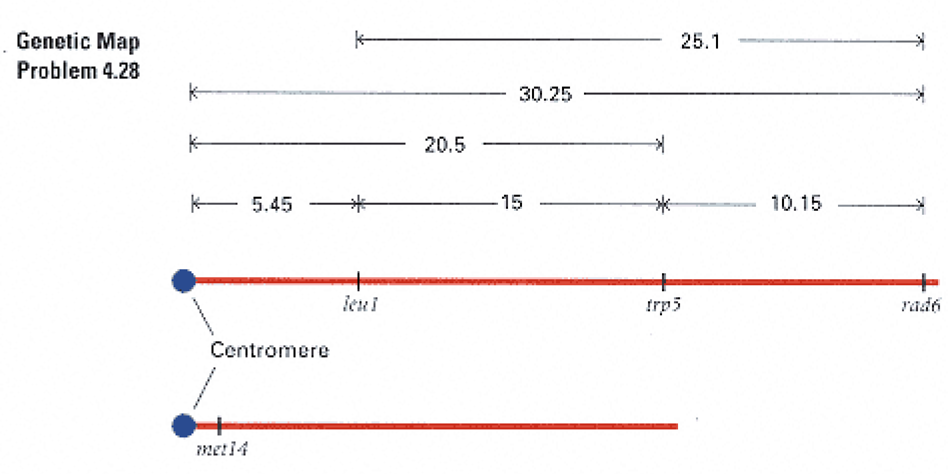
**4.26** Analysis of cross 1. Ascospore arrangements 1, 2, 4, and 5 result when there is a crossover between the gene and the centromere. Types 3 and 6 result when there is no crossing-over between the gene and the centromere. The map distance between the gene and its centromere is (1/2) (Frequency of second-division segregation) × 100 = (1/2) [(6 + 6 + 6 + 6)/120] × 100 = 10 centimorgans. Analysis of cross 2. There is no crossing-over between the gene and its centromere. Analysis of cross 3. The proportion of second-division segregation (asci types 1, 2, 4, and 5) equals 80/120 = 67 percent. This distribution of asci results when crossing-over is frequent enough to randomize the position of alleles with respect to their centromeres. The gene is said to be "unlinked" with its centromere.

**4.27** For each pair of genes, classify the tetrads as PD, NPD, or TT, and tabulate the results as follows:

| | |
|---|---|
| *leu2-trp1* | PD 230 + 235 = 465 |
| | NPD 215 + 220 = 435 |
| | TT 54 + 46 = 100 |
| *leu2-met14* | PD 235 + 220 = 455 |
| | NPD 230 + 215 = 445 |
| | TT 54 + 46 = 100 |
| *trp1-met14* | PD 235 + 215 + 46 = 496 |
| | NPD 230 + 220 + 54 = 504 |
| | TT 0 |

For *leu2-trp1*, PD = NPD; therefore, *leu2* and *trp1* are unlinked. Similarly, for *leu2-met14*, PD = NPD; these genes are also unlinked. Likewise, *trp1* and *met14* are unlinked because PD = NPD. However, for the *trp1-met14* gene pair, there are no tetratypes, which means that each gene is closely linked to its own centromere. Because they do not recombine with their centromeres, the segregation of *trp1* and *met14* can be used as genetic markers of centromere segregation; their presence marks the sister spores created by segregation in the first meiotic division. Using the *trp1* and *met14* markers in this fashion allows the remaining *leu2* gene to be mapped with respect to its centromere; all tetratype tetrads will result from crossing-over between *leu2* and its centromere because neither of the other two genes recombines with its centromere. The map distance between *leu2* and its centromere is therefore (1/2) (100/1000) × 100 = 5 centimorgans.

**4.28** For the *rad6-trp5* gene pair, PD = 188 + 206 = 140 + 154 + 154 + 109 = 797, NPD = 0, and TT = 105 + 92 + 3 + 2 + 1 = 203. Because PD >> NPD, the genes are linked. In this case, TT > 0 and NPD = 0, so the map distance is calculated as (1/2)(203/1000) × 100 = 10.15 centimorgans. For the *trp5-leu1* gene pair, PD = 188 + 206 + 105 + 92 + 109 = 700, NPD = 0, and TT = 140 + 154 + 3 + 2 + 1 = 300; here again, PD >> NPD, TT > 0, and NPD = 0, so the map distance is calculated as (1/2)(300/1000) × 100 = 15 centimorgans. For the *rad6-leu1* gene pair, PD = 188 + 206 + 109 = 503, NPD = 1, and TT = 105 + 92 + 140 + 154 + 3 + 2 = 496. In this case, PD >> NPD, but TT > 0 and NPD > 0. The map distance is calculated as (1/2)[(496 + 6 × 1)/1000] × 100 = 25.1 centimorgans. For the *leu1 - me14* gene pair, PD = 188 + 105 + 140 + 2 = 435, NPD = 206 + 92 +154 + 3 + 1 = 456, and TT = 109. In this case PD = NPD, so *leu1* is not linked to *met14.* However, we know from the previous problem that *met14* is tightly linked with its centromere, and this information can be used to calculate the map distance between *leu1* and its centromere. Because TT = 109 and also TT < 2/3, the map distance between *leu1* and its centromere is (1/2)(109/1000) × 100 = 5.45 centimorgans. For the *rad6-me14* gene pair, PD = 188 + 1 = 189, NPD = 206, and TT = 105 + 92 + 140 + 154 + 109 + 3 + 2 = 605. Here again, PD = NPD, which indicates that there is no linkage between the genes. Because *met14* is tightly linked to its centromere, and TT = 605 but TT < 2/3, the map distance between *rad6* and its centromere is (1/2)(605/1000) × 100 = 30.25 centimorgans. For the *trp5* - *met14* gene pair, PD = 188 + 105 = 293, NPD = 206 + 92 = 298, and TT = 140 + 154 + 109 + 3 + 2 + 1 = 409. Once again, PD = NPD, indicating no linkage between the genes. As before, we use *met14* as a centromere marker to calculate the distance between *trp5* and its centromere as (1/2) (409/1000) × 100 = 20.5 centimorgans. The overall conclusion is that *rad6*, *trp5*, and *leu1* are all linked to each other; *met14* is tightly linked to its centromere but not to the other three genes. The genetic map based on these data is shown below:



***Chapter 5***

**5.1** The complementary strand is 5'-T-C-C-G-A-G-3'.

**5.2** A nuclease is an enzyme capable of breaking a phosphodiester bond. An endonuclease can break any phosphodiester bond, whereas an exonuclease can only remove a terminal nucleotide.

**5.3** Movement along the template strand is from the 3' end to the 5' end, because consecutive nucleotides are added to the 3' end of the growing chain. In double-stranded DNA, one strand is replicated continuously, and the other strand is replicated in short segments that are later joined.

**5.4** DNA polymerase joins a 5'-triphosphate with a 3'-OH group, and the outermost two phosphates are released; the resulting phosphodiester bond contains one phosphate. DNA ligase joins a single 5'-P group with a 3'-OH group.

**5.5** The enzyme has polymerizing activity, it is a 3' $\rightarrow$ 5' exonuclease (the proofreading function), and it is a 5' $\rightarrow$ 3' exonuclease.

**5.6** Neither RNA polymerases nor primases require a primer to get synthesis started.

**5.7** Smaller molecules move faster because they can penetrate the pores of the gel more easily.

**5.8 (a)** In rolling-circle replication, one parental strand remains in the circular part and the other is at the terminus of the branch. Therefore, only half of the parental radioactivity will appear in progeny. **(b)** If we are assuming no recombination between progeny DNA molecules, one progeny phage will be radioactive. If a single recombination occurs, two particles will be radioactive. If recombination is frequent and occurs at random, the radioactivity will be distributed among virtually all the progeny.

**5.9** Rolling-circle replication must be initiated by a single-stranded break; θ replication does not need such a break.

**5.10 (a)** Because 18 percent is adenine, 18 percent is also thymine; so [T] = 18 percent. **(b)** Because [A] + [T] = 36 percent, [G] + [C] equals 64 percent, or [G] = [C] = 32 percent each. Overall, the base composition is [A] = 18 percent, [T] = 18 percent, [G] = 32 percent, and [C] = 32 percent; the [G] + [C] content is 64 percent.

**5.11 (a)** The distance between nucleotide pairs is 3.4 Å, or 0.34 nm. The length of the molecule is 34 μ, or $34 \times 10^3$ nm. Therefore, the number of nucleotide pairs equals $(34 \times 10^3)/0.34 = 10^5$. **(b)** There are ten nucleotide pairs per turn of the helix, so the total number of turns equals $10^5/10 = 10^4$.

**5.12 (a)** First, note that the convention for writing polynucleotide sequences is to put the 5' end at the left. Thus the sequence complementary to AGTC (which is 5'-AGTC-3') is 3'-TCAG-5', written in conventional format as GACT (that is, 5'-GACT-3'). Hence the frequency of CT is the same as that of its complement AG, which is given as 0.15. Similarly, AC = 0.03, TC = 0.08, and AA = 0.10. **(b)** If DNA had a parallel structure, then the 5' ends of the strands would be together, as would the 3' ends, so the sequence complementary to 5'-AGTC-3' would be 5'-TCAG-3'. Thus AG = 0.15 implies that TC = 0.15, and similarly, the other observed nearest neighbors would imply that CA = 0.03, CT = 0.08, and AA = 0.10.

**5.13** Remember that the chemical group at the growing end of the leading strand is always a 3'-OH group, because DNA polymerases can add nucleotides only to such a group. Because DNA is antiparallel, the opposite end of the leading strand is a 5'-P group. Therefore, **(a)** 3'-OH; **(b)** 3'-OH; **(c)** 5'-P.

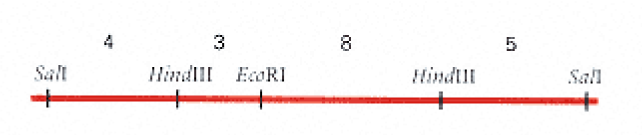
**5.14** Rolling-circle replication begins with a single-strand break that produces a 3'-OH group and a 5'-P group. The polymerase extends the 3'-OH terminus and displaces the 5'-P end. Because the DNA strands are antiparallel, the strand complementary to the displaced strand must be terminated by a 3'-OH group.

**5.15 (a)** The first base in the DNA that is copied is a T, so the first base in the RNA is an A. RNA grows by addition to the 3'-OH group, so the 5'-P end remains free. Therefore, the sequence of the eight-nucleotide primer is

```
5'-AGUCAUGC-3'
```

**(b)** The C at the 3' end has a free -OH; the A at the 5' end has a free *tri*phosphate. **(c)** Right to left, because this would require discontinuous replication of the strand shown.

**5.16 (a)** The cloned *Sal*I fragment is 20 kb in length, and there are no *Sal*I sites within it. **(b)** There is a single *Eco*RI site 7 kb from one end. **(c)** There are two *Hin*dIII sites within the fragment. **(d)** The *Hin*dIII sites must flank the *Eco*RI site, dividing the 7-kb *Eco*RI fragment into 3-kb and 4-kb subfragments and the 13-kb *Eco*RI fragment into 5-kb and 8-kb subfragments. The 3-kb and 8-kb fragments must be adjacent, because digestion with *Sal*I and *Hin*dIII alone produces an 11 -kb fragment. Except for the left-to-right orientation, the inferred restriction map must be as follows:




**5.17** The bands in the gel result from incomplete chains whose synthesis was terminated by incorporation of the dideoxynucleotide indicated at the top. The smaller fragments are at the bottom of the gel, the larger ones at the top. Because DNA strands elongate by addition to their 3' ends, the strand synthesized in the sequencing reactions has the sequence, from bottom to top, of

```
5'-ACTAGAGACCATGATCCTGTGATGAATAGC-3'
```

The template strand is complementary in sequence, and antiparallel, so its sequence is

```
3'-TGATCTCTGGTACTAGGACACTACTTATCG-5'
```

**5.18** For each site, the probability equals the product of the probability of each of the nucleotide pairs in turn, 1/4 for any of the specified pairs and 1/2 for R-Y. The average spacing is the reciprocal of the probability. Hence for *TaqI*, the average distance between sites is $4^4 = 256$ base pairs; for *Bam*HI, it is $4^6 = 4096$ base pairs; and for *Hae*II, it is $4^4$ $2^2 = 1024$ base pairs.

***Chapter 6***

**6.1** The 1.3-kb insertion is probably a transposable element. (It is a transposable element, called *mariner.)*

**6.2** Transposition in somatic cells could cause mutations that kill the host or decrease the ability of the host to reproduce, which reduces the chance of transmission of the transposable element to the next generation. Restriction of activity to the germ line lessens these potentially unfavorable effects.

**6.3** Insertion is initiated by a staggered cut in the host DNA sequence, and the overhanging single-stranded ends are later filled in by a DNA polymerase. This process creates a direct repeat.

**6.4** *E. coli:* 4,700,000 bp × 3.4 Å/bp × $10^{-7}$ mm/Å = 1.6 mm. Human beings: $3 \times 10^9$ bp × 3.4 Å/bp × $10^{-7}$ mm/Å = 1020 mm; in other words, there is approximately 1 *meter* of DNA in a human gamete.

**6.5** DNA structure is a long, thin thread, and its total length in most cells greatly exceeds the diameter of a nucleus, even allowing for a large number of fragments per cell. This forces you to conclude that each DNA molecule must be folded back on itself repeatedly.

**6.6** Matching of chromomeres, which are locally folded regions of chromatin. Polytene chromosomes do not divide because the cells in which they exist do not divide.

**6.7** No, because the *E. coli* chromosome is circular. There are no termini.

**6.8** The repeated sequences are located at the ends of the element. They may be in direct or inverted orientation, depending on the particular transposable element.

**6.9 (a)** There are ten base pairs for every turn of the double helix. With four turns of unwinding, 4 × 10 = 40 base pairs are broken, **(b)** Each twist compensates for the underwinding of one full turn of the helix; hence, there will be four twists.

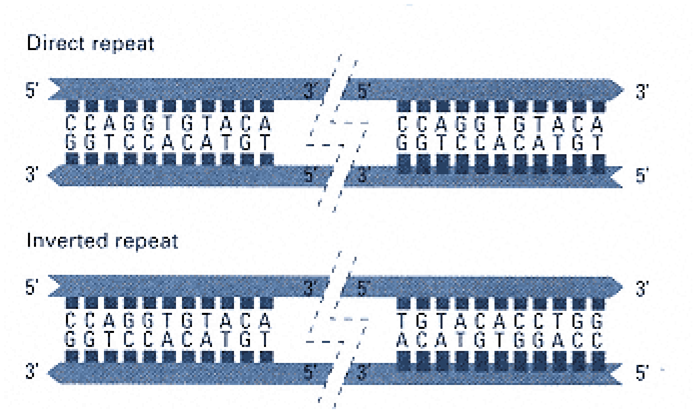
**6.10** The supercoiled form is in equilibrium with an underwound form having many unpaired bases. At the instant that a segment is single-stranded, S1 can attack it. Because a nicked circle lacks the strain of supercoiling, there is no tendency to have single-stranded regions, and so a nicked circle is resistant to S1 cleavage.

**6.11** The rate of movement in the gel is determined by the overall charge and by the ability of the molecule to penetrate the pores of the gel. Supercoiled and relaxed circles migrate at different rates because they have different conformations.

**6.12 (a)** Renaturation is concentration-dependent because it requires complementary molecules to collide by chance before the formation of base pairs can occur. **(b)** Here, the lengths of the GC tracts are important. Molecule 2 has a long GC segment, which will be the last region to separate. Hence molecule 1 has the lower temperature for strand separation.

**6.13** The hybrid molecules that can be produced from single-stranded DNA fragments from species A and B are AA, AB, BA, and BB, and these are expected in equal amounts. The molecules with a hybrid ($^{14}N/^{15}N$) density are AB and BA, and these account for 5 percent of the total. The same sequences can also renature as AA or BB which together must make up another 5 percent, so a total of 10 percent of the base sequences are common to the two species.

**6.14** In a direct repeat, the sequence is repeated in the same 5'-to-3' orientation. In an inverted repeat, the sequence is repeated in the reverse orientation and on the opposite strand, which is necessary to preserve the correct 5'-to-3' polarity in view of the antiparallel nature of DNA strands. The accompanying figure shows a direct and inverted repeat of the example sequence. The dashed lines represent unspecified DNA sequences between the repeats.




**6.15** All such mutations are probably lethal. The fact that the amino acid sequence of histones is virtually the same in all organisms suggests that functional histones are extremely intolerant of amino acid changes.

**6.16** The relatively large number of bands and the diverse locations in the genome among different flies suggests very strongly that the DNA sequence is a transposable element of some kind.

**6.17** In the first experiment, the nucleosomes form at random positions in different molecules. In the second experiment, the protein binds to a unique sequence in all the molecules, and the nucleosomes form by sequential addition of histone octamers on both sides of the protein-binding site. Because nuclease attacks only in the linker regions, the cuts are made in small regions that are highly localized.

***Chapter 7***

**7.1 (a)** The Klinefelter karyotype is 47,XXY; hence one Barr body. **(b)** The Turner karyotype is 45,X; hence no Barr bodies. **(c)** The Down syndrome karyotype is 47,+21; people with this condition have one or zero Barr bodies, depending on whether they are female (XX) or male (XY). **(d)** Males with the karyotype 47,XYY have no Barr bodies. **(e)** Females with the karyotype 47,XXX have two Barr bodies.

**7.2** An autopolyploid series is formed by the combination of identical sets of chromosomes. Therefore, the chromosome numbers must be even multiples of the monoploid number, or 10 (diploid), 20 (tetraploid), 30, 40, and 50.

**7.3** Species S is an allotetraploid of A and B formed by hybridization of A and B, after which the chromosomes in the hybrid became duplicated. The univalent chromosomes in the S × A cross are the 12 chromosomes from B, and the univalent chromosomes in the S × B cross are the 14 chromosomes from A.

**7.4** The chromosomes underwent replication with no cell division (endoreduplication), resulting in an autotetraploid.

**7.5** The only 45-chromosome karyotype found at appreciable frequencies in spontaneous abortions is 45,X, and so 45,X is the probable karyotype. Had the fetus survived, it would have been a 45,X female with Turner syndrome.

**7.6** Because the X chromosome in the 45,X daughter contains the color-blindness allele, the 45,X daughter must have received the X chromosome from her father through a normal X-bearing sperm. The nondisjunction must therefore have occurred in the mother, resulting in an egg cell lacking an X chromosome.

**7.7** The mother has a Robertsonian translocation that includes chromosome 21. The child with Down syndrome has 46 chromosomes, including two copies of the normal chromosome 21 plus an additional copy attached to another chromosome (the Robertsonian translocation). This differs from the usual situation, in which Down syndrome children have trisomy 21 (karyotype 47,+21).

**7.8** The inversion has the sequence *A B E D C F G,* the deletion *A B F G.* The possible translocated chromosomes are **(a)** *A B C D E T U V* and *M N 0 P Q R S F G* or **(b)***A B C D E S R Q P O N M* and *V U T F G.* One of these possibilities includes two monocentric chromosomes; the other includes a dicentric and an acentric. Only the translocation with two monocentrics is genetically stable.

**7.9** The order is *a e d f c b* or the other way around.

**7.10** Genes *b, a, c, e, d,* and *f* are located in bands 1, 2, 3, 4, 5, and 6, respectively. The reasoning is as follows. Because deletion 1 uncovers any gene in band 1 but the other deletions do not, the pattern -++++ observed for gene *b* puts *b* in band 1. Deletions 1–3 uncover band 2 but deletions 4–5 do not, so the pattern ---++ observed for gene *a* localizes gene *a* to band 2. Genes in band 3 are uncovered by deletion 1–4 but not deletion 5, so the pattern ----+ implies that gene *c* is in band 3. Band 4 is uncovered by deletions 3–5 but not deletions 1–2, which means that gene *e,* with pattern ++---, is in band 4. Genes in band 5 are uncovered by deletions 4–5 but not deletions 1–3, so the pattern +++-- observed for gene *d* places it in band 5. Genes in band 6 are uncovered by deletion 5 but not deletions 1–4, so the pattern ++++- puts gene *f* in band 6.

**7.11** The most probable explanation is an inversion. In the original strain the inversion is homozygous, so no problems arise in meiosis. The $F_1$ is heterozygous, and crossing-over within the inversion produces the dicentrics and acentrics. Because the crossover products are not recovered, the frequency of recombination is greatly reduced in the chromosome pair in which the inversion is heterozygous, so the inversion is probably in chromosome 6.

**7.12** Starting with chromosome (c), compare the chromosomes pairwise to find those that differ by a single inversion. In this manner, you can deduce that inversion of the i-d-c region in (c) gave rise to (d), inversion of the h-g-c-d region in (d) gave rise to (a), and inversion of the c-d-e-f region in (a) gave rise to (b). Because you were told that (c) is the ancestral sequence, the evolutionary ancestry is (c) → (d) → (a) → (b).

**7.13** The group of four synapsed chromosomes implies that a reciprocal translocation has taken place. The group of four includes both parts of the reciprocal translocation and their nontranslocated homologs.

**7.14** Translocation heterozygotes are semisterile because adjacent segregation from the four synapsed chromosomes in meiosis produces aneuploid gametes, which have large parts of chromosomes missing or present in excess. The aneuploid gametes fail to function in plants, and in animals they result in zygote lethality. The translocation homozygote is fully fertile because the translocated chromosomes undergo synapsis in pairs, and segregation is completely regular. In the cross of translocation homozygote × normal homozygote, all progeny are expected to be semisterile.

**7.15** In this male, there was a reciprocal translocation between the *Curly*-bearing chromosome and the Y chromosome. Because there is no crossing-over in the male, all Y-bearing sperm that give rise to viable progeny contain *Cy*, and all **X-**bearing sperm that give rise to viable progeny contain $y^+$.

**7.16** In this male, the tip of the X chromosome containing $y^+$ became attached to the Y chromosome, giving the genotype *y* **X**/$y^+$ Y. The gametes are *y* X and $y^+$ Y, so all of the offspring are either yellow females or wildtype males.

**7.17** The *A B c D E* chromosome results from two-strand double crossing-over in the *B*-*C* and *C*-*D* regions. The double recombinant chromosome still has the inversion.

**7.18** The unusual yeast strain is not completely haploid but is disomic for chromosome 2. Segregation of markers on this chromosome, like *his*7, is aberrant, but segregation of markers on the other chromosomes is completely normal.

**7.19** The parental classes are wildtype and *brachytic, fine-stripe* and the recombinant classes are *brachytic, fine-stripe*$^+$ and *brachytic*$^+$, *fine-stripe.* The recombination frequency between *brachytic* and *fine-stripe* is therefore equal to (17 + 1 + 6 + 8)/682 = 0.047.

Because semisterility is associated with the presence of the translocation, determining the recombination frequency between each of the mutations and the translocation breakpoint follows the same logic as mapping any other kind of genetic marker. For the parental classes, the mutations and the translocation are on different chromosomes. Members of the recombinant classes are those plants that show both the mutation in question and semisterility or neither semisterility nor the mutation in question. The recombination frequency between *brachytic* and the translocation breakpoint is therefore (17 + 25 + 19 + 8)/682 = 0.102; the recombination frequency between *fine*

*-stripe* and the translocation breakpoint is (19 + 6 + 1 + 25)/682 = 0.075.

***Chapter 8***

**8.1** Minimal medium containing galactose and biotin; the galactose would require the presence of *gal*$^+$, and the biotin would allow both *bio*$^+$ and *bio*$^-$ to grow.

**8.2** The *E. coli* genome contains approximately 4700 kilobase pairs, which implies approximately 47 kb per minute. Because the λ genome is 50 kb, the genetic length of the prophage is approximately 1 minute (more precisely, 47/50 = 0.94 minutes).

**8.3** Infect an *E. coli* λ lysogen with P1. The λ lysogen contains λ prophage, so some of the resulting P1 phage will include the part of the chromosome that contains the prophage.

**8.4** After circularization and integration at the bacterial *att* site, the map is . . . *att D E F A B C att* . . ..

**8.5** The specialized-transducing particles originate from aberrant prophage excision.

**8.6** Plate on minimal medium that lacks leucine (selects for Leu$^+$) but contains streptomycin (selects for Str-r). The *leu*$^+$ allele is the selected marker, and *str-r* the counterselected marker.

**8.7** One plaque per phage. One plaque, because the plating was done before lysis, so the progeny phage are confined to one tiny area.

**8.8** The T2 plaques will be turbid because T2 fails to lyse the resistant bacteria. Plaques made by the T2*h* mutant will be clear because the mutant can lyse both the normal and the resistant cells.

**8.9** The mean number of phage per cell equals 1. If you know the Poisson distribution, then it is apparent that the proportion of uninfected cells is $e^{-1}$ = 0.37. If you do not know the Poisson distribution, then the answer can be calculated as follows. The probability that a bacterial cell escapes infection by a particular phage is $1 - 10^{-6}$ = 0.999999, and the probability that it escapes infection by all one million phages is $P_0 = (0.999999)^{1000000}$. Therefore, In $P_0 = 1000000 \times \ln(0.999999) = -1$, so $P_0 = e^{-1} = 0.37$.

**8.10** Apparently, *h* and *tet* are closely linked, so recombinants containing the *h*$^+$ allele of the Hfr tend also to contain the *tet-s* allele of the Hfr, and these recombinants are eliminated by the counterselection for *tet-r.*

**8.11** All receive the *a*$^+$ allele, but whether or not a particular *b*$^+$ *str-r* recombinant contains *a*$^+$ depends on the positions of the genetic exchanges.

**8.12** Depending on the size of the F', it could be F' *g*, F' *g h*, F' *g h i,* and so forth, or F' *f*, F' *f e*, F' *f e d,* and so forth.

**8.13** The first selection is for Met$^+$ and so *lac*$^+$ must have been transferred. The probability that a marker that is transferred is incorporated into the recombinants is about 50 percent, so about 50 percent of the recombinants would be expected to be *lac*$^+$. The second selection is for Lac$^+$, and *met*$^+$ is a late marker. The great majority of mating pairs will spontaneously break apart before transfer of *met*$^+$, so the frequency of *met*$^+$ recombinants in this experiment will be close to zero.

**8.14** The order 500 *his*$^+$, 250 *leu*$^+$, 50 *trp*$^+$ implies that the genes are transferred in the order *his leu trp.* The *met* mutation is the counterselected marker that prevents growth of the Hfr parent. The medium containing histidine selects for *leu*$^+$ *trp*$^+$, and the number is small because both genes must be incorporated by recombination.

**8.15** The three possible orders are (1) *pur-pro-his,* (2) *pur-his-pro,* and (3) *pro-pur-his.* The predictions of the three orders follow. (1) Virtually all *pur*$^+$ *his*$^-$ transductants should be *pro*$^-$, but this is not true. (2) Virtually all *pur*$^+$*pro*$^-$ transductants should be *his*$^-$, but this is not true. (3) Some *pur*$^+$ *pro*$^-$ transductants will be *his*$^-$, and some *pur*$^+$ *his*$^-$ transductants will be *pro*$^-$ (depending on where the exchanges occur). Order (3) is the only one that is not contradicted by the data.

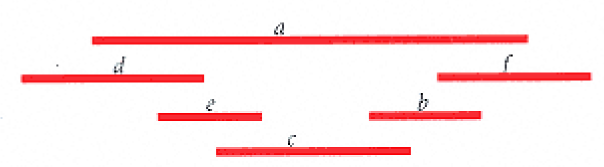
**8.16 (a)** Use $m = 2 - 2d^{1/3}$, in which *m* is map distance in minutes and *d* is cotransduction frequency. With the values of *d* given, the map distances are 0.74, 0.41, and 0.18 minutes, respectively. **(b)** Use $d = [1 - (m/2)]^3$. With the values of *m* given, cotransduction frequencies are 0.42, 0.12, and 0, respectively. For markers greater than 2 minutes apart, the frequency of cotransduction equals zero. **(c)** Map distance *a-b* equals 0.66 minutes, *b-c* equals 1.07 minutes, and these are additive, giving 1.73 minutes for the distance *a-c.* Predicted cotransduction frequency for this map distance is 0.002.

**8.17** It is probable that the *amp-r* gene in the natural isolate was present in a transposable element. Transposition into λ occurs in a small proportion of cases, and when these infect the laboratory strain, the transposable element can transpose into the chromosome before the infecting λ is lost.

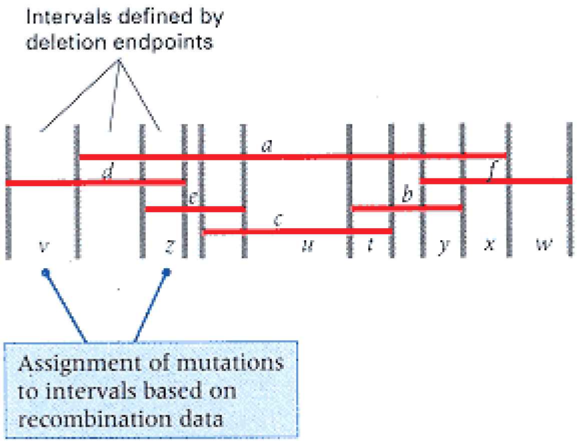
**8.18** The *tet-r* gene in the phage was included in a transposable element. Transposition occurred in the lysogen to another location in the *E. coli* chromosome, and when the prophage was lost, the antibiotic resistance remained.

**8.19** Both the order of times of entry and the level of the plateaus imply that the order of gene transfer is *a b c d*. The times of entry are obtained by plotting the number of recombinants of each type versus time and extrapolating back to the time axis. The values are *a*, 10 minutes; *b*, 15 minutes; *c*, 20 minutes; *d*, 30 minutes. The low plateau value for the *d* gene probably results from close linkage with *str-s*, because in this case, the *d* allele in the Hfr strain would tend to be inherited along with the *str-r* allele.

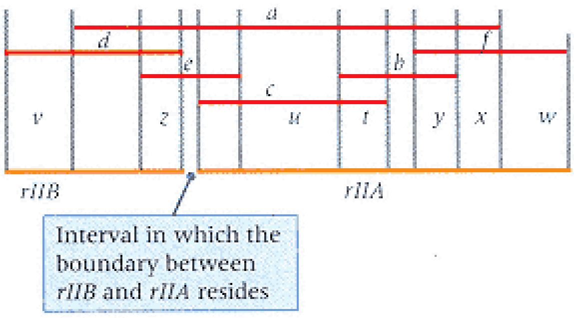
**8.20** The deletion map is shown in the accompanying illustration. A completely equivalent map can be drawn with the order of the genes reversed.




**8.21** The genetic map shows the genetic intervals defined by the deletion endpoints and the locations of the mutations with respect to these intervals.




**8.22** The mutations *v* and *z* are in the *rIIB* cistron, whereas the mutations *t*, *u*, *w*, *x*, and *y* are in the *rIIA* cistron. As indicated in the genetic map, the boundary between the cistrons lies between the left end of the *c* deletion and the right end of the *e* deletion.




The matrix for complementation among the deletions must be as follows:

| | a | b | c | d | e | f |
|---|---|---|---|---|---|---|
| a | – | – | – | – | – | – |
| b | | – | – | + | – | – |
| c | | | – | + | – | – |
| d | | | | – | – | + |
| e | | | | | – | – |
| f | | | | | | – |

**8.23** The λ prophage itself is being transduced as part of the chromosome. It maps next to *gal*. Hence when phage 363 is grown on strain D, 6 percent of the $Gal^+$ transductants are lysogenic, whereas none of the $Thr^+$ or $Lac^+$ transductants is lysogenic. The reciprocal cross gives the same basic result, only here λ is being removed from the chromosome and being replaced by phage-free DNA. Hence about 8 percent of the $Gal^+$ transductants are nonlysogenic. As expected from the close linkage of the phage λ attachment site and *gal*, no linkage is seen for the other two markers.

***Chapter 9***

**9.1** The DNA content per average band equals about $1.1 \times 10^8/5000 = 22$ kb, that per average lettered subdivision equals approximately $1.1 \times 10^8/600 = 183$ kb, and that per average numbered section equals about $1.1 \times 10^8/100 = 1.1$ Mb. A YAC with a 200-kb insert contains the DNA equivalent to about 9 salivary bands, 1.1 lettered subdivisions, and 0.2 numbered section. An 80-kb PI clone contains the DNA equivalent of about 4 salivary bands, 0.48 lettered subdivision, and 0.07 numbered section.

**9.2** The ends may be blunt or they may be cohesive (sticky) with either a 3' overhang or a 5' overhang.

**9.3** All restriction fragments have the same ends, because the restriction sites are all the same and they are cleaved in the same places. Opposite ends of the same fragment must also be identical, because restriction sites are palindromes.

**9.4** No. The complement of 5'-GGCC-3' is 3'-CCGG-5'.

**9.5** The insertion disrupts the gene, and sensitivity to the antibiotic shows that the vector has an inserted DNA sequence. The second antibiotic-resistance gene is needed to select transformants.

**9.6 (a)** The probability of a *Taq*I site is 1/6 × 1/3 × 1/3 × 1/6 = 1/324, so *Taq*I cleavage is expected every 324 base pairs; the probability of a *Mae*III site is 1/3 × 1/6 × 1 × 1/6 × 1/3 = 1/324, so *Mae*III cleavage is expected every 324 base pairs. **(b)** The answers are the same as in part (a), because for both *Taq*I and *Mae*III sites, the number of A + T = the number of G + C.

**9.7** Comparison of the genomic and cDNA sequences tells you where the introns are in the genomic sequence. The genomic sequence contains the introns; the cDNA does not.

**9.8** Bacterial cells do not normally recognize eukaryotic promoters, and if transcription does occur, the transcript usually contains introns that the bacterial cell cannot remove. If these problems are overcome, the protein may still not be produced because the mRNA lacks a bacterial ribosome-binding site, because the protein might require post-translational processing, or because the protein might be unstable in bacterial cells and subject to degradation.

**9.9** The finding is that the 3.6-kb and 5.3-kb fragments in the free phage are joined to form an 8.9-kb fragment in the intracellular form. This suggests that the the 3.6-kb and 5.3-kb fragments are terminal fragments of a linear molecule and that the intracellular form is circular.

**9.10** A base substitution may destroy a restriction site and prevent two potential fragments from being separated, or a deletion may eliminate a restriction site.

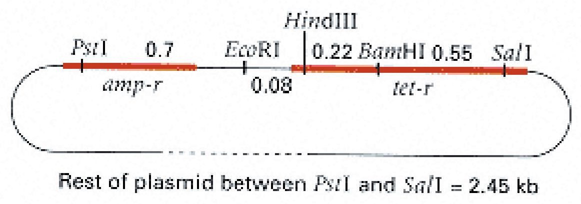
**9.11** The gene probably contains an *Eco*RI site that disrupts the gene in the process of cloning; the gene does not contain a *Hind*III site.

**9.12 (a)** The *tet-r* gene is not cleaved with *Bgl*I, so addition of tetracycline to the medium requires that the colonies be tetracycline-resistant (Tet-r) and hence contain the plasmid. **(b)** Cells with the phenotype Tet-r Kan-r or Tet-r Kan-s will form colonies. **(c)** Colonies with the phenotype Tet-r Kan-s contain inserts within the cleaved *kan* gene.

**9.13** An insertion of two bases is generated within the codon, resulting in a frameshift. Therefore, all colonies should be $Lac^-$.

**9.14** You must use cDNA under a suitable promoter that functions in the prokaryotic system. Eukaryotic genomic DNA includes regulatory elements that will not work in bacteria and introns that cannot be spliced out of RNA in prokaryotic cells.

**9.15** The restriction map is shown in the accompanying figure.




***Chapter 10***

**10.1** The start codon is AUG, which codes for methionine; the stop codons are UAA, UAG, and UGA. The probability of a start codon is 1/64, and the probability of a stop codon is 3/64. The average distance between stop codons is 64/3 = 21.3 codons.

**10.2** The mRNA sequence is

```
5'-AAAAUGCCCUUAAUCUCAGCGUCCUAC-3'
```

and translation begins with the first AUG, which codes for methionine. The amino acid sequence coded by this region is Met-Pro-Leu-Ile-Ser-Ala-Ser-Tyr.

**10.3** With a G at the 5' end, the first codon is GUU, which codes for valine; and with a G at the 3' end, the last codon is UUG, which codes for leucine.

**10.4** Codons are read from the 5' end of the mRNA, and the first amino acid in a polypeptide chain is at the amino terminus. If the G were at the 5' end, the first codon would be GAA, coding for glutamic acid at the amino terminus of the polypeptide. If the G were at the 3' end, the last codon would be AAG, coding for arginine at the carboxyl terminus. Thus the G was added at the 5' end.

**10.5** GUG codes for valine, and UGU codes for cysteine. Therefore, an alternating polypeptide containing only valine and cysteine would be made.

**10.6** The artificial mRNA can be read in any one of three reading frames, depending on where translation starts. The reading frames are

GUC GUC GUC . . ., UCG UCG UCG . . ., and CGU CGU CGU . . .

The first codes for polyvaline, the second for polyserine, and the third for polyarginine.

**10.7** The possible codons, their frequencies, and the amino acids coded are as follows:

| | | |
|---|---|---|
| AAA | $(3/4)^3 = 0.421$ | Lys |
| AAC | $(3/4)^2(1/4) = 0.141$ | Asn |
| ACA | $(3/4)^2(1/4) = 0.141$ | Thr |
| CAA | $(3/4)^2(1/4) = 0.141$ | Gln |
| CCA | $(3/4)(1/4)^2 = 0.141$ | Pro |
| CAC | $(3/4)(1/4)^2 = 0.047$ | His |
| ACC | $(3/4)(1/4)^2 = 0.047$ | Thr |
| CCC | $(1/4)^3 = 0.015$ | Pro |

Therefore, the amino acids in the random polymer and their frequencies are lysine (42.1 percent), asparagine (14.1 percent), threonine (18.8 percent), glutamine (14.1 percent), histidine (4.7 percent), and proline (6.2 percent).

**10.8** Methionine has 1 codon, histidine 2, and threonine 4, so the total number is 1 × 2 × 4 = 8. The sequence AUGCAYACN encompasses them all. Because arginine has six codons, the possible number of sequences coding for Met-Arg-Thr is 1 × 6 × 4 = 24. The sequences are AUGCGNACN (16 possibilities) and AUGAGRACN (8 possibilities).

**10.9** In an overlapping code, a single base change affects more than one codon, so changes in more than one amino acid should often occur (but not always, because single amino acid changes can result from redundancy in the code). For this reason, an overlapping code was eliminated from consideration as a likely possibility.

**10.10** Because the pairing of codon and anticodon is antiparallel, it is convenient to write the anticodon as 3'-UAI-5'. The first two codon positions are therefore 5'-AU-3', and the third is either A, U, or C (because each can pair with I). The possible codons are 5'-AUA-3', 5'-AUU-3', and 5'-AUC-3', all of which code for isoleucine.

**10.11** In theory, the codon 5'-UGG-3' (tryptophan) could pair with either 3'-ACC-5' or 3'-ACU-5'. However, 3'-ACU-5' can also pair with 5'-UGA-3', which is a chain-termination codon. If this anticodon were used for tryptophan, an amino acid would be inserted instead of terminating the chain. Therefore, the other anticodon (3'-ACC-5') is the one used.

**10.12 (a)** The deletion must have fused the amino-coding terminus of the *B* gene with the carboxyl-coding terminus of the *A* gene. The nontranscribed strand must therefore be oriented 5'-*B*-*A*-3'. **(b)** The number of bases deleted must be a multiple of 3; otherwise, the carboxyl terminus would not have the correct reading frame.

**10.13 (a)** The nontranscribed strand is given, so the wildtype reading frame of the mRNA is

AUG CAU CCG GGC UC**A** UUA GUC U...

which codes for Met-His-Pro-Gly-Ser-Leu-Val- . . . **(b)** The mutant X mRNA reading frame is

AUG GCA UCC GGG CUC **A**UU AGU CU...

which codes for Met-Ala-Ser-Gly-Leu-Ile-Ser-Leu- . . ..

**(c)** The mutant Y mRNA reading frame is

AUG CAU CCG GGC UCU UAG UCU...

which codes for Met-His-Pro-Gly-Ser (the UAG is a termination codon). **(d)** The double-mutant mRNA reading frame is

AUG GCA UCC GGG CUC UUA GUC U...

which codes for Met-Ala-Ser-Gly-Leu-Leu-Val- . . .. Note that the double mutant differs from wildtype only in the region between the insertion and deletion mutations, in which the reading frame is shifted.

**10.14** Mutation X can be explained by a single base change in the start codon for the first Met, in which case translation would begin with the second AUG codon. Mutation Y can be explained by a single base change that converts a codon for Tyr (UAU or UAC) into a chain-termination codon (UAA or UAG). The tripeptide sequence is Met-Leu-His.

**10.15** The probability of a correct amino acid at a particular site is 0.999, so the probability of all 300 amino acids being correctly translated is $(0.999)^{300} = 0.74$, or approximately 3/4.

**10.16** Each loop represents an intron, so the total number is seven introns.

**10.17** If transcribed from left to right, the transcribed strand is the bottom one, and the mRNA sequence is

AGA **CUU CAG GCU CAA CGU** GGU

which codes for Arg-Leu-Gln-Ala-Gln-Arg-Gly. To invert the molecule, the strands must be interchanged in order to preserve the correct polarity, and so the sequence of the inverted molecule is

```
5'-AGAACGTTGAGCCTGAAGGGT-3'
3'-TCTTGCAACTCGGACTTCCCA-5'
```

This codes for an mRNA with sequence

```
AGA ACG UUG AGC CUG AAG GGU
```

which codes for Arg-Thr-Leu-Ser-Leu-Lys-Gly.

**10.18** The DNA nontemplate strand is shown as sequence (1) in the table below. The corresponding mRMA for this region is shown as sequence (2). Translation in all three possible reading frames yields (in the single-letter amino acid codes) the sequences shown in (3).

**DNA nontemplate strand, mRNA region, and translation Problem 10.18**

| | |
|---|---|
| (1) | `5'-TAACGTATGCTTGACCTCCAAGCAATCGATGCCAGCTCAAGG-3'` |
| (2) | `5'-UAACGUAUGCUUGACCUCCAAGCAAUCGAUGCCAGCUCAAGG-3'` |
| (3) | `* R M L D L Q A I D A S S R`<br>`N V C L T S K Q S M P A Q`<br>`T Y A * P P S N R C Q L K` |

The sequence is from the middle of an exon, so you do not know the reading frame. However, two of the possible reading frames contain stop codons (*). Therefore, the only sequence without stop codons must be correct, because you know that the sequence comes from the middle of an exon of an active gene. Thus the correct amino acid sequence of the polypeptide is N V C L T S K Q S M P A Q.

**10.19** Unlike other 6-fold degenerate amino acids (Arg, Leu), is not possible to change between Ser codons with a single mutation. In particular, to get from the 4-fold degenerate class (UCN, where N is any of four bases) to the 2-fold degenerate class (AGY, where Y is any pyrimidine) requires a change in both the first and the second codon positions. It is highly unlikely that two point mutations would occur simultaneously in the same codon, so an organism would have temporarily had an amino acid other than Ser when changing from one codon class to the other. It is believed that the two classes of Ser codons may have evolved independently.

**10.20** Mutant 1 indicates the sequence AUGAARUAG, where R means any purine. Mutant 2 indicates the sequence AUGAU[U or C or A]GUNUAA, where N means any nucleotide. Because mutation 2 is a deletion in the second codon, the second codon in the wildtype gene must be either ?AU, A?U, or AU?, where the question mark indicates the deleted nucleotide; also, the third codon in the wildtype gene must be [U or C or A]GU, the fourth codon must be NUA, and the fifth must be A??. Similarly, because mutation 1 is an insertion in the second codon, the second codon in the wildtype gene must consist of three of the four bases AARU; also, the third codon in the wildtype gene must be AG?. The only consistent possibility for codon 2 is AAU, which codes for Asn. The third codon must be AGU, which codes for Ser. That leaves, for codon 4, the possibilities Val and Lys, and of these, only Val is coded by NUA, where N = G. Finally, the fifth codon must be AAR (Lys).

Hence the sequence of the nontranscribed strand of the wildtype gene, as nearly as can be determined from the data given, and separated into codons, is

```
5'-ATG-AAT-AGT-GTA-AAR-3'
```

Mutation 1 has a single nucleotide addition in codon 2 that yields a codon for Lys, so it must be an addition of a purine (R) at position 6 or of an A at either position 4 or position 5. Mutation 2 has a single nucleotide deletion that yields a codon for Ile, which means that the A at either position 3 or position 4 was deleted.

***Chapter 11***

**11.1** The enzymes are expressed constitutively because glucose is metabolized in virtually all cells. However, the levels of enzyme are regulated to prevent runaway synthesis or inadequate synthesis.

**11.2** The hemoglobin mRNA has a very long lifetime before being degraded.

**11.3** **(a)** Transcription, **(b)** RNA processing, **(c)** Translation.

**11.4** The repressor gene need not be near because the repressor is diffusible. The *trp* repressor is one example in which the repressor gene is located quite far from the structural genes.

**11.5** The mutant gene should bind more of the activator protein or bind it more efficiently. Therefore, the mutant gene should be induced with lower levels of the activator protein or expressed at higher levels than the wildtype gene, or both.

**11.6** **(a)** $\beta$-galactosidase (*lacZ* product), lactose permease (*lacY* product), and the transacetylase (*lacA* product), **(b)** The promoter region becomes inaccessible to RNA polymerase. **(c)** Repressor synthesis is constitutive.

**11.7** At 37°C, both *lacZ*$^-$ and *lacY*$^-$ are phenotyipcally Lac$^-$; at 30°C, the *lacY*$^-$ strain is Lac$^+$, and the *lacZ*$^-$ strain is Lac$^-$.

**11.8** Inducers combine with repressors and prevent them from binding with the operator regions of the respective operons.

**11.9** The cAMP concentration is low in the presence of glucose. Both types of mutations are phenotypically Lac$^-$. The cAMP-CRP binding does not interfere with repressor binding because the DNA binding sites are distinct.

**11.10** No; there is often a polar effect in which the relative amount of each protein synthesized decreases toward the 3' end of the mRNA.

**11.11** With a repressor, the inducer is usually an early (often the first) substrate in the pathway, and the repressor is inactivated by combining with the inducer. With an aporepressor, the effector molecule is usually the product of the pathway, and the aporepressor is activated by the binding.

**11.12** The attenuator is not a binding site for any protein, and RNA synthesis does not begin at an attenuator. An attenuator is strictly a potential termination site for transcription.

**11.13** Steroid hormones pass through the cell membrane and combine with specific receptor proteins; thereupon they are transported to the nucleus and act as transcriptional activator proteins that stimulate transcription of their target genes.

**11.14** The α cell would be unable to switch to the **a** mating type.

**11.15** The **a**1 protein functions only in combination with the α2 protein in diploids; hence the *MAT***a'** haploid cell would appear normal. However, the diploid *MAT***a'**/*MAT*α has the phenotype of an α haploid. This is because the genotype lacks an active **a**1-α2 complex to repress the haploid-specific genes as well as repressing α1, and the α1 gene product activates the α-specific genes.

**11.16** The mutant diploid *MAT***a**/*MAT*α has the phenotype of an α haploid. As in Problem 11.15, the genotype lacks an active **a**1-α2 complex to repress the haploid-specific genes as well as α1, and the α1 gene product activates the α-specific genes.

**11.17** In the absence of lactose or glucose, two proteins are bound: the *lac* repressor and CRP-cAMP. In the presence of glucose, only the repressor is bound.

**11.18 (a)** The strain is constitutive and has either the genotype *lacI*$^-$ or *lacO*$^c$. **(b)** The second mutant must be *lacZ*$^-$ because no β-galactosidase is made; and because permease synthesis is induced by lactose, the mutant must have a normal repressor-operator system. Therefore, the genotype is *lacI*$^+$ *lacO*$^+$ *lacZ*$^-$ *lacY*$^+$. **(c)** Because the partial diploid is regulated, the operator in the operon with a functional *lac*Z gene must be wildtype. Hence the mutant in part (a) must be *lacI*$^-$.

**11.19** The *lac* genes are transferred by the Hfr cell and enter the recipient. No repressor is present in the recipient initially, so *lac* mRNA is made. However, the *lacI* gene is also transferred to the recipient, and soon afterward the repressor is made and *lac* transcription stops.

**11.20** Amino acids with codons one step away from the nonsense codon, because their tRNA genes are the most likely to mutate to nonsense suppressors. For UGA, these are UGC (Cys), UGU (Cys), UGG (Trp), UUA (Leu), UCA (Ser), AGA (Arg), CGA (Arg), and GGA (Gly).

**11.21** The mutant clearly lacks a general system that regulates sugar metabolism, and the most likely possibility is the cAMP-dependent regulatory system. Several mutations can prevent activity of this system. For example, the mutant either could make a defective CRP protein (which either fails to bind to the promoter or is unresponsive to cAMP) or could have an inactive adenyl cyclase gene and be unable to synthesize cAMP.

**11.22 (a)** No enzyme synthesis; the mutation is *cis*-dominant because lack of RNA polymerase binding to the promoter prevents transcription of the adjacent genes. **(b)** Enzymes synthesized; this one is *cis*-dominant because failure to bind the repressor cannot prevent constitutive transcription of the adjacent genes. **(c)** Enzymes synthesized; this mutation is recessive because the defective repressor can be complemented by a wildtype repressor in the same cell. **(d)** Enzymes synthesized; the mutation is recessive because the mutant repressor can be complemented by a wildtype repressor in the same cell. **(e)** No enzyme synthesis; this mutation is dominant because the repressor is always activated and all *his* transcription is prevented.

**11.23** The wildtype mRNA reads

```
AAA CAC CAC CAU CAU CAC CAU CAU CCU GAC
```

which codes for Lys-His-His-His-His-His-His-His-Pro-Asp. The mutant mRNA reads AA**A** ACA CCA CCA UCA UCA CCA UCA UCC UGA C, which codes for Lys-Thr-Pro-Pro-Ser-Ser-Pro-Ser-Ser (the UGA is a normal stop codon). The expected phenotype of the mutant is His$^-$ because of lack of attenuation by low levels of histidine. That is, translation of the attenuator occurs (preventing efficient transcription of the operon) even when the level of histidine within the cell is very low. However, low levels of proline or serine will prevent translation of the attenuator and allow transcription of the histidine operon.

***Chapter 12***

**12.1** Autonomous developmental fates are controlled by genetically programmed changes within cells. Fates determined by positional information are controlled by interactions with morphogens or with other cells. Cell transplantation and cell ablation (destruction) are two kinds of surgical manipulation that help investigators distinguish between them.

**12.2** Lineage mutations affect the developmental fates of one or more cells within a group of cells related by descent that normally undergo a defined pattern of development. Lineage mutations are detected by the occurrence of abnormal types or numbers of cells at characteristic positions within the embryo.

**12.3** The *Drosophila* blastoderm is the flattened hollow ball of cells formed by cellularization of the syncytium formed by the early nuclear divisions. The fate map is a diagram of the blastoderm showing the locations of cells and their developmental fates. The existence of the fate map does not mean that development is autonomous, because many developmental decisions are made according to positional information in the blastoderm. In fact, most *Drosophila* developmental decisions are nonautonomous.

**12.4** A female that is homozygous for a maternal-effect lethal produces defective eggs resulting in inviable embryos.

However, the homozygous genotype itself is viable.

**12.5** Cells normally destined to die adopt some other developmental fate, with the result that there are supernumerary copies of one or more cell types.

**12.6** In a homeotic mutation, certain segments develop in a manner characteristic of that of other segments, and therefore the lineage effect is one of transformation.

**12.7** A sister-cell segregation defect would lead to both daughter cells differentiating as B or to both differentiating as C. A parent-offspring segregation defect would cause both daughter cells to differentiate as A.

**12.8** Heterochronic mutations affect the timing of developmental events relative to each other; hence a mutation that uniformly slows or accelerates development is not a heterochronic mutation.

**12.9** Because the *Drosophila* oocyte contains all the transcripts needed to support the earliest stages of development; the mouse oocyte does not.

**12.10** In a loss-of-function mutation, the genetic information in a gene is not expressed in some or all cells. In a gain-of-function mutation, the genetic information is expressed at inappropriate times or in inappropriate cells. Both mutations can occur in the same gene: Some alleles may be loss-of-function and others gain-of-function. However, a particular allele must be one or the other (or neither).

**12.11** Because the gene is necessary, a loss-of-function mutation will prevent the occurrence of the proper developmental fate. The allele will be recessive, however, because a normal allele in the homologous chromosome will allow the developmental pathway to occur.

**12.12** Because the gene is sufficient, a gain-of-function mutation will induce the developmental fate to occur. The allele will be dominant because a normal allele in the homologous chromosome will not prevent the developmental pathway from occurring.

**12.13** Developmental abnormalities in gap mutants occur across contiguous regions, resulting in a gap in the pattern of normal development. Developmental abnormalities in pairrule mutants are in even- or odd-numbered segments or parasegments, so the abnormalities appear in horizontal stripes along the embryo.

**12.14** The proteins required for cleavage and blastula formation are translated from mRNAs present in the mature oocyte, but transcription of zygotic genes is required for gastrulation.

**12.15** The material required for rescue is either mRNA transcribed from the maternal $o^+$ gene or a protein product of the gene that is localized in the nucleus.

**12.16** The transplanted nuclei respond to the cytoplasm of the oocyte and behave like the oocyte nucleus at each stage of development.

**12.17** XDH activity is needed only at an early stage in development for the eyes to have wildtype pigmentation. The data are explained by a maternal effect in which the $mal^+/mal$*;* $ry/ry$ females in the cross transmit enought $mal^+$ gene product to allow brief activity of XDH in the $ry^+/ry$ embryos. In theory, the $malU^+$ product could be a transcription factor that allows $ry^+$ transcription, but in fact the $mal^+$ gene product participates in the synthesis of a cofactor that combines with the $ry^+$ gene product to make an active XDH enzyme.

**12.18** First, examine embryos of strains (c) and (d) to determine the wildtype expression patterns of *kni* and *ftz.* Next, cross strains (a) and (d) and strains (b) and (c). Intercross the $F_1$ heterozygotes from the cross (a) × (d), and examine the embryos to detect changes in *ftz* expression. Intercross the $F_1$ heterozygotes from the cross (b) × (c), and examine the embryos to detect changes in *kni* expression. Some of the $F_2$ embryos from the (a) × (d) cross will show abnormal expression of *ftz,* indicating that wildtype *kni* is required for proper *ftz* expression. The $F_2$ embryos of (b) × (c) will show normal expression of *kni,* indicating that *ftz* is not required for proper *kni* expression.

***Chapter 13***

**13.1** Such people are somatic mosaics. This condition can be explained by somatic mutations in the pigmented cells of the iris of the eye or in their precursor cells.

**13.2** Causes of a nonreverting mutation include (1) deletion of the gene, (2) multiple nucleotide substitutions in the gene, and (3) complex DNA rearrangement with more than one breakpoint in the gene. Deletions do not revert because the genetic information in the gene is missing, and the others do not revert because the nucleotide substitutions or DNA rearrangements are unlikely to be reversed precisely in one step.

**13.3** Transitions are AT → GC, GC → AT, TA → CG and CG → TA (total four). Transversions are AT → TA, AT → CG, GC → CG, GC → TA, TA → AT, TA → GC, CG → GC, and CG → AT (total eight).

**13.4** All nucleotide substitutions in the second codon position result in either an amino acid replacement or a terminator codon.

**13.5** Both UUR and CUR code for leucine, and both CGR and AGR code for arginine (R stands for any purine, either A or G). Hence, in each of these codons, one of the possible substitutions in the first codon position does not result in an amino acid replacement. Even when an amino acid replacement does occur, the mutant protein may still be functional, because more than one amino acid at a given position in a protein molecule may be compatible with normal or nearly normal function.

**13.6** Any single nucleotide substitution creates a new codon with a random nucleotide sequence, so the probability that it is a chain-termination codon (UAA, UAG, or UGA) is 3/64, or approximately 5 percent.

**13.7** A frameshift mutagen will also revert frameshift mutations.

**13.8** Exactly three. Otherwise, there would be a frameshift, and all downstream amino acids would be different.

**13.9** No. Dominance and recessiveness are characteristics of the phenotype. They depend on the function of the gene and on which attributes of the phenotype are examined.

**13.10** Photoreactivation and excision repair.

**13.11 (a)** Development arrests at A at both the high and low temperatures. **(b)** Development arrests at B at both the high and low temperatures. **(c)** Development arrests at A at the high temperature and arrests at B at the low temperature. **(d)** Development arrests at B at the high temperature and arrests at A at the low temperature.

**13.12** The mutation frequency in the Lac$^-$ culture is at least $0.90 \times 10^{-8} = 9 \times 10^{-9}$. Because the mutation rate to a nonsense suppressor is independent of the presence of a suppressible mutation (in this case, in the Lac$^-$ mutant), the mutation rate in the original Lac$^+$ culture is also at least $9 \times 10^{-9}$. The qualifier *at least* is important because not all nonsense suppressors may suppress the mutation in the particular Lac$^-$ strain (that is, some suppressors may insert an amino acid that still results in a nonfunctional enzyme.)

**13.13 (a)** $5 \times 10^{-5}$ is the probability of a mutation per generation, and so $1 - (5 \times 10^{-5}) = 0.99995$ is the probability of no mutation per generation. The probability of no mutations in 10,000 generations equals $(0.99995)^{10,000} = 0.61$. **(b)** The average number of generations until a mutation occurs is $1/(5 \times 10^{-5}) = 20,000$ generations.

**13.14** At 1 Gy, the total mutation rate is double the spontaneous mutation rate. Because a dose of 1 Gy increases the rate by 100 percent, a dose of 0.5 Gy increases it by 50 percent and a dose of 0.1 Gy by 10 percent.

**13.15** Amino acid replacements at many positions in the A protein do not eliminate its ability to function; only amino acid replacements at critical positions that do eliminate tryptophan synthase activity are detectable as Trp$^-$ mutants.

**13.16** The possible lysine codons are AAA and AAG, and the possible glutamic acid codons are GAA and GAG. Therefore, the mutation results from an AT $\rightarrow$ GC transition in the first position of the codon.

**13.17** Six amino acid replacements are possible because the codons UAY (Y stands for either pyrimidine, C or U) can mutate in a single step to codons for Phe (UUY), Ser (UCY), Cys (UGY), His (CAY), Asn (AAY), or Asp (GAY).

**13.18** Single nucleotide substitutions that can account for the amino acid replacements are (1) Met (AUG) to Leu (UUG), (2) Met (AUG) to Lys (AAG), (3) Leu (CUN) to Pro (CCN), (4) Pro (CCN) to Thr (ACN), and (5) Thr (ACR) to Arg (AGR), where R stands for any purine (A or G) and N for any nucleotide. Substitutions (1), (2), (4), and (5) are transversion, and substitution (3) is a transition. Because 5-bromouracil primarily induces transitions, the Leu $\rightarrow$ Pro replacement is expected to be the most frequent.

**13.19** The sectors are explained by a nucleotide substitution in only one strand of the DNA duplex. Replication of this heteroduplex yields one daughter DNA molecule with the wildtype sequence and another with the mutant sequence. Cell division produces one Lac$^+$ and one Lac$^-$ cell, located side by side. Because the cells do not move on the agar surface, the Lac$^+$ cells form a purple half-colony, and the Lac$^-$ cells form a pink half-colony.

**13.20** The *a* and *b* gene products may be subunits of a multimeric protein. A few proteins with compensatory amino acid replacements can interact to give wildtype activity, but most mutant proteins cannot interact in this way with each other or with wildtype.

**13.21** Some of the mutagenized Pl particles acquire mutations in the *a* gene and become $a^- b^+$. Therefore, the next step is to select $b^+$ transductants and screen among these for ones that are $a^-$. The procedure depends on close linkage.

**13.22 (a)** All progeny would be able to grow in the absence of arginine. **(b)** The genotype of the diploid is $Su^+/Su^-$ $arg^+/arg^-$, where $Su^+$ indicates the ability to suppress, and $Su^-$ indicates inability to suppress. The resulting haploid spores are 1/4 $Su^+$ $arg^+$, 1/4 $Su^+$ $arg^-$, 1/4 $Su^-$ $arg^+$, and 1/4 $Su^-$ $arg^-$. The first three are Arg$^+$ and the last Arg$^-$, so the ratio is 3 Arg$^+$: 1 Arg$^-$. **c** The diploid genotype is $Su^+$ $arg^-/Su^-$ $arg^+$. The haploid spores are in the proportions 0.05 $Su^+$ $arg^+$, 0.45 $Su^+$ $arg^-$, 0.45 $Su^-$ $arg^+$, and 0.05 $Su^-$ $arg^-$. Therefore, 95 percent of the progeny are Arg$^+$ and 5 percent are Arg$^-$.

**13.23** Asci C, D, and F might arise from gene conversion at the *m* locus as follows. Branch migration through *m* results in two regions of heteroduplex DNA. In ascus type C, one region is replicated before repair, leaving + and *m* alleles intact; the other region is repaired before replication (+ to *m),* yielding two *m* alleles (therefore a 3 : 5 ratio). In ascus type D, both heteroduplex regions are replicated before repair, leaving all alleles intact but changing their order in the ascus (therefore a 3 : 1 : 1 : 3 ratio). In ascus type F, both heteroduplex regions are repaired (+ to *m)* before replication, yielding four *m* alleles (therefore a 2 : 6 ratio).

***Chapter 14***

**14.1** Inheritance is strictly maternal, so yellow is probably inherited cytoplasmically. A reasonable hypothesis is that the gene for yellow is contained in chloroplast DNA.

**14.2** Pollen is not totally devoid of cytoplasm, and occasionally it contains a chloroplast that is transmitted to the zygote.

**14.3** This trait is maternally inherited, so the phenotype is that of the mother. The progeny are **(a)** green, **(b)** white, **(c)** variegated, and **(d)** green.

**14.4** With X-linked inheritance, half of the $F_2$ males (1/4 of the total progeny) would be white. The observed result is that all of the $F_2$ progeny are green.

**14.5** If the daughter chloroplasts segregate at random, both of them will go to the same pole half the time. For all four products to contain a descendant chloroplast, the segregation has to be perfect in three divisions, which occurs with probability $(1/2)^3 = 1/8$. Therefore, the probability that at least one cell lacks a descendant chloroplast is $1 - 1/8 = 7/8$.

**14.6** The chloroplast from the $mt^-$ strain is preferentially lost, and the *mt* alleles segregate 1 : 1. The expected result is 1/2 $a^+$ $b^-$ $mt^+$ and 1/2 $a^+$ $b^-$ $mt^-$.

**14.7** Petite, because neither parent contributes functional mitochondria.

**14.8** Because every tetrad shows uniparental inheritance, it is likely that the trait is determined by factors transmitted through the cytoplasm, such as mitochondria. The 4 : 0 tetrads result from diploids containing only mutant factors, and the 0 : 4 tetrads result from diploids containing only wildtype factors.

**14.9** In autogamy, one product of meiosis becomes a homozygous diploid nucleus, so half of the *Aa* cells that undergo autogamy become *AA* and half become *aa.*

**14.10** The mating pairs are 1/4 *AA* × *AA,* 1/2 *AA* × *aa,* and 1/4 *aa* × *aa.* These produce only *AA, Aa,* and *aa* progeny, respectively, so the progeny are 1/4 *AA,* 1/2 *Aa,* and 1/4 *aa.* In autogamy, the *Aa* genotypes become homozygous for either *A* or *a,* so the expected result of autogamy is 1/2 *AA* and 1/2*aa.*

**14.11** The only functional pollen is *a B.* All the progeny of the *AA bb* plants are *Aa Bb,* and those of the *aa BB* plants are *aa BB.*

**14.12** The $F_1$ females will be heterozygous for nuclear genes, and possibilities (1) and (2) imply that these females should have some viable sons. The fact that all progeny are female supports the hypothesis of cytoplasmic inheritance.

**14.13** Because the progeny coiling is determined by the genotype of the mother, the rightward-coiling snail has genotype *ss.* To explain the rightward coiling, the mother of this parent snail would have had the + allele and must in fact have been +*/s.*

**14.14** Among Africans, the average time traversed in tracing the ancestry from a present-day mitochondrial DNA back to a common ancestor and forward again to another present-day mitochondrial DNA is between 78 × 1500 years and 78 × 3000 years, or 117,000 to 234,000 years; the backward and foward times are equal, so the person having the common ancestor of the mitochondrial DNA lived between 117,000/2 = 58,500 years ago and 234,000/2 = 117,000 years ago. Among Asians, similar calculations result in a range of 43,500 to 87,000 years for the common ancestor of mitochondrial DNA. Among Caucasians, the calculated range is 30,000 to 60,000 years.

**14.15** The accompanying table shows the percent of restriction sites shared by each pair of mitochondrial DNA molecules.

| | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 1 | 100 | 0 | 80 | 10 | 0 |
| 2 | | 100 | 0 | 10 | 100 |
| 3 | | | 100 | 11 | 0 |
| 4 | | | | 100 | 10 |

The entries in the table may be explained by example. Molecules 3 and 4 together have 9 restriction sites, of which 1 is shared, so the entry in the table is 1/9 = 11 percent. The diagonal entries of 100 mean that each molecule shares 100 percent of the restriction sites with itself. The two most closely related molecules are 2 and 5 (in fact, they are identical), so these also must be grouped together. The two next closest are 1 and 3, so these also must be grouped together. The 2–5 group shares an average of 10 percent restriction sites with molecule 4, and the 1-3 group shares an average of 10.5 percent restriction sites [that is, (10 + 11)/2] with molecule 4. Therefore, the 2–5 group and the 1–3 group are about equally distant from molecule 4. The diagram shows the inferred relationships among the species. Data of this type do not indicate where the "root" should be positioned in the tree diagram.

| | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 1 | 100 | 0 | 80 | 10 | 0 |
| 2 | | 100 | 0 | 10 | 100 |
| 3 | | | 100 | 11 | 0 |
| 4 | | | | 100 | 10 |

***Chapter 15***

**15.1** Adults 2/3 *Aa* and 1/3 *aa.* It is not necessary to assume random mating, because the only fertile matings are *Aa* × *Aa.* These produce 1/4 AA, 1/2 *Aa,* and 1/4 *aa* zygotes, but the *AA* zygotes die, leaving 2/3 *Aa* and 1/3 *aa* adults.

**15.2** The harmful recessive allele will have a smaller equilibrium frequency in the X-linked case, because the allele is exposed to selection in hemizygous males.

**15.3** The 10 *AA* individuals have 20 *A* alleles, the 15 *Aa* have 15 *A* and 15 *a,* and the 4 *aa* have 8 *a.* Altogether there 35 *A* and 23 *a* alleles, for a total of 58. The allele frequency of *A* is 35/58 = 0.603, and that of *a* is 23/58 = 0.397.

**15.4** The numbers of alleles of each type are *A,* 2 + 2 + 5 + 1 = 10; *B,* 5 + 12 + 12 + 10 = 39; *C,* 10 + 5 + 5 + 1 = 21. The total number is 70, so the allele frequencies are *A,* 10/70 = 0.14; *B,* 39/70 = 0.56; and *C,* 21/70 = 0.30.

**15.5 (a)** Two bands indicate heterozygosity for two different VNTR alleles. **(b)** A person who is homozygous for a VNTR allele will have a single VNTR band. (An alternative explanation is also possible: Some VNTR alleles are present in small DNA fragments that migrate so fast that they are lost from the gel; a heterozygote for one of these alleles will have only one band—the one corresponding to the larger DNA fragment—because the band corresponding to the smaller DNA fragment will not be detected. **(c)** VNTR alleles are codominant because both are detected in heterozygotes. **(d)** Because none of the bands in the four pairs of parents have the same electrophoretic mobility, the parents of each child can be identified as the persons who share a single VNTR band with the child. Child A has parents 7 and 2; child B has parents 4 and 1; child C has parents 6 and 8; and child D has parents 3 and 5.

**15.6** Allele 5.7 frequency = (2 × 9 + 21 + 15 + 6)/200 = 0.30; allele 6.0 frequency = (2 × 12 + 21 + 18 + 7)/200 = 0.35; allele 6.2 frequency = (2 × 6 + 15 + 18 + 5)/200 =

0.25; allele 6.5 frequency =(2 × 1 + 6 + 7 + 5) 200 = 0.10.

**15.7** If the genotype frequencies satisfy the Hardy-Weinberg principle, they should be in the proportions $p^2$, $2pq$, and $q^2$. In each case, $p$ equals the frequency of *AA* plus one-half the frequency of *Aa*, and $q = 1-p$. The allele frequencies and expected genotype frequencies with random mating are **(a)** $p = 0.50$, expected: 0.25, 0.50, 0.25; **(b)** $p = 0.635$, expected: 0.403, 0.464, 0.133; **(c)** $p = 0.7$, expected : 0.49, 0.42, 0.09; **(d)** $p = 0.775$, expected: 0.601, 0.349, 0.051; **(e)** $p = 0.5$, expected: 0.25, 0.50, 0.25. Therefore, only (a) and (c) fit the Hardy-Weinberg principle.

**15.8** $p^2 = 0.09$, and so $p = (0.09)^{1/2} = 0.30$. All other alleles have a combined frequency of 1 - 0.30 = 0.70.

**15.9** The frequency of the recessive allele is $q = (1/14{,}000)^{1/2} = 0.0085$. Hence $p = 1 - 0.0085 = 0.9915$, and the frequency of heterozygotes is $2pq = 0.017$ (about 1 in 60).

**15.10** Because the frequency of homozygous recessives is 0.10, the frequency of the recessive allele $q$ is $(0.10)^{1/2}$ =0.316. This means $p = 1 - 0.316 = 0.684$, so the frequency of heterozygotes is 2 × 0.316 × 0.684 = 0.43. Because the proportion 0.90 of the individuals are nondwarfs, the frequency of heterozygotes among nondwarfs is 0.43/0.90 = 0.48.

**15.11 (a)** Because the proportion 0.60 can germinate, the proportion 0.40 that cannot are the homozygous recessives. Thus the allele frequency $q$ of the recessive is $(0.4)^{1/2} = 0.63$. This implies that $p = 1\ 0.63 = 0.37$ is the frequency of the resistance allele. **(b)** The frequency of heterozygotes is $2pq = 2 \times 0.37 \times 0.63 = 0.46$. The frequency of homozygous dominants is $(0.37)^2 = 0.14$, and the proportion of the surviving genotypes that are homozygous is 0.14/0.60= 0.23.

**15.12** The bald and nonbald alleles have frequencies of 0.3 and 0.7, respectively. Bald females are homozygous and occur in the frequency $(0.3)^2 = 0.09$; nonbald females occur with frequency 0.91. Bald males are homozygous or heterozygous and occur with frequency $(0.3)^2 + 2(0.3)(0.7) = 0.51$; nonbald males occur with frequency 0.49.

**15.13** $I^AI^A$, $(0.16)^2 = 0.026$; $I^AI^O$, $2(0.16)(0.74) = 0.236$; $I^BI^B$, $(0.10)^2 = 0.01$; $I^BI^O$, $2(0.10)(0.74) = 0.148$;$I^Ol^O$, $(0.74)^2 = 0.548$; $l^Al^B$, $2(0.16)(0.10) = 0.032$. The phenotype frequencies are O, 0.548; A, 0.026 + 0.236 = 0.262;B, 0.01 + 0.148 = 0.158; AB, 0.032.

**15.14 (a)** There are 17 *A* and 53 *B* alleles, and the allele frequencies are *A*, 17/70 = 0.24; *B*, 53/70 = 0.76. **(b)** The expected genotype frequencies are *AA*, $(0.24)^2 = 0.058$;*AB*, $2(0.24)(0.76) = 0.365$; *BB*, $(0.76)^2 = 0.578$. The expected numbers are 2.0, 12.8, and 20.2, respectively.

**15.15** The frequency in males implies that $q = 0.2$. The expected frequency of the trait in females is $q^2 = 0.0004$. The expected frequency of heterozygous females is $2pq = 2(0.98)(0.02) = 0.0392$.

**15.16** The frequency of the yellow allele *y* is 0.2. In females, the expected frequencies of wildtype (+ + and *y* +) and yellow (*y y)* are $(0.8)^2 + 2(0.8)(0.2) = 0.96$ and $(0.2)^2 = 0.04$, respectively. Among 1000 females, there would be 960 wildtype and 40 yellow. The phenotype frequencies are very different in males. In males, the expected frequencies of wildtype (+ Y) and yellow (*y* Y) are 0.8 and 0.2, respectively, where Y represents the Y chromosome. Among 1000 males, there would be 800 wildtype and 200 yellow.

**15.17** The frequency of heterozygotes is smaller by the amount $2pqF$, in which $F$ is the inbreeding coefficient.

**15.18** Inbreeding is suggested by a deficiency of heterozygotes relative to the $2pq$ expected with random mating. The suggestive populations are (d) with expected heterozygote frequency 0.349 and (e) with expected heterozygote frequency 0.5.

**15.19** The allele frequency is the square root of the frequency of affected offspring among unrelated individuals, or $(8.5 \times 10^{-6})^{1\ 2} = 2.9 \times 10^{-3}$. With inbreeding, the expected frequency of homozygous recessives is $q^2(l - F) + qF$, in which $F$ is the inbreeding coefficient. When $F = 1/16$, the expected frequency of homozygous recessives is $(2.9 \times 10^{-3})^2(1 - 1/16) + (2.9 \times 10^{-3})(1/16) = 1.9 \times 10^{-4}$;when $F = 1/64$, the value is $5.3 \times 10^{-5}$.

**15.20 (a)** Here $F = 0.66$, $p = 0.43$, and $q = 0.57$. The expected genotype frequencies are *AA*, $(0.43)^2(1 - 0.66) + (0.43)(0.66) = 0.347$; *AB*, $2(0.43)(0.57)(1\ 0.66) = 0.167$;BB, $(0.57)^2(1 - 0.66) + (0.57)(0.66) = 0.487$. **(b)** With random mating, the expected genotype frequencies are *AA*, $(0.43)^2 = 0.185$; *AB*, $2(0.43)(0.57) = 0.490$; *BB*, $(0.57)^2 = 0.325$.

**15.21** Use Equation (11) with $n = 40$, $p_0q_0 = 1$ (equal amounts inoculated), and $p_{40}q_{40} = 0.35/0.65$. The solution for *w* is $w = 0.9846$, which is the fitness of strain B relative to strain A, and the selection coefficient against B is $s = 1 - 0.9846 = 0.0154$.

**15.22 (a)** *A* fixed; **(b)** *A* lost; **(c)** stable equilibrium; **(d)** *A* lost.

**15.23** An *Aa* female produces 1/2 *A* and 1/2 *a* gametes, and these yield heterozygous offspring if they are fertilized by a and by *A*, respectively. The overall probability of a heterozygous offspring is $(1/2)q + (1/2)p = (1/2)(p + q) = 1/2$.

**15.24** The frequency of each allele is 1 *n*, and there are *n* possible homozygotes. Therefore, the total frequency of homozygotes is $n \times (1/n)^2 = 1/n$, which implies that the total frequency of heterozygotes is $1 - 1/n$.

**15.25** The frequencies are calculated as in the HardyWeinberg principle except that, for apparently homozygous alleles, $2p$ is used instead of $p^2$. With regard to the genotype (6, 9.3) for *HUMTHO1*, (10, 11) for *HUMFES*, (6, 6) for

*D12S67*, and (18, 24) for *D1S80*, the calculation is 2 × 0.230 × 0.310 × 2 × 0.321 × 0.373 × 2 × 0.324 × 2 × 0.293 × 0.335 = 0.0043, or about one in 230 people. With regard to the genotype (8, 10) for *HUMTHO1*, (8, 13) for *HUMFES*, (1, 2) for *D12S67*, and (19, 19) for *D1S80*, the calculation is 2 × 0.105 × 0.002 × 2 × 0.007 × 0.066 × 2 × 0.015 × 0.005 × 2 × 0.011 = $1.28 \times 10^{-12}$, or about one in $8 \times 10^{11}$ people. The point is that DNA typing is at its most powerful in discriminating among people who carry rare alleles.

***Chapter 16***

**16.1** Genotype-environment interaction.

**16.2** Since the variance equals the square of the standard deviation, the only possibilities are that the variance equals either 1 or 0.

**16.3 (a)** The mean or average. **(b)** The one with the greater variance is broader. **(c)** 68 percent within one standard deviation of the mean, 95 percent within two.

**16.4** Broad-sense heritability is the proportion of the phenotypic variance attributable to differences in genotype, or $H^2 = \sigma_g^2 / \sigma_t^2$, which includes dominance effects and interactions between alleles. Narrow-sense heritability is the proportion of the phenotypic variance due only to additive effects, or $\sigma_e^2 / \sigma_t^2$, which is used to predict the resemblance between parents and offspring in artificial selection. If all allele frequencies equal 1 or 0, both heritabilities equal zero.

**16.5** The $F_1$ generation provides an estimate of the environmental variance. The variance of the $F_2$ generation is determined by the genotypic variance and the environmental variance.

**16.6** For fruit weight, $\sigma_e^2 / \sigma_t^2 = 0.13$ is given. Because $\sigma_t^2 = \sigma_g^2 + \sigma_e^2$, then $H^2 = \sigma_g^2 / \sigma_t^2 = 1 - 0.13 = 0.87$. Therefore, 87 percent is the broad-sense heritability of fruit weight. The values of $H^2$ for soluble-solid content and acidity are 91 percent and 89 percent, respectively.

**16.7** Estimates are as follows: mean, 104/10 = 10.4; variance, 24.4/(10 - 1) = 2.71; standard deviation $(2.71)^{1\ 2}$ = 1.65.

**16.8** An IQ of 130 is two standard deviations above the mean, so the proportion greater equals (1 - 0.95)/2 = 2.5 percent; 85 is one standard deviation below the mean, so the proportion smaller equals (1 - 0.68)/2 = 16 percent; the proportion above 85 equals 1 - 0.16 = 84 percent, which can be calculated alternatively as 0.50 + 0.68/2 = 0.84.

**16.9 (a)** To determine the mean, multiply each value for the pounds of milk produced (using the midpoint value) by the number of cows, and divide by 304 (the total number of cows) to obtain a value of the mean of 4032.9. The estimated variance is 693,633.8, and the estimated standard deviation, which is the square root of the variance, is 832.8. **(b)** The range that includes 68 percent of the cows is the mean minus the standard deviation to the mean plus the standard deviation, or 4033 - 833 = 3200 to 4033 + 833 = 4866. **(c)** Rounding to the nearest 500 yields 3000-5000. The observed number in this range is 239; the expected number is 0.68 × 304 = 207. **(d)** For 95 percent of the animals, the range is within two standard deviations, or 2367-5698. Rounding off to the nearest 500 yields 2500-6000. The observed number is 296, which compares well to the expected number of 0.95 × 304 = 289.

**16.10 (a)** Because 15 to 21 is the range of one standard deviation from the mean, the proportion is 68 percent. **(b)** Because 12 to 24 is the range of two standard deviations from the mean, the proportion is 95 percent. **(c)** A phenotype of 15 is one standard deviation below the mean; because a normal distribution is symmetrical about the mean, the proportion with phenotypes more than one standard deviation below the mean is (1 - 0.68)/2 =0.16. **(d)** Those with 24 leaves or greater are more than two standard deviations above the mean, so the proportion is (1 - 0.95)/2 = 0.025. **(e)** Those with 21 to 24 leaves are between one and two standard deviations above the mean, and the expected proportion is 0.16 - 0.025 = 0.135 (because 0.16 have more than 21 leaves, and 0.25 have more than 24 leaves).

**16.11 (a)** The genotypic variance is 0 because all $F_1$ mice have the same genotype. **(b)** With additive alleles, the mean of the $F_1$ generation will equal the average of the parental strains.

**16.12** Among the $F_1$, the total variance equals the environmental variance, which is given as 1.46. Among the $F_2$, the total variance, given as 5.97, is the sum of the environmental variance (1.46) and the genotypic variance. Thus the genotypic variance is 5.97 - 1.46 = 4.51. The broad-sense heritability is the genotypic variance (4.51) divided by the total variance (5.97), or 4.51/5.97 = 0.76.

**16.13** From the $F_1$ data, $\sigma_e^2$ = $(2)^2$ = 4; from the $F_2$ data, $\sigma_g^2$ + $\sigma_e^2$ = $(3)^2$ = 9; hence $\sigma_g^2$ = 9 - 4 = 5, and so $H^2$ = 5/9 = 56 percent.

**16.14** Using Equation (4), we find that the minimum number of genes affecting the trait is $D^2/8\sigma_g^2$ = $(17.6)^2/(8\times0.88)$=44.

**16.15** $\sigma_g^2$ = 0.0570 - 0.0144 = 0.0426 and *D* = 1.689 - ( - 0.137) = 1.826. The minimum number of genes is estimated as *n* = $(1.826)^2/(8 \times 0.0426)$ = 9.8.

**16.16** Use Equation (6) with *M** - *M* = 50 grams in each generation of selection, with *M* initially 700 grams. The expected gain in each generation of selection is 0.8 × 50 = 40 grams, so after five generations, the expected mean weight is 700 + 5 × 40 = 900 grams.

**16.17** Use Equation (6) with *M* = 10.8, *M** = 5.8, and *M*' = 8.8. The narrow-sense heritability is estimated as $h^2$ = (8.8 - 10.8)/(5.8 - 10.8) = 0.40.

**16.18 (a)** In this case, $M$ = 10.8, $M^*$ = 5.8, and $M'$ = 9.9. The estimate of $h^2$ is (9.9 - 10.8)/(5.8 - 10.8) = 0.18. **(b)** The result is consistent with the earlier result because of the differences in the variance. In the previous case, the additive variance was 0.4 × 4 = 1.6. In the present case, the phenotypic variance is 9. If the additive variance did not change, then the expected heritability would be 1.6/9 = 0.18, which is that observed. This problem has been idealized somewhat;in real examples, the additive variance would also be expected to change with a change in environment.

**16.19** The offspring are genetically related as half-siblings, and Table 16.3 says that the theoretical correlation coefficient is $h^2/4$.

**16.20** Table 16.3 gives the correlation coefficient between first cousins as $h^2/8$, and so $h^2$ = 8 × 0.09 = 72 percent.

**16.21** The mean weaning weight of the 20 individuals is 81 pounds, and the standard deviation is 13.8 pounds. The mean of the upper half of the distribution can be calculated by using the relationship given in the problem, which says that the mean of the upper half of the population is 81 + 0.8 × 13.8 = 92 pounds. Using Equation (6), we find that the expected weaning weight of the offspring is 81 + 0.2(92 - 81) = 83.2 pounds.

**16.22** One approach is to use Equation (4) with $D = 2n - 0 = 2n$, from which n = $(2n)^2/8\sigma_g^2$ , or $\sigma_g^2$ = n/2. A more direct approach uses the fact that with unlinked, additive genes, the total genetic variance is the summation of the genetic variance resulting from each gene. With 1 gene, the genetic variance equals 1/2, so with n genes, the total genetic variance equals $n/2$.

**16.23 (a)** This is a nonrecombinant class and so should have the genotype *Q/q* for the QTL affecting egg lay; the average should be (300 + 270)/2= 285 eggs per hen. **(b)** This is a crossover class of progeny. In view of the relative distances of the two intervals of 0.07 : 0.03, 7/10 of the crossovers should occur in the region *Cx-Q* and 3/10 in the region *Q-Dx.* The expected genotypes with respect to the QTL are 7/10 *q/q* (crossing-over in the *Cx-Q* region) and 3/10 *Q / q* (crossing-over in the *Q-Dx* region). The average phenotype is expected to be (7/10) × 270 + (3/10) × 285 = 274.5 eggs per hen. **(c)** This is also a crossover class, but it is the reciprocal of that in the previous part. In this case, the expected genotypes with respect to the QTL are 7/10 *Q/q* (crossing-over in the *Cx-Q* region) and 3/10 *q/q* (crossing-over in the *Q-Dx* region). The expected average egg lay is (7/10) × 285 + (3/10) × 270 = 280.5 eggs per hen. **(d)** These are again nonrecombinants with QTL genotype *q/q*, and the expected egg lay is 270 eggs per hen.

**16.24 (a)** The rare genetic disease is transmitted as a simple Mendelian dominant. Genotype *Q/q* is affected and *q/q* is not. Because the disease is rare, the frequency of *Q/Q* homozygous genotypes is negligible. **(b)** Male IV-2 has the genotype *Q/q*, so the probability that he has an affected daugher is 1/2. **(c)** Male IV-2 has the genotype *H/H* for gene 2; with random mating, the probability that his daughter will have the genotype *H/F* is equal to the probability that a randomly chosen egg contains the F allele, which is given in the population-frequency table as 0.05, or 5 percent. **(d)** The genotypes of the individuals in the pedigree are as shown in the accompanying table. At this stage in the analysis, we do not yet know anything about linkage, but we can, in some cases, specify the linkage phase in terms of which alleles were transmitted together in a single gamete. When known, the alleles transmitted in a single gamete are grouped together in parentheses. Also, the order in which the genes are listed is arbitrary. (The two columns on the right are referred to in the next part of the analysis.)

**Genotypes of individuals in pedigree**

| | | |
|---|---|---|
| I-1 | *(q A E)/(q A E)* | |
| I-2 | *Q/q ; C/D ; F/G* | |
| II-1 | *(q A E)/(Q D F)* | NRC |
| II-2 | *(q A H)/(q B H)* | |
| II-3 | *(q A E)/(q C G)* | NRC |
| II-4 | *(q A G)/(q A H)* | |

| | | | |
|---|---|---|---|
| II-5 | *(q A E)*/*(Q D G)* | NRC | |
| II-6 | *(q A H)*/*(q B H)* | | |
| II-7 | *(q A E)*/*(Q D F)* | NRC | |
| II-8 | *(q A E)*/*(q C G)* | NRC | |
| II-9 | *q*/*q ; A*/*B ; E*/*G* | | |
| III-1 | *(q B H)*/*(q B H)* | | |
| III-2 | *(q B H)*/*(Q D E)* | NRC | REC |
| III-3 | *(q A H)*/*(Q D E)* | NRC | REC |
| III-4 | *(q A H)*/*(q A F)* | NRC | REC |
| III-5 | *(q A G)*/*(q A E)* | | NRC |
| III-6 | *(q A H)*/*(q A G)* | | REC |
| III-7 | *(q B H)*/*(q A E)* | NRC | NRC |
| III-8 | *(q A H)*/*(Q D G)* | NRC | NRC |
| III-9 | *q/q ; B/C ; E/H* | | |
| III-10 | *(q B H)*/*(q A G)* | NRC | REC |
| III-11 | *(q A E)*/*(q A E)* | | NRC |
| III-12 | *(q B G)*/*(q A G)* | | REC |
| III-13 | *q/q ; B/C ; E/G* | | Undeterminable |
| IV-1 | *(q B H)*/*(Q D E)* | NRC | NRC |
| IV-2 | *(q B H)*/*(Q D H)* | NRC | REC |
| IV-3 | *(q B H)*/*(q B E)* | NRC | REC |
| IV-4 | *(q B E)*/*(Q D G)* | NRC | NRC |
| IV-5 | *(q C H)*/*(q A H)* | NRC | NRC |
| IV-6 | *(q C H)*/*(Q D G)* | NRC | NRC |

**e** There is evidence of linkage between genes 1 and 3, In particular, the allele *Q* is always transmitted with allele *D,*starting with male 1-2 at the top of the pedigree. If we assume that this male has the genotype (*Q D)*/(*q C),* then throughout the pedigree, there are 17 individuals in which recombination between genes 1 and 3 could have been detected. These cases are identified in the third column of the foregoing table. All 17 are nonrecombinant, so the recombination fraction between genes 1 and 3 in these data is 0/17. As far as one can discern from the pedigree alone, the *Q* and *D* alleles could be identical. However, the population-frequency table gives the allele frequency of *D* as 5 percent, and the text says that the *Q* allele is rare. Hence most D alleles in the population must be on chromosomes carrying *q,* so that *D* and *Q* are separable. Turning to genes 1 and 2, there are 17 people in whom recombination could be detected. These are identified in the fourth column in the foregoing table. Among the 17 cases, one is undeterminable, 8 are recombinant, and 8 are nonrecombinant. Genes 1 and 2 are therefore unlinked. They are probably on different chromosomes, but they could also be very far apart on the same chromosome.

### *Chapter 17*

**17.1** There are numerous chemoreceptors in *E. coli,* each of which interacts with a specific set of chemosensors for different small molecules. Mutations in a particular chemoreceptor affect the chemotactic response to all molecules whose chemosensors feed into the chemoreceptor. Hence cells with mutant chemosensors are multiply nonchemotactic.

**17.2** The mutant would be unable to make P-CheA and, therefore, P-CheY, so the phenotype would be that of continuous swimming.

**17.3** Extreme *cheB* mutants lack the CheB methylase and have high levels of chemoreceptor methylation; their phenotype is that of continuous tumbling, because the methylated chemoreceptors require high concentrations of attractant to be stimulated (which is why they swim in response to very strong attractant stimuli). Extreme *cheR* mutants cannot methylate the chemoreceptors and so are highly sensitive to stimuli; their phenotype is that of continuous swimming, but very strong repellent stimuli overcome the effects of undermethylation and result in tumbling.

**17.4** Phosphate groups are transferred among several key components (CheA, CheY, and CheB) in the chemotaxis-signaling pathway. Stimulated chemoreceptors promote autophosphorylation of CheA, in which a phosphate group is transferred from ATP to a specific amino acid residue in cheA. The phosphate group can be transferred readily from P-CheA to CheB or CheY. The P-CheY promotes clockwise flagellar rotation and causes the bacteria to tumble. If phosphate transfers were inhibited, the lack of P-CheY would result in failure to switch from swimming behavior to tumbling behavior, and chemotaxis would be impossible. Because protein phosphorylation is common to all of the chemoreceptors, inhibition of phosphorylation would result in a generally nonchemotactic phenotype.

**17.5** Male mating success is more likely to be affected by mutations in per because these mutations affect the courtship song, and it is the male who must sing acceptably to the female for successful courtship.

**17.6** Mate a mutagen-treated male with an attached-X female and select the male progeny. Each male progeny carries a different, mutagen-treated X chromosome. Mate each of the male progeny with an attached-X female; the progeny of this mating consist of attached-X daughters and XY sons, and the X chromosome in each son is a replica of the single mutagen-treated X chromosome present in the father.

**17.7** Either genetic or molecular tests would reveal whether the mutant allele is in *per.* The first step in genetic analysis is genetic mapping. If the mutant does not map near the chromosomal location of *per,* then it must be a different gene; if it does map near *per,* then a complementation test should be carried out with $per^0$. A molecular analysis would entail determining the DNA sequence of all or part of the *per* gene and comparing it with wildtype. If the sequences are identical, then the mutation must be in a gene different from *per;* if the sequences differ, then separate tests must be carried out to demonstrate that the mutant phenotype results from the difference in DNA sequence. The new mutation could easily be in a gene different from *per,* because courtship behavior is a complex trait influenced by many different genes.

**17.8 (a)** A short-period rhythm of approximately 40 seconds.**(b)** A long-period rhythm of approximately 90 seconds. **(c)** Because $per^S$ shortens the period and $per^L$ lengthens it, the $per^S/per^L$ heterozygote would be expected to exhibit an intermediate period of approximately (40 + 90)/2 = 65 seconds, which is very close to the period of the wildtype courtship song.

**17.9** A fly with a mutation that eliminates only one of the wavelength preferences would be classified as specifically nonphototactic. The mutation might affect the photopigments necessary for the detection of light of a particular wavelength, or it might affect the photoreceptor cells containing the particular visual pigment. A fly with a mutation that eliminates the phototactic response to both wavelengths is generally nonphototactic. The mutation might affect components of the visual pathway common to both 480-nm and 370-nm light—for example, a component common to all photoreceptor cells that is necessary for stimulation by light.

**17.10** It is an example of sensory adaptation, because intense light of a given wavelength results in a temporarily decreased response to light of the same wavelength.

**17.11** It is unlikely that an inbred strain would produce selected lines that differed significantly in maze-learning ability (or in any other trait), because genetic variation is essential for the success of artificial selection.

**17.12** Variability in the sense of smell in the initial population might result in variability in the ability of the rats to detect the food at the end of the maze. Therefore, differences in maze-learning ability may result from differences in the ability of the animals to detect the food used to motivate them to run through the maze. In this example, maze-learning ability is not a good measure of intelligence. You could control for variation in the sense of smell by using odorless food, masking the smell of the food, smothering the entire maze with the smell of the food, or using a nonfood goal.

**17.13** It is an example of genotype-environment interaction, because the relative performances of the two populations are reversed merely by changing the environment in which the test is carried out.

**17.14** The *waltzer* allele is recessive to the wildtype allele.

**17.15** Under conditions in which behavioral choices are strongly influenced by the ability to distinguish among different colors—for example, the ability to distinguish red and green in obeying traffic lights.

**17.16** The test would take advantage of the size differences between the wildtype and mutant alleles. The DNA of parents and child would be digested with a restriction enzyme that cleaves the DNA at sites flanking the CAG repeat. The resulting fragments would be separated by size with electrophoresis, and the fragments of interest would be identified by hybridization using a radioactive probe for the CAGcontaining fragment. The mutant allele should be longer than the wildtype allele because of the greater number of CAG repeats. Because the mutant allele is dominant, any mating between heterozygous mutant and homozygous wildtype is expected to result in 50 percent affected offspring.

**17.17** This example illustrates some of the complexities of behavior genetics in human beings. The correlation between the *MAO* gene and the behavioral disorder in the large pedigree is suggestive. However, single pedigrees may be misleading, because a correlation with a genetic marker may result from chance. Even if the disorder in this example is genetically determined, the cause need not be *MAO.* The disorder may result from a different gene that is genetically linked to *MAO*, or the large pedigree may also contain other genes influencing the disorder, and the *MAO* mutation may be only a contributing factor. The finding that unrelated men with similar behavioral disorders may have abnormal levels of monoamine oxidase is also suggestive. However, without data on the distribution of enzyme levels among normal men and among those with the behavioral disorder, it is impossible to evaluate the strength of the evidence.

## Supplementary Problems

### *Chapter 1— The Molecular Basis of Heredity and Variation*

**S1.1** Classify each of the following statements as true or false.

**(a)** Each gene is responsible for only one visible trait.

**(b)** Every trait is potentially affected by many genes.

**(c)** The sequence of nucleotides in a gene specifies the sequence of amino acids in a protein encoded by the gene.

**(d)** There is one-to-one correspondence between the set of codons in the genetic code and the set of amino acids encoded.

**S1.2** Is the following statement correct? " Heredity is solely responsible for the manifestation of anemia." Explain.

**S1.3** When a human trait is genetic in origin (for example, a disorder caused by an inborn error of metabolism), does this mean that the trait cannot be influenced by the environment?

**S1.4** What is meant by the term "pleiotropy" and what is a "pleiotropic effect"? What is the relevance of sickle-cell anemia to this phenomenon?

**S1.5** What facts known prior to Avery, MacLeod, and McCarty's experiment indirectly suggested that DNA might play a role in the genetic material?

**S1.6** Prior to the experiments of Avery, MacLeod, and McCarty and of Hershey and Chase, why did most biologists and biochemists believe that proteins were probably the genetic material?

x**S1.7** Each of the following statements pertains to the experiments of Avery, MacLeod, and McCarty on *Streptococcus pneumoniae*. Classify each statement as true or false. Which single statement is a conclusion that can be drawn from the experiments themselves?

**(a)** The polysaccharide capsule is required for virulence (the ability to infect and kill mice).

**(b)** The polysaccharide capsule is not important for virulence.

**(c)** DNA encodes the sequence of sugars in the polysaccharide capsule.

**(d)** DNA is the polysaccharide capsule.

**(e)** Proteins are not the molecules that carry the hereditary information for synthesis of the polysaccharide capsule.

**S1.8** What critical finding in Hershey and Chase's experiment enabled them to interpret the results as a proof that the genetic material in bacteriophage T2 is DNA?

**S1.9** Define the following terms: replication, transcription, translation, mutation, natural selection.

**S1.10** What can we infer about the function of DNA from its structure?

**S1.11** Is it correct to say that DNA is always the genetic material?

**S1.12** One of the early models for the structure of DNA was the so-called tetranucleotide hypothesis. According to this hypothesis, one of each of the four nucleotides containing adenine, thymine, guanine, and cytosine joined together to form a covalently linked planar unit (the "tetranucleotide"); these units were, in turn, linked together to yield a repeating polymer of the tetranucleotide. Why could a molecule with this structure never serve as the genetic material?

**S1.13** What are the major chemical components found in DNA?

**S1.14** The repeating unit of DNA is:

**(a)** a ribonucleotide.

**(b)** an amino acid.

**(c)** a nitrogenous base.

**(d)** a sugar, deoxyribose.

**(e)** none of the above.

**S1.15** The two strands of a duplex DNA molecule are said to be complementary in base sequence. Which of the following statements are implied by the complementary nature of the strands?

**(a)** The two strands have the same sequence of bases.

**(b)** The base A in one strand is paired with a T in the other strand.

**(c)** One strand consists of A's paired with T's; the other strand consists of G's paired with C's.

**(d)** The base G in one strand is paired with a C in the other strand.

**(e)** The two strands have the same sequence of bases when read in opposite directions.

**S1.16** In the following statements about double-stranded DNA, square brackets indicate the number of molecules. For example, [A] means the number of molecules of the base adenine, and [deoxyribose] means the number of molecules of 2' deoxyribose. Classify each of the statements as true or false.

**(a)** [A] = [G]

**(b)** [A] = [C]

**(c)** [A] = [T]

**(d)** [A] + [G] = [T] + [C]

**(e)** [deoxyribose] = [phosphate]

**S1.17** When the base composition of double-stranded DNA from *Micrococcus luteus* was determined, 37.5 percent of the bases were found to be cytosine. What is the percentage of adenine in the DNA of this organism?

**S1.18** Which nucleic acid base is found only in DNA? Which is found only in RNA?

**S1.19** What are three principal structural differences between RNA and DNA?

**S1.20** What is the most likely reason for the participation of RNA intermediates in the process of information flow from DNA to protein?

**S1.21** What are the consequences of the Glu $\rightarrow$ Val replacement at amino acid position 6 in the beta-globin chain?

**S1.22** Using the *vermilion* and *cinnabar* mutations affecting eye color in *D. melanogaster* as an example, explain what is meant by an epistatic interaction between genes.

**S1.23** Can all the existing diversity of life be explained as a result of adaptive evolution brought about by natural selection?

### *Chapter 2—*
### *Principles of Genetic Transmission*

**S2.1** Define the following terms: genotype, phenotype, allele, dominant, recessive.

**S2.2** How many different alleles of a particular gene may exist in a population of organisms? How many alleles of a particular gene can be present in an individual organism?

**S2.3** A breeder of Irish setters has a particularly valuable male show dog descended from a famous female named Rheona Didona, which was known to carry a harmful recessive gene for blindness caused by atrophy of the retina. Before she breeds the dog, she wants some assurance that it does not carry this harmful allele. How would you advise her to test the dog?

**S2.4** Any of a large number of different recessive mutations cause profound hereditary deafness. Children homozygous for such mutations are often found in small, isolated communities in which matings tend to be within the group. In these instances, a mutation that originates in one person can be transmitted to several members of the population in future generations and become homozygous through a marriage between group members who are carriers. A deaf man from one community marries a deaf woman from a different, unrelated community. Both of them have deaf parents. The marriage produces three children, all with normal hearing. How can you explain this result?

**S2.5** Consider a human family with four children, and remember that each birth is equally likely to result in a boy or a girl.

**(a)** What proportion of sibships will include at least one boy?

**(b)** What fraction of sibships will have the gender order GBGB?

**S2.6** Mendel's cross of homozygous round yellow with homozygous wrinkled green peas yielded the phenotypes 9/16 round yellow, 3/16 round green, 3/16 wrinkled yellow, and 1/16 wrinkled green progeny in the $F_2$ generation. If he had testcrossed the round green progeny individually, what proportion of them would have yielded some progeny with round yellow seeds?

**(a)** 1/4

**(b)** 1/3

**(c)** 1/2

**(d)** 3/4

**(e)** 0

**S2.7** Intestinal lactase deficiency (ILD) is a common inborn error of metabolism in humans; affected persons are homozygous for a recessive allele of a gene on chromosome 2. A woman and her husband, both phenotypically normal, had a daughter with ILD. She married a normal man and had three sons. One had ILD and two were normal. Recognizing that each succeeding child's genotype is not influenced by the genotypes of older siblings, determine the probability that the couple's next child will have ILD.

**(a)** 0.25

**(b)** 0.50

**(c)** 0.75

**(d)** 1.00

**(e)** 0.0625

**S2.8** Seborrheic keratosis is a rare hereditary skin condition due to an autosomal dominant mutation. Affected people have skin marked with numerous small, sharply margined, yellowish or brownish areas covered with a thin, greasy scale. A man with keratosis marries a normal woman, and they have three children. What is the chance that all three are normal? What is the chance that all three are affected?

**S2.9** In *Drosophila,* the dominant allele *Cy (Curly)* results in curly wings. The cross *Cy* + *X Cy* + (where + represents the wildtype allele of *Cy)* results in a ratio of 2 curly : 1 wildtype $F_1$ progeny. The cross between the curly $F_1$ progeny also gives a ratio of 2 curly : 1 wildtype $F_2$ progeny. How can this result be explained?

**S2.10** White Leghorn chickens are homozygous for a dominant allele, C, of a gene responsible for colored feathers, and also for a dominant allele, *I,* of an independently segregating gene that prevents the expression of *C.* The White Wyandotte breed is homozygous recessive for both genes *cc ii.* What proportion of the $F_2$ progeny obtained from mating White Leghorn × White Wyandotte $F_1$ hybrids would be expected to have colored feathers?

**S2.11** The tailless trait in the Manx cat is determined by the alleles of a single gene. In the cross Manx × Manx, both tailless and tailed progeny are produced, in a ratio of 2 tailless : 1 tailed. All tailless progeny from this cross, when mated with tailed, produce a 1 : 1 ratio of tailless to tailed progeny.

**(a)** Is the allele for the tailless trait dominant or recessive?

**(b)** What genetic hypothesis can account for the 2 : 1 ratio of tailless: tailed and the result of the testcrosses with the tailless cats?

**S2.12** Absence of the enzyme alpha-1-antitrypsin is associated with a very high risk of developing emphysema and early death due to damaged lungs. Enzyme deficiency is usually due to homozygosity for an allele denoted *PI-Z.* A phenotypically normal man whose father died of the emphysema associated with homozygosity for *PI-Z* marries a phenotypically normal woman with a negative family history for the trait. They consider having a child.

**(a)** Draw the pedigree as described above.

**(b)** If the frequency of heterozygous *PI-Z* in the population is 1 in 30, what is the chance that the first child will be homozygous *PI-Z*?

**(c)** If the first child is homozygous *PI-Z,* what is the probability that the second child will be either homozygous or heterozygous for the normal allele?

**S2.13** Plants of the genotypes *Aa Bb cc* and *aa Bb Cc* are crossed. The three genes are on different chromosomes. What proportion of the progeny will have the phenotype *A- B- C-*, and what fraction of the *A- B- C-* plants will have the genotype *Aa Bb Cc*?

**(a)** 1/16; all

**(b)** 3/8; 2/3

**(c)** 3/16; 2/3

**(d)** 1/2; 1/2

**(e)** 1/2; all

**S2.14** In the cross *Aa Bb Cc Dd* ×*Aa Bb Cc Dd,* in which all four genes undergo independent assortment, what proportion of offspring are expected to be heterozygous for all four genes?

What is the answer if the cross is *Aa Bb Cc Dd* × *Aa Bb cc dd*?

**S2.15** In the cross *AA Bb Cc* × *Aa Bb cc,* where all three genes assort independently and each uppercase allele is dominant, what proportion of the progeny are expected to have the phenotype *A- bb cc*?

**(a)** 1/64

**(b)** 1/32

**(c)** 1/16

**(d)** 1/8

**(e)** 1/4

**S2.16** In the cross *Aa Bb* ×*Aa Bb,* what fraction of the progeny will be homozygous for one gene and heterozygous for the other if the two genes assort independently?

**(a)** 1/16

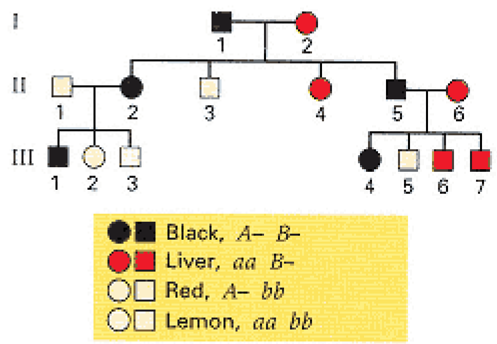
**(b)** 1/8

**(c)** 1/2

**(d)** 1/4

**(e)** 3/4

**S2.17** The pedigree in the accompanying illustration shows the inheritance of coat color in a group of cocker spaniels. The coat colors and genotypes are indicated in the key.




**(a)** Specify in as much detail as possible the genotype of each dog in the pedigree.

**(b)** What are the possible genotypes of animal III-4, and what is the probability of each genotype?

**(c)** If a single pup is produced from the mating of III-4 × III-3, what is the probability that the pup will be lemon?

**S2.18** In terms of the type of epistasis indicated, would you say that a 27 : 37 ratio has more in common with a 9 : 7 ratio or with a 15 : 1 ratio? Explain your argument.

**S2.19** A trihybrid cross *AA BB rr* × *aa bb RR* is made in a plant species in which *A* and *B* are dominant to their respective alleles but there is no dominance between *R* and *r*. Assuming independent assortment, consider the $F_2$ progeny from this cross.

**(a)** How many phenotypic classes are expected?

**(b)** What is the probability of the parental aa bb RR genotype?

**(c)** What proportion of the progeny would be expected to be homozygous for all three genes?

**S2.20** Four babies accidentally become mixed up in a maternity ward. The ABO types of the four babies are known to be O, A, B, and AB, and those of the four sets of parents are shown below. One of the attendants insists that DNA typing will be necessary to straighten out the mess, but another claims that the ABO blood types alone are sufficient. To determine who is correct, try to match each baby with its correct set of parents.

**(a)** AB x O

**(b)** A x O

**(c)** A x AB

**(d)** O x O

**S2.21** For each of the following $F_2$ ratios, some indicating epistasis, what phenotypic ratio would you expect in a testcross of the $F_1$ dihybrid?

**(a)** 15:1

**(b)** 13:3

**(c)** 9:4:3

**(d)** 9:6:1

**(e)** 12:3:1

**(f)** 9:3:3:1

**(g)** 3:6:3:1:2:1

**(h)** 1:2:1:2:4:2:1:2:1

**S2.22** In plants of the genus *Primula*, the *K* locus controls synthesis of a compound called malvidin. Two plants heterozygous at the *K* locus were crossed, producing the following distribution of progeny:

| | |
|---|---|
| 1010 | Make malvidin |
| 345 | Do not make malvidin |

You wish to determine if the *D* locus (located on a different chromosome from *K),* affects production of malvidin as well. True-breeding plants that make malvidin (*KK dd)* were crossed to true-breeding plants that do not make malvidin (*kk DD).* All $F_1$ plants failed to produce malvidin. $F_1$ heterozygotes were self-fertilized, and the $F_2$ progeny, assayed for malvidin synthesis, yielded the following distribution:

| | |
|---|---|
| 522 | Make malvidin |
| 2270 | Do not make malvidin |

**(a)** Write the genotypes and the corresponding phenotypes of all the $F_2$ progeny obtained.

**(b)** Does the *D* locus affect malvidin synthesis? What is the basis of your conclusion?

**S2.23** From the complementation data in the table below, assign the ten mutations *a1* through *a10* to complementation groups. Use the complementation groups to deduce the expected results of the complementation tests indicated as missing data (denoted by the question marks).

***Chapter 3—***
***Genes and Chromosomes***

**S3.1** Classify each statement as true or false as it applies to mitosis.

**(a)** DNA synthesis takes place in prophase; chromosome distribution to daughter cells takes place in metaphase.

**(b)** DNA synthesis takes place in metaphase; chromosome distribution to daughter cells takes place in anaphase.

**(c)** Chromosome condensation takes place in prophase; chromosome distribution to daughter cells takes place in anaphase.

**(d)** Chromosomes pair in metaphase.

**S3.2** Which of the following statements are true?

**(a)** A bivalent contains one centromere, two sister chromatids, and one homolog.

**(b)** A bivalent contains two centromeres, four sister chromatids, and two homologs.

**(c)** A bivalent contains two linear, duplex DNA molecules.

**(d)** A bivalent contains four linear, duplex DNA molecules.

**(e)** A bivalent contains eight linear, duplex DNA molecules.

**S3.3** Which of the following statements correctly describes a difference between mitosis and meiosis?

**(a)** Meiosis includes two nuclear divisions, mitosis only one.

**(b)** Bivalents appear in mitosis but not in meiosis.

**(c)** Sister chromatids appear in meiosis but not in mitosis.

**(d)** Centromeres do not divide in meiosis; they do in mitosis.

***Complementation Data Problem S2.23***

| | ***a1*** | ***a2*** | ***a3*** | ***a4*** | ***a5*** | ***a6*** | ***a7*** | ***a8*** | ***a9*** | ***a10*** |
|---|---|---|---|---|---|---|---|---|---|---|
| ***a1*** | – | + | + | + | – | + | + | + | ? | + |
| ***a2*** | | – | + | ? | + | + | + | – | ? | + |
| ***a3*** | | | – | + | + | – | ? | + | + | + |
| ***a4*** | | | | – | + | + | ? | – | + | + |
| ***a5*** | | | | | – | + | + | + | – | + |
| ***a6*** | | | | | | – | – | + | + | + |
| ***a7*** | | | | | | | – | + | + | + |
| ***a8*** | | | | | | | | – | + | + |
| ***a9*** | | | | | | | | | – | + |
| ***a10*** | | | | | | | | | | – |

**S3.4** In which of the following situations can the result of segregation be observed phenotypically?

**(a)** In the haploid products of meiosis or the cells derived from them by mitosis, provided that these can grow and divide (as in certain fungi).

**(b)** In the testcross progeny of a heterozygous diploid organism.

**(c)** In the $F_2$ progeny of genetically different diploid parents.

**(d)** Statements (a), (b), and (c).

**(e)** Statements (a) and (b) only.

**S3.5** Classify each of the following statements as true or false as it applies to the segregation of alleles for a genetic trait in a diploid organism.

**(a)** There is a 1 : 1 ratio of meiotic products.

**(b)** There is a 3 : 1 ratio of meiotic products.

**(c)** There is a 1 : 2 : 1 ratio of meiotic products.

**(d)** There is a 3 : 1 ratio of $F_2$ genotypes.

**(e)** All of the above.

**S3.6** What is a chiasma? In what stage of meiosis are chiasmata observed?

**S3.7** When in meiosis does the physical manifestation of independent assortment take place?

**S3.8** In the life cycle of corn, *Zea mays,* how many haploid chromosome sets are there in cells of the following tissues?

**(a)** Megasporocyte

**(b)** Megaspore

**(c)** Root tip

**(d)** Endosperm

**(e)** Embryo

**(e)** Microspore

**S3.9** Somatic cells of the red fox, *Vulpes vulpes,* normally have 38 chromosomes. What is the number of chromosomes present in the nucleus in a cell of this organism in each of the following stages? (For purposes of this problem, count each chromatid as a chromosome in its own right.)

**(a)** Metaphase of mitosis

**(b)** Metaphase I of meiosis

**(c)** Telophase I of meiosis

**(d)** Telophase II of meiosis

**S3.10** The terms "reductional division" and "equational division" are literally correct in describing the two meiotic divisions with respect to division of the centromeres. To assess the applicability of the terms to alleles along a chromosome, consider a chromosome arm in which exactly one exchange event takes place in prophase I. Denote as A the region between the centromere and the position of the exchange and as B the region between the position of the exchange and the tip of the chromosome arm. If "reductional" is defined as resulting in products that are genetically different and "equational" is defined as resulting in products that are genetically identical, what can you say about the "reductional" or "equational" nature of the two meiotic divisions with respect to the regions *A* and *B*?

**S3.11** What differences are there between the sexes in the pattern of transmission of genes located on the autosomes versus those on the sex chromosomes? (Assume that females are the homogametic sex and males the heterogametic sex.)

**S3.12** What observations about *Drosophila* provided proof of the chromosomal theory of heredity?

**S3.13** People with the chromosome constitution 47, XXY are phenotypically males. A normal woman whose father had hemophilia mates with a normal man and produces an XXY son who also has hemophilia. What kind of nondisjunction can explain this result?

**S3.14** Which of the following types of inheritance have the feature that an affected male has all affected daughters but no affected sons?

**(a)** Autosomal recessive

**(b)** Autosomal dominant

**(c)** Y-linked

**(d)** X-linked recessive

**(e)** X-linked dominant

**S3.15** Mendel studied the inheritance of phenotypic characters determined by seven pairs of alleles. It is an interesting coincidence that the pea plant also has seven pairs of chromosomes. What is the probability that, if seven loci are chosen at random in an organism that has seven chromosomes, each locus is in a different chromosome? (*Note:* Mendel's seven genes are actually located in only four chromosomes, but only two of the genes—tall versus short plant and smooth versus constricted pod—are close enough together that independent assortment would not be observed; he apparently never examined these two traits for independent assortment.)

**S3.16** *Drosophila montana* has a somatic chromosome number of 12. Say the centromeres of the six homologous pairs are designated as A/a, B/b, C/c, D/d, E/e, and F/f.

**(a)** How many different combinations of centromeres can be produced in meiosis?

**(b)** What is the probability that a gamete will contain only centromeres designated by capital letters?

**S3.17** How many different genotypes are possible for

**(a)** an autosomal gene with four alleles?

**(b)** an X-linked gene with four alleles?

**S3.18** For an autosomal gene with *m* alleles, how many different genotypes are there?

**S3.19** Consider the mating $Aa \times Aa$.

**(a)** What is the probability that a sibship of size 8 will contain no *aa* offspring?

**(b)** What is the probability that a sibship of size 8 will contain a perfect Mendelian ratio of 3 *A*-: 1 *aa*?

**S3.20** Matings between two types of guinea pigs with normal phenotypes produce 64 offspring, 11 of which showed a hyperactive behavioral abnormality. The other 53 offspring were behaviorally normal. Assume that the parents were heterozygous.

**(a)** Do these data fit the model that hyperactivity is caused by a recessive allele of a single gene?

**(b)** Do they fit the model that hyperactivity is caused by simultaneous homozygosity for recessive alleles of each of two unlinked genes?

**S3.21** In a $X^2$ test with 1 degree of freedom, the value of $X^2$ that results in significance at the 5 percent level ($P$ = 0.05) is approximately 4. If a genetic hypothesis predicts a 1 : 1 ratio of two progeny types, calculate what deviations from the expected 1 : 1 ratio yield $P = 0.05$ (and rejection of the genetic hypothesis) when

**(a)** the total number of progeny equals 50

**(b)** the total number of progeny equals 100

## *Chapter 4— Gene Linkage and Chromosome Mapping*

**S4.1** After a large number of genes have been mapped, how many linkage groups will be found in each of the following?

**(a)** The bacterial cell *Salmonella typhimurium,* with a single, circular DNA molecule

**(b)** The haploid yeast *Saccharomyces cerevisiae,* with 16 chromosomes

**(c)** The common wheat *Triticum vulgare,* with 42 chromosomes per somatic cell

**S4.2** Does physical distance always correlate with map distance?

**S4.3** In genetic linkage studies, what is the definition of a map unit?

**S4.4** Distinguish between the terms "chromosome interference" and "chromatid interference." Which is measured by the coincidence?

**S4.5** What does the coefficient of coincidence measure?

**(a)** The frequency of double crossovers

**(b)** The ratio of double crossovers to single crossovers

**(c)** The ratio of single crossovers in two adjacent regions

**(d)** The ratio of double crossovers to the number expected if there were no interference

**(e)** The ratio of recombinant to parental progeny

**S4.6** A coefficient of coincidence of 0.25 means that:

**(a)** the frequency of double crossovers is 1/4.

**(b)** the frequency of double crossovers is 1/4 of the number expected if there were no interference.

**(c)** there were four times as many single crossovers as double crossovers.

**(d)** there were four times as many single crossovers in one region as there were in an adjacent region.

**(e)** there were four times as many parental as recombinant progeny.

**S4.7** In wheat, normal plant height requires the presence of either or both of two dominant alleles, *A* and *B.* Plants homozygous for both recessive alleles (*a b*/*a b)* are dwarfed but otherwise normal. The two genes are on the same chromosome and recombine with a frequency of 16 percent. From the cross *A b*/*a B* × *A B*/*a b*, what is the expected frequency of dwarfed plants among the progeny?

**S4.8** Two yeast strains were mated; one was *red* with mating type **a;** the other was $red^+$ with mating type α. The resulting diploids were induced to undergo meiosis, and the meiotic products of the diploid were as follows (total progeny = 100):

| | | |
|---|---|---|
| 40 | $red^+$ | α |
| 44 | *red* | **a** |
| 9 | $red^+$ | **a** |
| 7 | *red* | α |

How many map units separate the *red* gene from that for mating type?

**S4.9** In meiosis, under what conditions does seconddivision segregation of a pair of alleles take place?

**S4.10** Explain why the most accurate measurement of map distance between two genes is obtained by summing shorter distances between the genes.

**S4.11** In *D. melanogaster,* the recessive allele *ap (apterous)* drastically reduces the size of the wings and halteres (flight balancers), and the recessive allele *cl (clot)* produces dark maroon eye color. Both genes are located in the second chromosome. You have a friend taking a genetics laboratory course. To obtain the genetic distance between these genes, she performs a cross of a double-heterozygous female and a maroon-eyed, wingless male. She examines 25 flies in the $F_1$ generation and calculates that the frequency of recombination is 56 percent. What is wrong with this number? What is a plausible explanation for the results obtained?

**S4.12** Your friend in Problem S4.11 logs onto FlyBase and learns that the *ap-cl* distance is 36 map units. She therefore decides to repeat the cross and to examine more $F_1$ flies. Because both genes are autosomal, she does not expect any difference between reciprocal crosses. Hence, in setting up the cross this time, she mates a double-heterozygous male with a maroon-eyed, wingless female. To her astonishment, among 1000 $F_1$ progeny she was unable to find even one recombinant progeny. How can this anomaly be explained?

**S4.13** The genetic map of an X chromosome of *Drosophila melanogaster* has a length of 73.1 map units. The X chromosome of the related species, *Drosophila virilis*, is even longer—170.5 map units. In view of the fact that the frequency of recombination between two genes cannot exceed 50 percent, how is it possible for the map distance between genes at opposite ends of a chromosome to exceed 50 map units?

**S4.14** In *D. virilis*, the mutations dusky body color (*dy)*, cut wings (*ct)*, and white eyes (*w)* are all recessive alleles located in the X chromosome. Females heterozygous for all three mutations were crossed to *dy ct w* males, and the following phenotypes were observed among the offspring.

| | |
|---|---|
| wildtype | 18 |
| dusky | 153 |
| cut | 6 |
| white | 76 |
| cut, white | 150 |
| dusky, cut | 72 |
| dusky, cut, white | 22 |
| dusky, white | 3 |

**(a)** What are the genotypes of two maternal X chromosomes?

**(b)** What is the map order of the genes?

**(c)** What are the two-factor recombination frequencies between each pair of genes?

**(d)** Are there as many observed double-crossover gametes as would be expected if there were no interference?

**(e)** What are the coefficient of coincidence and the interference across this region of the X chromosome?

**S4.15** The genes considered in Problem S4.14 are also located on the X chromosome in *D. melanogaster*. However, their order and the distances between them are completely different because of chromosome rearrangements that have happened since the divergence of two species approximately 40 million years ago. In *D.melanogaster*, the genetic map of this region is *dy*-16.2-*ct*-18.5-*w*. If you performed a cross identical to that in Problem S4.14 with *D. melanogaster* and observed 500 progeny, what types of progeny would be expected and in what numbers? Assume that the interference between the genes is the same in *D.melanogaster* as in *D. virilis*, and round the expected number of progeny to the nearest whole number. For convenience, list the progeny types in the same order as in Problem S4.14.

**S4.16** What experiment in *Drosophila* demonstrated that crossing-over takes place in the four-strand stage? How would the result of the experiment have been different if crossing-over took place in the two-strand stage?

**S4.17** *Neurospora crassa* is a haploid ascomycete fungus. A mutant strain that requires arginine for growth ($arg^-$) is crossed with a strain that requires tyrosine and phenylalanine ($try^-$ $phe^-$). Each requirement results from a single mutation. Tests on 200 randomly collected spores from the cross gave the following results, in which the + in any column indicates the presence of the wildtype allele of the gene.

| arg | tyr | phe | Number of spores |
|---|---|---|---|

| | | | |
|---|---|---|---|
| + | – | – | 43 |
| – | + | + | 42 |
| + | - | + | 43 |
| – | + | – | 42 |
| + | + | – | 8 |
| – | – | + | 9 |
| + | + | + | 6 |
| – | – | – | 7 |

**(a)** What are the three possible two-gene recombination frequencies?

**(b)** Is any linkage suggested by these data?

**S4.18** In the case of independently assorting genes, why are parental ditype (PD) and nonparental ditype (NPD) tetrads formed in equal numbers? Why are nonparental ditype tetrads rare for linked genes?

**S4.19** The yeast *Saccharomyces cerevisiae* has unordered tetrads. In a cross made to study the linkage relationship among three genes, the following tetrads were obtained. The cross was between a strain of genotype + *b c* and one of genotype*a* + +.

| Tetrad type | Genotypes of spores in tetrads | | | | Number of tetrads |
|---|---|---|---|---|---|
| 1 | *a* + + | *a* + + | + *b c* | + *b c* | 135 |
| 2 | *a b* + | *a b* + | + + *c* | + + *c* | 124 |
| 3 | *a* + + | *a* + *c* | + *b* + | + *b c* | 59 |
| 4 | *a b* + | a b c | + + + | + + *c* | 70 |
| 5 | *a* + *c* | *a* + *c* | + *b* + | + *b* + | 1 |
| 6 | a b c | a b c | + + + | + + + | 1 |
| | | | | Total | 390 |

**(a)** From these data, determine which (if any) of the genes are linked.

**(b)** For any linked genes, determine the map distances.

**S4.20** Consider a strain of *Neurospora* that exhibits complete chromosomal interference, even across the centromere. Two genes, *A* and *B*, are located in the same chromosome but on different sides of the centromere. Gene *A* is 10 map units from the centromere, and gene *B* is 5 map units from the centromere. Calculate the expected percentage of each of the following types of asci.

**(a)** Parental ditype

**(b)** Nonparental ditype

**(c)** Tetratype

**(d)** First-division segregation for *A*, first for *B*

**(e)** First-division segregation for *A*, second for *B*

**(f)** Second-division segregation for *A*, first for *B*

**(g)** Second-division segregation for *A*, second for *B*

**S4.21** You are a geneticist writing a set of test questions and wish to invent plausible numbers of the possible types of progeny in a three-point testcross to yield a frequency of recombination of 0.2 in "region I," a frequency of recombination of 0.3 in "region II," and an interference value of 2/3. Among 1000 total progeny, how many should be:

**(a)** nonrecombinant?

**(b)** recombinant in region I only?

**(c)** recombinant in region II only?

**(d)** recombinant in both regions (double crossovers)?

**(e)**In each class of progeny above, how should the total number be allocated between the two reciprocal products of recombination?

**S4.22** What is the answer to Problem S4.21 in the general case when the frequency of recombination in region I is $r_1$, that in region II is $r_2$, and the coincidence equals $c$?

**S4.23** In a testcross of a parent heterozygous for each of four linked genes, in which the order of the genes is known, the following numbers of progeny were obtained:

| | |
|---|---|
| Nonrecombinant | 486 |
| Recombinant in region I only | 61 |
| Recombinant in region II only | 143 |
| Recombinant in region III only | 226 |
| Recombinant in regions I and II only | 10 |
| Recombinant in regions II and III only | 45 |
| Recombinant in regions I and III only | 27 |
| Recombinant in regions I and II and III | 2 |

**(a)** What are the map distances in centimorgans of regions I, II, and III? (*Hint:* Solve the problem as in a three-point cross, but remember that recombinants in two or three regions are recombinant for each region separately.)

**(b)** Calculate the values of the interference for regions I + II, II + III, and I + III. Does the ranking of the interference values make intuitive sense?

**S4.24** In a fungal organism with ordered tetrads, deduce whether one observes first-division segregation or second-division segregation when the region between a gene and the centromere undergoes

**(a)** a two-strand double crossover

**(b)** either of the two types of three-strand double crossover

**(c)** a four-strand double crossover

If the four types of double crossover are equally frequent, what proportion of asci formed from double crossing-over will show second-division segregation?

**S4.25** Show that each of the following statements is true in a fungal organism with ordered tetrads,

**(a)** If a configuration of multiple crossovers will result in first-division segregation, then an additional crossover (distal to the existing ones) will convert the ascus to second-division segregation with probability 1.

**(b)** If a configuration of multiple crossovers will result in second-division segregation, then an additional crossover (distal to the existing ones) will convert the ascus to first-division segregation with probability 1/2.

## *Chapter 5— The Molecular Structure and Replication of the Genetic Material*

**S5.1** You have determined that one strand of a DNA double helix has the sequence 5'-AGCCTAG-3'. What is the sequence of the complementary strand?

**S5.2** In the replication of a bacterial chromosome, does replication begin at a random point?

**S5.3** The haploid genome of the wall cress, *Arabidopsis thaliana,* contains 100,000 kb and has five chromosomes. If a particular chromosome contains 10 percent of the DNA in the haploid genome, what is the approximate length of its DNA molecule in micrometers?

**S5.4** For the chromosome of *A. thaliana* in Problem S5.3, estimate the time of replication, assuming that there is only one origin of replication (exactly in the middle), that replication is bidirectional, and that the rate of DNA synthesis is

**(a)** 1500 nucleotide pairs per second (typical of bacterial cells)

**(b)** 50 nucleotide pairs per second (typical of eukaryotic cells)

**S5.5** In terms of the mechanism of DNA replication, explain why a linear chromosome without telomeres is expected to become progressively shorter in each generation.

**S5.6** A friend brings you three samples of nucleic acid and asks you to determine each sample's chemical identity (whether DNA or RNA) and whether the molecules are double-stranded or single-stranded. You use powerful nucleases to degrade each sample to its constituent nucleotides, and then you determine the approximate relative proportions of nucleotides. The results of your assays are presented below. What can you tell your friend about the nature of these samples?

***Assay results***
***Problem S5.6***

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Sample 1: | dGTP | 14% | dCTP | 15% | dATP | 36% | dTTP | 35% |
| Sample 2: | dGTP | 12% | dCTP | 36% | dATP | 47% | dTTP | 5% |
| Sample 3: | GTP | 22% | CTP | 47% | ATP | 17% | UTP | 14% |

**S5.7** What is meant by the statement that the DNA replication fork is asymmetrical?

**S5.8** What is the role of each of the following proteins in DNA replication?

**(a)** DNA helicase

**(b)** Single-stranded DNA-binding protein

**(c)** DNA polymerase

**(d)** DNA ligase

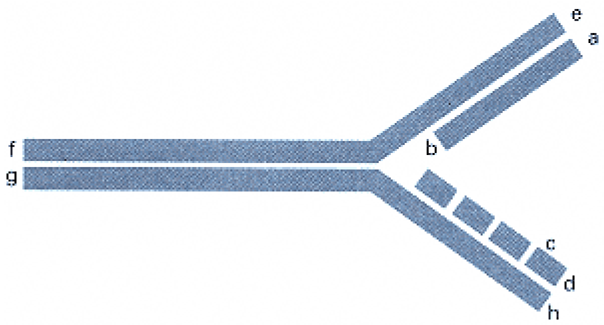
**S5.9** Which enzyme in DNA replication

**(a)** alters the helical winding of double-stranded DNA by causing a single-strand break, exchanging the relative position of the strands, and sealing the break?

**(b)** uses ribonucleoside triphosphates to synthesize short RNA primers?

**S5.10** Describe why topoisomerase is necessary during replication of the *E. coli* chromosome.

**S5.11** The accompanying drawing is of a replication fork. The ends of each strand are labeled with the letters a through h.




**(a)** Label the leading and lagging strands and, by the addition of arrowheads, show the direction of movement of the replication fork and the direction of synthesis of each DNA strand or fragment.

**(b)** Specify which of the letters a through h label 5' ends and which label 3' ends.

**S5.12** Some "thermophilic" bacteria can live in water so hot that it approaches the boiling point. Which of the following DNA sequences would be more likely to be found in such an organism? Explain.

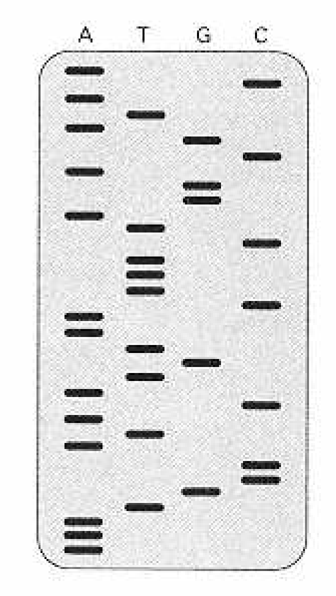
**(a)**
```
5'-TTATAAAATATATTTTTTATAT-3'
3'-AATATTTTATATAAAAATATA-5'
```

**(b)**
```
5'-CCCCCGCGCGGCCGGGCGCGCG-3'
3'-GGGGGCGCGCCGGCCCGCGCGC-5'
```

**S5.13** A 3.1-kilobase linear fragment of DNA was digested with *Pst*I and produced a 2-kb and a 1.1-kb fragment. When the same 3.1-kb fragment was cut with *Hind*III, it yielded a 1.5-kb fragment, a 1.3-kb fragment, and a 0.3-kb fragment. When the 3.1-kb molecule was cut with a mixture of the two enzymes, fragments of 1.5, 0.8, 0.5, and 0.3 kb resulted. Draw a map of the original 3.1-kb fragment, and label restriction sites and the distances between the sites.

**S5.14** The X-linked recessive mutation *dusky* in *Drosophila* causes small, dark wings. In a stock of wildtype flies, you find a single male that has the dusky phenotype. In wildtype flies, the *dusky* gene is contained within an 8-kb *Xho*I restriction fragment. When you digest genomic DNA from the mutant male with *Xho*I and probe with the 8-kb restriction fragment on a Southern blot, the size of the labeled fragment is 10 kb. You clone the 10-kb fragment and use it as a probe for a polytene chromosome *in situ* hybridization in a number of different wildtype strains, and you notice that this fragment hybridizes to multiple locations along the polytene chromosomes. Each wildtype strain has a different pattern of hybridization. What do these data suggest about the origin of the *dusky* mutation that you isolated?

**S5.15** A short region of an mRNA molecule is used as a primer in dideoxy sequencing. After transfer to a filter and autoradiography, the resulting film looks as follows, where the orientation is with the samples loaded at the top and the letters indicate the dideoxynucleotide present during synthesis.




**(a)** Write the sequence implied by the bands in the gel from the 5' end to the 3' end.

**(b)** Is this the sequence of the elongated RNA strand or the sequence of its complementary DNA strand?

**S5.16** For the following restriction endonucleases, calculate the average distance between restriction sites in an organism whose DNA has a random sequence and equal proportions of all four nucleotides. The symbol R means any purine (A or G) and Y means any pyrimidine (T or C), but an R-Y pair must be either A-T or G-C.

| | |
|---|---|
| *Taq*I | 5'-TCGA-3'<br>3'-AGCT-5' |
| *Bam*HI | 5'-GGATCC-3'<br>3'-CCTAGG-5' |
| *Hae*II | 5'-RGCGCY-3'<br>3'-YCGCGR-5' |

**S5.17** For each of the restriction enzymes in Problem S5.16, calculate the expected number of restriction sites in the genome of a nematode *Caenorhabditis elegans*, assuming the sequence is approximately random. The genome size of this organism is 100,000 kb.

**S5.18** In early studies of the properties of Okazaki fragments, it was shown that these fragments could hybridize to both strands of *E. coli* DNA. This was taken as an evidence that they are synthesized on both branches of the replication fork. However, one feature of the replication of *E. coli* DNA that was not known at the time invalidates this conclusion. What is this characteristic?

**S5.19** In experiments that confirmed that DNA replication is semiconservative, *E. coli* DNA was labeled by growing cells for a number of generations in a medium containing "heavy" $^{15}N$ isotope and then observing the densities of the molecules during subsequent generations of growth in a medium containing a great excess of "light" $^{14}N$. If DNA replication were conservative, what pattern of molecular densities would have been observed after one generation and after two generations of growth in the $^{14}N$-containing medium?

**S5.20** A covalently closed circular single-stranded DNA molecule is hybridized to a linear complementary strand that has a segment of 100 nucleotides missing. This duplex is exhaustively treated with various nucleases. List the final reaction products for each enzyme.

**(a)** An endonuclease that cuts only single-stranded DNA

**(b)** An exonuclease that degrades only single-stranded DNA

**(c)** An exonuclease that attacks only double-stranded DNA and degrades it in the 3' $\rightarrow$ 5' direction (removing the shorter strand from a paired duplex when the partners are of unequal length)

**(d)** An endonuclease that attacks and degrades only double-stranded DNA

**(e)** A mixture of enzymes (b) and (d)

**(f)** A mixture of enzymes (a) and (c)

**(g)** A mixture of enzymes (c) and (d)

## *Chapter 6—*
## *The Molecular Organization of Chromosomes*

**S6.1** What are the principal molecular constituents of the nucleosome?

**S6.2** Is it possible for a gene of average size to be present in a single nucleosome?

**S6.3** A geneticist wishes to isolate mutations that affect the amino acid sequence of any of the histone genes H2, H3A, H3B, and H4. What is the likely phenotype of such a mutation?

**S6.4** In spermatogenesis in many male animals, the normal somatic-cell histones are removed from the DNA and replaced with a still more highly basic (arginine-rich and lysine-rich) set of sperm-specific histones. Suggest one possible explanation.

**S6.5** What are principal structural levels of chromosome organization?

**S6.6** Why are centromeres and telomeres required for chromosomes to be genetically stable?

**S6.7** The nematode *C. elegans* has holocentric chromosomes. If the chromosomes in a cell are fragmented into many pieces by treatment with x rays, how would you expect each fragment to behave in cell division?

**S6.8** Name three features that distinguish euchromatin from heterochromatin.

**S6.9** Explain why topoisomerase is necessary during replication of the *E. coli* chromosome.

**S6.10** What is one possible reason why there is usually a single gene in the genome coding for a particular protein, whereas genes for rRNA and tRNA are repeated many times?

**S6.11** The X-linked recessive mutation *singed* (*sn)* causes gnarled bristles in *Drosophila.* In a stock of wildtype flies, you find a single male that has the singed phenotype.

**(a)** Diagram a crossing scheme that you will use to generate a true-breeding line of *sn* flies. Note that, other than the single singed male, you have only wildtype flies available.

**(b)** In wildtype flies, the *sn* gene is contained within an 8-kb *Bam*HI restriction fragment. When you digest genomic DNA from the mutant line with *Bam*HI and probe with the 8-kb restriction fragment on a Southern blot, the size of the labeled fragment on the filter is 13 kb. What possible explanations could account for this finding?

**S6.12** A genetically engineered strain of *Drosophila virilis* (strain 1) is homozygous for the recessive mutation *y* (yellow), an X-linked mutation that causes yellow body color. This strain also carries, at a site in the X chromosome, a genetically engineered transposable element called *Hermes* into which a copy of the $y^+$ gene was inserted; the engineered *Hermes* element is unable to produce the transposase protein needed for transposition of the element. Another genetically engineered strain, strain 2, is also homozygous *y* but contains a copy of *Hermes* that can produce the transposase in each homolog of chromosome 3 . A female of strain 1 is crossed with a male of strain 2. The phenotypically wildtype $F_1$ males are crossed with yellow females. About 5 percent of the $F_2$ progeny males have wildtype body color. Give two possible explanations of how these males might arise.

**S6.13** The term "satellite DNA" was originally coined for highly repetitive simple sequences that, in equilibrium density-gradient centrifugation in CsCl, formed a distinct band (a "satellite" band) different from the bulk of the genomic DNA, because of a difference in density of the DNA fractions. You analyze the genome organization of a new species by the kinetics of DNA renaturation and find that 10 percent of DNA is represented by a highly repetitive simple sequence. Does this mean that, after equilibrium density-gradient centrifugation in CsCl, you will find a satellite band?

**S6.14** DNA from species A that is labeled with $^{14}N$ and randomly fragmented is renatured with an equal concentration of DNA from species B that is labeled with $^{15}N$ and also randomly fragmented. The resulting mixture of molecules is centrifuged to equilibrium in CsCl. Ten percent of the total renatured DNA has a hybrid density. What fraction of the base sequences in the two species will be similar enough to renature?

**S6.15** You are studying the denaturation of two unusual DNA molecules, both consisting of 99 percent AT base pairs. Both molecules have the same melting temperature. However, observations in the electron microscope indicate that the strands of one of the molecules do not separate completely until raised to a temperature 12°C higher than that required for complete separation of the strands of the other molecule. Suggest an explanation.

**S6.16** A molecule of 5000 nucleotide pairs consists of the repeating sequence 5'-ACGTAG-3'. How would the melting temperature of this molecule compare to that of another molecule of the same length and base composition (50 percent G + C) but with a random sequence?

**S6.17** A Cot analysis is carried out with DNA obtained from a particular plant species. The chloroplast DNA is readily detected as a rapidly reannealing fraction because of its abundance in the DNA sample. Plants of the same species are maintained in the dark for several weeks until they lose all their chlorophyll and the leaves become white. DNA from these plants has the same Cot curve as observed in plants maintained in the light. What does this result tell you about the loss of chlorophyll?

### *Chapter 7— Variation in Chromosome Number and Structure*

**S7.1** *Drosophila* and human beings have very different mechanisms of sex determination. Which of the following statements correctly describes the phenotype of an XXY organism in both species?

**(a)** Fertile male in *Drosophila*, sterile female in humans.

**(b)** Sterile female in *Drosophila*, fertile male in humans.

**(c)** Fertile female in *Drosophila*, fertile male in humans.

**(d)** Fertile female in *Drosophila*, sterile male in humans.

**(e)** Fertile female in *Drosophila*, fertile female in humans.

**S7.2** What are the consequences of X-chromosome inactivation in mammals?

**(a)** The formation of Barr bodies

**(b)** A mutation

**(c)** A mosaic expression of different phenotypes in different cells of heterozygous X-linked genes in females

**(d)** A mosaic expression of heterozygous autosomal genes in females

**(e)** Turner syndrome

**S7.3** Nondisjunction of chromosomes in meiosis may take place at the first or the second meiotic division. If nondisjunction of the XY pair of chromosomes took place at the first division in the formation of sperm, which of the following chromosomal types of sperm could be formed?

**(a)** XY

**(b)** YY

**(c)** XX

**(d)** X

**(e)** Y

**S7.4** Which of the following human disorders can result from nondisjunction during the formation of the egg?

**(a)** Down syndrome

**(b)** Klinefelter syndrome

**(c)** XYY male

**S7.5** Why are abnormalities in the number of sex chromosomes in mammals usually less severe in their phenotypic effects than abnormalities in the number of autosomes?

**S7.6** What are the consequences of a single crossover within the inverted region of a pair of homologous chromosomes with the gene order is *A B C D* in one and *a c b d* in the other, in each of the following cases?

**(a)** The centromere is not included within the inversion.

**(b)** The centromere is included within the inversion.

**S7.7** What role might inversions play in evolution?

**S7.8** In human beings, trisomy 21 is the cause of Down syndrome. Ordinarily, only one child with Down syndrome is seen in a sibship. Occasionally, a sibship with multiple affected children is observed. In one such family, a pair of normal parents had seven children. Four were normal, and three had an atypical form of Down syndrome. In this atypical form, the slowed growth, mental handicap, transverse crease on the palm of the hand, and short broad hands were present, but the ear and heart defects often observed with Down syndrome were not present. One of the normal children gave rise to five offspring, two of whom had the same atypical form of Down syndrome. Provide a genetic explanation for this atypical form of Down syndrome and for its inheritance pattern.

**S7.9** Colchicine is used to cause a doubling of chromosome number in many plant species. Starting with a diploid strain of watermelon (*Citrullus lanatus),* outline a series of crosses you would use to create a pentaploid strain of this species.

**S7.10** Explain why, in *Drosophila,* triploid flies are much more likely to survive than trisomics.

**S7.11** In a certain fertile tetraploid plant, a locus with alleles *A* and *a* is situated very near the centromere of its chromosome. If a tetraploid plant of genotype *AAaa* is crossed with a diploid plant of genotype *aa,* what genotypic ratios are expected? What phenotypic ratios are expected if *A* is dominant to *a*? (Assume that homologous chromosomes in the tetraploid form pairs at random and that all gametes produced by the tetraploid are diploid. Assume also that the dominant phenotype is expressed whenever at least one *A* allele is present.)

**S7.12** Explain the apparent paradox that familial retinoblastoma is regarded as an example of a recessive oncogene (tumor suppressor gene) even though it is inherited in pedigrees as an autosomal dominant. Familial retinoblastomas are generally bilateral (that is, there are multiple independent tumors in both eyes of affected individuals), whereas sporadic retinoblastoma is generally limited to one eye and to fewer independent tumors. Account for these observations.

**S7.13** Two strains of a plant called shepherd's purse (*Capsella bursa-pastoris),* a widespread lawn and roadside weed of the mustard family, are compared. In strain 1, the recombination frequency between genes *A* and *B* is 6 percent, whereas in strain 2, it is 50 percent. The $F_1$ progeny from a cross between these strains produce gametes with the following properties: About half are viable and contain only non-recombinant chromosomes, and about half are nonviable. How can these results be explained?

**S7.14** True-breeding tetrasomic organisms are much easier to create than true-breeding trisomic organisms. Suggest a reason for this observation.

**S7.15** In plants, a cross between a diploid organism and a tetraploid organism of the same species is usually more fertile than that between a diploid organism and a triploid organism of the same species. Suggest a possible explanation.

**S7.16** In *Drosophila melanogaster,* the genes for *brown eyes* (*bw)* and *humpy thorax* (*hy)* are about 12 map units distant on the same arm of chromosome 2. A paracentric inversion spans about one-third of the region but does not include these genes. What recombinant frequency between *bw* and *hy* would you expect in each of the following?

**(a)** Females that are homozygous for the inversion

**(b)** Females that are heterozygous for the inversion

**S7.17** The sequence of genes in a normal human chromosome is *123 • 456789,* where the raised dot (•) represents the centromere. Aberrant chromosomes with the following arrangements of genes were observed.

I. *123 • 476589*

II. *123 • 46789*

III. *1654 • 32789*

IV. *123 • 4566789*

**(a)** Identify each chromosome as a paracentric inversion, pericentric inversion, deletion, or duplication, and diagram how each rearrangement would pair with the normal homolog in meiosis.

**(b)** Draw the meiotic products of a crossover between genes *5* and *6* when chromosome I pairs with a wildtype chromosome.

**(c)** Draw the products of a crossover between genes *5* and *6* when chromosome III pairs with a wildtype chromosome.

**S7.18** Semisterile tomato plants heterozygous for a reciprocal translocation between chromosomes 5 and 11 were crossed with chromosomally normal plants homozygous for the recessive mutant broad leaf on chromosome 11. When semisterile $F_1$ plants were crossed with the plants of broad-leaf parental type, the following phenotypes were found in the backcross progeny:

| | |
|---|---|
| semisterile broad leaf | 38 |
| fertile broad leaf | 242 |
| semisterile normal leaf | 282 |
| fertile normal leaf | 33 |

**(a)** What is the recombination frequency between the broad-leaf gene and the translocation breakpoint in chromosome 11?

**(b)** What ratio of phenotypes in the backcross progeny would have been expected if the broad-leaf gene had not been on the chromosome involved in the translocation?

**S7.19** Semisterile tomato plants heterozygous for the same translocation referred to in the previous problem were crossed with chromosomally normal plants homozygous for the recessive mutants *wilty* and *leafy* in chromosome 5. When semisterile $F_1$ plants were crossed with plants of *wilty, leafy* parental genotype, the following phenotypes were found in the backcross progeny.

| | Semisterile | Fertile | Total |
|---|---|---|---|
| wildtype | 333 | 19 | 352 |
| wilty | 17 | 6 | 23 |
| leafy | 1 | 8 | 9 |
| wilty and leafy | 25 | 273 | 298 |

What is the map distance between the two genes and the map distance between each of the genes and the translocation breakpoint in chromosome 5?

**S7.20** The accompanying diagrams depict wildtype chromosomes 2 and 3 from a certain plant. The filled circles represent the respective centromeres.


**(a)** Show how these chromosomes would pair in meiosis in an individual heterozygous for an inversion that includes genes *E, F,* and *G* as well as heterozygous for a reciprocal translocation between chromosomes 2 and 3 with breakpoints between *B* and *C* in 2 and between *K* and *L* in 3.

**(b)** Suppose that the breakpoints of the above inversion are sufficiently close to genes *D* and *H* that recombination between the genes is completely eliminated when the inversion is heterozygous. A plant heterozygous for the inversion and the translocation and of genotype *Dd Hh* is crossed with a plant with normal chromosomes and of genotype *dd hh.* The most frequent classes of offspring are *dd Hh* semisterile and *Dd hh* ertile. The rarest types are *Dd hh* semisterile and *dd Hh* fertile, which together have a combined frequency of 0.15. What is the recombination frequency between *D* and the translocation breakpoint?

**S7.21** Six mutations of bacteriophage T4 (*a* through *f)* are isolated and crossed in all pairwise combinations to three deletion mutations (numbered 1–3) with the results shown below. In the table, R means that wildtype recombinants were observed and 0 means that there were no wildtype recombinants.

| **Point mutations** | ***a*** | ***b*** | ***c*** | ***d*** | ***e*** | ***f*** |
|---|---|---|---|---|---|---|
| Deletions 1 | 0 | 0 | R | R | R | R |
| Deletions 2 | 0 | R | 0 | 0 | 0 | 0 |
| Deletions 3 | 0 | 0 | R | 0 | 0 | 0 |

In parallel with the above experiments, the mutations were tested in all possible pairwise combinations for complementation. The results are tabulated below, where + indicates complementation and - indicates no complementation.

| | ***a*** | ***b*** | ***c*** | ***d*** | ***e*** | ***f*** |
|---|---|---|---|---|---|---|
| ***a*** | – | + | + | – | + | + |
| ***b*** | + | – | + | + | + | + |
| ***c*** | + | + | – | + | – | + |
| ***d*** | – | + | + | – | + | + |
| ***e*** | – | + | + | – | + | + |
| ***f*** | + | + | + | + | + | – |

Assuming that complementation indicates that two mutations are in different genes, combine the deletion mapping and the complementation data to determine the most likely relative order of the mutations. Group the mutations into allelic classes, and determine the position and extent of the three deletions.

## *Chapter 8—*
## *The Genetics of Bacteria and Viruses*

**S8.1** Define the following terms:

**(a)** Transformation

**(b)** Conjugation

**(c)** Transduction

**(d)** Transposition

**(e)** Temperate phage

**S8.2** Describe the differences between phage λ and F-plasmid integration into bacterial chromosome.

**S8.3** Is it possible for a plasmid without any genes to exist?

**S8.4** What process is responsible for the conversion of an $F^+$ bacterial cell into an Hfr cell?

**S8.5** What is the major difference in bacteriophage between the lytic cycle and the lysogenic cycle?

**S8.6** Why are specialized transducing particles of phage λ generated only when a lysogen is induced to produce phage rather than in the process of lytic infection?

**S8.7** Why is generalized transduction not the preferred method for mapping two unknown mutants relative to one another on the bacterial chromosome?

**S8.8** A temperate phage has the gene order *a b c d e f g h,* whereas the prophage of the same phage has the gene order *g h a b c d e f.* What information does this circular permutation give you about the location of the phage attachment site?

**S8.9** Recombination in phage λ crosses reveals the order of certain genes to be


where *cos* represents the cohesive ends and *att* is the phage attachment site. What order of the genes would be found if the λ prophage were mapped by generalized P1 transduction of an *E. coli* strain lysogenic for phage λ?

**S8.10** You are studying a biochemical pathway in *Neurospora* that leads to the production of substance A. You isolate a set of mutations, each of which is unable to grow on minimal medium unless it is supplemented with A. By performing appropriate matings, you group all the mutants into four complementation groups (genes) designated *a1, a2, a3,* and *a4.* You know beforehand that the biochemical pathway for the production of A includes four intermediates: B, C, D, and E. You test the nutritional requirements of your mutants by growing them on minimal medium supplemented with each of these intermediates in turn. The results are summarized in the accompanying table, where the plus signs indicate growth and the minus signs indicate failure to grow.

| | A | B | C | D | E |
|---|---|---|---|---|---|
| *a1* | + | + | + | – | + |
| *a2* | + | – | + | – | + |
| *a3* | + | – | – | – | + |
| *a4* | + | – | – | – | – |

Determine in what order the substances A, B, C, D, and E are most likely to participate in the biochemical pathway, and indicate the enzymatic steps by arrows. Label each arrow with the name of the gene that codes for the corresponding enzyme.

**S8.11** A experiment was carried out in *E. coli* to map five genes around the chromosome using each of three different Hfr strains. The genetic markers were *bio, met, phe, his,* and *trp.* The Hfr strains were found to transfer the genetic markers at the times indicated in the accompanying table. Construct a genetic map of the *E. coli* chromosome that includes all five genetic markers, the genetic distances in minutes between adjacent gene pairs, and the origin and direction of transfer of each Hfr. Complete the missing entries in the table, indicated by the question marks.

| | | | | |
|---|---|---|---|---|
| Hfr1 markers | *bio* | *met* | *phe* | *his* |
| Time of entry | 26 | 44 | 66 | ? |
| Hfr2 markers | *phe* | *met* | *bio* | *trp* |
| Time of entry | ? | 26 | 44 | 75 |
| Hfr3 markers | *phe* | *his* | ? | *bio* |
| Time of entry | 6 | 27 | 35 | ? |

**S8.12** A first-year graduate student carried out transformation of an *E. coli* strain of genotype *trp*$^-$ *azi*$^-$ *phe*$^-$ *gal*$^-$ *pur*$^-$ *his*$^-$ *ser*$^-$ using a plasmid with the selectable markers *amp-r kan-r trp*$^+$ *azi*$^+$ *bio*$^+$ *phe*$^+$ *pur*$^+$ *tet-r.* After transformation, he plated the cells on minimal medium containing galactose as a carbon source along with histidine and serine and added ampicillin and kanamycin to select for the transformants. After incubating the plates overnight, the student returned to the lab to discover that no colonies had formed! Then he realized that he had made a stupid mistake. What was it?

**S8.13** The genes *A, B, G, H, I,* and *T* were tested in all possible pairs for cotransduction with bacteriophage P1. Only the following pairs were found able to be cotransduced: *G* and *H, G* and *I, T* and *A, I* and *B, A* and *H.* What is the order of the genes along the chromosome? Explain your logic.

**S8.14** Bacteriophage P1 was grown on a wildtype strain of *E. coli,* and the resulting progeny phage were used to infect a strain with three mutations, *arg*$^-$, *pro*$^-$, and *his*$^-$, each of which results in a requirement for an amino acid: arginine, proline, and histidine, respectively. Equal volumes of the infected culture were spread on plates containing minimal medium supplemented with various combinations of the amino acids, and the colonies appearing on each plate were counted. The results are shown here. In each row, the + signs indicate the amino acids present in the medium.

| **Plate** | | **Supplement** | | **Numer of** |
|---|---|---|---|---|
| **number** | **Arginine** | **Proline** | **Histidine** | **colonies** |
| 1 | + | + | 0 | 1000 |
| 2 | + | 0 | + | 1000 |
| 3 | 0 | + | + | 1000 |
| 4 | + | 0 | 0 | 200 |

| | | | | |
|---|---|---|---|---|
| 5 | 0 | + | 0 | 100 |
| 6 | 0 | 0 | + | 0 |

**(a)** For each of the six types of medium, list the donor markers that must be present to allow the transduced cell to form colonies.

**(b)** Using these results, construct a genetic map of the *arg, pro, his* region showing the relative order of the genes and the cotransduction frequencies between the markers.

**S8.15** The order of genes in the λ phage virion is

*A B C D E att int xis N cI O P Q S R*

**(a)** Given that the bacterial attachment site, *att*, is between *gal* and *bio* in the bacterial chromosome, what is the prophage gene order?

**(b)** A new phage is discovered whose gene order is identical in the virion and in the prophage. What does this say about the location of the *att* site with respect to the termini of the phage chromosome?

**(c)** A wildtype λ lysogen is infected with another λ phage carrying a genetic marker, *Z*, located between *E* and *att*. The superinfection gives rise to a rare, doubly lysogenic *E. coli* strain that carries both λ and λ-Z prophage. Assuming that the second phage also entered the chromosome at an *att* site, diagram two possible arrangements of the prophages in the bacterial chromosome, and indicate the locations of the bacterial genes *gal* and *bio*.

**S8.16** Four Hfr strains of *E. coli* transfer their genetic material in the following order. Construct a genetic map of the *E. coli* chromosome that includes all the markers listed here and the distances in minutes between adjacent gene pairs.

| | | | | | | |
|---|---|---|---|---|---|---|
| Hfr1 markers | *mal* | *met* | *thi* | *thr* | *trp* | |
| Time of entry | 10 | 17 | 22 | 33 | 57 | |
| Hfr2 markers | *his* | *phe* | *arg* | *mal* | | |
| Time of entry | 18 | 23 | 35 | 45 | | |
| Hfr3 markers | *phe* | *his* | *bio* | *azi* | *thr* | *thi* |
| Time of entry | 6 | 11 | 33 | 48 | 49 | 60 |
| Hfr4 markers | *arg* | *thy* | *met* | *thr* | | |
| Time of entry | 15 | 21 | 32 | 48 | | |

**S8.17** An F' factor consisting of the *E. coli* sex factor, F, and a segment of the bacterial chromosome that includes the wildtype tryptophan synthetase A gene ($trpA^+$) was isolated. The F' *trpA* factor was introduced by conjugation into an $F^-$ strain with a single nucleotide mutation in the *trpA* gene, and the resulting exconjugants were F' $trpA^+/trpA^-$ partial diploids. These cells were streaked to obtain single colonies, and each colony was inoculated into liquid nutrient medium. When mixed with female cells, most of the resulting cultures transferred only the original F' *trpA* factor. However, a very small percentage of the cultures transferred a variable length of the bacterial chromosome, starting with the *trpA* gene. These cultures fell into two classes. Class 1 transferred $trpA^+$ early, whereas class 2 transferred $trpA^-$ early.

**(a)** Draw a diagram of the interaction between the F' factor and the bacterial chromosome, showing how the cells of class 1 are likely to have arisen.

**(b)** Draw a similar diagram showing the probable origin of cells of class 2.

## *Chapter 9— Genetic Engineering and Genome Analysis*

**S9.1** Define the following terms:

**(a)** Vector

**(b)** Restriction site

**(c)** Palindrome

**(d)** Cohesive end

**(e)** Blunt end

**(f)** Complementary DNA

**(g)** Insertional inactivation

**(h)** Colony hybridization assay

**(i)** Transgenic organism

**S9.2** What are some of the principal practical applications of genetic engineering?

**S9.3** Reverse transcriptase, like most enzymes that make DNA, requires a primer. Explain why, when cDNA is to be made for the purpose of cloning a eukaryotic gene, a convenient primer is a short sequence of poly(dT)? Why does this method not work with a prokaryotic messenger RNA?

**S9.4** A DNA molecule has 23 occurrences of the sequence 5'-AATT-3' along one strand. How many times does the same sequence occur along the other strand?

**S9.5** After doing a restriction digest with the enzyme *Sse*I, which has the recognition site 5'-CCTGCA/GG-3' (the slash indicates the position of the cleavage), you wish to separate the fragments in an agarose gel. In order to choose the proper concentration of agarose, you need to know the expected size of the fragments. Assuming equivalent amounts of each of the four nucleotides in the target DNA, what average-size fragment would you expect?

**S9.6** How many clones are needed to establish a library of DNA from a species of monkey with a diploid genome size of $6 \times 10^9$ base pairs if (1) fragments whose sizes average $2 \times 10^4$ base pairs are used, and (2) one wishes 99 percent of the genomic sequences to be in the library? (*Hint:* If the genome is cloned at random with *x*-fold coverage, the probability that a particular sequence will be missing is $e^{-x}$.)

**S9.7** The yeast *Saccharomyces cerevisiae* has 16 chromosomes. All of them can be separated by pulse field gel electrophoresis. Strains with the following chromosome rearrangements were examined by pulse field electrophoresis along with DNA from a wildtype control:

**(a)** A reciprocal translocation in which about one-third of chromosome IV (the largest chromosome) was exchanged with about one-third of chromosome I (the smallest chromosome).

**(b)** An insertional translocation in which two-thirds of one chromosome was inserted into another chromosome.

**(c)** A pericentric inversion in chromosome IV.

**(d)** A paracentric inversion in chromosome IV.

What band pattern would you expect to see in each case?

**S9.8** As a laboratory exercise, a friend is required to express a cloned *Drosophila* gene in bacteria to obtain large amounts of the recombinant protein. She is given a recombinant phage containing a 12-kb fragment of *Drosophila* genomic DNA, which, when introduced into the germline of a mutant fly, is able to rescue the mutant phenotype.

Hence, the 12-kb fragment contains the entire gene. The fragment also hybridizes to a 4.7-kb RNA on Northern blots. (A Northern blot is one in which RNA is separated by electrophoresis and the bands are transferred to a filter and probed with a labeled complementary sequence). The protein encoded by the gene is approximately 110 amino acids in length and can be isolated by standard methods. Your friend excises the 12-kb fragment out of the phage and inserts it into a bacterial expression vector (a plasmid designed to allow transcription of inserts). Although molecular analysis indicates that the DNA was inserted correctly and that the insert is being transcribed, she is unable to isolate any of the gene product. She turns to you for advice. What explanations would you suggest and what would you recommend she do to obtain the protein?

**S9.9** Aliquots of a 7.8-kilobase (kb) linear piece of DNA are digested with the restriction enzymes *Pvu*II, *Hin*cII, *Cla*I, and *Ban*II, alone and in pairs. The digestion products are separated by gel electrophoresis, and the size of each fragment is determined by comparison to size standards. The fragment sizes obtained, in kilobase pairs, are given below. Draw a diagram of the 7.8-kb fragment, showing the location of the restriction sites for each enzyme.

**(a)** Digestion with *Pvu*II: 1.3 kb, 6.5 kb

**(b)** Digestion with *Hin*cII: 0.5 kb, 3.3 kb, 4 kb

**(c)** Digestion with *Cla*I: 1 kb, 6.8 kb

**(d)** Digestion with *Ban*II: 1.3 kb, 6.5 kb

**(e)** Digestion with *Pvu*II and *Cla*I: 1 kb, 5.5 kb, 1.3 kb

**(f)** Digestion with *Pvu*II and *Hin*cII: 0.5 kb, 0.8 kb, 2.5 kb, 4 kb

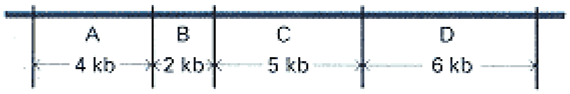
**(g)** Digestion with *Hin*cII and *Cla*I: 0.5 kb, 1 kb, 3 kb, 3.3 kb

**(h)** Digestion with *Hin*cII and *Ban*II: 0.5 kb, 0.8 kb, 2.5 kb, 4 kb

**(i)** Digestion with *Cla*I and *Ban*II: 1 kb, 1.3 kb, 5.5 kb

**S9.10** A 3.3-kb size fragment created after the *Hin*cII digestion in the previous problem is inserted into a unique *Hin*cII site of the plasmid pUC18 and transformed into *E. coli.* Individual transformed colonies are isolated, and the recombinant plasmids from each colony are purified and mapped with the restriction enzyme *Pvu*II. To the surprise of a novice molecular geneticist, the plasmids do not all give the same pattern of restriction fragments on a gel. The plasmids fall into two distinct groups. Both groups yield the same number of bands in the gel, the aggregate size of which is the same, but two of the fragment sizes are different. Explain the reason for this difference.

**S9.11** You have cloned a region of the genome of the nematode *C. elegans* that contains a mutation in a gene involved in controlling biorhythms. You make a restriction map of the cloned region and find that the following *Hin*dIII restriction map can be constructed:




Now you perform a *Hin*dIII digest of genomic DNA from individuals homozygous for the mutation and also from those that are homozygous for the wildtype allele. You subject both of these digests to electrophoresis in an agarose gel and transfer the DNA to a nylon membrane. When the membrane is probed with fragment A, B, or D, the probe hybridizes to fragments of 4, 2, and 6 kb, respectively, in both the mutant and the wildtype strain. When the membrane is probed with fragment C, the probe hybridizes to 11 different fragments in the mutant genome (including one fragment of 5 kb) and to 10 fragments in the wildtype genome (but not one of 5 kb). What could account for both the mutation and the pattern of bands seen with the C probe?

**S9.12** You are given a plasmid containing part of a gene of *D. melanogaster*. The gene fragment is 303 base pairs long. You would like to amplify it using the polymerase chain reaction (PCR). You design oligonucleotide primers 19 nucleotides in length that are complementary to the plasmid sequences immediately adjacent to both ends of the cloning site. What would be the exact size of the resulting PCR product?

**S9.13** Suppose that you digest the genomic DNA of a particular organism with *Sau*3A (/GATC). Then you ligate the resulting fragments into a unique *Bam*HI (G/GATCC) cloning site of a plasmid vector. Would it be possible to isolate the cloned fragments from the vector using *Bam*HI? From what proportion of clones would it be possible?

**S9.14** Digestion of a DNA molecule with *Hin*dIII yields two fragments of 2.2 kb and 2.8 kb. *Eco*RI cuts the molecule, creating 1.8 kb and 3.2 kb fragments. When treated with both enzymes, the same DNA molecule produces four fragments of 0.8 kb, 1.0 kb, 1.2 kb, and 2.0 kb. Draw a restriction map of this molecule.

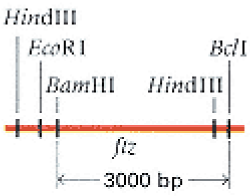
**S9.15** What are the main differences between cDNA and genomic DNA libraries?

**S9.16** You wish to amplify a unique 100-bp sequence from the human genome ($3 \times 10^9$ bp) using the polymerase chain reaction. At the end of the amplification, you want the amount of amplified DNA to represent not less than 99 percent of all the DNA present. Assume that the amplification is exponential.

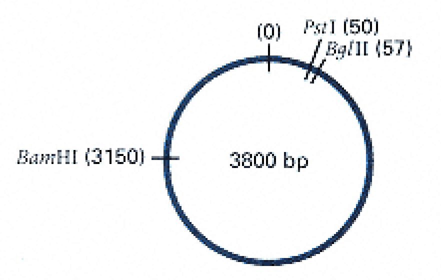
**(a)** How many cycles of amplification are necessary to achieve the 99 percent goal?

**(b)** How would the number of cycles change if you were to amplify a fragment of 1000 bp?

**S9.17** You are given a DNA fragment containing the *Drosophila* developmental gene *fushi tarazu (ftz)* along with the restriction map shown below. The restriction sites are *Hin*dIII (5'-A/AGCTT-3'), *Bcl*I (5'-T/GATCA-3'), *Eco*RI (5'-G/AATTC-3'), *Bam*HI (5'-G/GATCC-3').




You are instructed to subclone the gene into a 3.8-kb plasmid vector having relevant restriction sites *Pst*I (5'-CTGCA/G-3'), *Bgl*II (5'-A/GATCT-3'), and *Bam*HI (5'-G/GATCC-3') positioned as follows:



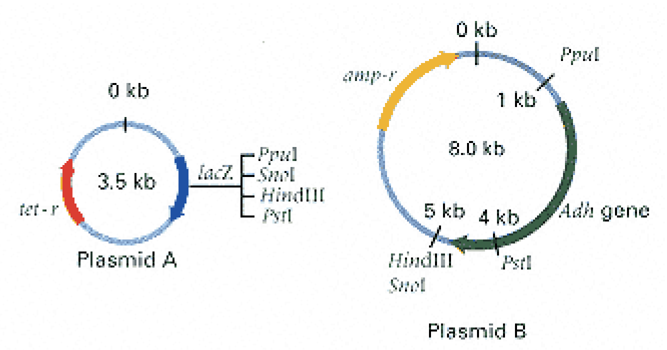

With what restriction enzymes should you cleave the *ftz*-containing region before ligating it with plasmid DNA that has been linearized with *Bgl*II?

**S9.18** Once you have subcloned the *ftz* region as in the previous problem, you want to determine the orientation in which the fragment is inserted in the resulting plasmid. You cleave the *ftz*-containing plasmid with *Bam*HI (5'-G/GATCC-3'), *Pst*I (5'-CTGCA/G-3'), and *Hin*dIII (5'-A/AGCTT-3'), and observe fragments of sizes approximately 3 kb and 1 kb.

**(a)** Can you determine the orientation of the inserted fragment from this result?

**(b)** Would your answer change if the cloning site used in the plasmid were *Bam*HI rather than *Bgl*II? Explain.

**S9.19** You are given two strains of *E. coli* containing plasmids whose restriction maps are




You are asked to subclone the gene for alcohol dehydrogenase (*Adh)* from plasmid B into plasmid A. Unsure of which cloning procedure would be the best, you try two different approaches.

**(a)** You digest both plasmids to completion with *Ppu*I (A/TGCAT) and *Sno* (G/TGCAC), isolate the 4-kb *Ppu*I-*Sno*I *Adh* gene fragment and the 3.5-kb *Ppu*I-*Sno*I vector A fragment, carry out a ligation reaction containing both isolated DNA fragments, transform *E. coli* with the resulting DNA, and plate the transformed cells on medium containing tetracycline. The plates also contain Xgal and IPTG, which makes it possible to identify cells that contain the original plasmid A, because they turn blue as a result of the functional *lacZ* gene. Colonies with an insertion that interrupts the *lacZ* gene are white because of insertional inactivation. You follow the same procedure except that you use *Hin*dIII (A/AGCTT) instead of *Sno*I. Using approach (a), you obtain many more blue colonies than using approach (b). Explain why this is so.

**S9.20** You purify plasmid DNA from white colonies obtained in both approaches in the previous problem. Predict the sizes of the resulting fragments when the isolated plasmids are digested with *Pst*I.

## *Chapter 10—*
## *Gene Expression*

**S10.1** Describe the main features of the genetic code.

**S10.2** If the triplet codons in the genetic code were read in an overlapping fashion, would there be any constraints on the amino acid sequences found in proteins?

**S10.3** If DNA consisted of only two nucleotides (say, A and T) in any sequence, what is the minimal number of adjacent nucleotides that would be needed to specify uniquely each of the 20 amino acids?

**S10.4** If DNA consisted of three possible base pairs (say, A–T, G-C, and X–Y), what is the minimal number of adjacent nucleotides that would be needed to specify uniquely each of the 20 amino acids?

**S10.5** A DNA strand consists of any sequence of four kinds of nucleotides. Suppose there were only 16 different amino acids instead of 20. Which of the following statements would be correct descriptions of the minimal number of nucleotides necessary to create a genetic code?

**(a)** 1

**(b)** 2, provided that chain termination does not require a special codon

**(c)** 3, provided that chain termination does require a special codon

**(d)** 2, no matter how chain termination is accomplished

**(e)** statements (b) and (c)

**S10.6** In the standard genetic code, synonymous codons often have the feature that the third nucleotide in the codon is either of the pyrimidines (T or C) or either of the purines (A or G) because of "wobble" in the base pairing at the third position of the codon. If this rule were followed consistently for all codons, would it be possible to have a triplet code that specifies 20 amino acids and one stop codon? Would there be any room for degeneracy?

**S10.7** Which of the following statements is true of polypeptide chains and nucleotide chains?

**(a)** Both have a sugar-phosphate backbone.

**(b)** Both are chains consisting of 20 types of repeating units.

**(c)** Both types of molecules contain unambiguous and interconvertible triplet codes.

**(d)** Both types of polymers are linear and unbranched.

**(e)** None of the above.

**S10.8** The notion that a strand of DNA serves as a template for transcription of an RNA that is translated into a polypeptide is known as the "central dogma" of gene expression. All three types of molecules have a polarity. In the DNA template and the RNA transcript, the polarity is determined by the free 3' and 5' groups at opposite ends of the polynucleotide chains; in a polypeptide, the polarity is determined by the free amino group (N terminal) and carboxyl group (C terminal) at opposite ends. Each statement below describes one possible polarity of the DNA template, the RNA transcript, and the polypeptide chain, respectively, in temporal order of use as a template or in synthesis. Which is correct?

**(a)** 5' to 3' DNA; 3' to 5' RNA; N terminal to C terminal.

**(b)** 3' to 5' DNA; 3' to 5' RNA; N terminal to C terminal.

**(c)** 3' to 5' DNA; 3' to 5' RNA; C terminal to N terminal.

**(d)** 3' to 5' DNA; 5' to 3' RNA; N terminal to C terminal.

**(e)** 5' to 3' DNA; 5' to 3' RNA; C terminal to N terminal.

**S10.9** Which of the following features of a protein structure is most directly determined by the genetic code?

**(a)** Its shape

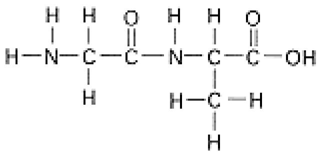
**(b)** Its secondary structure

**(c)** Its catalytic or structural role

**(d)** Its amino acid sequence

**(e)** Its subunit composition (quaternary structure)

**S10.10** The structure below is that of a dipeptide composed of alanine and glycine.



**(a)** Which end, left or right, is the amino terminal end?

**(b)** Which amino acid is at the amino terminal end?

**(c)** Which end, left or right, is the carboxyl terminal end?

**(d)** Which amino acid is at the carboxyl terminal end?

**(e)** Which bond is the peptide bond?

**S10.11** Summarize the studies in phage T4 that provided the first genetic evidence for a triplet code.

**S10.12** An alternating polymer AGAG is used as a messenger RNA in an *in vitro* protein-synthesizing system that does not need a start codon. What types of polypeptide chain are expected?

**S10.13** In a segment of DNA that contains overlapping genes, would you expect the two proteins to terminate at the same site or have the same number of amino acids? Explain.

**S10.14** What unique feature of the 5' end of a eukaryotic mRNA distinguishes it from a prokaryotic mRNA?

**S10.15** Consider the two codons AGA and AGG, which code for the amino acid arginine. A, C, U, G, or I is allowed in the 5' position of the anticodon.

**(a)** What is the minimum number of $tRNA^{Arg}$ types needed to translate these codons?

**(b)** For each of the $tRNA^{Arg}$ types in part (a), what is the anticodon?

**S10.16** A wildtype protein has glutamic acid (Glu) at amino acid position 56. A mutation is found in which glutamic acid is replaced with a different amino acid. Among the reverse mutations that restore protein function, some have thrreonine (Thr) at the position, some have isoleucine (Ile), and some have glutamic acid. Assuming that the original mutation and all the revertants result from single nucleotide substitutions, determine the wildtype codon for position 56, that of the original mutation, and those of the revertants.

**S10.17** What amino acid sequence would correspond to the following mRNA sequence?

```
5'-GCAUCAGAAUGUAGGGAGACA-3'
```

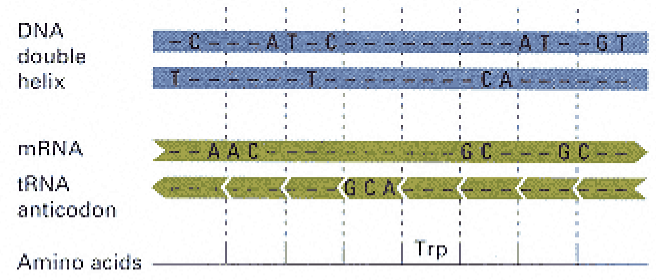
(If you use the conventional single-letter abbreviations for the amino acids, shown below, you will find a secret.)

| | | | | | | |
|---|---|---|---|---|---|---|
| Ala A | Arg R | Asn N | Asp D | Cys C | Gln Q | Glu E |
| Gly G | His H | Ile I | Leu L | Lys K | Met M | Phe F |
| Pro P | Ser S | Thr T | Trp W | Tyr Y | Val V | |

**S10.18** If the following double-stranded DNA molecule is transcribed from the bottom strand, does transcription start at the left or right end? What is the mRNA sequence? If translation starts at the first AUG, what is the polypeptide sequence?

```
5'-CGCTAGCATGGAGAACGACGAATTGAGCCCAGAAGCGTCATGAGG-3'
3'-GCCATCGTACCTCTTGCTGCTTAACTCGGGTCTTCGCAGTACTCC-5'
```

**S10.19** The following table shows matching regions of the DNA, mRNA, tRNA, and amino acids encoded in a particular gene. The mRNA is shown with its 5' end at the left, and the tRNA anticodon is shown with its 3' end at the left. The vertical lines define the reading frame.




**(a)** Complete the nucleic acid sequences, assuming Watson-Crick pairing between each codon and anticodon.

**(b)** Is the DNA strand transcribed the top or the bottom strand?

**(c)** Translate the mRNA in all three reading frames.

**(d)** Specify the nucleic acid strand(s) whose sequence could be used as a probe:

(i) In a Southern blot hybridization, in which the hybridization is carried out against genomic DNA.

(ii) In a Northern blot hybridization, in which the hybridization is carried out against mRNA.

**S10.20** Part of a DNA molecule containing a single short intron has the sequence

```
5'-ATAGCTCGACGGAAGGTCGACCAAATTTCGCAGGAGCTCGCT-3'
3'-TATCGAGCTGCCTTCCAGCTGGTTTAAAGCGTCCTCGAGCGA-5'
```

and the corresponding polypeptide has the sequence

Ile-Ala-Arg-Arg-Lys-Leu-Ala

Find the intron.

**S10.21** A double mutation produced by recombination contains two single nucleotide frameshifts separated by about 20 base pairs. The first is an insertion and the second a deletion. The amino acid sequence of the wildtype and that of the mutant polypeptide in this part of the protein are

Wildtype: Lys-Lys-Tyr-His-Gln-Trp-Thr-Cys-Asn
Mutant: Lys-Gln-Ile-Pro-Pro-Val-Asp-Met-Asn

What are the original and the double-mutant mRNA sequences? Which nucleotide in the wildtype sequence is the frameshift addition? Which nucleotide in the double-mutant sequence is the frameshift deletion? (In working this problem, it will be convenient to use the conventional symbols Y for unknown pyrimidine, R for unknown purine, N for unknown nucleotide, and H for A, C, T.)

**S10.22** The following RNA sequence is a fragment derived from a polycistronic messenger RNA including two genes.

```
5'-CUUAUGGAAGUAAACCCGAGC-3'
```

This fragment is known to include the initiation codon of one protein and the termination codon of the other.

**(a)** Determine the first four amino acids of the protein whose initiation codon is in the fragment.

**(b)** Determine the last two amino acids of the protein whose translation terminates in the fragment.

**(c)** What is unusual about these genes?

**(d)** What would be the consequence of a frameshift mutation that deleted the G from the initiation codon?

**S10.23** A strain of *E. coli* carries a mutation that completely inactivates the enzyme encoded in the gene. Several revertants with partly or fully restored activity were selected and the amino acid sequence of the enzyme determined. The only differences found were at position 10 in the polypeptide chain.

Revertant 1 had Thr.
Revertant 2 had Glu.
Revertant 3 had Met.
Revertant 4 had Arg.

Assume that the initial mutation itself, as well as each revertant, resulted from a single nucleotide substitution.

**(a)** What amino acid is present at position 10 in the mutant protein?

**(b)** What codon in the messenger RNA would encode this amino acid?

**S10.24** Make a sketch of a mature eukaryotic messenger RNA molecule hybridized to the transcribed strand of DNA of a gene that contains two introns, oriented with the promoter region of the DNA at the left. Clearly label the DNA and the mRNA. Use the following letters to label the location and/or boundaries of each segment. Some letters may be used several times, as appropriate; some, which are not applicable, may not be used at all.

**(a)** 5' end

**(b)** 3' end

**(c)** Promoter region

**(d)** Attenuator

**(e)** Intron

**(f)** Exon

**(g)** Polyadenylation signal

**(h)** Leader region

**(i)** Shine-Dalgarno sequence

**(j)** Translation start codon

**(k)** Translation stop codon

**(l)** 5' cap

**(m)** Poly-A tail

## *Chapter 11—*
## *The Regulation of Gene Activity*

**S11.1** Define the following terms:

**(a)** Enhancer

**(b)** Attenuator

**(c)** *Cis*-dominant mutation

**(d)** Autoregulation

**(e)** Combinatorial control

**(f)** Polycistronic mRNA

**(g)** Transcriptional activation by recruitment

**S11.2** How do gene and genome organization differ in eukaryotes and prokaryotes?

**S11.3** At what different levels can gene expression be regulated?

**S11.4** Give some examples of gene regulation by alterations in the DNA.

**S11.5** The regulation of transcription by attenuation cannot take place in eukaryotes. Why not?

**S11.6** Why is the *lac* operon of *E. coli* not inducible in the presence of glucose?

**S11.7** In eukaryotic organisms, the amounts of some proteins found in the cell are known to vary substantially with environmental conditions, whereas the amount of mRNA for these same proteins remains relatively constant. What mechanisms of regulation might be responsible for this phenomenon?

**S11.8** Two genotypes of *E. coli* are grown and assayed for levels of the enzymes of the *lac* operon. Using the information provided here for these two genotypes, predict the enzyme levels for the other genotypes listed in (a) through (d). (The levels of activity are expressed in arbitrary units relative to those observed under the induced conditions.)

| | **Uninduced level** | | **Induced level** | |
|---|---|---|---|---|
| **Genotype** | **Z** | **Y** | **Z** | **Y** |
| $I^+ O^+ Z^+ Y^+$ | 0.1 | 0.1 | 100 | 100 |
| $I^+ O^c Z^+ Y^+$ | 25 | 25 | 100 | 100 |

**(a)** $I^- O^+ Z^+ Y^+$

**(b)** F' $I^+ O^+ Z^- Y^-/I^- O^+ Z^+ Y^+$

**(c)** F' $I^+ O^+ Z^- Y^-/I^+ O^c Z^+ Y^+$

**(d)** F' $I^+ O^+ Z^- Y^-/I^- O^c Z^+ Y^+$

**S11.9** Imagine a bacterial species in which the methionine operon is regulated only by an attenuator and there is no repressor. In its mode of operation, the methionine attenuator is exactly analogous to the *trp* attenuator of *E. coli.* The relevant portion of the attenuator sequence is

5'-AA**A**<u>A</u>TGATGATGATGATGATGATGATGGACGAT-3'

The translation start site is located upstream from this sequence, and the AUG codons in the region shown cannot be used for translation initiation. What phenotype (constitutive, wildtype, or *met*⁻*)* would you expect of each of the following types of mutations? Explain your reasoning.

**(a)** The boldface A is deleted.

**(b)** Both the boldface and the underlined A are deleted.

**(c)** The first three As in the sequence are deleted.

**S11.10** How many proteins are bound to the *trp* operon when:

**(a)** Tryptophan and glucose are present?

**(b)** Tryptophan and glucose are absent?

**(c)** Tryptophan is present and glucose is absent?

**S11.11** Why are mutations of the *lac* operator often called *cis*- dominant? Why are some constitutive mutations of the *lac* repressor (*lacI)* called *trans*-recessive? Can you think of a way in which a noninducible mutation in the *lacI* gene might be *trans*-dominant?

**S11.12** An *E. coli* mutant strain synthesizes β-galactosidase whether or not the inducer is present. What genetic defect(s) might be responsible for this phenotype?

**S11.13** Temperature-sensitive mutations in the *lacI* gene of *E. coli* render the repressor nonfunctional (unable to bind the operator) at 42°C but leave it fully functional at 30°C. Under which of the following conditions would β-galactosidase be produced?

**(a)** In the presence of lactose at 30°C

**(b)** In the presence of lactose at 42°C

**(c)** In the absence of lactose at 30°C

**(d)** In the absence of lactose at 42°C

**S11.14** Many metabolic pathways are regulated by so-called feedback inhibition, in which the first enzyme in the pathway becomes inhibited by interacting with the final product of the entire pathway if the final product is present in a sufficient amount. Explain why a metabolic pathway should evolve in such a way that the first enzyme in the pathway is inhibited rather than the last enzyme in the pathway.

**S11.15** Among cells of a $lacI^+$ $lacO^+$ $lacZ^+$ $lacY^+$ strain of *E. coli*, a mutation was found in which the *lacY* gene underwent a precise inversion without any change in the coding region. However, the strain was found to be defective in the production of the permease. Why?

**S11.16** Cristina, a young immunologist, told her friend Jorge, who studied molecular biology, that there are $10^8$ different antibodies produced by human B cells. To her astonishment, Jorge strongly disagreed, saying that there are not enough genes in the human genome to code for such a great variety of immunoglobulins. Who is right? Is there any way to reconcile their views?

**S11.17** You are engaged in the study of a gene, the transcription of which is activated in a particular tissue by a steroid hormone. From cultured cells isolated from this tissue, you are able to isolate mutant cells that no longer respond to the hormone. What are some possible explanations?

**S11.18** A laboratory strain of *E. coli* produces β-galactosidase constitutively and permease inducibly, despite the fact that these two proteins are co-regulated in the *lac* operon in wildtype strains. What is the genotype of the *lac* operon in this strain?

## *Chapter 12—*
## *The Genetic Control of Development*

**S12.1** Define the following terms:

**(a)** Embryonic induction

**(b)** Gain-of-function mutation

**(c)** Fate map

**(d)** Pair-rule gene

**(e)** Segment-polarity gene

**(f)** Homeobox

**(g)** Homeotic mutation

**S12.2** What is the relationship between maternal-effect genes and zygotic genes?

**S12.3** In the early development of *C. elegans,* what is the role of the anchor cell in the differentiation of the vulva?

**S12.4** When in the development of most metazoan animals is the polarity of the embryo determined?

**S12.5** The same transmembrane receptor protein encoded by the *lin-12* gene is used in the determination of different developmental fates. What is the principal difference between the two types of target cells that develop ifferently in response to Lin-12 stimulation?

**S12.6** Programmed cell death (apoptosis) is responsible in part for shaping many organs and tissues in normal development. If a group of cells in the duck leg primordium that are destined to die are transplanted from their normal leg site to another part of the embryo just prior to the time they would normally die, they still die on schedule. The same operation performed a few hours earlier rescues the cells, and they do not die. How can you explain this observation?

**S12.7** Classify each of the following mutations:

**(a)** A mutation in *C. elegans* in which a cell that normally produces two distinct daughter cells gives rise to two identical cells.

**(b)** A mutation in *Drosophila* that causes an antenna to appear at the normal site of a leg.

**(c)** A lethal mutation in *Drosophila* that is responsible for abnormal gene expression in alternating segments of the embryo

**S12.8** What causes induction of the morphogenic events in the *Drosophila* imaginal disks? When does this induction happen?

**S12.9** If a small amount of cytoplasm is removed from the posterior end of the *Drosophila* embryo before blastoderm formation, the mature fly that develops from this embryo is normal in every way except it does not have germ cells. How can you explain this finding?

**S12.10** How would you prove that a newly discovered mutation has a maternal effect?

**S12.11** A recessive mutation *smooth* is found in a species of flower in which hairs (trichomes) normally present on the leaf surface are missing. Another mutation in the same species, called *hairy*, results in the presence of trichomes on the flower petals, where they are not normally formed. Cloning and sequencing indicate that these mutations are different alleles of the same gene. Molecular analysis reveals that *smooth* is a frameshift mutation near the start of the coding sequence and that *hairy* has an upstream insertion of a transposable element that allows transcription of the gene in petal tissue. What features of these observations suggest that the gene in question is an important gene for the regulation of trichome formation?

**S12.12** Two classes of genes involved in segmentation of the *Drosophila* embryo are gap genes, which are expressed in one region of the developing embryo, and pair-rule genes, which are expressed in seven stripes. Homozygotes for mutations in gap genes lack a continuous block of larval segments; homozygous mutations in pair-rule genes lack alternating segments. You examine gene expression by mRNA *in situ* hybridization and find that (1) the embryonic expression pattern of gap genes is normal in all pair-rule mutant homozygotes, and (2) the pair-rule gene expression pattern is abnormal in all gap gene mutant homozygotes. What do these observations tell you about the temporal hierarchy of gap genes and pair-rule genes in the developmental pathway of segmentation?

**S12.13** A paracentric inversion in *Drosophila* causes a dominant mutation that produces small wings. You mutagenize flies homozygous for this inversion with x rays and cross them to normal mates. Most progeny that arise have small wings, but a few rare progeny have normal wings. These rare progeny still have the original inversion but also contain a deletion spanning the more proximal of the two inversion breakpoints. (Proximal means closer to the centromere.) You correctly conclude from these observations that the dominant mutation is due to a genetic "gain of function" rather than to a reduction in gene activity. Explain how these results support this conclusion, and suggest an interpretation of how the gain-of-function phenotype arises.

**S12.14** One approach to the identification of important developmental genes in the mouse is to focus on genes whose DNA sequences are similar to those that have significant developmental roles in other organisms. Suppose that you want to study the mouse homolog of the *lin-3* gene of *C. elegans*, which is important in vulva formation in the worm. Homozygotes for a null mutation of *lin-3* lack a vulva. You propose that a homolog of *lin-3* exists in the mouse, and you want to examine the phenotype of a mutant mouse that lacks a functional form of this gene. Arrange the following steps in the order in which you would perform them to generate a mouse homozygous for a null mutation of the gene.

**(a)** Identify an open reading frame in a mouse cDNA that is similar to the *C. elegans lin-3* open reading frame.

**(b)** Create a deletion in a mouse genomic clone, and insert a gene resistant to the drug G418.

**(c)** Breed chimeric mice to donor strains to identify heterozygotes for a null mutation.

**(d)** Inject embryonic stem cells into differentially marked mouse blastocysts.

**(e)** Probe a mouse wildtype genomic library with a labeled mouse cDNA.

**(f)** Probe a mouse cDNA library with a labeled *C. elegans lin-3* cDNA.

**(g)** Select for embryonic stem cells that can grow on G418.

**(h)** Intercross null mutant heterozygotes to obtain a homozygous null mouse.

**(i)** Identify chimeric mice.

**(j)** Inject the DNA of an engineered mutation in the mouse *lin-3* homolog into embryonic stem cells.

**(k)** Identify embryonic stem cells that have a null mutation in the mouse *lin-3* homolog.

**S12.15** Edward B. Lewis, Christianne Nusslein-Volhard, and Eric Weichaus shared a 1995 Nobel Prize in Physiology or Medicine for their work on the developmental genetics of *Drosophila.* In their screen for developmental genes, NussleinVolhard and Weichaus initially identified 20 lines bearing maternal-effect mutations that produced embryos lacking anterior structures but having the posterior structures duplicated. When Nusslein-Volhard mentioned this result to a colleague, he was astonished to hear that mutations in 20 genes could give rise to this phenotype. Explain why his astonishment was completely unfounded.

**S12.16** The gene *bicoid* exemplifies a maternal-effect gene that, when mutated, produces embryos that lack anterior structures but have the posterior structures duplicated. The Bicoid protein is a morphogen that establishes the domain of expression of the gap gene *hunchback* in a concentration dependent manner. The Bicoid protein is usually present in an anterior-to-posterior gradient with the highest concentration at the anterior end of the early embryo. Diagrammed below is the major domain of expression of *hunchback* in an embryo from a wildtype mother.


What are the expected domains of *hunchback* expression in embryos from mothers with the following genotypes. Express your answers in the form of diagrams similar to the one here.

**(a)** Homozygous mutant for *bicoid*

**(b)** Homozygous for a duplication of a wildtype copy of *bicoid*

**(c)** Homozygous for transgene that causes bicoid to be localized at both poles of the embryo

## *Chapter 13— Mutation, DNA Repair, and Recombination*

**S13.1** Define the following terms:

**(a)** Gene conversion

**(b)** Mismatch repair

**(c)** Silent mutation

**(d)** Photoreactivation

**(e)** Transversion mutation

**S13.2** Prior to the early 1940s, some bacteriologists maintained that the exposure of a bacterial population to an antibiotic or other selective agent induced mutations that enabled the bacteria to survive. Which of the following statements describes what Joshua and Esther Lederberg demonstrated with the replica plating technique?

**(a)** Streptomycin caused the formation of streptomycinresistant bacteria.

**(b)** Streptomycin revealed the presence of streptomycinresistant bacteria.

**(c)** Mutations are usually beneficial.

**(d)** Mutations are usually deleterious.

**(e)** None of the above.

**S13.3** If base substitutions were random, what ratio of transversions to transitions would be expected?

**S13.4** In the real world, spontaneous base substitutions are biased in favor of transitions. What is the consequence of this bias for base substitutions in the third position of codons?

**S13.5** How many different mutations can result from a single base substitution in the unique tryptophan codon TGG? Classify each as silent, missense, or nonsense.

**S13.6** What is the minimum number of single-nucleotide substitutions that would be necessary for each of the following amino acid replacements?

**(a)** Trp $\rightarrow$ Lys

**(b)** Tyr $\rightarrow$ Gly

**(c)** Met $\rightarrow$ His

**(d)** Ala $\rightarrow$ Asp

**S13.7** Every human gamete contains, very approximately, 50,000 genes. If a mutation rate is between $10^{-5}$ and $10^{-6}$ new mutations per gene per generation, what percentage of all gametes will carry a gene that has undergone a spontaneous mutation?

**S13.8** If a mutation rate is between $10^{-5}$ and $10^{-6}$ per gene per generation, what does this imply in terms of the number of mutations of a particular gene per million gametes per generation?

**S13.9** If the spontaneous mutation rate of a gene is $10^{-6}$ mutations per generation, and a dose of ionizing radiation of 10 grays doubles the rate of mutation, what mutation rate would be expected with 1 gray of radiation? Assume that the rate of induced mutation is proportional to the radiation dose.

**S13.10** In theory, deamination of both cytosine and 5-methyl-cytosine results in a mutation. However, nucleotide substitutions produced by loss of the amino group from cytosine are rarely observed, although those produced by loss of the amino group from 5-methylcytosine are relatively frequent. What is a possible explanation for this phenomenon?

**S13.11** The mismatch repair system can detect mismatches that occur in DNA replication and can correct them after replication. In *E. coli*, how does the system use DNA methylation to determine which base in a pair is the incorrect one?

**S13.12** If one of the many copies of the tRNA gene for Glu underwent a mutation in which the anticodon became CUA instead of CUC, what would be the result? (*Hint:* The key is that only one of many tRNA copies is mutated.)

**S13.13** A strain of *E. coli* contains a mutant Leu tRNA. Instead of recognizing the codon 5'-CUG-3', as it does in nonmutant cells, the mutant tRNA now recognizes the codon 5'-GUG-3'. A missense mutation of another gene, affecting amino acid number 28 along the chain, is suppressed in cells with the mutant Leu tRNA.

**(a)** What are the anticodons of the wildtype and mutant Leu tRNA molecules?

**(b)** What kind of mutation is present in the mutant Leu tRNA gene?

**(c)** What amino acid would be inserted at position 28 of the mutant polypeptide chain if the missense mutation were not suppressed?

**(d)** What amino acid is inserted at position 28 when the missense mutation is suppressed?

**S13.14** A mutation is selected after treatment with a particular mutagen, and organisms containing the mutation are then mutagenized to find revertants.

**(a)** Can a mutation caused by hydroxylamine be induced to revert to the wildtype sequence by re-exposure to hydroxylamine?

**(b)** Can a mutation caused by hydroxylamine be induced to revert by nitrous acid?

(Hydroxylamine reacts specifically with cytosine and causes GC R AT transitions; nitrous acid causes deamination of A and C).

**S13.15** The tryptophan synthase of *E. coli* is a tetrameric enzyme formed by the polypeptide products of *trpB* and *trpA* genes. The wildtype subunit encoded by *trpB* has a glycine in position 38, whereas two mutants, B13 and B32, have Arg and Glu at this position, respectively. Plating these mutants on minimal medium sometimes yields spontaneous revertants to prototrophy. Four independent revertants of B13 have Ile, Thr, Ser, and Gly in position 38, and three independent revertants of B32 have Gly, Ala, and Val in the same position. What codon is in position 38 in the wildtype *trpB* gene?

**S13.16** In the *rIIB* gene of bacteriophage T4, there is a small coding region near the beginning of the gene in which the particular amino acid sequence that is present is not essential for protein function. (Single-base insertions and deletions in this region were used to prove the triplet nature of the genetic code.) Under what condition would a + 1 frameshift mutation at the beginning of this region not be suppressed by a - 1 frameshift mutation at the end of the region?

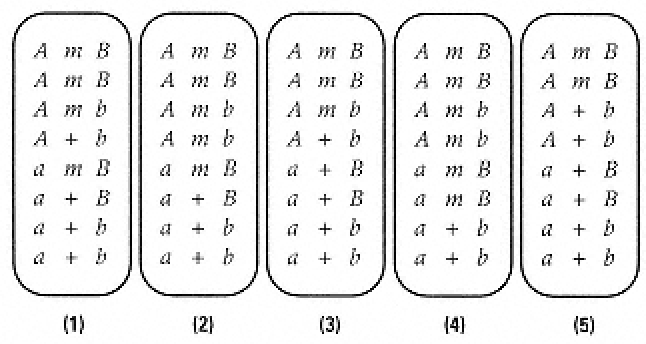
**S13.17** *E. coli* mutation *lacZ-1* was induced by acridine treatment, whereas *lacZ-2* was induced by 5-bromouracil. What kind of mutations are they likely to be? What sort of mutant β-galactosidase will these mutations produce?

**S13.18** Is it correct to say that a Holliday structure and a chiasma are the same?

**S13.19** When two mutations are closely linked (for example, two nucleotide substitutions in the same gene), a single gene conversion event may include both sites. This phenomenon is called co-conversion. How would you expect the frequency of co-conversion to depend on the distance between mutant nucleotide sites?

**S13.20** In spite of gene conversion, among gametes chosen at random, one still observes the Mendelian segregation ratio of 1 : 1 in crosses. Why?

**S13.21** You carry out a large-scale cross of genotypes *A m B* × *a* + *b* of two strains of the mold *Neurospora crassa* and observe a number of aberrant asci, some of which are shown below.




The collegue who provided the strains insists that they are both deficient in the same gene in the DNA mismatch repair pathway. The results of your cross seem to contradict this assertion.

**(a)** Which of the asci depicted above would you exhibit as evidence that your colleague is incorrect?

**(b)** Which ascus (or asci) would you exhibit as definitive evidence that DNA mismatch repair in these strains is not 100 percent efficient?

**(c)** Among all asci resulting from meioses in which heteroduplexes form across the *m* + region, what proportion of acsi will be tetratype for the *A* and *B* markers and also show 4 : 4 segregation of *m:* +, under the assumption that 30 percent of heteroduplexes in the region are not corrected, 40 percent are corrected to *m.* and 30 percent are corrected to +?

**S13.22** In molecular evolutionary studies, biologists often regard serine as though it were two different amino acids, one type of "serine" corresponding to the fourfold-degenerate UCN codon and the other type of "serine" corresponding to the twofold-degenerate AGY codon. Why does it make sense to regard serine in this way but not the other amino acids encoded by six codons: leucine (UUR and CUN) and arginine (AGR and CGN)? (In this symbolism for the nucleotides, Y is any pyrimidine, R any purine, and N any nucleotide.)

### *Chapter 14—*
### *Extranuclear Inheritance*

**S14.1** Define the following terms:

**(a)** Maternal inheritance

**(b)** Maternal effect

**(c)** Cytoplasmic male sterility

**(d)** Heteroplasmy

**(e)** RNA editing

**(f)** Phylogenetic tree

**S14.2** A woman with the mild form of a mitochondrial disease, Leber's hereditary optic neuropathy (LHON), has four children: two with severe disease expression, one unaffected, and one with the same level of expression as the mother. Explain this pattern of inheritance.

**S14.3** How do mitochondrial and nuclear genomes interact at the functional level?

**S14.4** A dwarfed corn plant is crossed as a female parent to a normal plant, and the $F_1$ progeny are found to be dwarfed. The $F_1$ plants are self-fertilized, and the $F_2$ plants are found to be normal. Each $F_2$ plant is self-fertilized, and 3/4 of $F_3$ generation are found to be normal whereas 1/4 of the plants are dwarfed. Which of the following hypotheses are consistent with these results? Explain your reasoning.

**(a)** Cytoplasmic inheritance

**(b)** Nuclear inheritance

**(c)** Maternal effect

**(d)** Cytoplasmic inheritance plus mutation in a nuclear gene

**S14.5** A strain of yeast with a mutation *ery-r* conferring resistance to the antibiotic erycetin is crossed with a strain of opposite mating type with the genotype *ery-s.* After meiosis and sporulation, three tetrads are isolated. The spores have the following genotypes:

| **I** | **II** | **III** |
|---|---|---|
| **a** *ery-r* | α *ery-s* | α *ery-r* |
| **a** *ery-r* | α *ery-s* | **a** *ery-r* |
| α *ery-r* | **a** *ery-s* | **a** *ery-r* |
| α *ery-r* | **a** *ery-s* | α *ery-r* |

**(a)** What genetic hypothesis can explain these results?

**(b)** If an *ery-r* strain were used to generate petites, would you expect some of the petites to be sensitive to erycetin? Explain.

**S14.6** In a fungal organism closely related to *Saccharomyces cerevisiae*, a strain resistant to three antibiotics is used to induce petite colonies. The antibiotics are erycetin, chlorpromanin, and paralysin, and the resistance genes are denoted *ery-r. ch-r,* and *par-r,* respectively. Each of the resistance genes is known to be present in mitochondrial DNA. Each row in the accompanying table is the result of an experiment in which petite colonies selected for loss of genetic marker A (one of the antibiotic resistances) were tested for simultaneous loss of genetic marker B (another of the antibiotic resistances).

| Marker A | Marker B | Loss of maker A only | Loss of makers A and B |
|---|---|---|---|
| *ery-r* | *chl-r* | *170* | 19 |
| *ery-r* | *par-r* | 54 | 81 |
| *chl-r* | *par-r* | 116 | 8 |

From these data, deduce the order of the genes and the relative distances between them in the mitochondrial DNA.

**S14.7** Two haploid strains of *Neurospora* are isolated that have genetic defects in the ATPase complex. The mutants are designated $A1^-$ and $A2^-$. The crosses $A1^- \times A1^+$ and $A2^- \times A2^+$ were carried out, and one ascus was analyzed from each cross. The cross $A1^- \times A1^+$ yielded a 4 : 4 ratio of $A1^+ : A1^-$ ascospores, whereas the cross $A2^- \times A2^+$ yielded an 8 : 0 ratio of $A2^+ : A2^-$ ascospores.

**(a)** What difference between the mutations can explain these discordant results?

**(b)** What other types of tetrads would one expect from these crosses?

**S14.8** Mothers addicted to cocaine who use the drug during pregnancy can give birth to babies who are also addicted. Suggest a mechanism for this maternal effect.

**S14.9** If organelles were originally symbionts, why don't we call them symbionts instead of organelles?

**S14.10** How would you prove the existence of RNA editing in mitochondria?

**S14.11** What is controversial about the hypothesis of a mitochondrial Eve who lived in Africa? Is the controversial part the conclusion that there is a common ancestor of human mitochondrial DNA, or is it the inference of the geographical location of the common ancestor? Explain.

### *Chapter 15—*
### *Population Genetics and Evolution*

**S15.1** Define the following terms:

**(a)** Population

**(b)** Natural selection

**(c)** Random genetic drift

**(d)** Polymorphic gene

**(e)** Selectively neutral mutation

**(f)** Selection-mutation balance

**S15.2** In a large, randomly mating herd of cattle, 16 percent of the newborn calves have a certain type of dwarfism because of a homozygous recessive allele. Assume Hardy-Weinberg frequencies.

**(a)** What is the frequency of the recessive dwarfing allele?

**(b)** What is the frequency of heterozygous carriers in the herd?

**(c)** Among the cattle that are nondwarfs, what is the frequency of carriers?

**S15.3** A population of fish includes 1 percent albinos resulting from a homozygous recessive allele that eliminates the normal pigmentation. Assuming Hardy-Weinberg proportions, determine the frequency of heterozygous genotypes.

**S15.4** Hereditary hemochromatosis, a defect in iron metabolism resulting in impaired liver function, arthritis, and other symptoms, is one of the most common genetic diseases in northern Europe. The disease is relatively mild and easily treated when recognized. About one person in 400 is affected. Assuming Hardy-Weinberg proportions, determine the expected frequency of carriers.

**S15.5** An inherited defect in a certain receptor protein confers resistance to infection with HIV1 (human immunodeficiency virus 1, the cause of AIDS). Resistant people are homozygous recessive and are found at a frequency of 1 percent in the population. Assuming Hardy-Weinberg proportions, determine the frequency of heterozygous genotypes.

**S15.6** Organisms were sampled from three different populations polymorphic for the alleles *A* and *a* and yielded these observed numbers of each of the possible genotypes:

| | | | |
|---|---|---|---|
| **(a)** *AA:* 5 | *Aa:* 60 | *aa:* 75 | Total: 140 |
| **(b)** *AA:* 10 | *Aa:* 50 | *aa:* 80 | Total: 140 |
| **(c)** *AA:* 15 | *Aa:* 40 | *aa:* 85 | Total: 140 |

In terms of the magnitude of the chi-square obtained in a test of goodness of fit to Hardy-Weinberg proportions, which of these samples is closest to Hardy-Weinberg proportions? Do any of the samples deviate significantly from Hardy-Weinberg proportions?

**S15.7** In a population in Hardy-Weinberg equilibrium for a recessive allele at frequency $q$, what value of $q$ is necessary for the ratio of carriers to affected organisms to equal each of the following?

**(a)** 10

**(b)** 100

**(c)** 1000

**S15.8** A man is known to be a carrier of the cystic fibrosis allele. He marries a phenotypically normal woman. In the general population, the incidence of cystic fibrosis at birth is approximately one in 1700. Assume Hardy-Weinberg proportions.

**(a)** What is the probability that the wife is also a carrier?

**(b)** What is the probability that their first child will be affected?

**S15.9** A man with normal parents whose brother has phenylketonuria marries a phenotypically normal woman. In the general population, the incidence of phenylketonuria at birth is approximately one in 10,000. Assume Hardy-Weinberg proportions.

**(a)** What is the probability that the man is a carrier?

**(b)** What is the probability that the wife is also a carrier?

**(c)** What is the probability that their first child will be affected?

**S15.10** What genotype frequencies constitute the Hardy-Weinberg principle for a gene in a large, randomly mating allotetraploid population with two alleles at frequencies *p* and *q*?

**S15.11** A large population includes 10 alleles of a gene. The ratio of the allele frequencies forms the arithmetic progression 1 : 2 : 3 : . . .: 8 : 9 : 10. The genotypes are in Hardy-Weinberg proportions.

**(a)** What ratio is formed by the frequencies of the homozygous genotypes?

**(b)** Among all heterozygous genotypes that carry the most common allele, what ratio is formed by the frequencies of the heterozygous genotypes?

**S15.12** For an X-linked gene with two alleles in a large, randomly mating population, the frequency of carrier females equals one-half of the frequency of the males carrying the recessive allele. What are the allele frequencies?

**S15.13** A trait due to a harmful recessive X-linked allele in a large, randomly mating population affects one male in 50. What is the frequency of carrier females? What is the expected frequency of affected females?

**S15.14** In DNA typing, how many genes must be examined to prove that evidence left at a crime scene comes from a suspect? How many genes must be examined to prove that the evidence does not come from a suspect?

**S15.15** The DNA type of a suspect and that of blood left at the scene of a crime match for each of *n* genes, and each gene is heterozygous and so yields two bands. The frequency of each of the bands in the general population is 0.1. Use the product rule to calculate:

**(a)** The number of genes for which the probability of a match by chance is less than one in a million.

**(b)** The number of genes for which the probability of a match by chance is less than one in ten billion.

**S15.16** Two strains of bacteria, A and B, are placed into direct competition in a chemostat. A is favored over B. What is the value of the selection coefficient (*s)* if, after one generation, the ratio of the number of A cells to the number of B cells:

**(a)** Increases by 10 percent?

**(b)** Increases by 90 percent?

**(c)** Increases by a factor of 2?

**S15.17** Two strains of bacteria, A and B, are placed into direct competition in a chemostat. A is favored over B. If the selection coefficient per generation is constant, what is its value if, in an interval of 100 generations:

**(a)** The ratio of A cells to B cells increases by 10 percent?

**(b)** The ratio of A cells to B cells increases by 90 percent?

**(c)** The ratio of A cells to B cells increases by a factor of 2?

**S15.18** A strain of pathogenic bacteria undergoes cell division exactly two times as often as a nonpathogenic mutant strain of the same bacteria. What is the relative fitness of the pathogenic strain compared to the nonpathogenic strain? You may regard the generation time as the length of time for one division of a nonpathogenic cell.

**S15.19** An allele *A* undergoes mutation to the allele *a* at the rate of $10^{-5}$ per generation. If a very large population is fixed for *A* (generation 0), what is the expected frequency of *A* in the following generation (generation 1)? What is the expected frequency of *A* in generation 2? Deduce the rule for the frequency of *A* in generation *n.*

**S15.20** Large samples are taken from a single population of each of four related species of the flowering plant *Phlox.* The genotype frequencies for a gene with two alleles are estimated from each of the samples, and this yields the genotype frequencies in the accompanying table. For each species, calculate the allele frequencies and the genotype frequencies expected using the Hardy-Weinberg principle. Identify which of the populations give evidence of inbreeding, and for each apparently inbreeding population, calculate the inbreeding coefficient.

| | **Genotype frequency** | | |
|---|---|---|---|
| **Species** | ***AA*** | ***Aa*** | ***aa*** |
| (a) | 0.0720 | 0.2560 | 0.6720 |
| (b) | 0.0400 | 0.3200 | 0.6400 |
| (c) | 0.0080 | 0.3840 | 0.6080 |
| (d) | 0.880 | 0.2240 | 0.6880 |

**S15.21** The considerations in this problem illustrate the principle that for a rare, harmful, recessive allele, matings between relatives can account for a majority of affected individuals even though they account for a small proportion of all matings. Suppose that the frequency of the rare homozygous recessive genotype among the progeny of nonrelatives is one in 4 million and that first-cousin matings constitute 1 percent of all matings in the population.

**(a)** What is the frequency of the homozygous recessive among the offspring of first-cousin matings?

**(a)** Among all homozygous recessive genotypes, what proportion come from first-cousin matings?

**S15.22** A diploid population consists of 50 breeding organisms, of which one organism is heterozygous for a new mutation. If the population is maintained at a constant size of 50, what is the probability that the new mutation will be lost by chance in the first generation as a result of random genetic drift? (You may assume that the species is monoecious.) What is the answer in the more general case in which the breeding population consists of *N* diploid organisms?

**S15.23** A population segregating for the alleles *A* and *a* is sampled, and genotypes are found in the proportions 1/3 *AA*, 11/3 *Aa*, and 1/3 *aa*. These are not Hardy-Weinberg proportions. How big would the size of the sample have to be for the resulting *P* value to equal 0.05 in a chi-square test for goodness of fit?

**S15.24** An observed sample of a gene with two alleles from a population conforms almost exactly to the genotype frequencies expected for a population with an inbreeding coefficient *F* and allele frequencies 0.2 and 0.8. What is the smallest value of *F* that will lead to rejection of HardyWeinberg proportions at the 5 percent level of significance in each of the following cases? (*Hint:* Set chi-square equal to 4.0.)

**(a)** A sample size of 200 organisms.

**(b)** A sample size of 500 organisms.

**(c)** A sample size of 1000 organisms.

## *Chapter 16—*<br>*Quantitative Genetics*

**S16.1** Define the following terms:

**(a)** Multifactorial trait

**(b)** Quantitative-trait locus (QTL)

**(c)** Threshold trait

**(d)** Individual selection

**(e)** Broad-sense heritability

**(f)** Genotype-environment association

**S16.2** A population of sweet clover includes 20 percent with two leaves, 70 percent with three leaves, and 10 percent with four leaves. What are the mean and variance in leaf number in this population?

**S16.3** In an experimental population of the flour beetle *Tribolium castaneum*, the pupal weight is distributed normally with a mean of 2.0 mg and a standard deviation of 0.2 mg. What proportion of the population is expected to have a pupal weight between 1.8 and 2.2 mg? Between 1.6 and 2.4 mg? Would you expect to find an occasional pupa weighing 3.0 mg or more? Explain your answer.

**S16.4** Tabulated below are the numbers of eggs laid by 50 hens over a 2-month period. The hens were selected at random from a much larger population. Estimate the mean, variance, and standard deviation of the distribution of egg number in the entire population from which the sample was drawn.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 48 | 50 | 51 | 47 | 54 | 45 | 50 | 38 | 40 | 52 |
| 58 | 47 | 55 | 53 | 54 | 41 | 59 | 48 | 53 | 49 |
| 51 | 37 | 31 | 47 | 55 | 46 | 49 | 48 | 43 | 68 |
| 59 | 51 | 52 | 66 | 54 | 37 | 46 | 55 | 59 | 45 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 44 | 44 | 57 | 51 | 50 | 57 | 50 | 40 | 63 | 33 |

**S16.5** Suppose that the 50 hens in the previous problem constituted the entire population of hens present in a flock instead of a sample from a larger flock. What are the mean, variance, and standard deviation of egg number in the population? Why are some of these numbers different from those calculated in the previous problem?

**S16.6** Two homozygous genotypes of *Drosophila* differ in the number of abdominal bristles. In genotype *AA*, the mean bristle number is 20 with a standard deviation of 2. In genotype *aa*, the mean bristle number is 23 with a standard deviation of 3. Both distributions conform to the normal curve, in which the proportions of the population that have a phenotype within an interval defined by the mean ± 1, ± 1.5, ± 2, and ± 3 standard deviations are 68, 87, 95, and 99.7 percent, respectively.

**(a)** In genotype *AA*, what is the proportion of flies with a bristle number between 20 and 23?

**(b)** In genotype *aa*, what is the proportion of flies with a bristle number between 20 and 23?

**(c)** What proportion of *AA* flies have a bristle number greater than the mean of *aa* flies?

**(d)** What proportion of *aa* flies have a bristle number greater than the mean of *AA* flies?

**S16.7** If the genotypes in the previous problem were crossed and the $F_1$ flies mated to produce an $F_2$ generation, would you expect the distribution of bristle number among $F_2$ flies to show segregation of the *A* and *a* alleles? Explain your answer.

**S16.8** A meristic trait is determined by a pair of alleles at each of four unlinked loci: *A, a* and *B, b* and *C, c* and *D, d.* The alleles act in an additive fashion such that each capital letter adds one unit to the phenotypic score. Thus the phenotype ranges from a score of 0 (genotype *aa bb cc dd)* to 8 (genotype *AA BB CC DD).* What is the expected distribution of phenotype score in the $F_2$ generation of the cross *aa bb cc dd* × *AA BB CC DD*? Draw a histogram of the distribution.

**S16.9** With respect to the trait in the previous problem, what is the expected distribution of phenotypes in a randomly mating population in which the frequency of each allele indicated with a capital letter is 0.8 and in which the alleles of the different loci are combined at random to produce the genotypes?

**S16.10** The $F_1$ generation of a cross between two inbred strains of maize has a standard deviation in mature plant height of 15 cm. When grown under the same conditions, the $F_2$ generation produced by the $F_1 \times F_1$ cross has a standard deviation in plant height of 25 cm. What are the genotypic variance and the broad-sense heritability of mature plant height in the $F_2$ population?

**S16.11** Two inbred strains of *Drosophila* are crossed and the $F_1$ generation is brother-sister mated to produce a large, genetically heterogeneous $F_2$ population that is split into two parts. One part is reared at a constant temperature of 25°C, and the other part is reared under conditions in which the temperature fluctuates among 18°C, 25°C, and 29°C. The broadsense heritability of body size at the constant temperature is 45 percent, but under fluctuating temperatures it is 15 percent. If the genotypic variance is the same under both conditions, by what factor does the temperature fluctuation increase the environmental variance?

**S16.12** The narrow-sense heritability of withers height in a population of quarterhorses is 30 percent. (Withers height is the height at the highest point of the back, between the shoulder blades.) The average withers height in the population is 17 hands. (A "hand" is a traditional measure equal to the breadth of the human hand, now taken to equal 4 inches.) From this population, studs and mares with an average withers height of 16 hands are selected and mated at random. What is the expected withers height of the progeny? How does the value of the narrow-sense heritability change if withers height is measured in meters rather than hands?

**S16.13** The narrow-sense heritability of adult stature in people is about 25 percent. What does this value imply about the correlation coefficient in adult height between father and son?

**S16.14** Suppose that the narrow-sense heritability of adult stature in people equals 25 percent. A man whose height is 5 cm above the average height for men marries a woman whose average height equals that for women. Among their progeny, what is the expected average deviation from the mean adult height in the entire population?

**S16.15** From a population of *Drosophila* with an average abdominal bristle number of 23.2, flies with an average bristle number of 27.6 were selected and mated at random to produce the next generation. The average bristle number among the progeny was 25.4. What is the realized heritability? Is this a narrow-sense or a broad-sense heritability? Explain your answer.

**S16.16** In human beings, there is about a 90 percent correlation coefficient between identical twins in the total number of raised skin ridges forming the fingerprints. What is the broad-sense heritability of total fingerprint ridge count? What is the maximum value of the narrow-sense heritability?

**S16.17** In human beings, the correlation coefficient between first cousins in the total fingerprint ridge count is 10 percent. On the basis of this value, what is the narrow-sense heritability of this trait?

**S16.18** The narrow-sense heritability of 6-week body weight in mice is 25 percent. From a population whose average 6-week weight is 20 grams, individual mice weighing an average of 24 grams are selected and mated at random. What is the expected weight of the progeny mice in the next generation?

**S16.19** In the mouse population in the problem above, the selection for greater 6-week weight is carried out for a total of five generations, and in each generation, parents are chosen with an average 6-week weight that is, on average, 4 grams heavier than the population mean in their generation. What is the expected average 6-week weight after the five generations of selection? Make a graph of average 6week weight against generation number.

**S16.20** Suppose that the selection for 6-week weight in the mouse population above were carried out in both directions for five generations. In each generation, the parents in the "up" selection are, on average, 4 grams heavier than the population mean, and the parents in the "down direction" are, on average, 4 grams lighter than the population mean.

**(a)** In each generation, how different are the expected means of the "up" and "down" populations?

**(b)** Draw a graph illustrating the expected course of selection in both populations.

**(c)** In the "down" population, is the realized heritability of 6-week weight expected to remain 25 percent indefinitely?

**S16.21** Two inbred lines of maize differ in ear length. In one, the average length of ear is 30 cm, and in the other it is 15 cm. In both lines the variance in ear length is 2 $cm^2$. In the $F_2$ generation of a cross between the lines, the average ear length is 22.5 cm with a variance of 5 $cm^2$. What is the minimum number of genes required to account for these data?

**S16.22** A threshold trait affects 2.5 percent of the organisms in a population. The underlying liability is a multifactorial trait that is normally distributed with mean 0 and variance 1.

**(a)** What is the value of the threshold, a liability above which causes the appearance of the condition?

**(b)** If the underlying liability were normally distributed with a mean of 1 and a variance of 4, what would be the corresponding value of the threshold?

**(c)** Is there any real difference between these two situations?

## *Chapter 17— Genetics of Biorhythms and Behavior*

**S17.1** Define the following terms:

**(a)** Specific nonchemotaxis

**(b)** Multiple nonchemotaxis

**(c)** General nonchemotaxis

**(d)** Sensory adaptation

**(e)** Circadian rhythm

**(f)** Genotype-enviroment interaction

**S17.2** What is the target of the molecular pathway common to all forms of chemotaxis?

**S17.3** What is the role of methylation in sensory adaptation? What is the expected phenotype of a mutant in *cheR* that cannot transfer methyl groups?

**S17.4** What period of courtship song would you expect of *D. melanogaster* males hemizygous for the $per^0$ allele?

**S17.5** What kind of an experiment would enable you to demonstrate that *per* controls the expression of other genes?

**S17.6** How does the circadian system of *Drosophila* become reset each day so that the fly's intrinsic period becomes synchronized to the 24-hour day?

**S17.7** You are studying two groups of laboratory mice raised in similar environments. One is a highly inbred strain, and the other is derived from a large randomly mating population. Which group would you expect to have a higher value of the broad-sense heritability of a particular behavioral trait? Explain your reasoning.

**S17.8** Suppose that the broad-sense heritability of IQ score is 0.5. Can you conclude that genes are responsible for 50 percent of the IQ score of a particular person? Why or why not?

**S17.9** Explain why twins and adoption studies are useful in genetic research on behavioral traits in human beings. Why are ordinary family studies not sufficient?

**S17.10** For a given actual population size, how would the magnitude of random genetic drift differ in two animal populations, one with a strong social hierarchy and another lacking an established social structure?

**S17.11** It has been reported that males of *Drosophila* with a genotype that is uncommon in the population have greater reproductive success than males with more common genotypes. What effect would a rare-male mating advantage have on the genetic composition of a population of flies?

**S17.12** Humans have changed from a species living in small hunting populations to a species many members of which live in large and highly organized communities. Have those changes also led to changes in the relative contribution of different evolutionary factors (such as, for instance, random genetic drift) to the shaping of human genotypic structure?

**S17.13** Consider a complex behavioral trait in which the environmental variance consists solely of the variance that results from errors in measurement. In other words, the behavioral trait is completely determined genetically but is difficult to measure because of random fluctuations in measurement. For a single measurement of the trait, let the environmental variance be represented as $\sigma_e^2$ If errors in measurement are the only cause of environmental variance, then it can be shown that the environmental variance for the mean of $n$ independent measurements is $\sigma_e^2$ /n For such a trait with a broad-sense heritability of $H^2 = \sigma_g^2 / (\sigma_g^2 + \sigma_e^2)$ for a single measurement, show that the broad-sense heritability of the mean of $n$ measurements, $H'^2$, equals $nH^2/[1 + (n - 1)H^2]$. (*Hint:* Find an expression for $\sigma_e^2 / \sigma_g^2$ in terms of $H^2$.)

## Concise Dictionary of Genetics

### A

**aberrant 4 : 4 segregation** The presence of equal numbers of alleles among the spores in a single ascus, yet with a spore pair in which the two spores have different genotypes.

**acentric chromosome** A chromosome with no centromere. (p. 260)1

**acridine** A chemical mutagen that intercalates between the bases of a DNA molecule, causing single-base insertions or deletions. (p. 571)

**acrocentric chromosome** A chromosome with the centromere near one end. (p. 261)

**active site** The part of an enzyme at which substrate molecules bind and are converted into their reaction products.

**acylated tRNA** A tRNA molecule to which an amino acid is linked. (p.431)

**adaptation** Any characteristic of an organism that improves its chance of survival and reproduction in its environment; the evolutionary process by which a species undergoes progressive modification favoring its survival and reproduction in a given environment. (p. 652)

**addition rule** The principle that the probability that any one of a set of mutually exclusive events is realized equals the sum of the probabilities of the separate events. (p. 49)

**additive variance** The magnitude of the genetic variance that results from the additive action of genes and that accounts for the genetic component of the parent-offspring correlation; the value that the genetic variance would assume if there were no dominance or interaction of alleles affecting the trait.

**adenine (A)** A nitrogenous purine base found in DNA and RNA. (p. 9)

**adenyl cyclase** The enzyme that catalyzes the synthesis of cyclic AMP. (p. 468)

**adjacent segregation** Segregation of a heterozygous reciprocal translocation in which a translocated chromosome and a normal chromosome segregate together, producing an aneuploid gamete.

**adjacent-1 segregation** Segregation from a heterozygous reciprocal translocation in which homologous centromeres go to opposite poles of the first-division spindle. (p. 289)

**adjacent-2 segregation** Segregation from a heterozygous reciprocal translocation in which homologous centromeres go to the same pole of the first-division spindle. (p. 290)

**agarose** A component of agar used as a gelling agent in gel electrophoresis; its value is that few molecules bind to it, so it does not interfere with electrophoretic movement. (p. 206)

**agglutination** The clumping or aggregation, as of viruses or blood cells, caused by an antigen-antibody interaction. (p. 70)

**albinism** Absence of melanin pigment in the iris, skin, and hair of an animal; absence of chlorophyll in plants. (p. 53)

**alkaptonuria** A recessively inherited metabolic disorder in which a defect in the breakdown of tyrosine leads to excretion of homogentisic acid (alkapton) in the urine.

**alkylating agent** An organic compound capable of transferring an alkyl group to other molecules. (p. 570)

**allele** Any of the alternative forms of a given gene. (p. 40)

**allele frequency** The relative proportion of all alleles of a gene that are of a designated type. (p. 628)

**allopolyploid** A polyploid formed by hybridization between two different species. (p. 263)

**allosteric protein** Any protein whose activity is altered by a change of shape induced by binding a small molecule.

**allozyme** Any of the alternative electrophoretic forms of a protein coded by different alleles of a single gene. (p. 630)

α **helix** A fundamental unit of protein folding in which successive amino acids form a right-handed helical structure held together by hydrogen bonding between the amino and carboxyl components of the peptide bonds in successive loops of the helix. (p. 414)

**alpha satellite** Highly repetitive DNA sequences associated with mammalian centromeres. (p. 247)

**alternate segregation** Segregation from a heterozygous reciprocal translocation in which both parts of the reciprocal translocation separate from both nontranslocated chromosomes in the first meiotic division. (p. 292)

**amber codon** Common jargon for the UAG stop codon; an amber mutation is a mutation in which a sense codon has been altered to UAG. (p. 447)

**Ames test** A bacterial test for mutagenicity; also used to screen for potential carcinogens. (p. 586)

**amino acid** Any one of a class of organic molecules that have an amino group and a carboxyl group; 20 different amino acids are the usual components of proteins. (pp. 12, 412)

**amino acid attachment site** The 3' terminus of a tRNA molecule at which an amino acid is attached.

**aminoacylated tRNA** A tRNA covalently attached to its amino acid; charged tRNA. (p. 431)

**aminoacyl site** The tRNA-binding site on the ribosome to which each incoming charged tRNA is initially bound. (p. 431)

**A site** *See* **aminoacyl site.**

**aminoacyl tRNA synthetase** The enzyme that attaches the correct amino acid to a tRNA molecule. (p. 431)

**amino terminus** The end of a polypeptide chain at which the amino acid bears a free amino group ($-NH_2$). (p. 412)

**amniocentesis** A procedure for obtaining fetal cells from the amniotic fluid for the diagnosis of genetic abnormalities. (p. 273)

**amorph** A mutation in which the function of the gene product is completely abolished. (p. 66)

**analog** *See* **base analog.**

**anaphase** The stage of mitosis or meiosis in which chromosomes move to opposite ends of the spindle. In anaphase I of meiosis, homologous centromeres separate; in anaphase II, sister centromeres separate. (pp. 86, 94)

**aneuploid** A cell or organism in which the chromosome number is not an exact multiple of the haploid number; more generally, aneuploidy refers to a condition in which particular genes or chromosomal regions are present in extra or fewer copies compared with wildtype. (p. 269)

**annealing** The coming together of two strands of nucleic acid that have complementary base sequences to form a duplex molecule. (p. 200)

**antibiotic-resistant mutant** A cell or organism that carries a mutation conferring resistance to an antibiotic. (p. 311)

**antibiotic-resistant mutation** A mutation conferring resistance to one or more antibiotics. (p. 311)

**antibody** A blood protein produced in response to a specific antigen and capable of binding with the antigen. (pp. 69, 483)

**anticodon** The three bases in a tRNA molecule that are complementary to the three-base codon in mRNA. (pp. 431, 432)

**antigen** A substance able to stimulate the production of antibodies. (p. 69)

**antigen presentation** An immune process in which a cell carrying an antigen bound to an MHC protein stimulates a responsive cell.

**antigen-presenting cell** Any cell, such as a macrophage, expressing certain products of the major histocompatibility complex and capable of stimulating T cells that have appropriate antigen receptors.

**antimorph** A mutation whose phenotypic effects are opposite to those of the wildtype allele. (p. 67)

**antiparallel** The chemical orientation of the two strands of a double-stranded nucleic acid molecule; the 5'-to-3' orientations of the two strands are opposite one another. (p. 181)

**AP endonuclease** An endonuclease that cleaves a DNA strand at any site at which the deoxyribose lacks a base. (p. 566)

**apoptosis** Genetically programmed cell death, especially in embryonic development. (p. 524)

**aporepressor** A protein converted into a repressor by binding with a particular molecule. (p. 462, 472)

**Archaea** One of the three major classes of organisms; also called archaebacteria, they are unicellular microorganisms, usually found in extreme environments, that differ as much from bacteria as either group differs from eukaryotes. *See also* **Bacteria.** (p. 22)

**artificial selection** Selection, imposed by a breeder, in which organisms of only certain phenotypes are allowed to breed. (p. 683)

**ascospore** *See* **ascus.**

**ascus** A sac containing the spores (ascospores) produced by meiosis in certain groups of fungi, including *Neurospora* and yeast. (p. 150)

**assortative mating** Nonrandom selection of mating partners with respect to one or more traits; it is positive when like phenotypes mate more frequently than would be expected by chance and is negative when the reverse occurs. (p. 633)

**ATP** Adenosine triphosphate, the primary molecule for storing chemical energy in a living cell.

**attached-X chromosome** A chromosome in which two X chromosomes are joined to a common centromere; also called a compound-X chromosome. (p. 134)

**attachment site** The base sequence in a bacterial chromosome at which bacteriophage DNA can integrate to form a prophage; either of the two attachment sites that flank an integrated prophage. (p. 341)

**attenuation** *See* **attenuator.**

**attenuator** A regulatory base sequence near the beginning of an mRNA molecule at which transcription can be terminated; when an attenuator is present, it precedes the coding sequences. (p. 473)

**attractant** A substance that draws organisms to it through chemotaxis. (p. 704)

**AUG** The usual initiation codon for polypeptide synthesis as well as the internal codon for methionine. (pp. 432, 443)

**autogamy** Reproductive process in ciliates in which the genotype is reconstituted from the fusion of two identical haploid nuclei resulting from mitosis of a single product of meiosis; the result is homozygosity for all genes. (p. 614)

**autonomous determination** Cellular differentiation determined intrinsically and not dependent on external signals or interactions with other cells. (p. 515)

**autopolyploidy** Type of polyploidy in which there are more than two sets of homologous chromosomes. (p. 263)

**autoradiography** A process for the production of a photographic image of the distribution of a radioactive substance in a cell or large cellular molecule; the image is produced on a photographic emulsion by decay emission from the radioactive material.

**autoregulation** Regulation of gene expression by the product of the gene itself. (p. 462)

**autosomes** All chromosomes other than the sex chromosomes. (p. 97)

**auxotroph** A mutant microorganism unable to synthesize a compound required for its growth but able to grow if the compound is provided. (p. 312)

## B

**backcross** The cross of an $F_1$ heterozygote with a partner that has the same genotype as one of its parents. (p. 41)

**Bacteria** One of the major kingdoms of living things; includes most bacteria. *See also* **Archaea.** (p. 22)

**bacterial attachment site** *See* **attachment site**

**bacterial transformation** *See* **transformation**

**bacteriophage** A virus that infects bacterial cells; commonly called a phage. (pp. 6. 308)

**band** In gel electrophoresis, a compact region of heavy staining due to the accumulation of molecules of a given size; in *Drosophila* genetics, one of the striations found in the giant polytene chromosomes in the larval salivary glands. (p. 206)

**Barr body** A darkly staining body found in the interphase nucleus of certain cells of female mammals; consists of the condensed, inactivated X chromosome. (p. 277)

**base** Single-ring (pyrimidine) or double-ring (purine) component of a nucleic acid. (p. 9)

**base analog** A chemical so similar to one of the normal bases that it can be incorporated into DNA. (p. 567)

**base composition** The relative proportions of the bases in a nucleic acid or of A + T versus G + C in duplex DNA. (p. 175)

**base pair** A pair of nitrogenous bases, most commonly one purine and one pyrimidine, held together by hydrogen bonds in a double-stranded region of a nucleic acid molecule; commonly abbreviated bp, the term is often used interchangeably with the term *nucleotide pair.* The normal base pairs in DNA are AT and GC. (pp. 9. 13)

**base-substitution mutation** Incorporation of an incorrect base into a DNA duplex. (p. 557)

**B cells** A class of white blood cells, derived from bone marrow, with the potential for producing antibodies.

**becquerel** A unit of radiation equal to 1 disintegration second, abbreviated Bq; equal to $2.7 \times 10^{11}$ curie.

**behavior** The observed response of an organism to stimulation or environmental change. (p. 704)

**B-form DNA** The right-handed helical structure of DNA proposed by Watson and Crick. (p. 177) **β-galactosidase** An enzyme that cleaves lactose into its glucose and galactose constituents; produced by a gene in the *lac* operon. (p. 462) **βsheet** A fundamental structure formed in protein folding in which two polypeptide segments are held together in an antiparallel array by hydrogen bonds. (p. 414)

**bidirectional replication** DNA replication proceeding in opposite directions from an origin of replication. *See also* ***θ* replication.** (p. 189)

**binding site** A DNA or RNA base sequence that serves as the target for binding by a protein; the site on the protein that does the binding.

**biochemical pathway** A diagram showing the order in which intermediate molecules are produced in the synthesis or degradation of a metabolite in a cell. (p. 19)

**bivalent** A pair of homologous chromosomes, each consisting of two chromatids, associated in meiosis I. (p. 92)

**blastoderm** Structure formed in the early development of an insect larva; the syncytial blastoderm is formed from repeated cleavage of the zygote nucleus without cytoplasmic division; the cellular blastoderm is formed by migration of the nuclei to the surface and their inclusion in separate cell membranes. (p. 529)

**blastula** A hollow sphere of cells formed early in development. (p. 514)

**blood-group system** A set of antigens on red blood cells resulting from the action of a series of alleles of a single gene, such as the ABO, MN, or Rh blood-group systems. (p. 68)

**blunt ends** Ends of a DNA molecule in which all terminal bases are paired; the term usually refers to termini formed by a restriction enzyme that does not produce single-stranded ends. (pp. 204, 362)

**branched pathway** A metabolic pathway in which an intermediate serves as a precursor for more than one product.

**branch migration** In a DNA duplex invaded by a single strand from another molecule, the process in which the size of the heteroduplex region increases by progressive displacement of the original strand. (p. 138)

**broad-sense heritability** The ratio of genotypic variance to total phenoty variance. (p. 682)

## C

**cAMP-CRP complex** A regulatory complex consisting of cyclic AMP (cAMP) and the CAP protein; the complex is needed for transcription of certain operons. (p. 468)

**candidate gene** A gene proposed to be involved in the genetic determination of a trait because of the role of the gene product in the cell or organism. (p. 732)

**cap** A complex structure at the 5' termini of most eukaryotic mRNA molecules, having a 5'-5' linkage instead of the usual 3'-5' linkage. (p. 425)

**CAP protein** Acronym for catabolite activator protein, the protein that binds cAMP and that regulates the activity of inducible operons in prokaryotes; also called CRP, for cAMP receptor protein. (p. 469)

**carbon-source mutant** A cell or organism that carries a mutation preventing the use of a particular molecule or class of molecules as a source of carbon. (p. 312)

**carboxyl terminus** The end of a polypeptide chain at which the amino acid has a free carboxyl group (-COOH). (p. 412)

**carcinogen** A physical agent or chemical reagent that causes cancer.

**carrier** A heterozygote for a recessive allele. (p. 53)

**cascade** A series of enzyme activations serving to amplify a weak chemical signal. (p. 537)

**cassette** In yeast, either of two sets of inactive mating-type genes that can become active by relocating to the *MAT* locus. (p. 482)

**catabolite activator protein** *See* **CAP protein.**

**cDNA** *See* **complementary DNA.**

**cell cycle** The growth cycle of a cell; in eukaryotes, it is subdivided into $G_1$ (gap 1), S (DNA synthesis), $G_2$ (gap 2), and M (mitosis). (p. 83)

**cell fate** The pathway of differentiation that a cell normally undergoes. (p. 514)

**cell hybrid** Product of fusion of two cells in culture, often from different species.

**cell lineage** The ancestor-descendant relationships of a group of cells in development. (p. 519)

**cellular oncogene** A gene coding for a cellular growth factor whose abnormal expression predisposes to malignancy. *See also* **oncogene.** (p. 298)

**centimorgan** A unit of distance in the genetic map equal to 1 percent recombination; also called a map unit. (p. 128)

**central dogma** The concept that genetic information is transferred from the nucleotide sequence in DNA to the nucleotide sequence in an RNA transcript to the amino acid sequence of a polypeptide chain. (p. 12)

**centromere** The region of the chromosome that is associated with spindle fibers and that participates in normal chromosome movement during mitosis and meiosis. (p. 86)

**chain elongation** The process of addition of successive amino acids to the growing end of a polypeptide chain. (p. 421)

**chain initiation** The process by which polypeptide synthesis is begun. (p. 421)

**chain termination** The process of ending polypeptide synthesis and releasing the polypeptide from the ribosome; a chain-termination mutation creates a new stop codon, resulting in premature termination of synthesis of the polypeptide chain. (p. 421)

**chaperone** A protein that assists in the three-dimensional folding of another protein. (p. 414)

**Chargaff's rules** In double-stranded DNA, the amount of A equals that of T, and the amount of G equals that of C. (p. 176)

**charged tRNA** A tRNA molecule to which an amino acid is linked; acylated tRNA. (p. 431)

**chemoreceptor** Any molecule, usually a protein, that binds with other molecules and stimulates a behavioral response, such as chemotaxis, sensation, etc. *See also* **chemosensor.** (p. 709)

**chemosensor** In bacteria, a class of proteins that bind with particular attractants or repellents. Chemosensors interact with chemoreceptors to stimulate chemotaxis. *See also* **chemotaxis.** (p. 709)

**chemotaxis** Behavioral response in which organisms move toward chemicals that attract them and away from chemicals that repel them. (p. 704)

**chiasma** The cytological manifestation of crossing-over; the cross-shaped exchange configuration between nonsister chromatids of homologous chromosomes that is visible in prophase I of meiosis. The plural can be either *chiasmata* or *chiasmas.* (p. 93)

**chimeric gene** A gene produced by recombination, or by genetic engineering, that is a mosaic of DNA sequences from two or more different genes. (p. 285)

**chi-square** ($X^2$) A statistical quantity calculated to assess the goodness of fit between a set of observed numbers and the theoretically expected numbers. (p. 109)

**chromatid** Either of the longitudinal subunits produced by chromosome replication. (p. 86)

**chromatid interference** In meiosis, the effect that crossing-over between one pair of nonsister chromatids may have on the probability that a second crossing-over in the same chromosome will involve the same or different chromatids; chromatid interference does not generally occur. (p. 141)

**chromatin** The aggregate of DNA and histone proteins that makes up a eukaryotic chromosome. (p. 228)

**chromocenter** The aggregate of centromeres and adjacent heterochromatin in nuclei of *Drosophila* larval salivary gland cells. (p. 234)

**chromomere** A tightly coiled, bead-like region of a chromosome most readily seen during pachytene of meiosis; the beads are in register in a polytene chromosome, resulting in the banded appearance of the chromosome. (pp. 92)

**chromosome** In eukaryotes, a DNA molecule that contains genes in linear order to which numerous proteins are bound and that has a telomere at each end and a centromere; in prokaryotes, the DNA is associated with fewer proteins, lacks telomeres and a centromere, and is often circular; in viruses, the chromosome is DNA or RNA, singles-stranded or double-stranded, linear or circular, and often free of bound proteins. (p. 2, 82)

**chromosome complement** The set of chromosomes in a cell or organism. (p. 82)

**chromosome interference** In meiosis, the phenomenon by which a crossing-over at one position inhibits the occurrence of another crossing-over at a nearby position. (p. 143)

**chromosome map** A diagram showing the locations and relative spacing of genes along a chromosome. (p. 127)

**chromosome painting** Use of differentially labeled, chromosome-specific DNA strands for hybridization with chromosomes to label each chromosome with a different color. (p. 264)

**chromosome theory of heredity** The theory that chromosomes are the cellular objects that contain the genes. (p. 104)

**circadian rhythm** A biological cycle with an approximate 24-hour period synchronized with the cycle of day and night. (p. 714)

**circular permutation** A permutation of a group of elements in which the elements are in the same order but the beginning of the sequence differs. (p. 335)

***cis* configuration** The arrangement of linked genes in a double heterozygote in which both mutations are present in the same chromosome—for example, $a_1\ a_2/++$; also called coupling. (pp. 126, 451)

***cis*-dominant mutation** A mutation that affects the expression of only those genes on the same DNA molecule as that containing the mutation. (p. 464)

***cis* heterozygote** *See* ***cis* configuration.**

***cis-trans* test** *See* **complementation test.**

**cistron** A DNA sequence specifying a single genetic function as defined by a complementation test; a nucleotide sequence coding for a single polypeptide. (p. 339)

***CIB* method** A genetic procedure used to detect X-linked recessive lethal mutations in *Drosophila melanogaster;* so named because one X chromosome in the female parent is marked with an inversion (*C),* a recessive lethal allele (*l),* and the dominant allele for Bar eyes (*B*). (p. 563)

**cleavage division** Mitosis in the early embryo. (p. 514)

**clone** A collection of organisms derived from a single parent and, except for new mutations, genetically identical to that parent; in genetic engineering, the linking of a specific gene or DNA fragment to a replicable DNA molecule, such as a plasmid or phage DNA. (p. 308)

**cloned gene** A DNA sequence incorporated into a vector molecule capable of replication in the same or a different organism. (pp. 205, 360)

**cloning** The process of producing cloned genes. (p. 360)

**coding region** The part of a DNA sequence that codes for the amino acids in a protein.

**coding sequence** A region of a DNA strand with the same sequence as is found in the coding region of a messenger RNA, except that T is present in DNA instead of U. (p. 424)

**coding strand** In a transcribed region of a DNA duplex, the strand that is not transcribed.

**codominance** The expression of both alleles in a heterozygote. (p. 68)

**codon** A sequence of three adjacent nucleotides in an mRNA molecule, specifying either an amino acid or a stop signal in protein synthesis. (pp. 14, 431)

**coefficient of coincidence** An experimental value obtained by dividing the observed number of double recombinants by the expected number calculated under the assumption that the two events take place independently. (p. 143)

**cohesive end** A single-stranded region at the end of a double-stranded DNA molecule that can adhere to a complementary single-stranded sequence at the other end or in another molecule. (p. 340)

**colchicine** A chemical that prevents formation of the spindle in nuclear division. (p. 267)

**colinearity** The linear correspondence between the order of amino acids in a polypeptide chain and the corresponding sequence of nucleotides in the DNA molecule. (p. 416)

**colony** A visible cluster of cells formed on a solid growth medium by repeated division of a single parental cell and its daughter cells. (p. 3)

**colony hybridization assay** A technique for identifying colonies containing a particular cloned gene; many colonies are transferred to a filter, lysed, and exposed to radioactive DNA or RNA complementary to the DNA sequence of interest, after which colonies that contain a sequence complementary to the probe are located by autoradiography. (p. 372)

**color blindness** In human beings, the usual form of color blindness is X-linked red-green color blindness. Unequal crossing-over between the adjacent red and green opsin pigment genes results in chimeric opsin genes causing mild or severe green-vision defects (deuteranomaly or deuteranopia, respectively) and mild or severe red-vision defects (protanomaly or protanopia, respectively). (p. 284)

**combinatorial control** Strategy of gene regulation in which a relatively small number of time- and tissue-specific positive and negative regulatory elements are used in various combinations to control the expression of a much larger number of genes. (p. 499)

**combinatorial joining** The DNA splicing mechanism by which antibody variability is produced, resulting in many possible combinations of V and J regions in the formation of a light-chain gene and many possible combinations of V, D, and J regions in the formation of a heavy-chain gene. (p. 485)

**common ancestor** Any ancestor of both one's father and one's mother. (p. 634)

**compartment** In *Drosophila* development, a group of descendants of a small number of founder cells with a determined pattern of development. (p. 540)

**compatible** Blood or tissue that can be transfused or transplanted without rejection. (p. 70)

**complementary DNA** A DNA molecule made by copying RNA with reverse transcriptase; usually abbreviated cDNA. (p. 369)

**complementation** The phenomenon in which two recessive mutations with similar phenotypes result in a wildtype phenotype when both are heterozygous in the same genotype; complementation means that the mutations are in different genes. (p. 55)

**complementation group** A group of mutations that fail to complement one another. (p. 58)

**complementation test** A genetic test to determine whether two mutations are alleles (are present in the same functional gene). (p. 55)

**complex** A term used to refer to an ordered aggregate of molecules, as in an enzyme-substrate complex or a DNA-histone complex.

**compound-X chromosome** *See* **attached-X chromosome.**

**condensation** The coiling process by which chromosomes become shorter and thicker during prophase of mitosis or meiosis. (p. 137)

**conditional mutation** A mutation that results in a mutant phenotype under certain (restrictive) environmental conditions but results in a wildtype phenotype under other (permissive) conditions. (p. 556)

**congenital** Present at birth.

**conjugation** A process of DNA transfer in sexual reproduction in certain bacteria; in *E. coli*, the transfer is unidirectional, from donor cell to recipient cell. Also, a mating between cells of *Paramecium.* (p. 314, 614)

**consanguineous mating** A mating between relatives. (p. 634)

**consensus sequence** A generalized base sequence derived from closely related sequences found in many locations in a genome or in many organisms; each position in the consensus sequence consists of the base found in the majority of sequences at that position. (p. 419)

**conserved sequence** A base or amino acid sequence that changes very slowly in the course of evolution. (p. 229)

**constant region** The part of the heavy and light chains of an antibody molecule that has the same amino acid sequence among all antibodies derived from the same heavy-chain and light-chain genes. (p. 484)

**constitutive mutation** A mutation that causes synthesis of a particular mRNA molecule (and the protein that it encodes) to take place at a constant rate, independent of the presence or absence of any inducer or repressor molecule. (p. 464)

**contig** A set of cloned DNA fragments overlapping in such a way as to provide unbroken coverage of a contiguous region of the genome; a contig contains no gaps. (p. 373)

**continuous trait** A trait in which the possible phenotypes have a continuous range from one extreme to the other rather than falling into discrete classes. (p. 671)

**coordinate gene** Any of a group of genes that establish the basic anterior-posterior and dorsal-ventral axes of the early embryo. (pp. 532, 534)

**coordinate regulation** Control of synthesis of several proteins by a single regulatory element; in prokaryotes, the proteins are usually translated from a single mRNA molecule. (pp. 436, 461)

**core particle** The aggregate of histones and DNA in a nucleosome, without the linking DNA. (p. 230)

**co-repressor** A small molecule that binds with an aporepressor to create a functional repressor molecule. (p. 461)

**correlated response** Change of the mean in one trait in a population accompanying selection for another trait. (p. 687)

**correlation coefficient** A measure of association between pairs of numbers, equaling the covariance divided by the product of the standard deviations. (p. 688)

**Cot** Mathematical product of the initial concentration of nucleic acid ($C_0$) and the time ($t$). (p. 237)

**Cot curve** A graph of percent nucleic acid renaturation as a function of the initial concentration ($C_0$) multiplied by time. (p. 237)

**cotransduction** Transduction of two or more linked genetic markers by one transducing particle. (p. 326)

**cotransformation** Transformation in bacteria of two genetic markers carried on a single DNA fragment. (p. 313)

**counterselected marker** A mutation used to prevent growth of a donor cell in an Hfr × $F^-$ bacterial mating. (p. 319)

**coupled transcription-translation** In prokaryotes, the translation of an mRNA molecule before its transcription is completed.

**coupling** *See* ***cis*** **configuration.**

**courtship song** In *Drosophila*, the regular oscillation in the time interval between pulses in successive episodes of wing vibration that the male directs toward the female during mating behavior.

**covalent bond** A chemical bond in which electrons are shared.

**covalent circle** *See* **covalently closed circle.**

**covalently closed circle** A ring-shaped duplex DNA molecule whose ends are joined by covalent bonds. (p. 224)

**covariance** A measure of association between pairs of numbers that is defined as the average product of the deviations from the respective means. (p. 688)

**C region** *See* **constant region.**

**crossing-over** A process of exchange between nonsister chromatids of a pair of homologous chromosomes that results in the recombination of linked genes. (p. 93)

**crown gall tumor** Plant tumor caused by infection with *Agrobacterium tumefaciens.* (p. 381)

**CRP protein** *See* **CAP protein.**

**curie** A unit of radiation equal to 3. 7 × $10^{10}$ disintegrations/second, abbreviated Ci; equal to 3. 7 × $10^{10}$ becquerels.

**cryptic splice site** A potential splice site not normally used in RNA processing unless a normal splice site is blocked or mutated. (p. 430)

**cyclic adenosine monophosphate (cAMP)** *See* **cyclic AMP.**

**cyclic AMP** Cyclic adenosine monophosphate, usually abbreviated cAMP, used in the regulation of cellular processes; cAMP synthesis is regulated by glucose metabolism; its action is mediated by the CAP protein in prokaryotes and by protein kinases (enzymes that phosphorylate proteins) in eukaryotes. (p. 468)

**cyclic AMP receptor protein (CRP)** *See* **CAP protein.**

**cyclin** One of a group of proteins that participates in controlling the cell cycle. Different types of cyclins interact with the p34 kinase subunit and regulate the $G_1$-S and $G_2$-M transitions. The proteins are called cyclins because their abundance rises and falls rhythmically in the cell cycle. (p. 84)

**cytogenetics** The study of the genetic implications of chromosome structure and behavior.

**cytokinesis** Division of the cytoplasm. (p. 83)

**cytological map** Diagrammatic representation of a chromosome. (p. 234)

**cytoplasmic inheritance** Transmission of hereditary traits through self-replicating factors in the cytoplasm—for example, mitochondria and chroloplasts. (p. 600)

**cytoplasmic male sterility** Type of pollen abortion in maize with cytoplasmic inheritance through the mitochondria. (p. 612)

**cytosine (C)** A nitrogenous pyrimidine base found in DNA and RNA. (p. 9)

**cytotoxic T cell** Type of white blood cell that participates directly in attacking cells marked for destruction by the immune system.

## D

**daughter strand** A newly synthesized DNA or chromosome strand. (p. 182)

**deamination** Removal of an amino group ($-NH_2$) from a molecule.

**decaploid** An organism with ten sets of chromosomes. (p. 261)

**deficiency** *See* **deletion.**

**degeneracy** The feature of the genetic code in which an amino acid corresponds to more than one codon; also called redundancy. (p. 442)

**degenerate code** *See* **degeneracy.**

**degree of dominance** The extent to which the phenotype of a heterozygous genotype resembles one of the homozygous genotypes. (p. 656)

**degrees of freedom** An integer that determines the significance level of a particular statistical test. In the goodness of-fit type of chi-square test in which the expected numbers are not based on any quantities estimated from the data themselves, the number of degrees of freedom is one less than the number of classes of data. (p. 112)

**deletion** Loss of a segment of the genetic material from a chromosome; also called deficiency. (p. 281)

**deletion mapping** The use of overlapping deletions to locate a gene on a chromosome or a genetic map. (p. 336)

**deme** *See* **local population.**

**denaturation** Loss of the normal three-dimensional shape of a macromolecule without breaking covalent bonds, usually accompanied by loss of its biological activity; conversion of DNA from the double-stranded into the single-stranded form; unfolding of a polypeptide chain. (p. 199)

**denaturation mapping** An electron-microscopic technique for localizing regions of a double-stranded DNA molecule by noting the positions at which the individual strands separate in the early stages of denaturation.

**deoxyribonuclease** An enzyme that breaks sugar-phosphate bonds in DNA, forming either fragments or the component nucleotides; abbreviated DNase. (p. 225)

**deoxyribonucleic acid** *See* **DNA.**

**deoxyribose** The five-carbon sugar present in DNA. (p. 174)

**depurination** Removal of purine bases from DNA. (p. 570)

**derepression** Activation of a gene or set of genes by inactivation of a repressor.

**deuteranomaly** *See* **color blindness.**

**deuteranopia** *See* **color blindness.**

**deviation** In statistics, a difference from an expected value. (p. 672)

**diakinesis** The substage of meiotic prophase I, immediately preceding metaphase I, in which the bivalents attain maximum shortening and condensation. (pp. 92, 93)

**dicentric chromosome** A chromosome with two centromeres. (p. 260)

**dideoxyribose** A deoxyribose sugar lacking the 3' hydroxyl group; when incorporated into a polynucleotide chain, it blocks further chain elongation. (p. 211)

**dideoxy sequencing method** Procedure for DNA sequencing in which a template strand is replicated from a particular primer sequence and terminated by the incorporation of a nucleotide that contains dideoxyribose instead of deoxyribose; the resulting fragments are separated by size via electrophoresis. (p. 211)

**differentiation** The complex changes of progressive diversification in cellular structure and function that take place in the development of an organism; for a particular line of cells, this results in a continual restriction in the types of transcription and protein synthesis of which each cell is capable.

**diffuse centromere** A centromere that is dispersed throughout the chromosome, as indicated by the dispersed binding of spindle fibers, rather than being concentrated at a single point. (p. 246)

**dihybrid** Heterozygous at each of two loci; progeny of a cross between true-breeding or homozygous strains that differ genetically at two loci. (p. 42)

**dimer** A protein formed by the association of two polypeptide subunits.

**diploid** A cell or organism with two complete sets of homologous chromosomes. (p. 82)

**diplotene** The substage of meiotic prophase I, immediately following pachytene and preceding diakinesis, in which pairs of sister chromatids that make up a bivalent (tetrad) begin to separate from each other and chiasmata become visible. (pp. 92, 93)

**direct repeat** Copies of an identical or very similar DNA or RNA base sequence in the same molecule and in the same orientation. (p. 243)

**discontinuous variation** Variation in which the phenotypic differences for a trait fall into two or more discrete classes.

**disjunction** Separation of homologous chromosomes to opposite poles of a division spindle during anaphase of a mitotic or meiotic nuclear division.

**displacement loop** Loop formed when part of a DNA strand is dislodged from a duplex molecule because of the partner strand's pairing with another molecule. (p. 590)

**distribution** In quantitative genetics, the mathematical relation that gives the proportion of members in a population that have each possible phenotype. (p. 671)

**disulfide bond** Two sulfur atoms covalently linked; found in proteins when the sulfur atoms in two different cysteines are joined.

**dizygotic twins** Twins that result from the fertilization of separate ova and that are genetically related as siblings; also called fraternal twins. (p. 682)

**D loop** *See* **displacement loop.**

**DNA** Deoxyribonucleic acid, the macromolecule, usually composed of two polynucleotide chains in a double helix, that is the carrier of the genetic information in all cells and many virsues. (p. 2)

**DNA cloning** *See* **cloned gene.**

**DNA gyrase** One of a class of enzymes called topoisomerases, which function during DNA replication to relax positive supercoiling of the DNA molecule and which introduce negative supercoiling into nonsupercoiled molecules early in the life cycle of many phages. (p. 225)

**DNA ligase** An enzyme that catalyzes formation of a covalent bond between adjacent 5'-P and 3'-OH termini in a broken polynucleotide strand of double-stranded DNA. (p. 196)

**DNA looping** A mechanism by which enhancers that are distant from the immediate proximity of a promoter can still regulate transcription; the enhancer and promoter, both bound with suitable protein factors, come into indirect physical contact by means of the looping out of the DNA between them. The physical interaction stimulates transcription. (p. 494)

**DNA methylase** An enzyme that adds methyl groups ($-CH_3$) to certain bases, particularly cytosine. (p. 487)

**DNA polymerase** Any enzyme that catalyzes the synthesis of DNA from deoxynucleoside 5'-triphosphates, using a template strand. (p. 191)

**DNA polymerase I (Pol I)** In *E. coli* DNA replication, the enzyme that fills in gaps in the lagging strand and replaces the RNA primers with DNA. (p. 192)

**DNA polymerase III (Pol III)** The major DNA replication enzyme in *E. coli.* (p.192)

**DNA repair** Any of several different processes for restoration of the correct base sequence of a DNA molecule into which incorrect bases have been incorporated or whose bases have been chemically modified. (p. 577)

**DNA replication** The copying of a DNA molecule. (p. 10)

**DNase** *See* **deoxyribonuclease.**

**DNA typing** Electrophoretic identification of individual persons by the use of DNA probes for highly polymorphic regions of the genome, such that the genome of virtually every person exhibits a unique pattern of bands; sometimes called DNA fingerprinting. (p. 640)

**DNA uracyl glycosylase** An enzyme that removes uracil bases when they occur in double-stranded DNA. (p. 565)

**domain** A folded region of a polypeptide chain that is spatially somewhat isolated from other folded regions. (p. 430)

**dominance** Condition in which a heterozygote expresses a trait in the same manner as the homozygote for one of the alleles. The allele or the corresponding phenotypic trait expressed in the heterozygote is said to be dominant. (p. 35)

**dominance variance** The part of the genotypic variance that results from the dominance effects of alleles affecting the trait.

**dosage compensation** A mechanism regulating X-linked genes such that their activities are equal in males and females; in mammals, random inactivation of one X chromosome in females results in equal amounts of the products of X-linked genes in males and females. (p. 275)

**double-stranded DNA** A DNA molecule consisting of two antiparallel strands that are complementary in nucleotide sequence. (p. 10)

**double-Y syndrome** The clinical features of the karyotype 47,XYY. (p. 278)

**Down syndrome** The clinical features of the karyotype 47,+21 (trisomy 21). (p. 273)

**drug-resistance plasmid** A plasmid that contains genes whose products inactivate certain antibiotics.

**duplex DNA** A double-stranded DNA molecule. (p. 10)

**duplication** A chromosome aberration in which a chromosome segment is present more than once in the haploid genome; if the two segments are adjacent, the duplication is a tandem duplication. (p. 282)

## E

**ectopic expression** Gene expression that occurs in the wrong tissues or the wrong cell types. (p. 67)

**editing function** The activity of DNA polymerases that removes incorrectly incorporated nucleotides; also called the proofreading function. (p. 193)

**effector** A molecule that brings about a regulatory change in a cell, as by induction.

**electrophoresis** A technique used to separate molecules on the basis of their different rates of movement in response to an applied electric field, typically through a gel. (pp. 205, 629)

**electroporation** Introduction of DNA fragments into a cell by means of an electric field. (p. 363)

**elongation** Addition of amino acids to a growing polypeptide chain or of nucleotides to a growing nucleic acid chain. (p.433)

**embryo** An organism in the early stages of development (the second through seventh weeks in human beings).

**embryoid** A small mass of dividing cells formed from haploid cells in anthers that can give rise to a mature haploid plant. (p. 266)

**embryonic induction** Developmental process in which the fate of a group of cells is determined by interactions with nearby cells in the embryo. (p. 517)

**embryonic stem cells** Cells in the blastocyst that give rise to the body of the embryo, (p. 378)

**endonuclease** An enzyme that breaks internal phosphodiester bonds in a single- or double-stranded nucleic acid molecule; usually specific for either DNA or RNA. (p.192)

**endoreduplication** Doubling of the chromosome complement because of chromosome replication and centromere division without nuclear or cytoplasmic division. (p. 263)

**endosperm** Nutritive tissues formed adjacent to the embryo in most flowering plants; in most diploid plants, the endosperm is triploid.

**endosymbiont theory** Theory that mitochondria and chloroplasts were originally free-living organisms that invaded ancestral eukaryotes, first perhaps as parasites, later becoming symbionts. (p. 613)

**enhancer** A base sequence in eukaryotes and eukaryotic viruses that increases the rate of transcription of nearby genes; the defining characteristics are that it need not be adjacent to the transcribed gene and that the enhancing activity is independent of orientation with respect to the gene. (p. 494)

**enhancer trap** Mutagenesis strategy in which a reporter gene with a weak promoter is introduced at many random locations in the genome; tissue-specific expression of the reporter gene identifies an insertion near a tissue-specific enhancer, (p. 499)

**environmental variance** The part of the phenotypic variance that is attributable to differences in environment. (p. 677)

**enzyme** A protein that catalyzes a specific biochemical reaction and is not itself altered in the process. (p. 12)

**episome** A DNA element that can exist in the cell either as an autonomously replicating entity or become incorporated into the genome. (p. 314)

**epistasis** A term referring to an interaction between non-allelic genes in their effects on a trait. Generally, *epistasis* means any type of interaction in which the genotype at one locus affects the phenotypic expression of the genotype at another locus. In a more restricted sense, *epistasis* refers to a situation in which the genotype at one locus determines the phenotype in such a way as to mask the genotype present at a second locus.

(p. 20)

**equational division** Term applied to the second meiotic division because the haploid chromosome complement is retained throughout. (p.94)

**equilibrium** *See* **genetic equilibrium.**

**equilibrium centrifugation in a density gradient** A method for separating macromolecules of differing density by high-speed centrifugation in a solution of approximately the same density as the molecules to be separated; the centrifugation produces a gradient of density inside the centrifuge tube, and each macromolecule migrates to a position in the gradient that matches its own density. (p. 183)

**equilibrium density-gradient centrifugation** *See* **equilibrium centrifugation in a density gradient.**

**erythroblastosis fetalis** Hemolytic disease of the new-born; blood cell destruction that occurs when anti-$Rh^+$ antibodies in a mother cross the placenta and attack $Rh^+$ cells in a fetus.

**E site** *See* **exit site.**

**estrogen** A female sex hormone; of great interest in the study of cellular regulation in eukaryotes because the number of types of target cells is large. (p. 493)

**ethidium bromide** A fluorescent molecule that binds to DNA and changes its density; used to purify supercoiled DNA molecules and to localize DNA in gel electrophoresis.

**euchromatin** A region of a chromosome that has normal staining properties and undergoes the normal cycle of condensation; relatively uncoiled in the interphase nucleus (compared with condensed chromosomes), it apparently contains most of the genes.(pp. 145, 241)

**Eukarya** One of the major kingdoms of living organisms, in which the cells have a true nucleus and divide by mitosis or meiosis. (p. 22)

**eukaryote** A cell with a true nucleus (DNA enclosed in a membranous envelope) in which cell division takes place by mitosis or meiosis; an organism composed of eukaryotic cells, (p. 23)

**euploid** A cell or an organism having a chromosome number that is an exact multiple of the haploid number. (p.269)

**evolution** Cumulative change in the genetic characteristics of a species through time, resulting in greater adaptation. (pp. 22, 649)

**excision** Removal of a DNA fragment from a molecule.(p. 345)

**excisionase** An enzyme that is needed for prophage excision; works together with an integrase. (p. 345)

**excision repair** Type of DNA repair in which segments of a DNA strand that are chemically damaged are removed enzymatically and then resynthesized, using the other strand as a template, (p. 581)

**exit site** The tRNA-binding site on the ribosome that binds each uncharged tRNA just prior to its release. (p. 431)

**exon** The sequences in a gene that are retained in the messenger RNA after the introns are removed from the primary transcript. (p. 425)

**exon shuffle** The theory that new genes can evolve by the assembly of separate exons from preexisting genes, each coding for a discrete functional domain in the new protein. (p. 430)

**exonuclease** An enzyme that removes a terminal nucleotide in a polynucleotide chain by cleavage of the terminal phosphodiester bond; nucleotides are removed successively, one by one; usually specific for either DNA or RNA and for either single-stranded or double-stranded nucleic acids. A 5'-to-3' exonuclease cleaves successive nucleotides from the 5' end of the molecule; a 3'-to-5' exonuclease cleaves successive nucleotides from the 3' end. (pp. 192, 196)

**expressivity** The degree of phenotypic expression of agene.

**extranuclear inheritance** Inheritance mediated by self-replicating cellular factors located outside the nucleus—for example, in mitochondria or chloroplasts. (p. 600)

**F**

**fate** In development, the final product of differentiation of a cell or of a group of cells. (p. 514)

**fate map** A diagram of the insect blastoderm identifying the regions from which particular adult structures derive. (p. 530)

**favism** Destruction of red blood cells (anemia) triggered by eating raw fava beans or by coming into contact with certain other environmental agents; associated with a deficiency of the enzyme glucose-6-phosphate dehydrogenase. (p. 20)

**feedback inhibition** Inhibition of an enzyme by the product of the enzyme or, in a metabolic pathway, by a product of the pathway.

**fertility** The ability to mate and produce offspring.(p. 655)

**fertility factor** *See* **F plasmid.**

**fertility restorer** A suppressor of cytoplasmic male sterility. (p. 612)

**F factor** *See* **F plasmid**

**$F_1$ generation** The first generation of descent from a given mating. (p. 33)

**$F_2$ generation** The second generation of descent, produced by intercrossing or self-fertilizing $F_1$ organisms. (p. 36)

**first-division segregation** Separation of a pair of alleles into different nuclei in the first meiotic division; happens when there is no crossing-over between the gene and the centromere in a particular cell. (p. 155)

**first meiotic division** The meiotic division that reduces the chromosome number; sometimes called the reduction division. (p. 89)

**fitness** A measure of the average ability of organisms with a given genotype to survive and reproduce.

**5'-P group** The end of a DNA or RNA strand that terminates in a free phosphate group not connected to a sugar farther along. (p. 175)

**five prime end (5' end)** The end of a polynucleotide chain that terminates with a 5' carbon unattached to another nucleotide. (p. 175)

**fixed allele** An allele whose allele frequency equals 1.0. (p. 629)

**flagella** The whip-like projections from the surface of bacterial cells that serve as the organs of locomotion.(p. 705)

**fluctuation test** A statistical test used to determine whether bacterial mutations occur at random or are produced in response to selective agents.

**folded chromosome** The form of DNA in a bacterial cell in which the circular DNA is folded to have a compact structure; contains protein. (p. 225)

**folding domain** A short region of a polypeptide chain within which interactions between amino acids result in a three-dimensional conformation that is attained relatively independently of the folding of the rest of the molecule.(p. 430)

**forward mutation** A change from a wildtype allele to a mutant allele. (pp. 582, 652)

**founder effect** Random genetic drift that results when a group of founders of a population are not genetically representative of the population from which they were derived.

**F plasmid** A bacterial plasmid—often called the F factor, fertility factor, or sex plasmid—that is capable of transferring itself from a host ($F^+$) cell to a cell not carrying an F factor ($F^-$ cell); when an F factor is integrated into the bacterial chromosome (in an Hfr cell), the chromosome becomes transferrable to an $F^-$ cell during conjugation. (p. 314)

**F'plasmid** An F plasmid that contains genes obtained from the bacterial chromosome in addition to plasmid genes; formed by aberrant excision of an integrated F factor, taking along adjacent bacterial DNA. (p. 324)

**fragile-X chromosome** A type of X chromosome containing a site toward the end of the long arm that tends to break in cultured cells that are starved for DNA precursors; causes fragile-X syndrome. (p. 279)

**fragile-X syndrome** A common form of inherited mental retardation associated with a type of X chromosome prone to breakage (the fragile-X chromosome). (p. 279)

**frameshift mutation** A mutational event caused by the insertion or deletion of one or more nucleotide pairs in a gene, resulting in a shift in the reading frame of all codons following the mutational site. (p. 440)

**fraternal twins** Twins that result from the fertilization of separate ova and are genetically related as siblings; also called dizygotic twins. (p. 682)

**free radical** A highly reactive molecule produced when ionizing radiation interacts with water; free radicals are potent oxidizing agents. (p. 573)

**frequency of recombination** The proportion of gametes carrying combinations of alleles that are not present in either parental chromosome, (p. 125)

## G

**gain-of-function mutation** Mutation in which a gene is overexpressed or inappropriately expressed. (p. 526)

**galactosidase** *See* **β-galactosidase.**

**gamete** A mature reproductive cell, such as sperm or egg in animals.

**gametophyte** In plants, the haploid part of the life cycle that produces the gametes by mitosis. (p. 89)

**gap gene** Any of a group of genes that control the development of contiguous segments or parasegments in *Drosophila* such that mutations result in gaps in the pattern of segmentation. (pp. 533, 538)

**gastrula** Stage in early animal development marked by extensive cell migration. (p. 514)

**G-E association** *See* **genotype-environment association**

**G-E interaction** *See* **genotype-environment interaction**

**gel electrophoresis** *See* **electrophoresis.**

**gene** The hereditary unit containing genetic information transcribed into an RNA molecule that is processed and either functions directly or is translated into a polypeptide chain; a gene can mutate to various forms called alleles. (pp. 2, 40)

**gene amplification** A process in which certain genes undergo differential replication either within the chromosome or extrachromosomally, increasing the number of copies of the gene. (p. 481)

**gene cloning** *See* **cloned gene.**

**gene conversion** The phenomenon in which the products of a meiotic division in an *Aa* heterozygous genotype are in some ratio other than the expected 1*A* : 1*a*—for example, 3*A* : 1*a*, 1*A* : 3*a*, 5*A* : 3*a*, or 3*A* : 5*a*. (p. 586)

**gene dosage** Number of gene copies. (p. 481)

**gene expression** The multistep process by which a gene is regulated and its product synthesized. (p. 412)

**gene flow** Exchange of genes among populations resulting from either dispersal of gametes or migration of individuals; also called migration. (p. 650)

**gene library** A large collection of cloning vectors containing a complete (or nearly complete) set of fragments of the genome of an organism. (p. 372)

**gene pool** The totality of genetic information in a population of organisms. (p. 628)

**gene product** A term used for the polypeptide chain translated from an mRNA molecule transcribed from a gene; if the RNA is not translated (for example, ribosomal RNA), the RNA molecule is the gene product. (p. 412)

**generalized transduction** *See* **transducing phage.**

**general transcription factor** A protein molecule needed to bind with a promoter before transcription can proceed; transcription factors are necessary, but not sufficient, for transcription, and they are shared among many different promoters. (p. 494)

**gene regulation** Processes by which gene expression is controlled in response to external or internal signals. (p. 460)

**gene targeting** Disruption or mutation of a designated gene by homologous recombination. (p. 379)

**gene therapy** Deliberate alteration of the human genome for alleviation of disease. (p. 387)

**genetic code** The set of 64 triplets of bases (codons) corresponding to the twenty amino acids in proteins and the signals for initiation and termination of polypeptide synthesis. (p. 438)

**genetic differentiation** Accumulation of differences in allele frequency between isolated or partially isolated populations.

**genetic divergence** *See* **genetic differentiation.**

**genetic engineering** The linking of two DNA molecules by *in vitro* manipulations for the purpose of generating a novel organism with desired characteristics. (p. 360)

**genetic equilibrium** In population genetics, a situation in which the allele frequencies remain constant from one generation to the next. (p. 652)

**genetic map** *See* **linkage map.**

**genetic marker** Any pair of alleles whose inheritance can be traced through a mating or through a pedigree. (p. 129)

**genetics** The study of biological heredity. (p. 2)

**genome** The total complement of genes contained in a cell or virus; commonly used to refer to all genes present in one complete haploid set of chromosomes in eukaryotes. (p. 222)

**genome equivalent** The number of clones of a given size that contain the same aggregate amount of DNA as the haploid genome size of an organism. (p. 389)

**genotype** The genetic constitution of an organism or virus, typically with respect to one or a few genes of interest, as distinguished from its appearance, or phenotype. (p. 40)

**genotype-environment association** The condition in which genotypes and environments are not in random combinations. (p. 680)

**genotype-environment interaction** The condition in which genetic and environmental effects on a trait are not additive. (p. 678)

**genotype frequency** The proportion of members of a population that are of a prescribed genotype. (p. 628)

**genotypic variance** The part of the phenotypic variance that is attributable to differences in genotype. (p. 676)

**germ cell** A cell that gives rise to reproductive cells. (p. 82)

**germinal mutation** A mutation in a cell from which gametes are derived, as distinguished from a somatic mutation. (p. 556)

**germ line** Cell lineage consisting of germ cells.

**germ-line mutation** *See* **germinal mutation.**

**glucose-6-phosphate dehydrogenase (G6PD)** An enzyme used in one biochemical pathway for the metabolism of glucose. In human beings, a defective G6PD enzyme (G6PD deficiency) is associated with a form of anemia in which the red blood cells are prone to burst. (p. 20)

**glyphosate** A widely used herbicide. (p. 386)

**goodness of fit** The extent to which observed numbers agree with the numbers expected on the basis of some specified genetic hypothesis. (p. 109)

**G6PD** *See* **glucose-6-phosphate dehydrogenase.**

**G6PD deficiency** *See* **glucose-6-phosphate dehydrogenase.**

**$G_1$ period** *See* **cell cycle.**

**$G_2$ period** *See* **cell cycle.**

**gray** A unit of absorbed radiation equal to 1 joule of radiation energy absorbed per kilogram of tissue, abbreviated Gy; equal to 100 rad.

**guanine (G)** A nitrogenous purine base found in DNA and RNA. (p. 9)

**guide RNA** The RNA template present in telomerase. (p. 248)

**gynandromorph** A sexual mosaic; an individual organism that exhibits both male and female sexual differentiation.

## H

**Haldane's mapping function** A mapping function based on the assumption of no chromosome interference. (p. 145)

**haploid** A cell or organism of a species containing the set of chromosomes normally found in gametes. (pp. 83, 262)

**haplotype** The allelic form of each of a set of linked genes present in a single chromosome, particularly genes of the major histocompatibility complex.

**Hardy-Weinberg principle** The genotype frequencies expected with random mating. (p. 634)

**H chain** *See* **heavy chain.**

**heavy (H) chain** One of the large polypeptide chains in an antibody molecule. (p. 483)

**helicase** A protein that separates the strands of double-stranded DNA (198)

**helix** *See* α **helix.**

**helix-loop-helix** A DNA-binding motif found in many regulatory proteins.

**helper T cell** A type of white blood cell involved in activating other cells in the immune system.

**hemizygous gene** A gene present in only one dose, such as the genes on the X chromosome in XY males.

**hemoglobin** The oxygen-carrying protein in red blood cells. (p. 15)

**hemophilia A** One of two X-linked forms of hemophilia; patients are deficient in blood-clotting factor VIII. (p. 101)

**heritability** A measure of the degree to which a phenotypic trait can be modified by selection. *See also* **broad-sense heritability** and **narrow-sense heritability.**

**heterochromatin** Chromatin that remains condensed and heavily stained during interphase; commonly present adjacent to the centromere and in the telomeres of chromosomes. Some chromosomes are composed primarily of heterochromatin. (pp. 145, 240)

**heterochronic mutation** A mutation in which an otherwise normal gene is expressed at the wrong time. (p. 525)

**heteroduplex region** Region of a double-stranded nucleic acid molecule in which the two strands have different hereditary origins; produced either as an intermediate in recombination or by the *in vitro* annealing of single-stranded complementary molecules. (p. 587)

**heterogametic** Refers to the production of dissimilar gametes with respect to the sex chromosomes; in most animals, the male is the heterogametic sex, but in birds, moths, butterflies, and some reptiles, it is the female. *See also* **homogametic.** (p. 97)

**heterogeneous nuclear RNA (hnRNA)** The collection of primary RNA transcripts and incompletely processed products found in the nucleus of a eukaryotic cell.

**heterokaryon** A cell or individual having nuclei from genetically different sources, the result of cell fusion not accompanied by nuclear fusion.

**heteroplasmy** The presence of two or more genetically different types of the same organelle in a single organism. (p. 602)

**heterosis** The superiority of hybrids over either inbred parent with respect to one or more traits; also called hybrid vigor. (p. 687)

**heterozygote superiority** The condition in which a heterozygous genotype has greater fitness than either of the homozygotes. (p. 657)

**heterozygous** Carrying dissimilar alleles of one or more genes; not homozygous. (p. 40)

**hexaploid** A cell or organism with six complete sets of chromosomes. (p. 261)

**H4 histone** *See* **histone.**

**Hfr** An *E. coli* cell in which an F plasmid is integrated into the chromosome, making possible the transfer of part or all of the chromosome to an $F^-$ cell. (p. 317)

**high-copy-number plasmid** A plasmid for which there are usually considerably more than 2 (and often more than 20) copies per cell.

**highly repetitive sequence** A DNA sequence present in thousands of copies in a genome. (p. 239)

**highly significant** *See* **statistically significant.**

**histocompatability** Acceptance by a recipient of transplanted tissue from a donor.

**histocompatibility antigens** Tissue antigens that determine transplant compatibility or incompatibility.

**histocompatibility genes** Genes coding for histocompatibility antigens.

**histone** Any of the small basic proteins bound to DNA in chromatin; the five major histones are designated H1, H2A, H2B, H3, and H4. Each nucleosome core particle contains two molecules each of H2A, H2B, H3, and H4. The H1 histone forms connecting links between nucleosome core particles. (p. 229)

**Holliday junction resolving enzyme** An enzyme that catalyzes the breakage and rejoining of two DNA strands in a Holliday junction to generate two independent duplex molecules. (p. 590)

**Holliday model** A molecular model of genetic recombination in which the participating duplexes contain heteroduplex regions of the same length. (p. 587)

**Holliday structure** A cross-shaped configuration of two DNA duplexes formed as an intermediate in recombination. (p. 588)

**holocentric chromosome** A chromosome with a diffuse centromere. *See* **diffuse centromere.** (p. 246)

**holoenzyme** *See* **RNA polymerase holoenzyme.**

**homeobox** A DNA sequence motif found in the coding region of many regulatory genes; the amino acid sequence corresponding to the homeobox has a helix-loop-helix structure. (p.543)

**homeotic gene** Any of a group of genes that determine the fundamental patterns of development. *See also* **homeotic mutation.**(p. 541)

**homeotic mutation** A developmental mutation that results in the replacement of one body structure by another body structure. (p.542)

**homeotic selector gene** *See* **homeotic gene.**

**homogametic** Producing only one kind of gamete with respect to the sex chromosomes. *See also* **heterogametic.** (p.97)

**homologous** In reference to DNA, having the same or nearly the same nucleotide sequence. (p. 88)

**homologous chromosomes** Chromosomes that pair in meiosis and have the same genetic loci and structure; also called homologs. (p. 88)

**homologous recombination** Genetic exchange between identical or nearly identical DNA sequences.

**homothallism** The capacity of cells in certain fungi to undergo a conversion in mating type to make possible mating between cells produced by the same parental organism.(p. 482)

**homozygous** Having the same allele of a gene in homologous chromosomes. (p. 40)

**HI histone** *See* **histone.**

**Hoogstein base pairing** An unusual type of base pairing in which four guanine bases form a hydrogen-bonded quartet. (p. 250)

**hormone** A small molecule in higher eukaryotes, synthesized in specialized tissue, that regulates the activity of other specialized cells. In animals, hormones are transported from their source to a target tissue in the blood. (p. 493)

**hot spot** A site in a DNA molecule at which the mutation rate is much higher than the rate for most other sites. (p.339, 565)

**housekeeping gene** A gene that is expressed at the same level in virtually all cells and whose product participates in basic metabolic processes. (p. 488)

**H substance** The carbohydrate precursor of the A and B red-blood-cell antigens.

**H3 histone** *See* **histone.**

**H2A histone** *See* **histone.**

**H2B histone** *See* **histone.**

**human genome project** A worldwide project to map genetically and sequence the human genome. (p. 392)

**Huntington disease** Dominantly inherited degeneration of the neuromuscular system, with onset in middle age. (p. 52)

**hybrid** An organism produced by the mating of genetically unlike parents; also, a duplex nucleic acid molecule produced of strands derived from different sources. (p. 32)

**hybrid vigor** *See* **heterosis.**

**hydrogen bond** A weak noncovalent linkage between two negatively charged atoms in which a hydrogen atom is shared.

**hydrophobic interaction** A noncovalent interaction between nonpolar molecules or nonpolar groups, causing the molecules or groups to cluster when water is present.

**hypermorph** A mutation in which the gene product is produced in greater abundance than wildtype. (p. 66)

**hypomorph** A mutation in which the gene product is produced in less abundance than wildtype. (p. 66)

**I**

**identical twins** *See* **monozygotic twins.**

**identity by descent** The condition in which two alleles are both replicas of a single ancestral allele in the recent past. (p. 647)

**imaginal disk** Structures present in the body of insect larvae from which the adult structures develop during pupation. (p. 540)

**immune response** Tendency to develop resistance to a disease-causing agent after an initial infection.

**immunity** A general term for resistance of an organism to specific substances, particularly agents of disease. (pp. 344, 483)

**immunoglobulin** One of several classes of antibody proteins. (p. 483)

**immunoglobulin class** The category in which an immunoglobulin is placed on the basis of its chemical characteristics, including the identity of its heavy chain. (p. 483)

**imprinting** A process of DNA modification in gametogenesis that affects gene expression in the zygote; a probable mechanism is the methylation of certain bases in the DNA. (p. 280)

**inborn error of metabolism** A genetically determined biochemical disorder, usually in the form of an enzyme defect that produces a metabolic block. (p. 12)

**inbreeding** Mating between relatives. (pp. 634, 645)

**inbreeding coefficient** A measure of the genetic effects of inbreeding in terms of the proportionate reduction in heterozygosity in an inbred organism compared with the heterozygosity expected with random mating. (p. 646)

**inbreeding depression** The deterioration in a population's fitness or performance that accompanies inbreeding. (p. 687)

**incompatible** Refers to blood or tissue whose transfusion or transplantation results in rejection.

**incomplete dominance** A situation in which the heterozygous genotype has a phenotype that falls somewhere in the range between the two homozygous genotypes. (p. 67)

**incomplete penetrance** Condition in which a mutant phenotype is not expressed in all organisms with the mutant genotype. (p. 71)

**independent assortment** Random distribution of unlinked genes into gametes, as with genes in different (nonhomologous) chromosomes or genes that are so far apart on a single chromosome that the recombination frequency between them is 1/2. (p. 45)

**individual selection** Selection based on each organism's own phenotype. (p. 684)

**induced mutation** A mutation formed under the influence of a chemical mutagen or radiation. (p. 557)

**inducer** A small molecule that inactivates a repressor, usually by binding to it and thereby altering the ability of the repressor to bind to an operator. (p. 461)

**inducible** A gene that is expressed, or an enzyme that is synthesized, only in the presence of an inducer molecule. *See also* **constitutive.** (pp. 461, 463)

**inducible transcription** *See* **inducible.**

**induction** Activation of an inducible gene; prophage induction is the derepression of a prophage that initiates a lytic cycle of phage development. (p. 461)

**initiation** The beginning of protein synthesis. (pp. 187, 431)

**inosine (I)** One of a number of unusual bases found in transfer RNA.

**insertional inactivation** Inactivation of a gene by interruption of its coding sequence; used in genetic engineering as a means of detecting insertion of a foreign DNA sequence into the coding region of a gene.

**insertion sequence** A DNA sequence capable of transposition in a prokaryotic genome; such sequences usually code for their own transposase. (p. 347)

***in situ* hybridization** Renaturation of a radioactive probe nucleic acid to the DNA or RNA in a cell; used to localize particular DNA molecules to chromosomes or parts of chromosomes and to identify the time and tissue distribution of RNA transcripts. (p. 234)

**integrase** An enzyme that catalyzes the site-specific exchange when a prophage is inserted into or excised from a bacterial chromosome; in the excision process, an accessory protein, excisionase, also is needed. (p. 343)

**integration** The process by which one DNA molecule is inserted intact into another replicable DNA molecule, as in prophage integration and the integration of plasmid or tumor viral DNA into a chromosome. (p. 314)

**intercalation** Insertion of a planar molecule between the stacked bases in duplex DNA. (p. 571)

**interference** The tendency for crossing-over to inhibit the formation of another crossover nearby.

**intergenic complementation** Complementation between mutations in different genes. *See also* **complementation test.**

**intergenic suppression** Suppression of a mutant phenotype because of a mutation in a different gene; often refers specifically to a tRNA molecule able to recognize a stop codon or two different sense codons. (p. 585)

**interphase** The interval between nuclear divisions in the cell cycle, extending from the end of telophase of one division to the beginning of prophase of the next division. (p. 83)

**interrupted-mating technique** In an Hfr X $F^-$ cross, a technique by which donor and recipient cells are broken apart at specific times, allowing only a particular length of DNA to be transferred. (p. 319)

**intervening sequence** *See* **intron.**

**intron** A noncoding DNA sequence in a gene that is transcribed but is then excised from the primary transcript in forming a mature mRNA molecule; found primarily in eukaryotic cells. *See also* **exon.** (p. 425)

**inversion** A structural aberration in a chromosome in which the order of several genes is reversed from the normal order. A pericentric inversion includes the centromere within the inverted region, and a paracentric inversion does not include the centromere. (p. 286)

**inversion loop** Loop structure formed by synapsis of homologous genes in a pair of chromosomes, one of which contains an inversion. (p. 287)

**inverted repeat** Either of a pair of base sequences present in the same molecule that are identical or nearly identical but are oriented in opposite directions; often found at the ends of transposable elements. (p. 243)

*in vitro* **experiment** An experiment carried out with components isolated from cells.

*in vivo* **experiment** An experiment performed with intact cells.

**ionizing radiation** Electromagnetic or particulate radiation that produces ion pairs when dissipating its energy in matter. (p. 572)

**IS element** *See* **insertion sequence.**

**isochromosome** A chromosome with two identical arms containing homologous genes.

**isotopes** The forms of a chemical element that have the same number of electrons and the same number of protons but differ in the number of neutrons in the atomic nucleus;unstable isotopes undergo transitions to a more stable state and, in so doing, emit radioactivity.

## J

**J (joining) region** Any of multiple DNA sequences that code for alternative amino acid sequences of part of the variable region of an antibody molecule; the J regions of heavy and light chains are different. (p. 485)

**just-so story** Made-up stories, not supported by hard evidence, about the presumed adaptive significance of a trait; named after the 1902 book *Just So Stories* by Rudyard Kipling. (p. 24)

## K

**kappa particle** An intracellular parasite in *Paramecium* that releases a substance capable of killing sensitive cells. (p. 614)

**karyotype** The chromosome complement of a cell or organism; often represented by an arrangement of metaphase chromosomes according to their lengths and the positions of their centromeres. (p. 269)

**kilobase pair (kb)** Unit of length of a duplex DNA molecule; equal to 1000 base pairs. (p. 222)

**kindred** A group of relatives.

**kinetochore** The cellular structure, formed in association with the centromere, to which the spindle fibers become attached in cell division. (p. 86)

**Klinefelter syndrome** The clinical features of human males with the karyotype 47,XXY. (p. 279)

**Kosambi's mapping function** A mapping function based on the assumption that the coincidence between crossovers is proportional to the distance between them. (p. 145)

**kuru** A neurological disease caused by a prion protein. (p. 724)

## L

***lac* operon**The set of genes required to metabolize lactose in bacteria. (p. 466)

**Lac repressor** A protein coded by the *lacI* gene that, in the absence of lactose, binds with the operator and prevents transcription. (p. 464)

**lactose** Milk sugar; each molecule consists of a joined glucose and galactose. (p. 462)

**lactose permease** An enzyme responsible for transport of lactose from the environment into bacteria. (p. 462)

**lagging strand** The DNA strand whose complement is synthesized in short fragments that are ultimately joined together. (p. 195)

**lariat structure** Structure of an intron immediately after excision in which the 5' end loops back and forms a 5'-2' linkage with another nucleotide. (p. 426)

**L chain** *See* **light chain.**

**leader** *See* **leader polypeptide, leader sequence.**

**leader polypeptide** A short polypeptide encoded in the leader sequence of some operons coding for enzymes in amino acid biosynthesis; translation of the leader polypeptide participates in regulation of the operon through attenuation. (pp. 424, 473)

**leader sequence** The region of an mRNA molecule from the 5' end to the beginning of the coding sequence, sometimes containing regulatory sequences; in prokaryotic mRNA, it contains the ribosomal binding site.

**leading strand** The DNA strand whose complement is synthesized as a continuous unit. (p. 195)

**leptotene** The initial substage of meiotic prophase I during which the chromosomes become visible in the light microscope as unpaired thread-like structures. (p. 92)

**lethal mutation** A mutation that results in the death of an affected organism.

**liability** Risk, particularly toward a threshold type of quantitative trait. (p. 690)

**library** *See* **gene library.**

**ligand**The molecule that binds to a specific receptor. (p.528)

**light (L) chain** One of the small polypeptide chains in an antibody molecule. (p. 483)

**lineage** The ancestral history of a cell in development or of a species in evolution.(p. 519)

**lineage diagram** A diagram of cell lineages and their developmental fates. (p. 519)

**linkage** The tendency of genes located in the same chromosome to be associated in inheritance more frequently than expected from their independent assortment in meiosis. (p. 124)

**linkage equilibrium** The condition in which the alleles of different genes in a population are present in gametes in proportion to the product of the frequencies of the alleles.

**linkage group** The set of genes present together in a chromosome. (p. 132)

**linkage map** A diagram of the order of genes in a chromosome in which the distance between adjacent genes is proportional to the rate of recombination between them; also called a genetic map. (pp. 124, 127)

**linker** In genetic engineering, synthetic DNA fragments that contain restriction-enzyme cleavage sites that are used to join two DNA molecules. (p. 230)

**local population** A group of organisms of the same species occupying an area within which most individual members find their mates; synonomous terms are *deme* and *Mendelian population.* (p. 628)

**localized centromere** The type of centromere organization found in most eukaryotic cells in which the centromere is located in one small region of the chromosome. (p. 246)

**locomotor activity** Movement of organisms from one place to another under their own power. (p. 714)

**locus** The site or position of a particular gene on a chromosome. (p. 132)

**loss-of-functionmutation** A mutation that eliminates gene function; also called a null mutation. (p. 526)

**lost allele** An allele no longer present in a population; its frequency is 0. (p. 629)

**lysis** Breakage of a cell caused by rupture of its cell membrane and cell wall. (p. 330)

**lysogen** Clone of bacterial cells that have acquired a prophage. (p. 341)

**lysogenic cycle** In temperate bacteriophage, the phenomenon in which the DNA of an infecting phage becomes part of the genetic material of the cell. (p. 329)

**lysozyme** One of a class of enzymes that dissolves the cell wall of bacteria; found in chicken egg white, human tears, and coded by many phages.

**lytic cycle** The life cycle of a phage, in which progeny phage are produced and the host bacterial cell is lysed. (p. 329)

## M

**macrophage** One of a class of white blood cells that processes antigens for presentation as a necessary step in stimulation of the immune response.

**major groove** In B-form DNA, the larger of two continuous indentations running along the outside of the double helix. (p. 179)

**major histocompatibility complex (MHC)** The group of closely linked genes coding for antigens that play a major role in tissue incompatibility and that function in regulation and other aspects of the immune response.

**map-based cloning** A strategy of gene cloning based on the position of a gene in the genetic map; also called positional cloning. (p. 374)

**mapping function** The mathematical relation between the genetic map distance across an interval and the observed percentage of recombination in the interval. (p. 144)

**map unit** A unit of distance in a linkage map that corresponds to a recombination frequency of 1 percent. Technically, the map distance across an interval in map units equals one-half the average number of crossovers in the interval expressed as a percentage. Map units are sometimes denoted centimorgans (cM). (p. 128)

**masked mRNA** Messenger RNA that cannot be translated until specific regulatory substances are available; present in eukaryotic cells, particularly eggs; storage mRNA. (p. 503)

**maternal effect** A phenomenon in which the genotype of a mother affects the phenotype of the offspring through substances present in the cytoplasm of the egg. (p. 603)

**maternal effect gene** A gene that influences early development through its expression in the mother and the presence of the gene product in the oocyte. (p. 532)

**maternal inheritance** Extranuclear inheritance of a trait through cytoplasmic factors or organelles contributed by the female gamete. (p. 601)

**maternal sex ratio** Aberrant sex ratio in certain *Drosophila* species caused by cytoplasmic transmission of a parasite that is lethal to male embryos. (p. 616)

**mating system** The norms by which members of a population choose their mates; important systems of mating include random mating, assortative mating, and inbreeding. (p. 632)

**mating-type interconversion** Phenomenon in homothallic yeast in which cells switch mating type as a result of the transposition of genetic information from an unexpressed cassette into the active mating-type locus. (p. 481)

**Maxam-Gilbert method** A technique for determining the nucleotide sequence of DNA by means of strand cleavage at the positions of particular nucleotides.

**MCS** Multiple cloning site. *See* **polylinker.**

**MCP** *See* **methyl-accepting chemotaxis protein.**

**mean** The arithmetic average. (p. 672)

**megabase pair** Unit of length of a duplex nucleic acid molecule; equal to 1 million base pairs. (p. 222)

**meiocyte** A germ cell that undergoes meiosis to yield gametes in animals or spores in plants. (p.88)

**meiosis** The process of nuclear division in gametogenesis or sporogenesis in which one replication of the chromosomes is followed by two successive divisions of the nucleus to produce four haploid nuclei. (p. 87)

**melting curve** A graph of the amount of denatured DNA present in a solution (measured by UV light adsorption) as a function of increasing temperature. (p. 199)

**melting temperature** The midpoint of the narrow temperature range at which the strands of duplex DNA denature. (p. 199)

**Mendelian genetics** The mechanism of inheritance in which the statistical relations between the distribution of traits in successive generations result from (1) pariculate hereditary determinants (genes), (2) random union of gametes, and (3) segregation of unchanged hereditary determinants in the reproductive cells. (p. 32)

**Mendelian population** *See* **local population.**

**meristem** The mitotically active growing point of plant tissue. (p. 543)

**meristic trait** A trait in which the phenotype is determined by counting, such as number of ears on a stalk of corn or number of eggs laid by a hen. (p. 671)

**merodiploid** A bacterial cell carrying two copies of a region of the genome and therefore diploid for the genes in this region. (p. 324)

**messenger RNA (mRNA)** An RNA molecule transcribed from a DNA sequence and translated into the amino acid sequence of a polypeptide. In eukaryotes, the primary transcript undergoes elaborate processing to become the mRNA. (pp. 12, 423)

**metabolic pathway** A set of chemical reactions that take place in a definite order to convert a particular starting molecule into one or more specific products.

**metacentric chromosome** A chromosome with its centromere about in the middle so that the arms are equal or almost equal in length. (p. 260)

**metaphase** In mitosis, meiosis I, or meiosis II, the stage of nuclear division in which the centromeres of the condensed chromosomes are arranged in a plane between the two poles of the spindle. (pp. 86, 93)

**metaphase plate** Imaginary plane, equidistant from the spindle poles in a metaphase cell, on which the centromeres of the chromosomes are aligned by the spindle fibers. (p. 86)

**methyl-accepting chemotaxis protein** Alternative designation of chemoreceptors in bacterial chemotaxis; abbreviated MCP. (p. 712)

**methylation** The modification of a protein or a DNA or RNA base by the addition of a methyl group ($-CH_3$). (p. 487)

**MHC** *See* **major histocompatibility complex.**

**micrococcal nuclease** A nuclease that is isolated from a bacterium and that cleaves double-stranded DNA without regard to sequence. (p. 230)

**middle repetitive DNA sequence** A DNA sequence present tens to hundreds of times per genome. (p. 239)

**migration** Movement of organisms among subpopulations; also, the movement of molecules in electrophoresis. (pp. 205, 650)

**minimal medium** A growth medium consisting of simple inorganic salts, a carbohydrate, vitamins, organic bases, essential amino acids, and other essential compounds; its composition is precisely known. Minimal medium contrasts with complex medium or broth, which is an extract of biological material (vegetables, milk, meat) that contains a large number of compounds and the precise composition of which is unknown. (p. 312)

**minor groove** In B-form DNA, the smaller of two continuous indentations running along the outside of the double helix. (p. 179)

**mismatch** An arrangement in which two nucleotides opposite one another in double-stranded DNA are unable to form hydrogen bonds. (p. 578)

**mismatch repair** Removal of one nucleotide from a pair that cannot properly hydrogen-bond, followed by replacement with a nucleotide that can hydrogen-bond. (p. 578)

**missense mutation** An alteration in a coding sequence of DNA that results in an amino acid replacement in the polypeptide. (pp. 444, 558)

**mitosis** The process of nuclear division in which the replicated chromosomes divide and the daughter nuclei have the same chromosome number and genetic composition as the parent nucleus. (p. 83)

**mitotic spindle** The set of fibers that arches through the cell during mitosis and to which the chromosomes are attached. *See also* **spindle.** (p. 86)

**monocistronic mRNA** A mRNA molecule that codes for a single polypeptide chain.

**monoclonal antibody** Antibody directed against a single antigen and produced by a single clone of B cells or a single cell line of hybridoma cells.

**monohybrid** A genotype that is heterozygous for one pair of alleles; the offspring of a cross between genotypes that are homozygous for different alleles of a gene. (p. 33)

**monomorphic gene** A gene for which the most common allele is virtually fixed.

**monoploid** The basic chromosome set that is reduplicated to form the genomes of the species in a polyploid series; the smallest haploid chromosome number in a polyploid series. (p. 261)

**monosomic** Condition of an otherwise diploid organism in which one member of a pair of chromosomes is missing. (p. 269)

**monozygotic twins** Twins developed from a single fertilized egg that splits into two embryos at an early division; also called identical twins. (p. 682)

**morphogen** Substance that induces differentiation. (p. 515)

**mosaic** An organism composed of two or more genetically different types of cells. (p. 276)

**M period** *See* **cell cycle.**

**mRNA** *See* **messenger RNA.**

**multifactorial trait** A trait determined by the combined action of many factors, typically some genetic and some environmental. (p. 670)

**multiple alleles** The presence in a population of more than two alleles of a gene. (p. 60)

**multiple cloning site** *See* **polylinker.**

**multiplication rule** The principle that the probability that all of a set of independent events are realized simultaneously equals the product of the probabilities of the separate events. (p. 50)

**multivalent** An association of more than two homologous chromosomes resulting from synapsis during meiosis in a polysomic or polyploid individual.

**mutagen** An agent that is capable of increasing the rate of mutation. (p. 556)

**mutagenesis** The process by which a gene undergoes a heritable alteration; also, the treatment of a cell or organism with a known mutagen.

**mutant** Any heritable biological entity that differs from wildtype, such as a mutant DNA molecule, mutant allele, mutant gene, mutant chromosome, mutant cell, mutant organism, or mutant heritable phenotype; also, a cell or organism in which a mutant allele is expressed.

**mutant screen** A type of genetic experiment in which the geneticist seeks to isolate multiple new mutations that affect a particular trait. (p. 55)

**mutation** A heritable alteration in a gene or chromosome; also, the process by which such an alteration happens. Used incorrectly, but with increasing frequency, as a synonym for mutant, even in some excellent textbooks. (pp. 15, 556, 650)

**mutation pressure** The generally very weak tendency of mutation to change allele frequency.

**mutation rate** The probability of a new mutation in a particular gene, either per gamete or per generation. (p. 563)

**N**

**narrow-sense heritability** The fraction of the phenotypic variance revealed as resemblance between parents and offspring; technically, the ratio of the additive genetic variance to the total phenotypic variance. (p. 684)

**natural selection** The process of evolutionary adaptation in which the genotypes genetically best suited to survive and reproduce in a particular environment give rise to a disproportionate share of the offspring and so gradually increase the overall ability of the population to survive and reproduce in that environment. (pp. 24, 650)

**nature versus nurture** Archaic term for genetics versus environment; usually a false dichotomy because most traits are influenced by both genetic and environmental factors.

**negative assortative mating** *See* **assortative mating**

**negative chromatid interference** In meiosis, when two or more crossovers occur in a chromosome, a tendency for the nonsister chromatids that participate in one event to also participate in another event. (p. 141)

**negative regulation** Regulation of gene expression in which mRNA is not synthesized until a represser is removed from the DNA of the gene.

**negative supercoiling** Supercoiling of DNA that counteracts underwinding of the DNA. (p. 225)

**neomorph** A mutation in which the mutant allele expresses either a novel gene product or a mutant gene product that has a novel phenotypic effect. (p. 66)

**neutral allele** An allele that has a negligible effect on the ability of the organism to survive and reproduce.

**nick** A single-strand break in a DNA molecule. (p. 196)

**nicked circle** A circular DNA molecule containing one or more single-strand breaks.

**nitrous acid** $HNO^2$, a chemical mutagen. (p. 569)

**nondisjunction** Failure of chromosomes to separate (disjoin) and move to opposite poles of the division spindle; the result is loss or gain of a chromosome. (p. 103)

**nonhistone chromosomal proteins** A large class of proteins, not of the histone class, found in isolated chromosomes.

**nonparental ditype** An ascus containing two pairs of recombinant spores. (p. 150)

**nonpermissive conditions** Environmental conditions that result in expression of the phenotype of a conditional mutation.

**nonselective medium** A growth medium that allows growth of wildtype and of one or more mutant genotypes. (p. 312)

**nonsense mutation** A mutation that changes a codon specifying an amino acid into a stop codon, resulting in premature polypeptide chain termination; also called a chaintermination mutation. (pp. 444, 558)

**nonsense suppression** Suppression of the phenotype of a polypeptide chain-termination mutation mediated by a mutant tRNA that inserts an amino acid at the site of the termination codon. (pp. 447, 585)

**nontaster** Individual with the inherited inability to taste the substance phenylthiocarbamide (PTC); phenotype of the homozygous recessive.

**normal distribution** A symmetrical bell-shaped distribution characterized by the mean and the variance; in a normal distribution, approximately 68 percent of the observations are within 1 standard deviation from the mean, and approximately 95 percent are within 2 standard deviations. (p. 673)

**nuclease** An enzyme that breaks phosphodiester bonds in nucleic acid molecules. (p. 192)

**nucleic acid** A polymer composed of repeating units of phosphate-linked five-carbon sugars to which nitrogenous bases are attached. *See also* **DNA** and **RNA.** (p. 175)

**nucleic acid hybridization** The formation of duplex nucleic acid from complementary single strands. (p. 202)

**nucleoid** A DNA mass, not bounded by a membrane, within the cytoplasm of a prokaryotic cell, chloroplast, or mitochondrion; often refers to the major DNA unit in a bacterium. (pp. 225, 308)

**nucleolar organizer region** A chromosome region containing the genes for ribosomal RNA; abbreviated NOR.

**nucleolus (*pl.* nucleoli)** Nucleoli organelle in which ribosomal RNA is made and ribosomes are partially synthesized; usually associated with the nucleolar organizer region. A nucleus may contain several nucleoli. (p. 86)

**nucleoside** A purine or pyrimidine base covalently linked to a sugar. (p. 174)

**nucleosome** The basic repeating subunit of chromatin, consisting of a core particle composed of two molecules each of four different histones around which a length of DNA containing about 145 nucleotide pairs is wound, joined to an adjacent core particle by about 55 nucleotide pairs of linker DNA associated with a fifth type of histone. (p. 230)

**nucleotide** A nucleoside phosphate. (pp. 9, 174)

**nucleotide analog** A molecule that is structurally similar to a normal nucleotide and that is incorporated into DNA. (p. 568)

**nucleus** The organelle, bounded by a membranous envelope, that contains the chromosomes in a eukaryotic cell.

**nullisomic** A cell or individual that contains no copies of a particular chromosome.

**nutritional mutation** A mutation in a metabolic pathway that creates a need for a substance to be present in the growth medium or that eliminates the ability to utilize a substance present in the growth medium. (p. 311)

## O

**ochre codon** Jargon for the UAA stop codon; an ochre mutation is a UAA codon formed from a sense codon. (p. 447)

**octoploid** A cell or organism with eight complete sets of chromosomes. (p. 261)

**Okazaki fragment** Any of the short strands of DNA produced during discontinuous replication of the lagging strand; also called a precursor fragment. (p. 194)

**oligonucleotide** A short, single-stranded nucleic acid, usually synthesized for use in DNA sequencing, as a primer in the polymerase chain reaction, as a probe, or for oligonucleotide site-directed mutagenesis.

**oligonucleotide primer** A short, single-stranded nucleic acid, synthesized for use in DNA sequencing or as a primer in the polymerase chain reaction. (p. 207)

**oligonucleotide site-directed mutagenesis** Replacement of a small region in a gene with an oligonucleotide that contains a specified mutation. (p. 375)

**oncogene** A gene that can initiate tumor formation. (p. 298)

**open reading frame** In the coding strand of DNA or in mRNA, a region containing a series of codons uninterrupted by stop codons and therefore capable of coding for a polypeptide chain; abbreviated ORF. (pp. 400, 424)

**operator** A regulatory region in DNA that interacts with a specific repressor protein in controlling the transcription of adjacent structural genes. (p. 465)

**operon** A collection of adjacent structural genes regulated by an operator and a repressor. (p. 600)

**ORF** *See* **open reading frame.**

**organelle** A membrane-bounded cytoplasmic structure that has a specialized function, such as a chloroplast or mitochondrion. (p. 600)

**origin of replication** A DNA base sequence at which replication of a molecule is initiated. (p. 187)

**overdominance** A condition in which the fitness of a heterozygote is greater than the fitness of both homozygotes. (p. 657)

**overlapping genes** Genes that share part of their coding sequences. (p. 450)

**ovule** The structure in seed plants that contains the embryo sac (female gametophyte) and that develops into a seed after fertilization of the egg.

**P**

**pachytene** The middle substage of meiotic prophase I in which the homologous chromosomes are closely synapsed. (p. 92)

**pair-rule gene** Any of a group of genes active early in *Drosophila* development that specifies the fates of alternating segments or parasegments. Mutations in pair-rule genes result in loss of even-numbered or odd-numbered segments or parasegments. (pp. 533, 538)

**palindrome** In nucleic acids, a segment of DNA in which the sequence of bases on complementary strands reads the same from a central point of symmetry—for example 5'-GAATTC-3'; the sites of recognition and cleavage by restriction endonucleases are frequently palindromic. (pp. 204, 361)

**paracentric inversion** An inversion that does not include the centromere. (p. 287)

**parasegment** Developmental unit in *Drosophila* consisting of the posterior part of one segment and the anterior part of the next segment in line. (p. 532)

**parental combination** Alleles present in an offspring chromosome in the same combination as that found in one of the parental chromosomes. (p. 124)

**parental ditype** An ascus containing two pairs of nonrecombinant spores. (p. 150)

**parental strand** In DNA replication, the strand that served as the template in a newly formed duplex. (p. 182)

**partial digestion** Condition in which a restriction enzyme cleaves some, but not all, of the restriction sites present in a DNA molecule. (p. 391)

**partial diploid** A cell in which a segment of the genome is duplicated, usually in a plasmid. (p. 324)

**partial dominance** A condition in which the phenotype of the heterozygote is intermediate between the corresponding homozygotes but resembles one more closely than the other.

**Pascal's triangle** Triangular configuration of integers in which the *n*th row gives the binomial coefficients in the expansion of $(x + y)^{n-1}$. The first and last numbers in each row equal 1, and the others equal the sum of the adjacent numbers in the row immediately above. (p. 107)

**paternal inheritance** Extranuclear inheritance of a trait through cytoplasmic factors or organelles contributed by the male gamete. (p. 601)

**path in a pedigree** A closed loop in a pedigree, with only one change of direction, that passes through a common ancestor of the parents of an individual. (p. 646)

**pathogen** An organism that causes disease. (p. 348)

**pattern formation** The creation of a spatially ordered and differentiated embryo from a seemingly homogeneous egg cell. (p. 513)

**PCR** *See* **polymerase chain reaction.**

**pedigree** A diagram representing the familial relationships among relatives. (p. 51)

**penetrance** The proportion of organisms having a particular genotype that actually express the corresponding phenotype. If the phenotype is always expressed, penetrance is complete; otherwise, it is incomplete. (p. 72)

**peptide bond** A covalent bond between the amino group ($-NH^2$) of one amino acid and the carboxyl group (-COOH) of another. (p. 412)

**peptidyl site** The tRNA-binding site on the ribosome to which the tRNA bearing the nascent polypeptide becomes bound immediately after formation of the peptide bond. (p. 431)

**peptidyl transferase** The enzymatic activity of ribosomes responsible for forming a peptide bond. (p. 433)

**percent G + C** The proportion of base pairs in duplex DNA that consist of GC pairs. (p. 175)

**pericentric inversion** An inversion that includes the centromere. (p. 287)

**permease** A membrane-associated protein that allows a specific small molecule to enter the cell.

**permissive condition** An environmental condition in which the phenotype of a conditional mutation is not expressed; contrasts with nonpermissive or restrictive condition. (p. 556)

**petite mutant** A mutation in yeast resulting in slow growth and small colonies. (p. 610)

**PEV** *See* **position effect variegation.**

**PFGE** *See* **pulse-field gel electrophoresis.**

**$P_1$ generation** The parents used in a cross, or the original parents in a series of generations; also called the P generation if there is no chance of confusion with the grandparents or more remote ancestors. (p. 35)

**P group** *See* **5'-P group.**

**phage** *See* **bacteriophage.**

**phage-attachment site** The base sequence in a bacterial chromosome at which bacteriophage DNA can integrate to form a prophage. (p. 341)

**phage repressor** Regulatory protein that prevents transcription of genes in a prophage.

**phenotype** The observable properties of a cell or an organism, which result from the interaction of the genotype and the environment. (p. 41)

**phenotypic variance** The total variance in a phenotypic trait in a population.

**phenylketonuria** A hereditary human condition resulting from inability to convert phenylalanine into tyrosine; causes severe mental retardation unless treated in infancy and childhood by a low-phenylalanine diet; abbreviated PKU. (p. 730)

**phenylthiocarbamide** *See* **PTC tasting.**

**Philadelphia chromosome** Abnormal human chromosome 22, resulting from reciprocal translocation and often associated with a type of leukemia. (p. 298)

**phosphodiester bond** In nucleic acids, the covalent bond formed between the 5'-phosphate group (5'-P) of one nucleotide and the 3'-hydroxyl group (3'-OH) of the next nucleotide in line; these bonds form the backbone of a nucleic acid molecule. (p. 175)

**photoreactivation** The enzymatic splitting of pyrimidine dimers produced in DNA by ultraviolet light; requires visible light and the photoreactivation enzyme. (p. 580)

**phylogenetic tree** A diagram showing the genealogical relationships among a set of genes or species. (p. 619)

**physical map** A diagram showing the relative positions of physical landmarks in a DNA molecule; common landmarks include the positions of restriction sites and particular DNA sequences. (p. 392)

**plaque** A clear area in an otherwise turbid layer of bacteria growing on a solid medium, caused by the infection and killing of the cells by a phage; because each plaque results from the growth of one phage, plaque counting is a way of counting viable phage particles. The term is also used for animal viruses that cause clear areas in layers of animal cells grown in culture. (p. 331)

**plasmid** An extrachromosomal genetic element that replicates independently of the host chromosome; it may exist in one or many copies per cell and may segregate in cell division to daughter cells in either a controlled or a random fashion. Some plasmids, such as the F factor, may become integrated into the host chromosome. (p. 314)

**pleiotropic effect** Any phenotypic effect that is a secondary manifestation of a mutant gene. (p. 17)

**pleiotropy** The condition in which a single mutant gene affects two or more distinct and seemingly unrelated traits. (p. 17)

**point mutation** A mutation caused by the substitution, deletion, or addition of a single nucleotide pair.

**polarity** The 5'-to-3' orientation of a strand of nucleic acid. (p. 175)

**pole cell** Any of a group of cells, set off at the posterior end of the *Drosophila* embryo, from which the germ cells are derived. (p. 529)

**poly-A tail** The sequence of adenines added to the 3' end of many eukaryotic mRNA molecules in processing. (p. 425)

**polycistronic mRNA** An mRNA molecule from which two or more polypeptides are translated; found primarily in prokaryotes. (pp. 436, 449)

**polygenic inheritance** The determination of a trait by alleles of two or more genes.

**polylinker** A short DNA sequence that is present in a vector and that contains a number of unique restriction sites suitable for gene cloning. (p. 372)

**polymer** A regular, covalently bonded arrangement of basic subunits or monomers into a large molecule, such as a polynucleotide or polypeptide chain.

**polymerase** An enzyme that catalyzes the covalent joining of nucleotides—for example, DNA polymerase and RNA polymerase. (p. 191)

**polymerase α** The major DNA replication enzyme in eukaryotic cells. (p. 192)

**polymerase chain reaction** Repeated cycles of DNA denaturation, renaturation with primer oligonucleotide sequences, and replication, resulting in exponential growth in the number of copies of the DNA sequence located between the primers. (pp. 207, 369)

**polymerase λ** The enzyme responsible for replication of mitochondrial DNA. (p. 192)

**polymorphic gene** A gene for which there is more than one relatively common allele in a population. (p. 630)

**polymorphism** The presence in a population of two or more relatively common forms of a gene, chromosome, or genetically determined trait.

**polynucleotide chain** A polymer of covalently linked nucleotides. (p. 175)

**polypeptide** *See* **polypeptide chain.**

**polypeptide chain** A polymer of amino acids linked together by peptide bonds. (pp. 15, 412)

**polyploidy** The condition of a cell or organism with more than two complete sets of chromosomes. (p. 261)

**polyprotein** A protein molecule that can be cleaved to form two or more finished protein molecules.

**polyribosome** *See* **polysome.**

**polysome** A complex of two or more ribosomes associated with an mRNA molecule and actively engaged in polypeptide synthesis; a polyribosome. (p. 450)

**polysomy** The condition of a diploid cell or organism that has three or more copies of a particular chromosome. (p. 267)

**polytene chromosome** A giant chromosome consisting of many identical strands laterally apposed and in register, exhibiting a characteristic pattern of transverse banding. (p. 234)

**Pl bacteriophage** Temperate virus that infects *E. coli;* in cells lysogenic for Pl, the phage DNA is not integrated into the chromosome and replicates as a plasmid.

**population** A group of organisms of the same species. (p. 628)

**population genetics** Application of Mendel's laws and other principles of genetics to entire populations of organisms. (p. 628)

**population structure** *See* **population substructure.**

**population substructure** Organization of a population into smaller breeding groups between which migration is restricted. Also called population subdivision. (p. 642)

**positional cloning** A strategy of gene cloning based on the position of a gene in the genetic map; also called mapbased cloning. (p. 374)

**positional information** Developmental signals transmitted to a cell by virtue of its position in the embryo. (p. 515)

**position effect** A change in the expression of a gene depending on its position within the genome. (p. 296)

**position-effect variegation** Mosaic phenotype (variegation) due to variation in the level of expression of a gene in different cell lineages owing to its position in the genome. (p. 296)

**positive assortative mating See assortative mating.**

**positive chromatid interference** In meiosis, when two or more crossovers occur in a chromosome, a tendency for the nonsister chromatids that participate in one event not to participate in another event. (p. 141)

**positive regulation** Mechanism of gene regulation in which an element must be bound to DNA in an active form to allow transcription. Positive regulation contrasts with negative regulation, in which a regulatory element must be removed from DNA. (p. 461)

**postmeiotic segregation** Segregation of genetically different products in a mitotic division following meiosis, as in the formation in *Neurospora* of a pair of ascospores that have different genotypes. (p. 150)

**postreplication repair** DNA repair that takes place in nonreplicating DNA or after the replication fork is some distance beyond a damaged region. (p. 582)

**posttranslocation state** The state of the ribosome immediately following the movement of the small subunit one codon farther along the mRNA; the aminoacyl (A) site is unoccupied. (p. 434)

**procursor fragment** *See* **Okazaki fragment.**

**prediction equation** In quantitative genetics, an equation used to predict the improvement in mean performance of a population brought about by artifical selection; it always includes heritability as one component.

**pretranslocation state** The state of the ribosome immediately prior to the movement of the small subunit one codon farther along the mRNA; the exit (E) site is unoccupied. (p. 434)

**Pribnow box** A base sequence in prokaryotic promoters to which RNA polymerase binds in an early step of initiating transcription.

**primary transcript** An RNA copy of a gene; in eukaryotes, the transcript must be processed to form a translatable mRNA molecule. (p. 424)

**primase** The enzyme responsible for synthesizing the RNA primer needed for initiating DNA synthesis. (p. 196)

**primer** In nucleic acids, a short RNA or single-stranded DNA segment that functions as a growing point in polymerization. (p. 191)

**primer oligonucleotide** A single-stranded DNA molecule, typically from 18 to 22 nucleotides in length, that can hybridize with a longer DNA strand and serve as a primer for replication. In the polymerase chain reaction, two primers anneal to opposite strands of a DNA duplex with their 3'-OH ends facing. *See also* **polymerase chain reaction.** (p. 207)

**primosome** The enzyme complex that forms the RNA primer for DNA replication in eukaryotic cells. (p. 196)

**prion protein** Protein infectious agent; a protein that can transmit disease. (p. 723)

**probability** A mathematical expression of the degree of confidence that certain events will or will not be realized. (p. 106)

**probe** A radioactive DNA or RNA molecule used in DNA RNA or DNA-DNA hybridization assays. (pp. 206, 372)

**processing** A series of chemical reactions in which primary RNA transcripts are converted into mature mRNA, rRNA, or tRNA molecules or in which polypeptide chains are converted into finished proteins. (p. 412)

**programmed cell death** Cell death that happens as part of the normal developmental process. *See also* **apoptosis.** (p. 524)

**prokaryote** An organism that lacks a nucleus; prokaryotic cells divide by fission. (p. 23)

**promoter** A DNA sequence at which RNA polymerase binds and initiates transcription. (pp. 419, 465)

**promoter mutation** A mutation that increases or decreases the ability of a promoter to initiate transcription. (p. 422)

**promoter recognition** The first step in transcription. (p. 418)

**proofreading function** *See* **editing function.**

**prophage** The form of phage DNA in a lysogenic bacterium; the phage DNA is repressed and usually integrated into the bacterial chromosome, but some prophages are in plasmid form. (p. 341)

**prophage induction** Activation of a prophage to undergo the lytic cycle. (p. 345)

**prophase** The initial stage of mitosis or meiosis, beginning after interphase and terminating with the alignment of the chromosomes at metaphase; often absent or abbreviated between meiosis I and meiosis II. (p. 86, 92)

**protanomaly** *See* **color blindness.**

**protanopia** *See* **color blindness.**

**protein** A molecule composed of one or more polypeptide chains. (pp. 15, 412)

**protein subunit** Any of the polypeptide chains in a protein molecule made up of multiple polypeptide chains. (p. 414)

**prototroph** Microbial strain capable of growth in a defined minimal medium that ideally contains only a carbon source and inorganic compounds. The wildtype genotype is usually regarded as a prototroph. (p. 312)

**pseudoautosomal region** In mammals, a small region of the X and Y chromosome containing homologous genes. (p. 273)

**pseudogene** A DNA sequence that is not expressed because of one or more mutations but that has a functional counterpart in the same organism; pseudogenes are regarded as mutated forms of ancient gene duplications.

**P site** *See* **peptidyl site.**

**PTC tasting** Genetic polymorphism in the ability to taste phenylthiocarbamide. (p. 729)

***P* transposable element** A *Drosophila* transposable element used for the induction of mutations, germ-line transformation, and other types of genetic engineering.

**pulse-field gel electrophoresis** A type of electrophoresis in which the electric field is manipulated to make possible the separation of large DNA fragments, (p. 228)

**Punnet square** A cross-multiplication square used for determining the expected genetic outcome of a mating. (p. 43)

**purine** An organic base found in nucleic acids; the predominant purines are adenine and guanine. (p. 174)

**pyrimidine** An organic base found in nucleic acids; the predominant pyrimidines are cytosine, uracil (in RNA only), and thymine (in DNA only). (p. 174)

**pyrimidine dimer** Two adjacent pyrimidine bases, typically a pair of thymines, in the same polynucleotide strand, between which chemical bonds have formed; the most common lesion formed in DNA by exposure to ultraviolet light. (p. 571)

## Q

**QTL** *See* **quantitative trait locus.**

**quantitative trait** A trait—typically measured on a continuous scale, such as height or weight—that results from the combined action of several or many genes in conjunction with environmental factors. (p. 670)

**quantitative trait locus** A locus segregating for alleles that have different, measurable effects on the expression of a quantitative trait. (p. 692)

## R

**race** A genetically or geographically distinct subgroup of a species.

**rad** Radiation absorbed dose; an amount of ionizing radiation resulting in the dissipation of 100 ergs of energy in 1 gram of matter; equal to 0.01 gray.

**random genetic drift** Fluctuation in allele frequency from generation to generation resulting from restricted population size. (p. 650, 658)

**random mating** System of mating in which mating pairs are formed independently of genotype and phenotype. (p. 632)

**random spore analysis** In fungi, the genetic analysis of spores collected at random rather than from individual tetrads. (p. 154)

**reading frame** The phase in which successive triplets of nucleotides in mRNA form codons; depending on the reading frame, a particular nucleotide in an mRNA could be in the first, second, or third position of a codon. The reading frame actually used is defined by the AUG codon that is selected for chain initiation. (p. 440)

**reannealing** Reassociation of dissociated single strands of DNA to form a duplex molecule. (p. 200)

**recessive** Refers to an allele, or the corresponding phenotypic trait, expressed only in homozygotes. (p. 35)

**reciprocal cross** A cross in which the sexes of the parents are the reverse of another cross. (p. 35)

**reciprocal translocation** Interchange of parts between nonhomologous chromosomes. (p. 288)

**recombinant** A chromosome that results from crossing-over and that carries a combination of alleles differing from that of either chromosome participating in the crossover; the cell or organism containing a recombinant chromosome. (p. 124)

**recombinant DNA technology** Procedures for creating DNA molecules composed of one or more segments from other DNA molecules. (p. 360)

**recombination** Exchange of parts between DNA molecules or chromosomes; recombination in eukaryotes usually entails a reciprocal exchange of parts, but in prokaryotes it is often nonreciprocal. (p. 124)

**recombination repair** Repair of damaged DNA by exchange of good for bad segments between two damaged molecules. (p. 586)

**recruitment** The process in which a transcriptional activator protein interacts with one or more components of the transcription complex and attracts it to the promoter. (p. 495)

**red-green color blindness** *See* **color blindness.**

**reductional division** Term applied to the first meiotic division because the chromosome number (counted as the number of centromeres) is reduced from diploid to haploid. (p. 89)

**redundancy** The feature of the genetic code in which an amino acid corresponds to more than one codon; also called degeneracy. (p. 443)

**regulatory gene** A gene with the primary function of controlling the expression of one or more other genes.

**rejection** An immune response against transfused blood or transplanted tissue.

**relative fitness** The fitness of a genotype expressed as a proportion of the fitness of another genotype. (p. 693)

**relaxed DNA** A DNA circle whose supercoiling has been removed either by introduction of a single-strand break or by the activity of a topoisomerase. (p. 224)

**rem** The quantity of any kind of ionizing radiation that has the same biological effect as one rad of high-energy gamma rays; rem stands for roentgen equivalent man.

**renaturation** Restoration of the normal three-dimensional structure of a macromolecule; in reference to nucleic acids, the term means the formation of a double-stranded molecule by complementary base pairing between two single-stranded molecules. (p. 200)

**repair** *See* **DNA repair.**

**repair synthesis** The enzymatic filling of a gap in a DNA molecule at the site of excision of a damaged DNA segment. (p. 581)

**repellent** Any substance that elicits chemotaxis of organisms in a direction away from itself. (p. 704)

**repetitive sequence** A DNA sequence present more than once per haploid genome. (p. 235)

**replica plating** Procedure in which a particular spatial pattern of colonies on an agar surface is reproduced on a series of agar surfaces by stamping them with a template that contains an image of the pattern; the template is often produced by pressing a piece of sterile velvet upon the original surface, which transfers cells from each colony to the cloth. (p. 561)

**replication** *See* **DNA replication; θ replication.**

**replication fork** In a replicating DNA molecule, the region in which

nucleotides are added to growing strands. (p. 187)

**replication origin** The base sequence at which DNA synthesis begins. (p. 187)

**replication slippage** The process in which the number of copies of a small tandem repeat can increase or decrease during replication. (p. 559)

**replicon** A DNA molecule that has a replication origin.

**reporter gene** A gene whose expression can be monitored. (p. 500)

**repression** A regulatory process in which a gene is temporarily rendered unable to be expressed. (p. 461)

**repressor** A protein that binds specifically to a regulatory sequence adjacent to a gene and blocks transcription of the gene. (pp. 344, 461, 464)

**repulsion** *See* ***trans*** **configuration.**

**restriction endonuclease** A nuclease that recognizes a short nucleotide sequence (restriction site) in a DNA molecule and cleaves the molecule at that site; also called a restriction enzyme. (pp. 147, 202, 360)

**restriction enzyme** *See* **restriction endonuclease.**

**restriction fragment** A segment of duplex DNA produced by cleavage of a larger molecule by a restriction enzyme. (p. 205)

**restriction fragment length polymorphism (RFLP)** Genetic variation in a population associated with the size of restriction fragments that contain sequences homologous to a particular probe DNA; the polymorphism results from the positions of restriction sites flanking the probe, and each variant is essentially a different allele. (p. 147)

**restriction map** A diagram of a DNA molecule showing the positions of cleavage by one or more restriction endonucleases. (p. 205)

**restriction site** The base sequence at which a particular restriction endonuclease makes a cut. (pp. 204, 360)

**restrictive condition** A growth condition in which the phenotype of a conditional mutation is expressed. (p. 556)

**retinoblastoma** An inherited cancer caused by a mutation in the tumor-suppressor gene located in chromosome band *13q14*. Inheritance of one copy of the mutation results in multiple malignancies in retinal cells of the eyes in which the mutation becomes homozygous—for example, through a new mutation or mitotic recombination. (p. 298)

**retrovirus** One of a class of RNA animal viruses that cause the synthesis of DNA complementary to their RNA genomes on infection. (p. 378)

**reverse genetics** Procedure in which mutations are deliberately produced in cloned genes and introduced back into cells or the germ line of an organism. (p. 377)

**reverse mutation** A mutation that undoes the effect of a preceding mutation. (pp. 583, 652)

**reverse transcriptase** An enzyme that makes complementary DNA from a single-stranded RNA template. (p. 369)

**reverse transcriptase PCR** Amplification of a duplex DNA molecule originally produced by reverse transcriptase using an RNA template. (p. 370)

**reversion** Restoration of a mutant phenotype to the wildtype phenotype by a second mutation. (p. 583)

**RFLP** *See* **restriction fragment length polymorphism.**

**R group** *See* **side chain.**

**Rh** Rhesus blood-group system in human beings; maternal-fetal incompatibility of this system may result in hemolytic disease of the newborn.

**Rh-negative** Refers to the phenotype in which the red blood cells lack the D antigen of the Rh blood-group system.

**rhodopsin** One of a class of proteins that function as visual pigments. (p. 283)

**Rh-positive** Refers to the phenotype in which the red blood cells possess the D antigen of the Rh blood-group system.

**ribonuclease** Any enzyme that cleaves phosphodiester bonds in RNA; abbreviated RNase.

**ribonucleic acid** *See* **RNA.**

**ribonucleoprotein particle** Nuclear particle containing a short RNA molecule and several proteins; involved in intron excision and splicing and other aspects of RNA processing.

**ribose** The five-carbon sugar in RNA. (p. 195)

**ribosomal RNA (rRNA)** RNA molecules that are components of the ribosomal subunits; in eukaryotes, there are four rRNA molecules—5S, 5.8S, 18S, and 28S; in prokaryotes, there are three—5S, 16S, and 23S. (p. 12)

**ribosome** The cellular organelle on which the codons of mRNA are translated amino acids in protein synthesis. Ribosomes consist of two subunits, each composed of RNA and proteins. In prokaryotes, the subunits are 30S and 50S particles; in eukaryotes, they are 40S and 60S particles. (p. 13)

**ribosome-binding site** The base sequence in a prokaryotic mRNA molecule to which a ribosome can bind to initiate protein synthesis; also called the Shine-Dalgarno sequence. (p. 432)

**ribozyme** An RNA molecule able to catalyze one or more biochemical reactions. (p. 427)

**ring chromosome** A chromosome whose ends are joined; one that lacks telomeres. (p. 260)

**RNA** Ribonucleic acid, a nucleic acid in which the sugar constituent is ribose; typically, RNA is single-stranded and contains the four bases adenine, cytosine, guanine, and uracil. (p. 12)

**RNA editing** Process in which certain nucleotides in RNA are chemically changed to other nucleotides after transcription. (p. 605)

**RNA polymerase** An enzyme that makes RNA by copying the base sequence of a DNA strand. (pp. 196, 418)

**RNA polymerase holoenzyme** A large protein complex in eukaryotic nuclei consisting of Pol II in combination with at least 9 other subunits; Pol II itself includes 12 subunits (p. 495)

**RNA processing** The conversion of a primary transcript into an mRNA, rRNA, or tRNA molecule; includes splicing, cleavage, modification of termini, and (in tRNA) modification of internal bases. (p. 424)

**RNase** Any enzyme that cleaves RNA.

**RNA splicing** Excision of introns and joining of exons. (p. 425)

**Robertsonian translocation** A chromosomal aberration in which the long arms of two acrocentric chromosomes become joined to a common centromere. (p. 293)

**roentgen** A unit of ionizing radiation, defined as the amount of radiation resulting in $2.083 \times 10^9$ ion pairs per $cm^3$ of dry air at 0°C and 1 atm pressure; abbreviated R; 1 R produces 1 electrostatic unit of charge per $cm^3$ of dry air at 0°C and 1 atm pressure.

**rolling-circle replication** A mode of replication in which a circular parent molecule produces a linear branch of newly formed DNA. (p. 189)

**Rous sarcoma virus** A type of retrovirus that infects the chicken.

**R plasmid** A bacterial plasmid that carries drug-resistance genes; commonly used in genetic engineering. (p. 348)

**rRNA** *See* **ribosomal RNA.**

**RT-PCR** *See* **reverse transcriptase PCR.**

**S**

**S** *See* **cell cycle.**

**salvage pathway** A minor source of deoxythymidine triphosphate (dTTP) for DNA synthesis that uses the enzyme thymidine kinase (TK).

**satellite DNA** Eukaryotic DNA that forms a minor band at a different density from that of most of the cellular DNA in equilibrium density gradient centrifugation; consists of short sequences repeated many times in the genome (highly repetitive DNA) or of mitochondrial or chloroplast DNA. (p. 235)

**scaffold** A protein-containing material in chromosomes, believed to be responsible in part for the compaction of chromatin. (p. 233)

**second-division segregation** Segregation of a pair of alleles into different nuclei in the second meiotic division, the result of crossing-over between the gene and the centromere of the pair of homologous chromosomes. (p. 156)

**second meiotic division** The meiotic division in which the centromeres split and the chromosome number is not reduced; also called the equational division. (p. 89)

**segment** Any of a series of repeating morphological units in a body plan. (p. 532)

**segmentation gene** Any of a group of genes that determines the spatial pattern of segments and parasegments in *Drosophila* development. (p. 532)

**segment-polarity gene** Any of a group of genes that determines the spatial pattern of development within the segments of *Drosophila* larvae. (pp. 533, 538)

**segregation** Separation of the members of a pair of alleles into different gametes in meiosis. (p. 40)

**segregational petites** Slow-growing yeast colonies deficient in respiration because of mutation in a chromosomal gene; inheritance is Mendelian. (p. 610)

**segregation mutation** Developmental mutation in which sister cells express the same fate when they should express different fates. (p. 523)

**selected marker** A genetic mutation that allows growth in selective medium. (p. 319)

**selection** In evolution, intrinsic differences in the ability of genotypes to survive and reproduce; in plant and animal breeding, the choosing of organisms with certain phenotypes to be parents of the next generation; in mutation studies, a procedure designed in such a way that only a desired type of cell can survive, as in selection for resistance to an antibiotic.

**selection coefficient** The amount by which relative fitness is reduced or increased. (p. 653)

**selection differential** In artificial selection, the difference between the mean of the selected organisms and the mean of the population from which they were chosen.

**selection limit** The condition in which a population no longer responds to artificial selection for a trait. (p. 686)

**selection-mutation balance** Equilibrium determined by the opposing effects of mutation tending to increase the frequency of a deleterious allele and selection tending to decrease its frequency.

**selection pressure** The tendency of natural or artificial selection to change allele frequency.

**selectively neutral mutation** A mutation that has no (or negligible) effects on fitness. (p. 650)

**selective medium** A medium that allows growth only of cells with particular genotypes. (p. 312)

**self-fertilization** The union of male and female gametes produced by the same organism.

**selfish DNA** DNA sequences that do not contribute to the fitness of an organism but are maintained in the genome through their ability to replicate and transpose. (p. 246)

**semiconservative replication** The usual mode of DNA replication, in which each strand of a duplex molecule serves as a template for the synthesis of a new complementary strand, and the daughter molecules are composed of one old (parental) and one newly synthesized strand. (p. 183)

**semisterility** A condition in which a significant proportion of the gametophytes produced by a plant or of the zygotes produced by an animal are inviable, as in the case of a translocation heterozygote. (p. 288)

**sense strand** Originally defined as the template strand in duplex DNA from which the mRNA is transcribed. Some later authors used the term in the opposite sense to mean the nontemplate DNA strand that has the same nucleotide sequence as the mRNA. Because both usages persist, the term is ambiguous and is best avoided. Use *template strand* or *transcribed strand* instead.

**sequence-tagged site (STS)** A DNA sequence, present once per haploid genome, that can be amplified by the use of suitable oligonucleotide primers in the polymerase chain reaction in order to identify clones that contain the sequence. (p. 395)

**sex chromosome** A chromosome, such as the human X or Y, that has a role in the determination of sex. (p. 96)

**sex-influenced trait** A trait whose expression depends on the sex of the individual.

**sex-limited trait** A trait expressed in one sex and not in the other.

**sex-linked** Refers to a trait determined by a gene on a sex chromosome, usually the X. (p. 97)

**Shine-Dalgarno sequence** *See* **ribosome-binding site.**

**shuttle vector** A vector capable of replication in two or more organisms—for example, yeast and *E. coli.* (p. 392)

**sib** *See* **sibling.**

**sibling** A brother or sister, each having the same parents. (p. 51)

**sibship** A group of brothers and sisters. (p. 49)

**sickle-cell anemia** A severe anemia in human beings inherited as an autosomal recessive and caused by an amino acid replacement in the β-globin chain; heterozygotes tend to be more resistant to falciparum malaria than are normal homozygotes. (pp. 17, 558)

**side chain** In protein structure, the chemical group attached to the α-carbon atom of an amino acid by which the chemical properties of each amino acid are determined; also called R group. (p. 412)

**sievert** A unit of radiation exposure of biological tissue equal to the dose in gray multiplied by a quality factor that measures the relative biological effectiveness of the type of radiation; abbreviated Sv. Roughly speaking, 1 sievert is the amount of radiation equivalent in its biological effects to one gray of gamma rays; 1 Sv equals 100 rem.

**sigma (σ) subunit** The subunit of RNA polymerase needed for promoter recognition.

**signal transduction** The process by which a regulatory signal ultimately determines cell fate by activation of a series of regulatory proteins.

**significant** *See* **statistically significant.**

**silent mutation** A mutation that has no phenotypic effect. (pp. 444, 557)

**simple tandem repeat polymorphism (STRP)** A DNA polymorphism in a population in which the alleles differ in the number of copies of a short, tandemly repeated nucleotide sequence. (p. 150)

**single-active-X principle** In mammals, the genetic inactivation of all X chromosomes except one in each cell lineage, except in the very early embryo. (p. 275)

**single-copy sequence** A DNA sequence present only once in each haploid genome.

**single-strand binding protein** A protein able to bind single-stranded DNA. (p. 198)

**single-stranded DNA** A DNA molecule that consists of a single polynucleotide chain. (p. 10)

**sister chromatids** Chromatids produced by replication of a single chromosome. (p. 86)

**site-specific exchange** *See* **site-specific recombination.**

**site-specific recombination** Recombination that is catalyzed by a particular enzyme and that always takes place at the same site between the same DNA sequences. (p. 330)

**small ribonucleoprotein particles** Small nuclear particles that contain short RNA molecules and several proteins. They are involved in intron excision and splicing and other aspects of RNA processing.

**somatic cell** Any cell of a multicellular organism other than the gametes and the germ cells from which gametes develop. (p. 82)

**somatic-cell genetics** Study of somatic cells in culture.

**somatic mosaic** An organism that contains two genetically different types of cells. (p. 160)

**somatic mutation** A mutation arising in a somatic cell. (pp. 487, 556)

**SOS repair** An inducible, error-prone system for repair of DNA damage in *E. coli.* (p. 582)

**Southern blot** A nucleic acid hybridization method in which, after electrophoretic separation, denatured DNA is transferred from a gel to a membrane filter and then exposed to radioactive DNA or RNA under conditions of renaturation; the radioactive regions locate the homologous DNA fragments on the filter. (p. 206)

**spacer sequence** A noncoding base sequence between the coding segments of polycistronic mRNA or between genes in DNA.

**specialized transduction** *See* **transducing phage.**

**species** Genetically, a group of actually or potentially inbreeding organisms that is reproductively isolated from other such groups.

**S period** *See* **cell cycle.**

**spindle** A structure composed of fibrous proteins on which chromosomes align during metaphase and move during anaphase. (p. 86)

**splice acceptor** The 5' end of an exon. (p. 526)

**splice donor** The 3' end of an exon. (p. 526)

**spliceosome** An RNA-protein particle in the nucleus in which introns are removed from RNA transcripts. (p. 426)

**spontaneous mutation** A mutation that happens in the absence of any known mutagenic agent. (p. 557)

**spore** A unicellular reproductive entity that becomes detached from the parent and can develop into a new organism upon germination; in plants, spores are the haploid products of meiosis. (p. 89)

**sporophyte** The diploid, spore-forming generation in plants, which alternates with the haploid, gamete-producing generation (the gametophyte). (p. 89)

**standard deviation** The square root of the variance. (p. 673)

**start codon** An mRNA codon, usually AUG, at which polypeptide synthesis begins. (p. 443)

**statistically significant** Said of the result of an experiment or study that has only a small probability of happening by chance on the assumption that some hypothesis is true. Conventionally, if results as bad or worse would be expected less than 5 percent of the time, the result is said to be statistically significant; if less than 1 percent of the time, the result is called statistically highly significant; both outcomes cast the hypothesis into serious doubt. (p. 111)

**sticky end** A single-stranded end of a DNA fragment produced by certain restriction enzymes capable of reannealing with a complementary sequence in another such strand. (pp. 204, 361)

**stop codon** One of three mRNA codons—UAG, UAA, and UGA—at which polypeptide synthesis stops. (p. 436)

**structural gene** A gene that encodes the amino acid sequence of a polypeptide chain. (p. 415)

**STRP** *See* **simple tandem repeat polymorphism.**

**STS** *See* **sequence-tagged site.**

**submetacentric chromosome** A chromosome whose centromere divides it into arms of unequal length. (p. 261)

**subpopulation** Any of the breeding groups within a larger population between which migration is restricted. (p. 628)

**subspecies** A relatively isolated population or group of populations distinguishable from other populations in the same species by allele frequencies or chromosomal arrangements and sometimes exhibiting incipient reproductive isolation.

**substrate** A substance acted on by an enzyme.

**subunit** A component of an ordered aggregrate of macro-molecules—for example, a single polypeptide chain in a protein containing several chains. (p. 414)

**supercoiled DNA** *See* **supercoiling.**

**supercoiling** Coiling of double-stranded DNA in which strain caused by overwinding or underwinding of the duplex makes the double helix itself twist; a supercoiled circle is also called a twisted circle or a superhelix. (pp. 224, 225)

**suppressive petites** A class of cytoplasmically inherited, small-colony mutants in yeast that are not corrected by the addition of normal mitochondria.

**suppressor mutation** A mutation that partially or completely restores the function impaired by another mutation at a different site in the same gene (intragenic suppression) or in a different gene (intergenic suppression). (p. 583)

**suppressor tRNA** Usually, a tRNA molecule capable of translating a stop codon by inserting an amino acid in its place, but a few suppressor tRNAs replace one amino acid with another. (p. 447)

**synapsis** The pairing of homologous chromosomes or chromosome regions in zygotene of the first meiotic prophase. (p. 92)

**synaptonemal complex** A complex protein structure that forms between synapsed homologous chromosomes in the pachytene substage of the first meiotic prophase.

**syncytial blastoderm** Stage in early *Drosophila* development formed by successive nuclear divisions without division of the cytoplasm. (p. 529)

**syndrome** A group of symptoms that appear together with sufficient regularity to warrant designation by a special name; also, a disorder, disease, or anomaly.

**synteny** The presence of two different genes on the same chromosome.

**synteny group** A group of genes present in a single chromosome in two or more species. (p. 398)

**T**

**tandem duplication** A pair of identical or closely related DNA sequences that are adjacent and in the same orientation. (p. 283)

**taster** One who has the inherited ability to taste the substance phenylthiocarbamide (PTC); the phenotype of the homozygous dominant and heterozygote.

**TATA binding protein (TBP)** A protein that binds to the TATA motif in the promoter region of a gene. (p. 495)

**TATA box** The base sequence 5 -TATA-3' in the DNA of a promoter. (p. 420)

**tautomeric shift** A reversible change in the location of a hydrogen atom in a molecule, altering the molecule from one isomeric form to another. In nucleic acids, the shift is typically between a keto group (keto form) and a hydroxyl group (enol form).

**TBP** TATA-box binding protein. *See* **TATA binding protein.**

**TBP-associated factor (TAF)** Any protein found in close association with TATA binding protein. (p. 495)

**T cells** A class of white blood cells instrumental in various aspects of the immune reponse. (p. 487)

***T* DNA** Transposable element found in *Agrobacterium tumefaciens,* which produces crown gall tumors in a wide variety of dicotyledonous plants. (p. 381

**telomerase** An enzyme that adds specific nucleotides to the tips of the chromosomes to form the telomeres. (p. 248)

**telomere** The tip of a chromosome, containing a DNA sequence required for stability of the chromosome end. (p. 247)

**telophase** The final stage of mitotic or meiotic nuclear division. (p. 87, 94)

**temperate phage** A phage capable of both a lysogenic and a lytic cycle. (p. 329)

**temperature-sensitive mutation** A conditional mutation that causes a phenotypic change only at certain temperatures. (p. 556)

**template strand** A nucleic acid strand whose base sequence is copied in a polymerization reaction to produce either a complementary DNA or RNA strand. (pp. 10, 191, 421)

**terminal redundancy** In bacteriophage T4, a short duplication found at opposite ends of the linear molecule; the duplicated region differs from one phage to the next. (p. 335)

**termination** Final stage in polypeptide synthesis when the ribosome encounters a stop codon and the finished polypeptide is released from the ribosome. (p. 436)

**testcross** A cross between a heterozygote and a recessive homozygote, resulting in progeny in which each phenotypic class represents a different genotype. (p. 41)

**testis-determining factor (TDF)** Genetic element on the mammalian Y chromosome that determines maleness. (p. 273)

**tetrad** The four chromatids that make up a pair of homologous chromosomes in meiotic prophase I and metaphase I; also, the four haploid products of a single meiosis. (p. 93)

**tetrad analysis** A method for the analysis of linkage and recombination using the four haploid products of single meiotic divisions. (p. 150)

**tetraploid** A cell or organism with four complete sets of chromosomes; in an autotetraploid, the chromosome sets are homologous; in an allotetraploid, the chromosome sets consist of a complete diploid complement from each of two distinct ancestral species. (p. 261)

**tetratype** An ascus containing spores of four different genotypes—one each of the four genotypes possible with two alleles of each of two genes. (p. 150)

**thermophile** An organism that normally lives at an unusually high temperature. (p. 208)

*θ* **replication** Bidirectional replication of a circular DNA molecule, starting from a single origin of replication. (p. 187)

**30-nm fiber** The level of compaction of eukaryotic chromatin resulting from coiling of the extended, nucleosome-bound DNA fiber. (p. 233)

**three-point cross** Cross in which three genes are segregating; used to obtain unambiguous evidence of gene order. (p. 141)

**3'-OH group** The end of a DNA or RNA strand that terminates in a sugar and so has a free hydroxyl group on the number-3' carbon. (p. 175)

**threshold trait** A trait with a continuously distributed liability or risk; organisms with a liability greater than a critical value (the threshold) exhibit the phenotype of interest, such as a disorder. (p. 671)

**thymine (T)** A nitrogenous pyrimidine base found in DNA. (p. 9)

**thymine dimer** *See* **pyrimidine dimer.**

**time of entry** In an Hfr × $F^-$ bacterial mating, the earliest time that a particular gene in the Hfr parent is transferred to the $F^-$ recipient. (p. 320)

***Ti* plasmid** A plasmid present in *Agrobacterium tumefaciens* and used in genetic engineering in plants. (p. 381)

**topoisomerase** An enzyme that introduces or eliminates either underwinding or overwinding of double-stranded DNA. It acts by introducing a single-strand break, changing the relative positions of the strands, and sealing the break. (p. 187)

**topoisomerase I** A nick-closing type of topoisomerase enzyme in which one supercoil is removed from DNA per reaction cycle. (p. 225)

**topoisomerase II** A topoisomerase enzyme able to remove both positive and negative supercoils from DNA. These enzymes can also cleave duplex DNA and pass another duplex DNA molecule through the gap. (p. 225)

**total variance** Summation of all sources of genetic and environmental variation. (p. 678)

**totipotent cell** A cell capable of differentiation into a complete organism; the zygote is totipotent.

**trait** Any aspect of the appearance, behavior, development, biochemistry, or other feature of an organism. (p. 2)

***trans* configuration** The arrangement in linked inheritance in which a genotype heterozygous for two mutant sites has received one of the mutant sites from each parent—that is, $a_1$+ +$a_2$. (p. 126)

**transcript** An RNA strand that is produced from, and is complementary in base sequence to, a DNA template strand. (p. 13)

**transcription** The process by which the information contained in a template strand of DNA is copied into a single-stranded RNA molecule of complementary base sequence. (pp. 13, 412)

**transcriptional activator protein** Positive control element that stimulates transcription by binding with particular sites in DNA. (pp. 462, 491)

**transcription complex** An aggregate of RNA polymerase (consisting of its own subunits) along with other polypeptide subunits that makes transcription possible.

**transducing phage** A phage type capable of producing particles that contain bacterial DNA (transducing particles). A specialized transducing phage produces particles that carry only specific regions of chromosomal DNA; a generalized transducing phage produces particles that may carry any region of the genome. (p. 325)

**transduction** The carrying of genetic information from one bacterium to another by a phage. (p. 325)

**transfer RNA (tRNA)** A small RNA molecule that translates a codon into an amino acid in protein synthesis; it has a three-base sequence, called the anticodon, complementary to a specific codon in mRNA, and a site to which a specific amino acid is bound. (p. 13)

**transformation** Change in the genotype of a cell or organism resulting from exposure of the cell or organism to DNA isolated from a different genotype; also, the conversion of an animal cell, whose growth is limited in culture, into a tumor-like cell whose pattern of growth is different from that of a normal cell. (p. 3, 363)

**transformation mutation** A mutation in which affected cells undergo a developmental fate that is characteristic of other cells. (p. 522)

**transgenic** Refers to animals or plants in which novel DNA has been incorporated into the germ line. (p. 378)

***trans* heterozygote** *See* ***trans*** **configuration.**

**transition mutation** A mutation resulting from the substitution of one purine for another purine or that of one pyrimidine for another pyrimidine. (p. 557)

**translation** The process by which the amino acid sequence of a polypeptide is synthesized on a ribosome according to the nucleotide sequence of an mRNA molecule. (pp. 14, 412, 430)

**translocation** Interchange of parts between nonhomologous chromosomes; also, the movement of mRNA with respect to a ribosome during protein synthesis. *See also* **reciprocal translocation.** (pp. 288, 434)

**transmembrane receptor** A receptor protein containing amino acid sequences that span the cell membrane. (p. 514)

**transposable element** A DNA sequence capable of moving (transposing) from one location to another in a genome. (pp. 242, 243)

**transposase** Protein necessary for transposition. (p. 246, 347)

**transposition** The movement of a transposable element. (pp. 243, 346)

**transposon** A transposable element that contains bacterial genes—for example, for antibiotic resistance; also loosely used as a synonym for transposable element. (p. 347)

**transposon tagging** Insertion of a transposable element that contains a genetic marker into a gene of interest. (p. 349)

**transversion mutation** A mutation resulting from the substitution of a purine for a pyrimidine or that of a pyrimidine for a purine. (p. 557)

**trinucleotide repeat** A tandemly repeated sequence of three nucleotides; genetic instability in trinucleotide repeats is the cause of a number of human hereditary diseases. (p. 279)

**triplet code** A code in which each codon consists of three bases. (p. 439)

**triplication** The presence of three copies of a DNA sequence that is ordinarily present only once.

**triploid** A cell or organism with three complete sets of chromosomes. (p. 263)

**trisomic** A diploid organism with an extra copy of one of the chromosomes. (p. 267)

**trisomy-X syndrome** The clinical features of the karyotype 47,XXX. (p. 278)

**trivalent** Structure formed by three homologous chromosomes in meiosis I in a triploid or trisomic chromosome when each homolog is paired along part of its length with first one and then the other of the homologs. (p. 267)

**tRNA** *See* **transfer RNA.**

**true-breeding** Refers to a strain, breed, or variety of organism that yields progeny like itself; homozygous. (p. 32)

**truncation point** In artificial selection, the value of the phenotype that determines which organisms will be retained for breeding and which will be culled. (p. 684)

**tumor-suppressor gene** A gene whose absence predisposes to malignancy; also called an anti-oncogene. (p. 298)

**Turner syndrome** The clinical features of human females with the karyotype 45,X. (p. 279)

**twin spot** A pair of adjacent, genetically different regions of tissue, each derived from one daughter cell of a mitosis in which mitotic recombination took place. (p. 159)

**U**

**ultracentrifuge** A centrifuge capable of speeds of rotation sufficiently high to separate molecules of different density in a suitable density gradient. (p. 183)

**uncharged tRNA** A tRNA molecule that lacks an amino acid. (p. 434)

**uncovering** The expression of a recessive allele present in a region of a structurally normal chromosome in which the homologous chromosome has a deletion. (p. 281)

**underwound DNA** A DNA molecule whose strands are untwisted somewhat; hence, some of its bases are unpaired. (p. 187)

**unequal crossing-over** Crossing-over between nonallelic copies of duplicated or other repetitive sequences—for example, in a tandem duplication, between the upstream copy in one chromosome and the downstream copy in the homologous chromosome. (p. 283)

**uniparental inheritance** Extranuclear inheritance of a trait through cytoplasmic factors or organelles contributed by only one parent. *See also* **maternal inheritance.** (p. 600)

**unique sequence** A DNA sequence that is present in only one copy per haploid genome, in contrast with repetitive sequences. (p. 239)

**univalent** Structure formed in meiosis I in a monoploid or a monosomic when a chromosome has no pairing partner. (p. 269)

**uracil (U)** A nitrogenous pyrimidine base found in RNA. (p. 12)

**V**

**variable expressivity** Differences in the severity of expression of a particular genotype. (p. 71)

**variable number of tandem repeats (VNTR)** Any region of the genome that, among the members of a population, is highly poplymorphic in the number of tandem repeats of a short nucleotide sequence; the result is polymorphism in the size of the restriction fragment that contains the repeated sequence. (p. 148)

**variable region** The portion of an immunoglobulin molecule that varies greatly in amino acid sequence among antibodies in the same subclass. *See also* **V region.** (p. 484)

**variance** A measure of the spread of a statistical distribution; the mean of the squares of the deviations from the mean. (p. 672)

**variegation** Mottled or mosaic expression. (p. 296)

**vector** A DNA molecule, capable of replication, into which a gene or DNA segment is inserted by recombinant DNA techniques; a cloning vehicle. (p. 362)

**viability** The probability of survival to reproductive age. (p. 655)

**viral oncogene** A class of genes found in certain viruses that predispose to cancer. Viral oncogenes are the viral counterparts of cellular oncogenes. *See also* **cellular oncogene, oncogene.** (p. 298)

**virulent phage** A phage or virus capable only of a lytic cycle; contrasts with temperate phage. (p. 329)

**virus** An infectious intracellular parasite able to reproduce only inside living cells. (p. 308)

**VNTR** *See* **variable number of tandem repeats.**

**Von Hippel-Lindau disease** Hereditary disease marked by tumors in the retina, brain, other parts of the central nervous system, and various organs throughout the body. The disease is caused by an autosomal dominant mutation in chromosome 3.

**V region** One of multiple DNA sequences coding for alternative amino acid sequences of part of the variable region of an antibody molecule. *See also* **variable region.** (p. 485)

**V-type position effect** A type of position effect in *Drosophila* that is characterized by discontinuous expression of one or more genes during development; usually results from chromosome breakage and rejoining such that euchromatic genes are repositioned in or near centromeric heterochromatin.

**W**

**Watson-Crick pairing** Base pairing in DNA or RNA in which A pairs with T (or U in RNA) and G pairs with C. (p. 10)

**wildtype** The most common phenotype or genotype in a natural population; also, a phenotype or genotype arbitrarily designated as a standard for comparison. (p. 19)

**wobble** The acceptable pairing of several possible bases in an anticodon with the base present in the third position of a codon. (p. 446)

**X**

$x^2$ *See* **chi-square.** (p. 109)

**X chromosome** A chromosome that plays a role in sex determination and that is present in two copies in the homogametic sex and in one copy in the heterogametic sex. (p. 97)

**xeroderma pigmentosum** An inherited defect in the repair of ultraviolet-light damage to DNA, associated with extreme sensitivity to sunlight and multiple skin cancers. (p. 572)

**X inactivation** *See* **single-active-X principle.**

**X-linked inheritance** The pattern of hereditary transmission of genes located in the X chromosome; usually evident from the production of nonidentical classes of progeny from reciprocal crosses. (p. 99)

**Y**

**YAC** *See* **yeast artificial chromosome.**

**Y chromosome** The sex chromosome present only in the heterogametic sex; in mammals, the male-determining sex chromosome. (p. 97)

**yeast artificial chromosome (YAC)** In yeast, a cloning vector that can accept very large fragments of DNA; a chromosome introduced into yeast derived from such a vector and containing DNA from another organism. (p. 391)

**Z**

**Z-form DNA** One of the unusual three-dimensional structures of duplex DNA in which the helix is left-handed. (p. 181)

**zinc finger** A structural motif, found in many DNA-binding proteins, in which finger-like projections entrap a zinc ion. (p. 492)

**zygote** The product of the fusion of a female and a male gamete in sexual reproduction; a fertilized egg.

**zygotene** The substage of meiotic phrophase I in which homologous chromosomes synapse. (p. 92)

**zygotic gene** Any of a group of genes that control early development through their expression in the zygote. (p. 532)

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